CHAPTER

ATA-52

DOORS

THIS MANUAL IS INTENDED FOR TRAINING PURPOSE ONLY

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Contents

Doors, General

Introduction

The doors give access to the aircraft for the crew, passengers and maintenance and service personnel. Emergency exits are also supplied for emergency escape. An internal door separates the flight and passenger compartments.

General Description

Refer Figure 52-1 & Refer Figure 52-2

The aircraft has two passenger and two baggage doors. A Type II/III emergency exit and a flight compartment escape hatch is supplied for emergency use. Service doors are supplied for ground service.

The forward passenger door is on the front left side of the passenger compartment. It includes air stairs and a seal inflated by the airframe deice system. The aft passenger door is on the rear left side of the passenger compartment. This door has a passive type seal inflated by cabin pressure. As an optional configuration the aft passenger door is provided with an aft entry folding stairway.

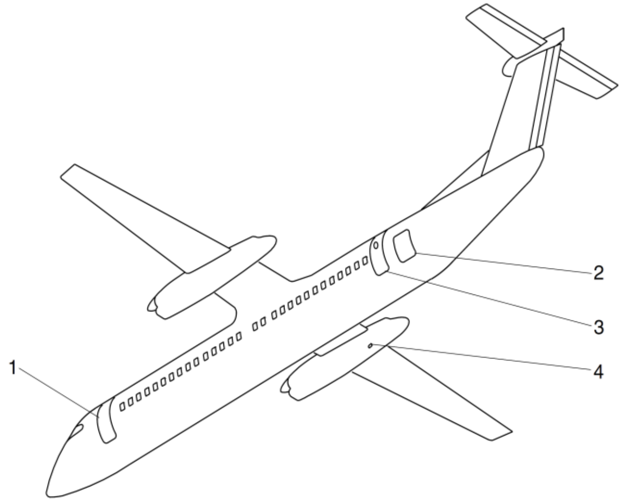
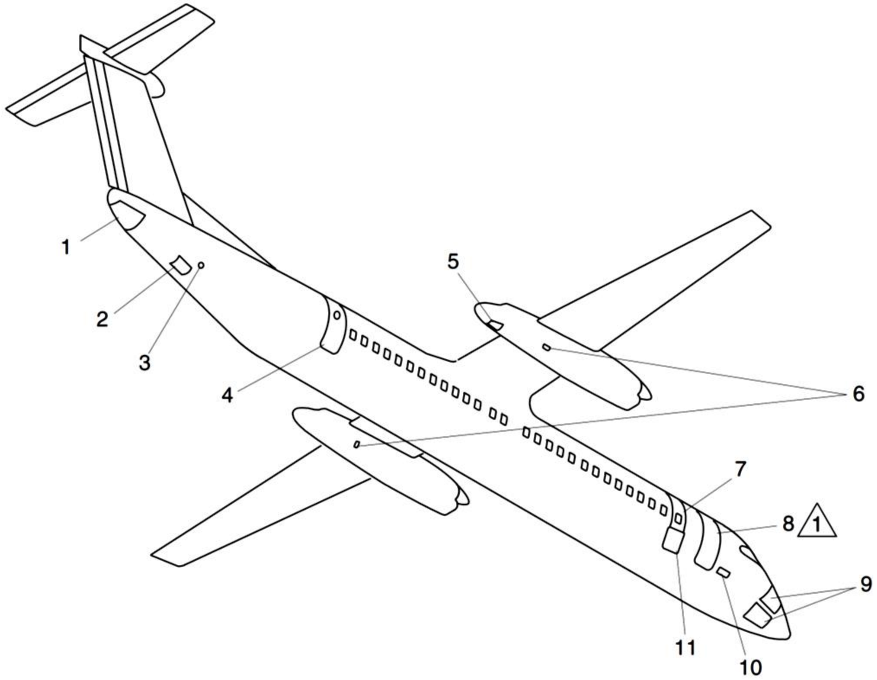


Figure 52-1 Fuselage Doors – Detailed 1 of 3



LEGEND

1. APU Access Door.
2. Aft Fuselage Access Door.
3. Ground Air Conditioning Door.
4. Aft Service Door.
5. Refuel/Defuel Door.
6. Hydraulic Ground Power.
7. Upper Emergency Type II Exit Door.
8. Forward Baggage Door.
9. Nose Compartment Doors.
10. Lavatory Service Door.
11. Lower Emergency Type II Exit Door.

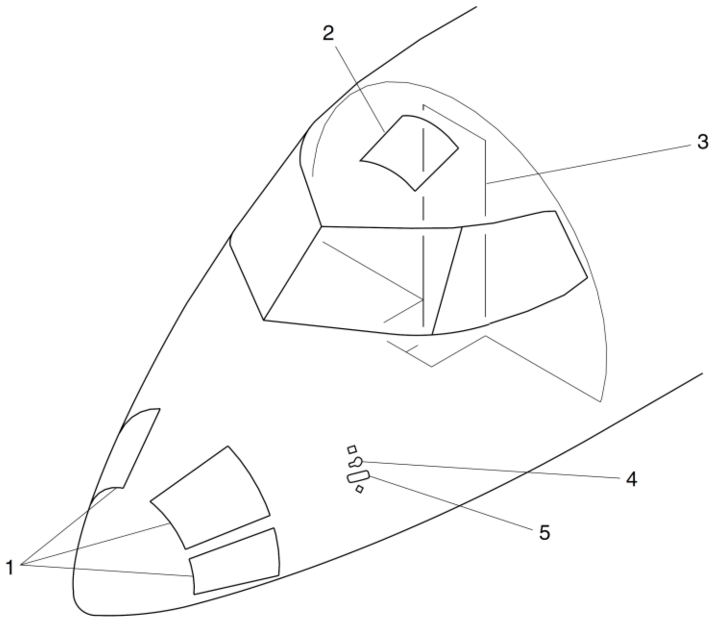
NOTE

1 On aircraft with ModSum 4−459339 incorporated, the forward baggage door is reworked and converted into Type I Emergency Exit Door.

Figure 52-2 Fuselage Doors – Detailed 2 of 3

Refer Figure 52-3

Emergency exit from the flight compartment is through a flight compartment escape hatch in the roof or the flight compartment door. Both passenger doors and the aft service door are used as Type I emergency exit doors. One Type II/III emergency exit is located on the forward right side of the passenger compartment. The Type II/III exit has a removable upper door and a hinged lower door.



LEGEND

1. Nose compartment doors.
2. Flight compartment escape hatch.
3. Flight compartment door.
4. Ground jack point door.
5. DC ground power door.

Figure 52-3 Fuselage Doors – Detailed 3 of 3

Refer Figure 52-4 & *Refer Figure 52-5*

Aft baggage compartment access is through an outward opening door on the left rear fuselage. This door has a seal inflated by the airframe deice system. Forward baggage compartment external access is through a door on the right front fuselage. This door has a passive type seal.

The aft service door is located on the right rear of the passenger compartment. The aft fuselage access door is installed on the bottom of the aft fuselage.

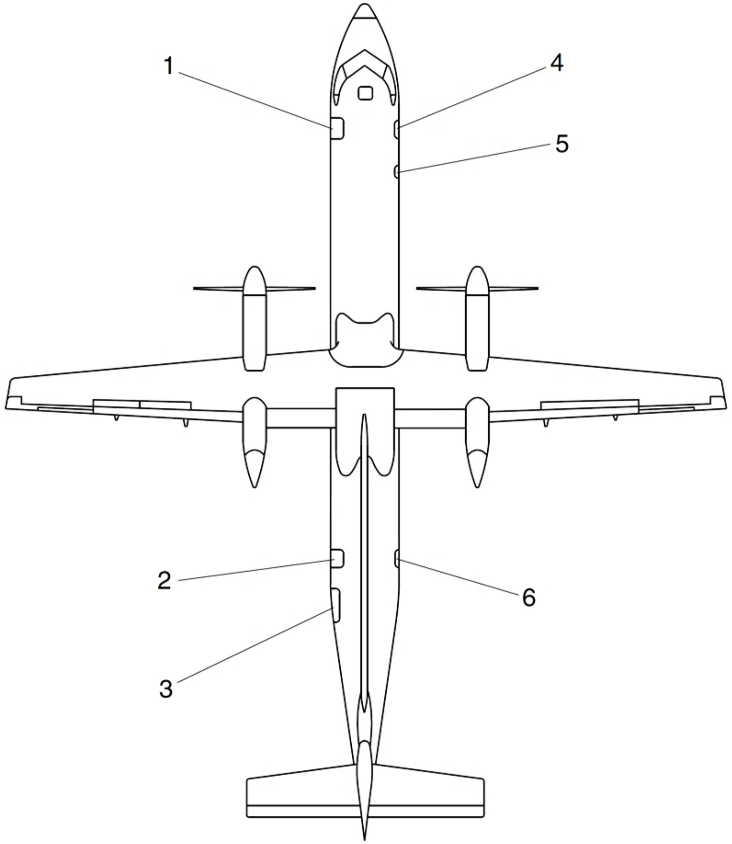
Flight compartment access is through a lockable door in the flight compartment bulkhead.

The door warning system uses the Proximity Sensor Electronic Unit (PSEU) to monitor all pressurized fuselage doors. Door status is shown on the:

- Electronic Flight Information System (EFIS)
- Caution and Warning Control Panel

The Doors System includes:

- Passenger Doors
- Emergency Exits
- Baggage Doors
- Service Doors
- Interior Doors

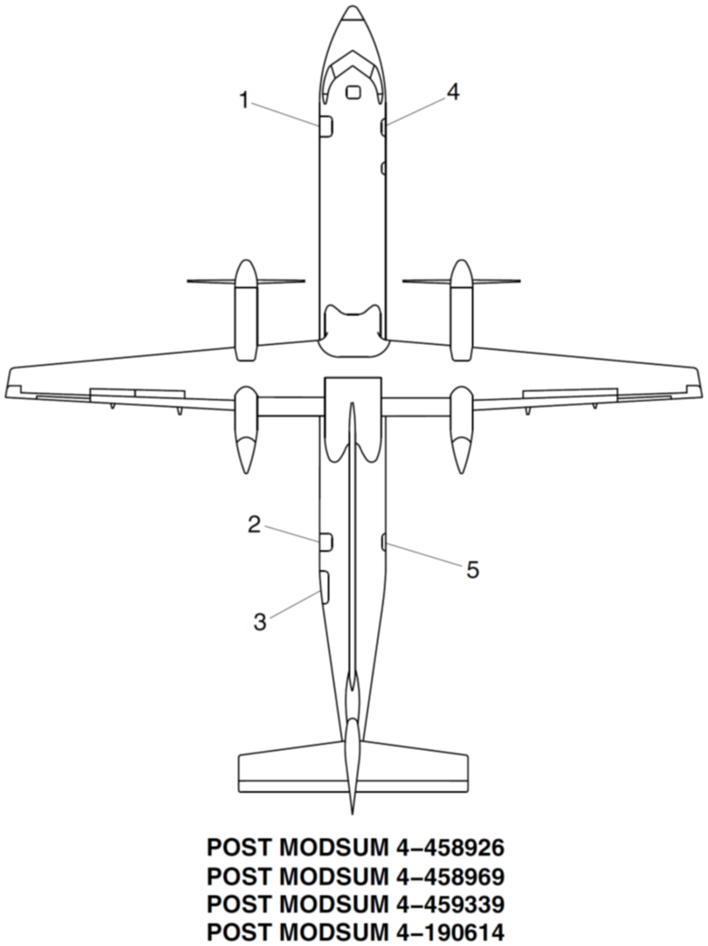


LEGEND

1. Forward Passenger Door.
   "Closed" Sensor and "Locked" Sensor.
2. Aft Passenger Door.
   "Closed" Sensor and "Locked" Sensor.
3. Aft Baggage Door.
   "Closed" Sensor and "Locked" Sensor.
4. Forward Baggage Door.
   "Closed" Sensor and "Locked" Sensor.
5. Upper (Type II) Emergency Exit.
   "Locked" Sensor.
6. Aft Service Door.
   "Closed" Sensor and "Locked" Sensor.

PRE MODSUM 4–458926
PRE MODSUM 4–458969
PRE MODSUM 4–459339
PRE MODSUM 4–190614

Figure 52-4 Emergency Exits – Detailed 1 of 2



LEGEND

1. Forward passenger door
   "Closed" sensor and "Locked" sensor.
2. Aft passenger door
   "Closed" sensor and "Locked" sensor.
3. Aft baggage door
   "Closed" sensor and "Locked" sensor.
4. Forward RHS type I
   "Closed" sensor and "Locked" sensor.
5. Aft service door
   "Closed" sensor and "Locked" sensor.

Figure 52-5 Emergency Exits – Detailed 2 of 2

Detailed Description

The forward and aft passenger doors are located on the left side of the fuselage and are also used as emergency exits. Both passenger doors can be opened from either the inside or outside using internal or external handles. The forward passenger door is used as the main access to the passenger compartment. As an optional configuration the aft passenger door is provided with an aft entry folding stairway. The aft entry folding stairway is used as the second entry door stairs for the entry and exit of passengers.

A flight compartment escape hatch is located in the flight compartment roof, aft of the pilot seats. It is completely detachable, or can be partially opened to increase the airflow in the flight compartment when the aircraft is on the ground.

There is a type II/III emergency exit doors located on the forward right side of the passenger compartment. The emergency exit has an upper and lower door. The upper door is a plug type door which is removed completely for opening. The lower door is hinged at the bottom, and opens outward and downward from the floor level.

All the emergency exits in the passenger compartment have both internal and external handles. The flight compartment escape hatch is the only emergency exit which does not have an external handle.

On aircraft with ModSum 4−459339 incorporated, the forward baggage door is reworked and converted into Type I emergency exit door.

The aft baggage compartment is accessible through an outward and upward opening baggage door in the left side rear fuselage. The forward baggage compartment is accessible externally through a door in the right front fuselage. The door opens inward and upward, and then translates outward and forward. Both baggage doors can be opened or closed manually by an external handle.

Internal access to the forward baggage compartment is through a hinged door installed in the passenger compartment divider. The door has a key operated lock. There is no internal access to the aft baggage compartment.

The service door system includes an aft service door, an aft fuselage access door an APU clamshell door. It also includes several ground service and nose compartment doors.

The four nose compartment doors are located on the nose. The aft service door is located on the right side of the aircraft at the rear of the passenger compartment. The Aft Fuselage Access Door is located flush with the underside of the rear fuselage, through which the unpressurized equipment compartment is accessed. The APU access doors are located on the bottom of the tailcone section on either side of the fuselage centerline. The doors are hinged horizontally and kept in a locked and closed position by the two pin latches, hook latches and keepers, and the three shear pins. The Refuel/Defuel Panel door is located on the lower side of the No. 2 engine nacelle, aft of the right main landing gear rear doors. The dc electrical ground power door gives access to the 28 Vdc ground power receptacle on the left side of the aircraft nose. The ac electrical ground power door gives access to the ac ground power receptacle. The hydraulic ground power doors, on the right side of both engine nacelles, give access to the hydraulic ground power connections. The ground air conditioning door is located flush with the lower right side of the rear fuselage. The lavatory service door is located flush with the lower right side of the forward fuselage.

A flight compartment door separates the passenger compartment from the flight compartment. The door is installed to the bulkhead separating the passenger and flight compartments. The door can be locked to prevent passenger entry. It is hinged on the left side and swings aft to open.

Forward Passenger Door

Refer Figure 52-6

The forward passenger door is located at the left side of the fuselage at the front of the passenger compartment.

The forward passenger door is hinged at the bottom and has five steps built into the inboard side of the door structure. The steps also strengthen the door. This door has an external aluminum alloy skin braced by cross beams and intercostals. Door pressurization loads are carried by ten adjustable pressure stop bolts, five on each side. The stop bolts engage with stop brackets on the door surround structure. The door pressure seal inflates to prevent air leakage.

The forward passenger door is the semi−plug type. On opening, the initial movement of the door is upward and inward. The door then rotates outward about the door’s hinge.

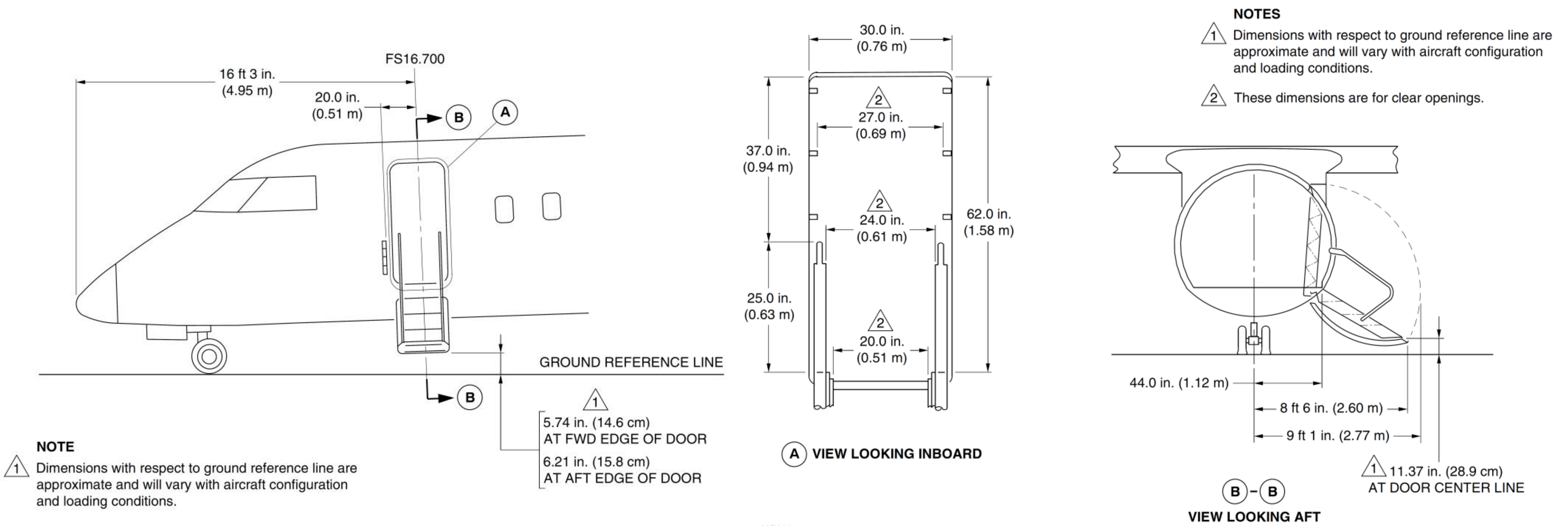


Figure 52-6 Forward Passenger Door

Aft Passenger Door

Refer Figure 52-7

The aft passenger door is a translating type located at the left side of -the fuselage at the rear of the passenger compartment.

This door rotates outwards about the two hinges (main and secondary). A stabilizer rod prevents the yawing movement of the door during opening and closing. Door pressurization loads are carried by 12 stops that engage with their related stop brackets on the door surround structure. In unpressurized flight, the door is prevented from sliding upward by the cam rollers on each side of the lift/latch shaft. If one cam roller assembly fails, the remaining cam rollers prevent the door from sliding. The lift/latch shaft is secured in the closed position by both the handle lock mechanism and over−center springs.

The aft passenger door has the components that follow:

- Stabilizer bar
- Main Hinge
- Secondary Hinge
- Vent door.

Operation of either the internal or external spring loaded handle unlocks the door. The door opens or closes with the door face parallel with the fuselage.

As an optional configuration the aft passenger door is provided with an aft entry folding stairway. The aft entry folding stairway is used as the second entry door stairs for the entry and exit of passengers.

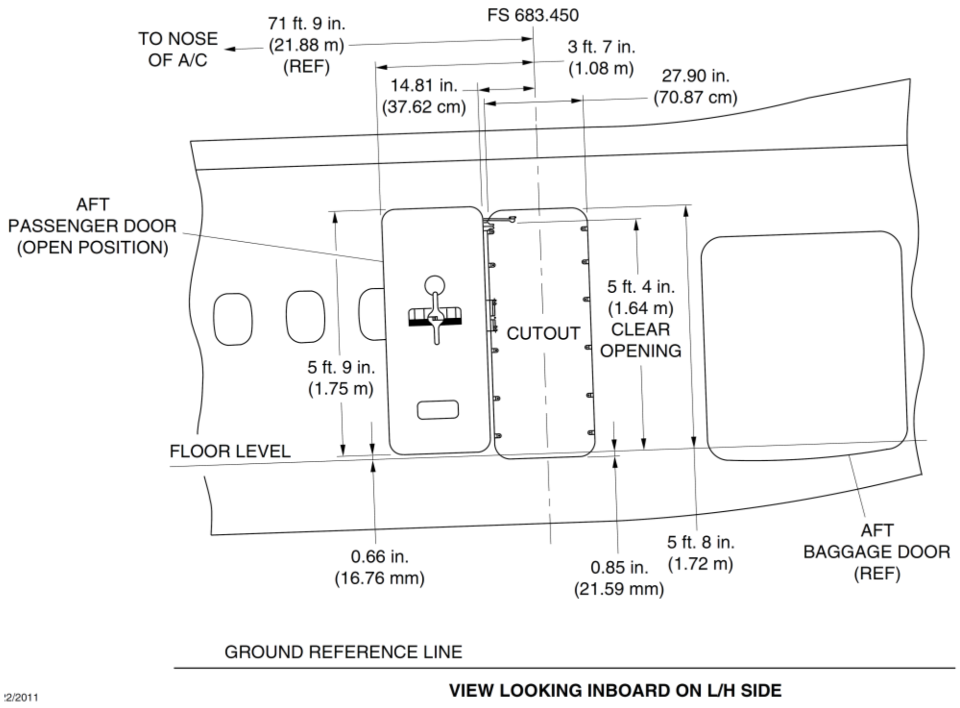


Figure 52-7 Aft Passenger Door – Detailed Dimensions

Flight Compartment Escape Hatch

Refer Figure 52-8

The flight compartment escape hatch is opened by turning the handle counterclockwise to the open position. The hatch is opened to permit depressurization or to increase airflow in the flight compartment when the aircraft is on the ground.

The door movement is controlled by the geometry of the torque tube and rollers. The hatch pivots about the rear fittings and opens approximately one inch at the front.

A pull force on the handle of approximately 40 lb. (18.1 kg), releases the rollers against the action of the locking and release springs, and the hatch can be removed from the flight compartment roof.

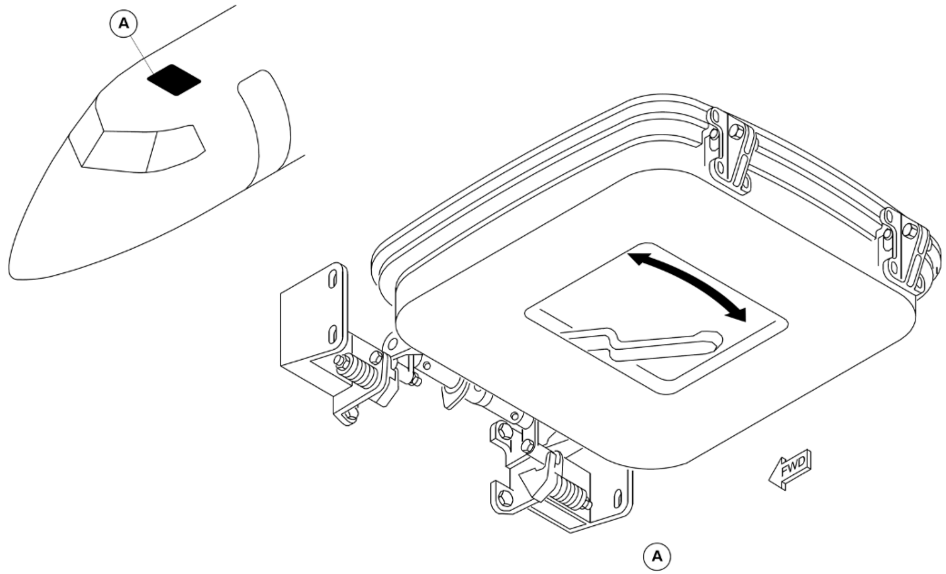


Figure 52-8 Flight Compartment Escape Hatch

Type II/III Exit Doors

Refer Figure 52-9

The upper emergency exit door is secured by a lock pin and three adjustable stops on each vertical edge that engage with three stop brackets installed on the sides of the door to secure it against cabin pressure loads. The lower emergency exit door is installed on a piano hinge at the lower edge. It is kept in the closed position by four stops on the upper edge of the door that engage with their related stop brackets on the lower edge of the upper door. The lower door opens automatically when the upper door is removed, and the lower door handle is in the GROUND position. The lower door stays closed when the lower door handle is in the WATER position.

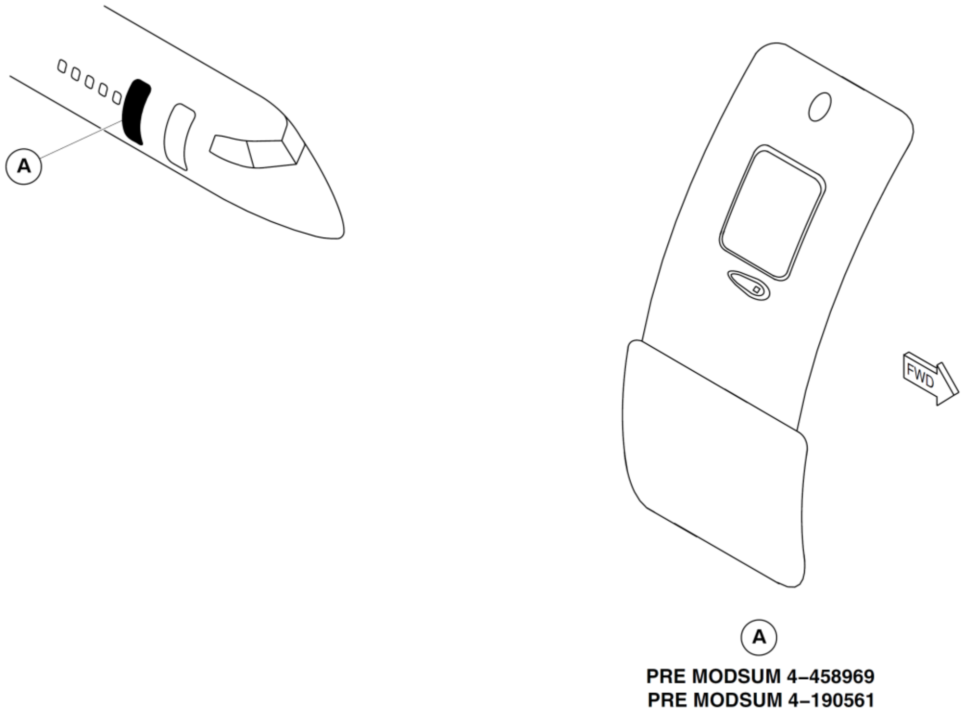


Figure 52-9 Type II/III Exit Doors

Aft Baggage Door

Refer Figure 52-10

The aft baggage door opens outward and upward about an upper hinge, and is held open by a telescopic strut. An external tee handle is used to lock and unlock the door from the outside. There is no internal handle for the baggage door. The door pressure seal inflates to prevent air leakage. The aft baggage door has an external, aluminum alloy skin braced by cross beams. The door warning system monitors the door status and supplies a visual indication and warning in the flight compartment.

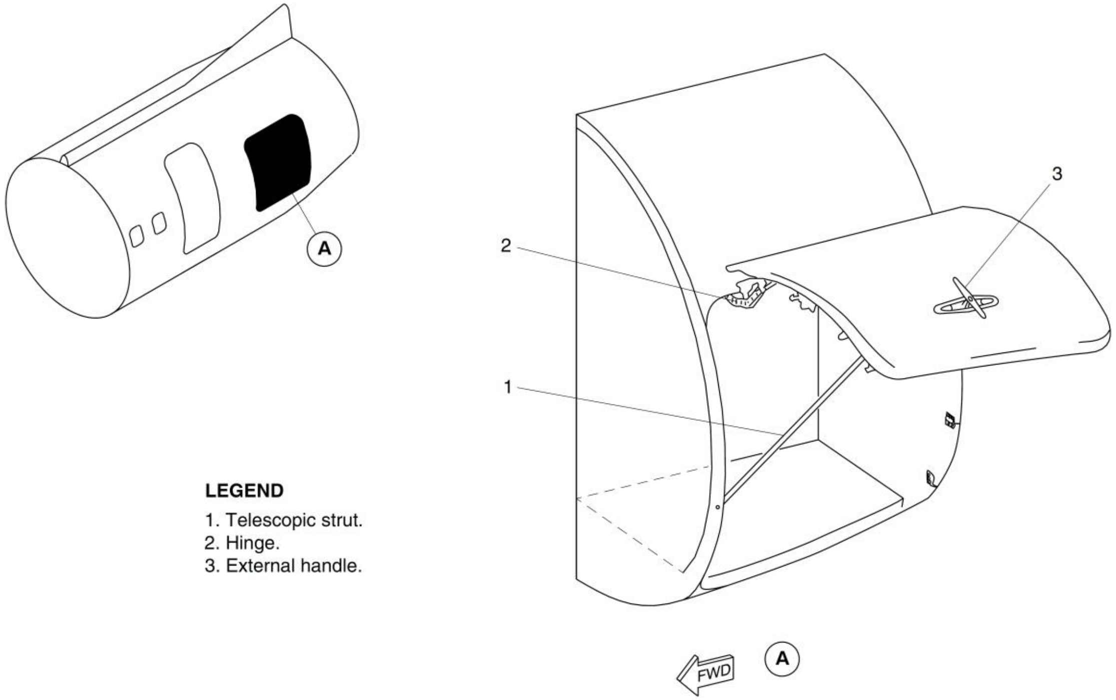


Figure 52-10 Aft Baggage Door

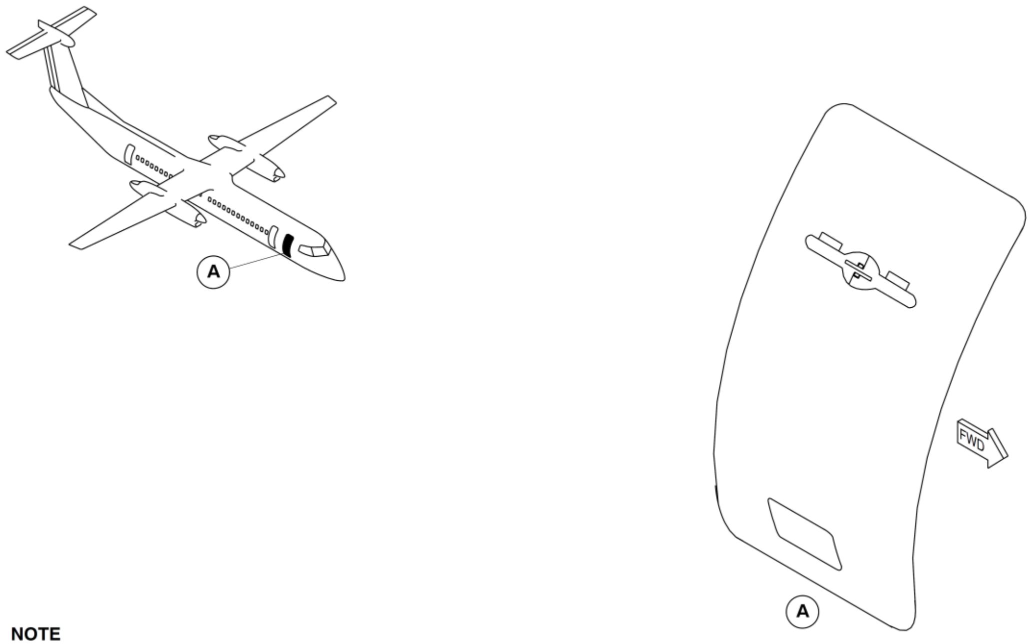
Forward Baggage Door

Refer Figure 52-11

On aircraft with ModSum 4–459339 incorporated, the forward baggage door is reworked and converted into Type I emergency exit door.

The forward baggage door is unlocked by the operation of an external tee handle. The door opens or closes with the door face parallel with the fuselage skin. This is done by two hinges (main and secondary) and a stabilizer.

Proximity sensors monitor the position of the upper fixed rollers and the door handle. A red warning light comes on in the flight compartment when the door is open or unlocked. The door warning system monitors door status and supplies visual indication and warning in the flight compartment.



NOTE

On aircraft with ModSum 4–459339 incorporated, the forward baggage door is reworked and converted into Type I Emergency Exit Door.

Figure 52-11 Forward Baggage Door

Aft Service Door

Refer Figure 52-12

The aft service door is located on the right side of the aircraft at the rear of the passenger compartment. The aft service door is classified as a Type I emergency exit. It is a translating type secured against pressurization loads by five stops on each vertical edge that engage with five related stop brackets installed on the door surround.

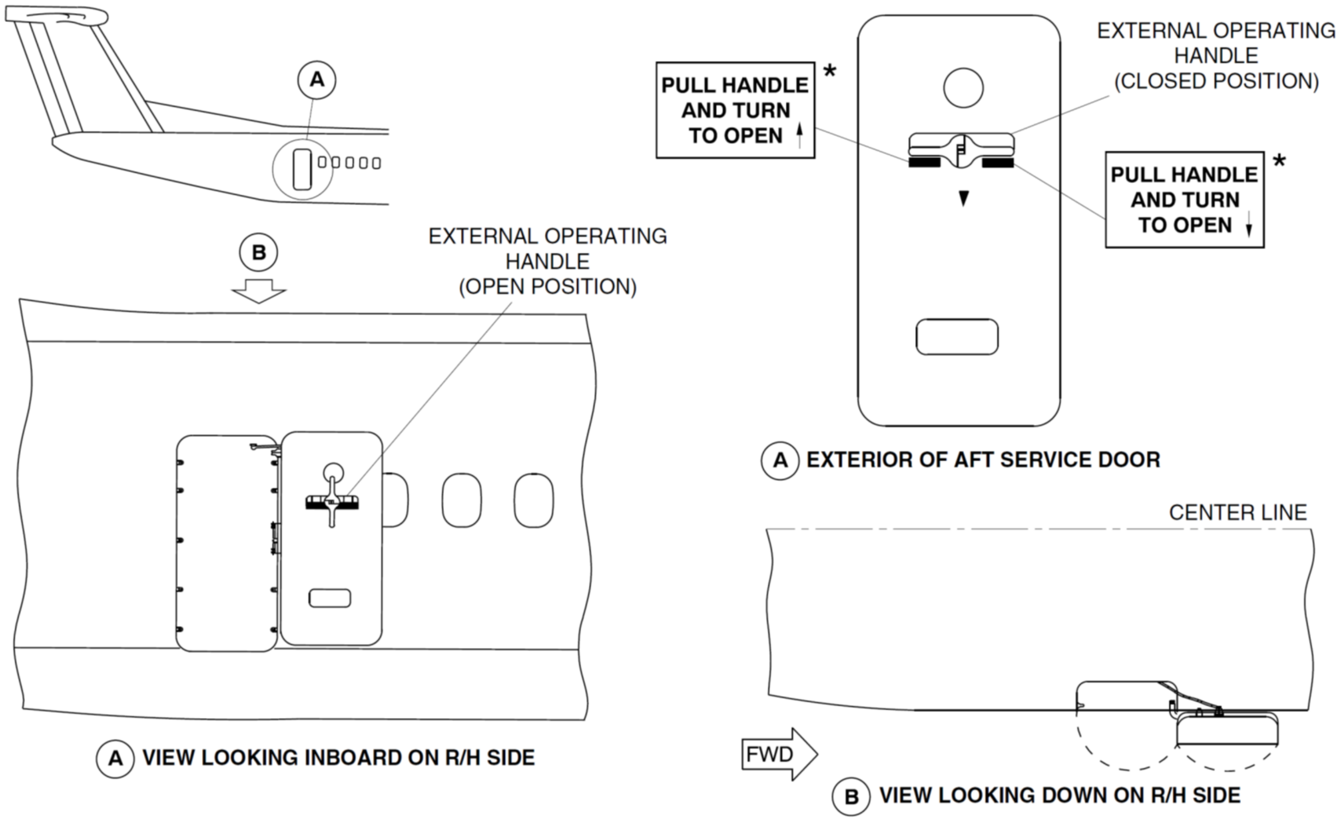


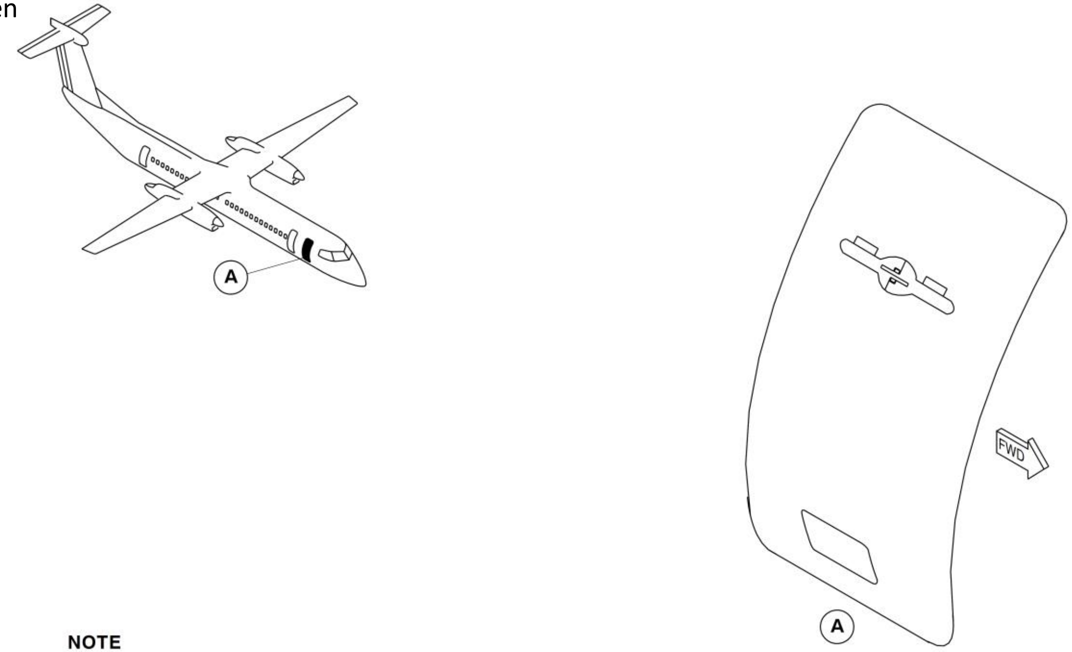
Figure 52-12 Aft Service Door

Forward Type I Emergency Exit Door

Refer Figure 52-13

On aircraft with ModSum 4–459339 incorporated, the forward baggage door is reworked and converted into Type I emergency exit door.

The forward emergency exit type I door gives access to the forward passenger compartment. The forward emergency exit type I door is located on the right side of the forward fuselage. It is a translating type door and is secured against pressurization loads by five stops on each vertical edge that engage with five related stop brackets installed on the door surround structure. The proximity sensor monitors the position of the upper fixed rollers and the door handle. A red warning light comes on in the flight compartment when the door is open and unlocked.



NOTE

On aircraft with ModSum 4–459339 incorporated, the forward baggage door is reworked and converted into Type I Emergency Exit Door.

Figure 52-13 Forward Type I Emergency Exit Door

Door Seal Pressurization System

Refer Figure 52-14

The door seal pressurization system prevents cabin air leakage through the forward passenger and aft baggage doors. Pressurized air is supplied from the airframe deicing system, and from the system reservoir tank when the engines or APU are not operating.

The door seal pressurization system has the components that follow:

- Heated check valve
- Reservoir tank
- Control valves
- Door seals
- Drain valve
- Charging valve
- Electro pneumatic shut–off valve
- Plumbing lines

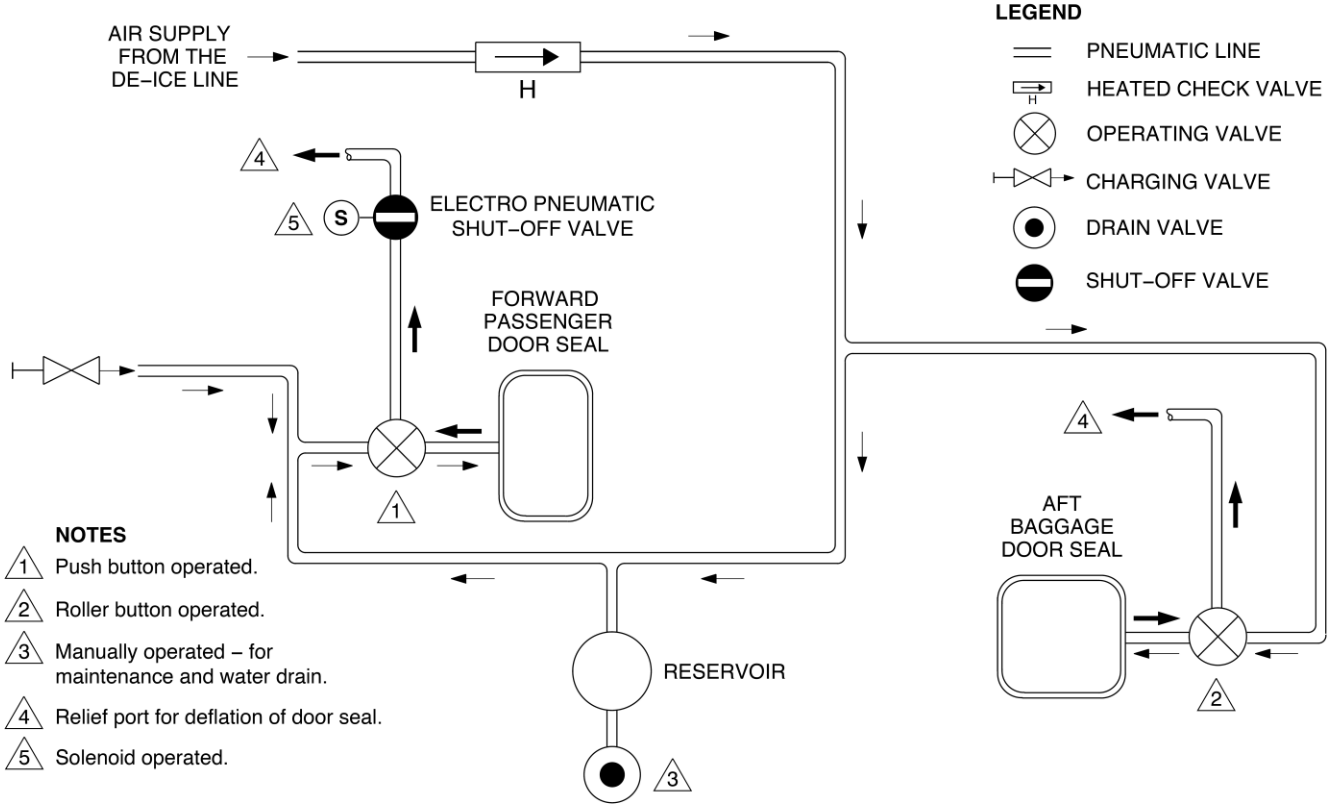


Figure 52-14 Door Seal Pressurization System

Door Warning System

Refer Figure 52-15

The door warning system sends door sensor outputs to the Proximity Sensor Electronic Unit (PSEU).

The PSEU outputs data to the Input Output Processors (IOP 1 and IOP 2). This data is displayed in the flight compartment and is used by the Cabin Pressure Control System (CPCS).

The door warning system monitors the position of the fuselage doors that follow:

- Forward Passenger Door
- Aft Passenger Door
- Type II/III Emergency Exit Door
- Forward Baggage Door
- Aft Baggage Door
- Aft Service Door

On aircraft with ModSum 4–459339 incorporated, the door warning system monitors the position of the fuselage doors that follow:

- Forward Passenger Door
- Aft Passenger Door
- Forward Emergency Exit Type I Door
- Aft Baggage Door
- Aft Service Door

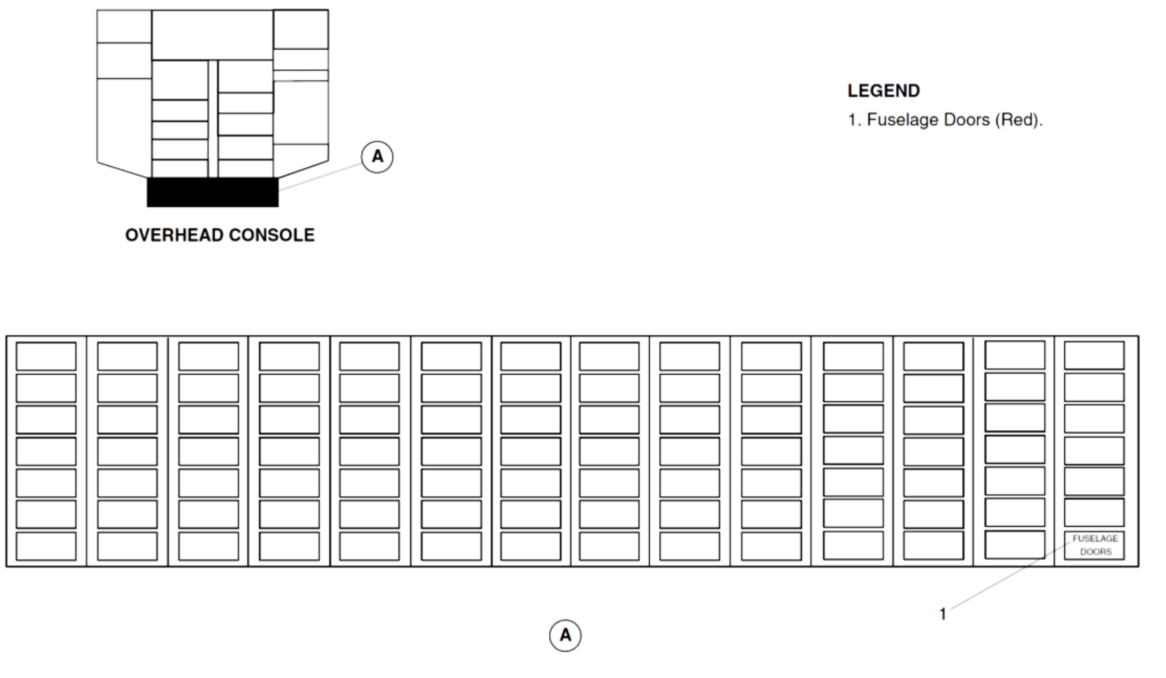
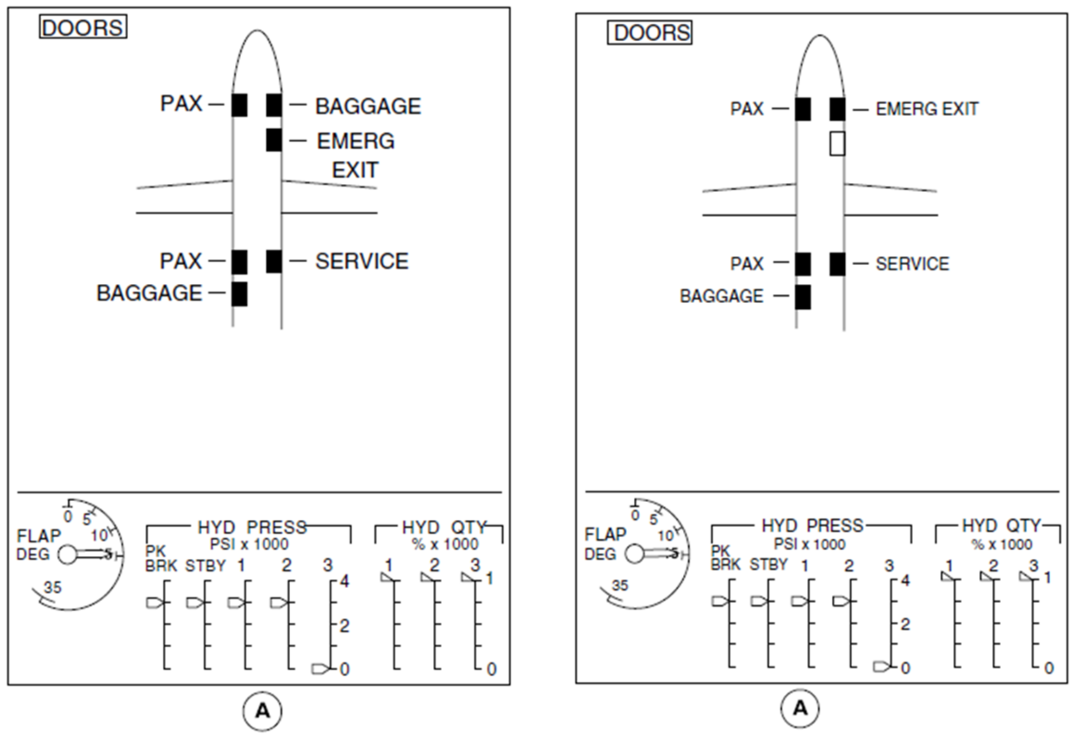


Figure 52-15 Fuselage Door Warning Light

Refer Figure 52-16

The door warning system does not monitor the flight compartment emergency hatch.

When any of the fuselage doors are unlocked, both the FUSELAGE DOORS warning light and the MASTER WARNING light flash. If a fuselage door is not fully closed and locked, the FUSELAGE DOORS warning light and the MASTER WARNING switch light flash. Three tones also sound over the flight compartment speakers. When the MASTER WARNING switch light is pushed, the light turns off. The FUSELAGE DOORS warning light stays on until the door is fully closed and locked.



PRE MODSUM 4–458926
PRE MODSUM 4–458969
PRE MODSUM 4–459339
PRE MODSUM 4–190614

POST MODSUM 4–458926
POST MODSUM 4–458969
POST MODSUM 4–459339
POST MODSUM 4–190614

Figure 52-16 Door System Page- Door Opened

Refer Figure 52-17

The DOORS system page on the Multi–Function Display (MFD) shows all the fuselage doors and identifies them as:

- PAX for each of the passenger doors
- BAGGAGE for each of the baggage doors
- SERVICE for the aft service door
- EMERG EXIT for the type II emergency exit door.

If a fuselage door is not fully closed and locked, the DOORS system page identifies the door and shows it as a red filled rectangle. If the door is fully closed and locked, it is shown only as a green empty rectangle. If the data from the PSEU is invalid, the DOORS system page shows a global white INVALID DATA message.

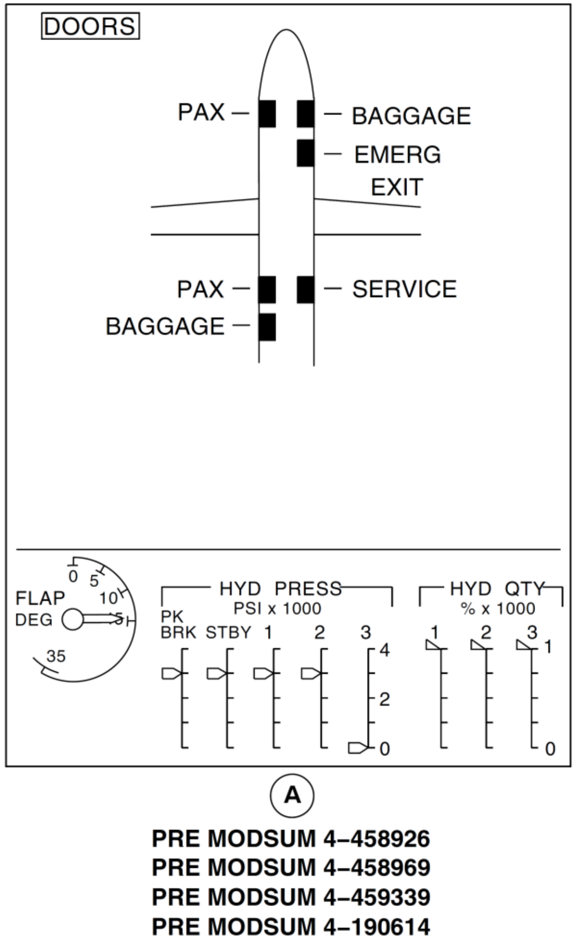


Figure 52-17 Door System Page- Door Open

Forward Passenger Door

Introduction

The forward passenger door gives access to the passenger compartment and also functions as a type I emergency exit.

General Description

The forward passenger door is located on the left side of the fuselage at the front of the passenger compartment, and opens outward and down. The forward passenger door is classified as a type I emergency exit.

This door has five steps on the inboard side of the door structure. Internal and external door handles permit manual opening and closing from either the inside or outside of the aircraft. There are two automatically folding handrails, one on each side of the door. The door has four tread lights controlled by a switch on the flight attendant's control panel. Main landing gear lock pins are stored on the forward side of the door structure. An inflatable pressure seal prevents cabin air leakage past the door. The door warning system monitors door status and supplies visual indication and warning in the flight compartment.

The Forward Passenger Door System has the components that follow:

- Airstair
- Shell, Door
- Mechanism, Door Lift
- Mechanism, Door Balance
- Handrail, Door
- Struts
- Seals, Door
- Valve, Control−Door Seal
- Electro pneumatic shut−off valve
- Drain Plugs

The Forward passenger door interfaces with:

- Door Seal Pressurization System
- Door Warning System
- Proximity Sensors

Detailed Description

Refer Figure 52-18

The forward passenger door gives access to the passenger compartment and also functions as a type I emergency exit. The forward passenger door opens and closes manually from either the inside or outside.

LEGEND

1. Internal handle.
2. Door lift actuating rod.
3. Door hinge shaft.
4. External handle.
5. Inflatable seal.
6. Handrail.
7. Step.
8. Tread light.
9. Weather seal.
10. Main landing gear locking pin storage.
11. Drain plug.

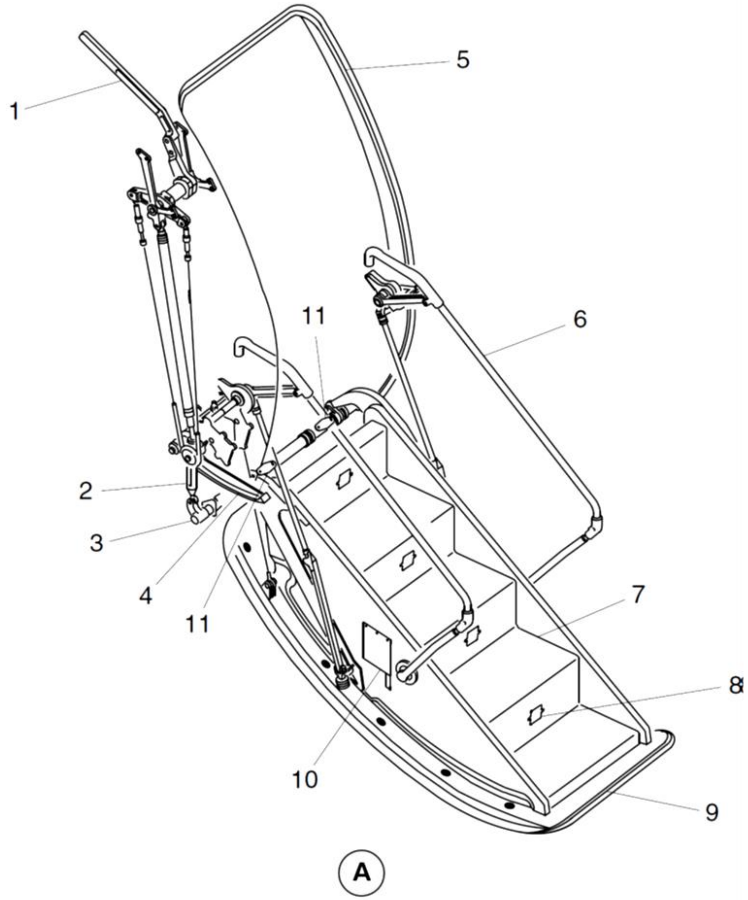


Figure 52-18 Forward Passenger Door – Detailed

Refer Figure 52-19

An internal handle is located in a recess above the forward flight attendant seat. An external handle is located forward of the door, in a recess in the fuselage.

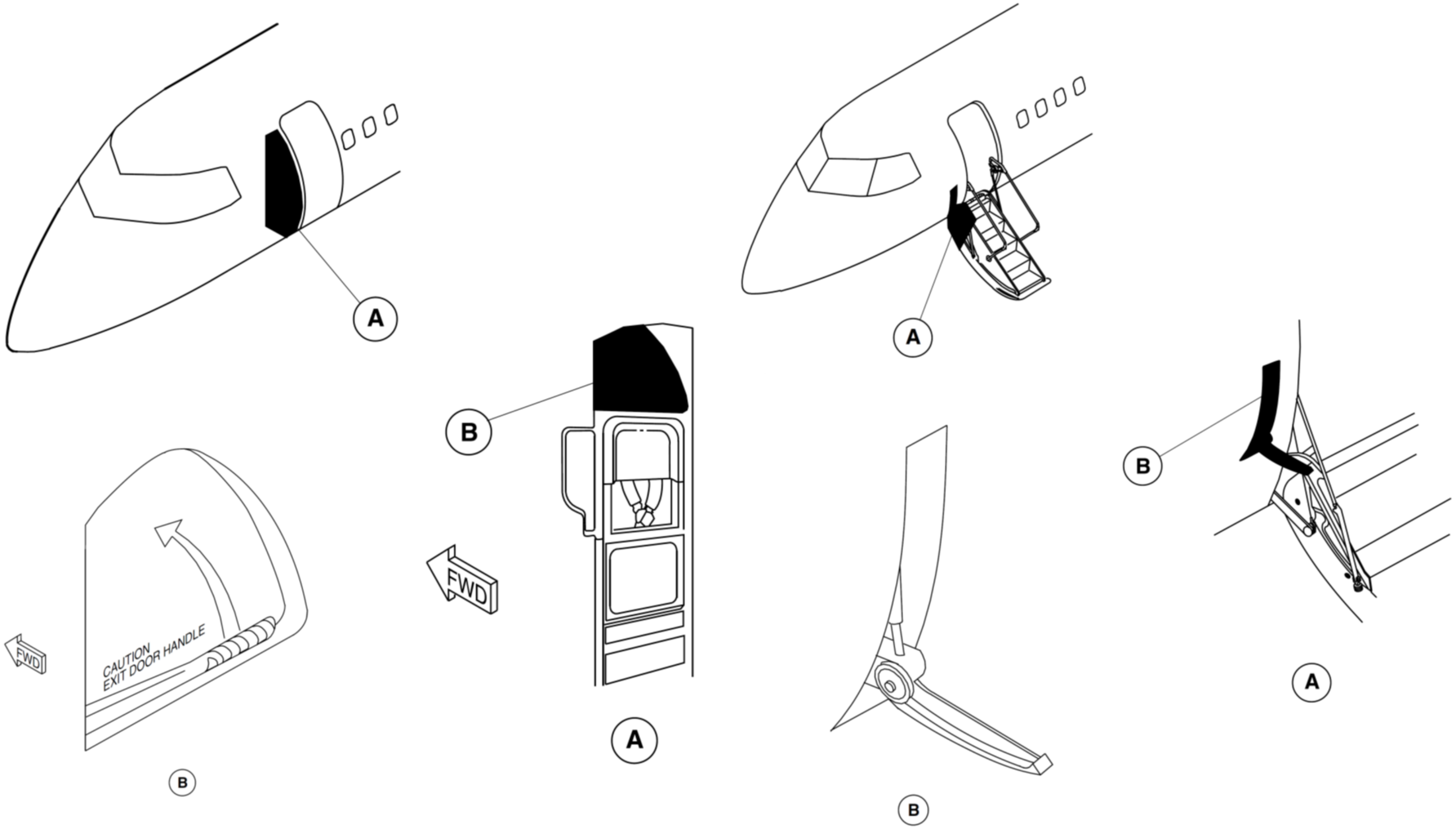


Figure 52-19 Forward Passenger Door Handle

Refer Figure 52-20

The internal and external handles are interconnected by a cable and chain system. A flush handhold, on the door outer skin, is used to pull the door open from the outside. Handrails on each side of the stairs automatically fold during door closing and unfold during opening. On opening, the initial movement is upward and inward. The door then rotates about a hinge at the bottom of the door.

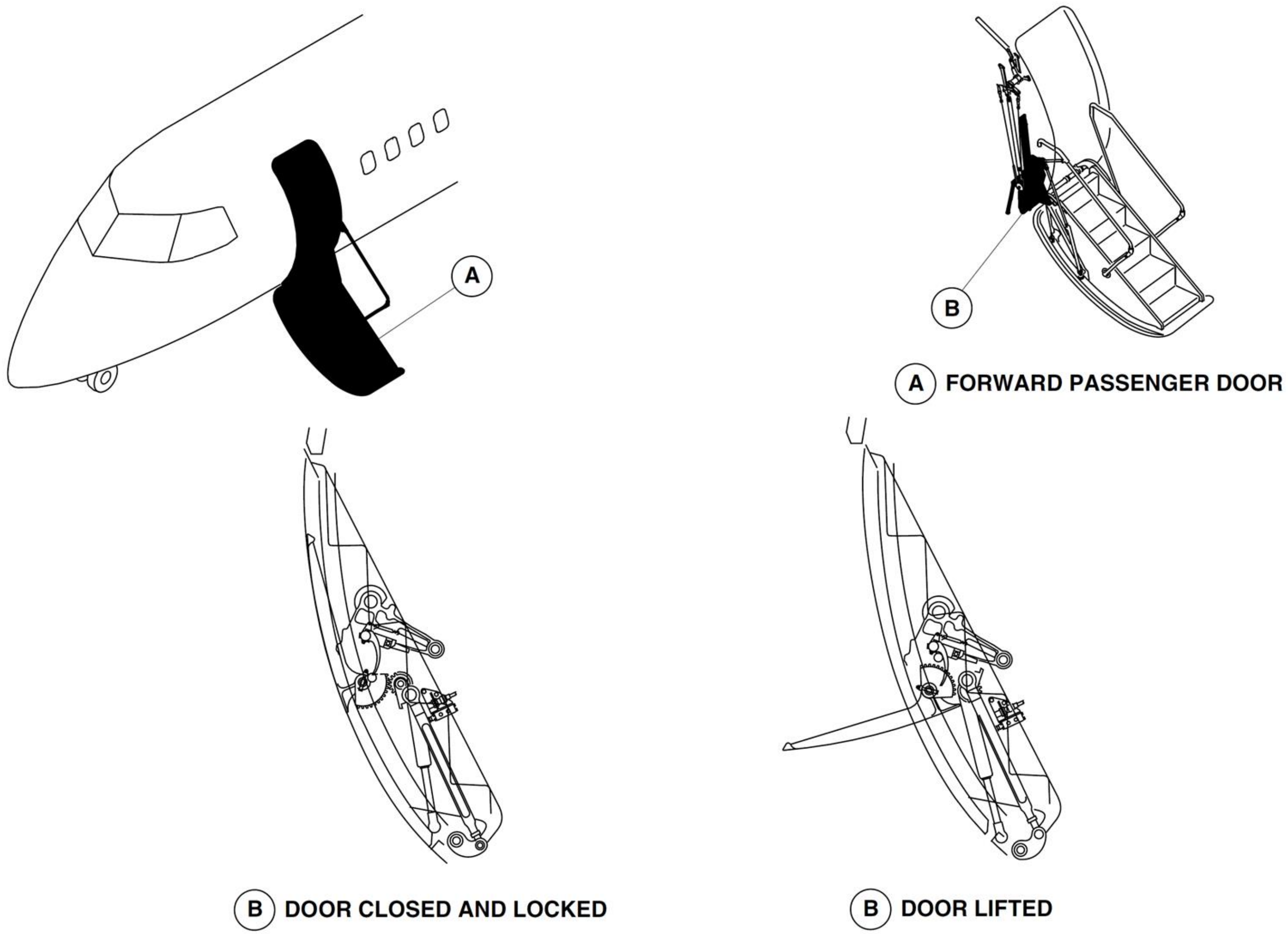


Figure 52-20 Forward Passenger Door Closed and Lifted Positions

Refer Figure 52-21

Two hinge arms attach the door to a door hinge shaft installed in the fuselage below the door surround bottom sill. The door hinge shaft is connected through a crank and actuating rod to a door lift mechanism. The lift mechanism has an upper crank and a pinion assembly, and is operated by the interconnected internal and external handles. Operation of either handle rotates a gear segment to drive a pinion, rotating the door hinge shaft through a pushrod, raising or lowering the door hinges.

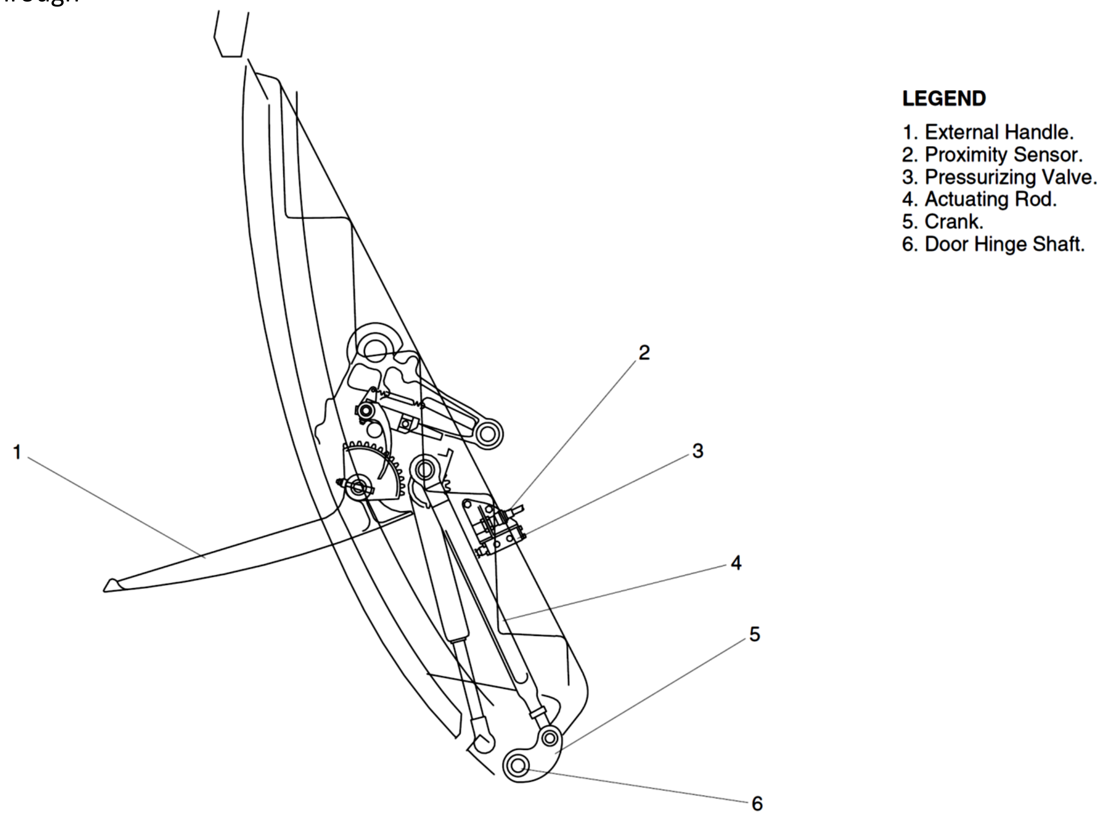
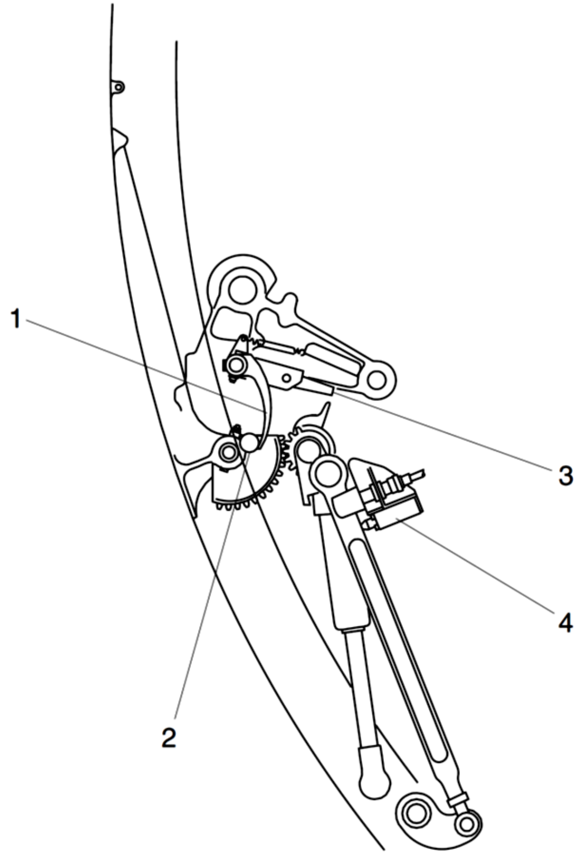


Figure 52-21 Forward Passenger Door Lift Mechanism

Refer Figure 52-22

The upper crank and pinion assembly is attached to a gas spring assembly attached to the fuselage. A spring controlled over centering trip lever and locking bar, are operated by a roller on the forward side of the door. The trip lever and spring, in conjunction with the gas spring, gives a positive over center lock of the lift mechanism when the door is either in the opened or closed position.



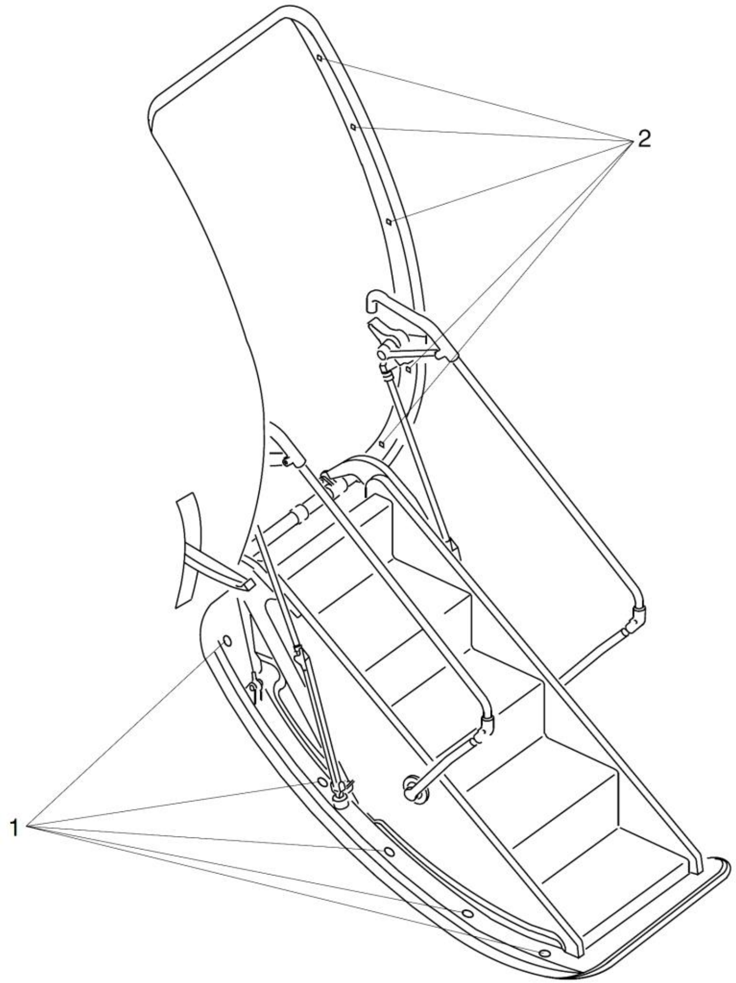
LEGEND

1. Trip Lever.
2. Roller (Door Down Position).
3. Locking Bar.
4. Pressurization Valve.

Figure 52-22 Forward Passenger Door Trip Lever and Locking Bar

Refer Figure 52-23

When the door is in the closed position, door pressurization loads are carried by ten adjustable stop bolts, five on each side of the door. These stop bolts engage with related stop brackets on the door surround structure. Each door stop is inclined downward, so that pressure on the door loads the door downwards towards the closed position. The geometry of the door transporting system is such that, if the door moves away from the fully closed position, cabin pressurization loading applies a downward force on the door, preventing any further movement towards the disengaged position.



LEGEND

1. Door Stop Bolts (5 Per Side).
2. Door Stop Brackets (5 Per Side).

Figure 52-23 Forward Passenger Door Stop Bolts

Refer Figure 52-24

A door counterbalance system controls the rate the door opens and helps to close the door. The system is controlled by two balance gas springs, attached at the top to a slide block and pulley assembly and at the bottom to a fixed bracket. The fixed bracket is bolted to the counterbalance system mounting.

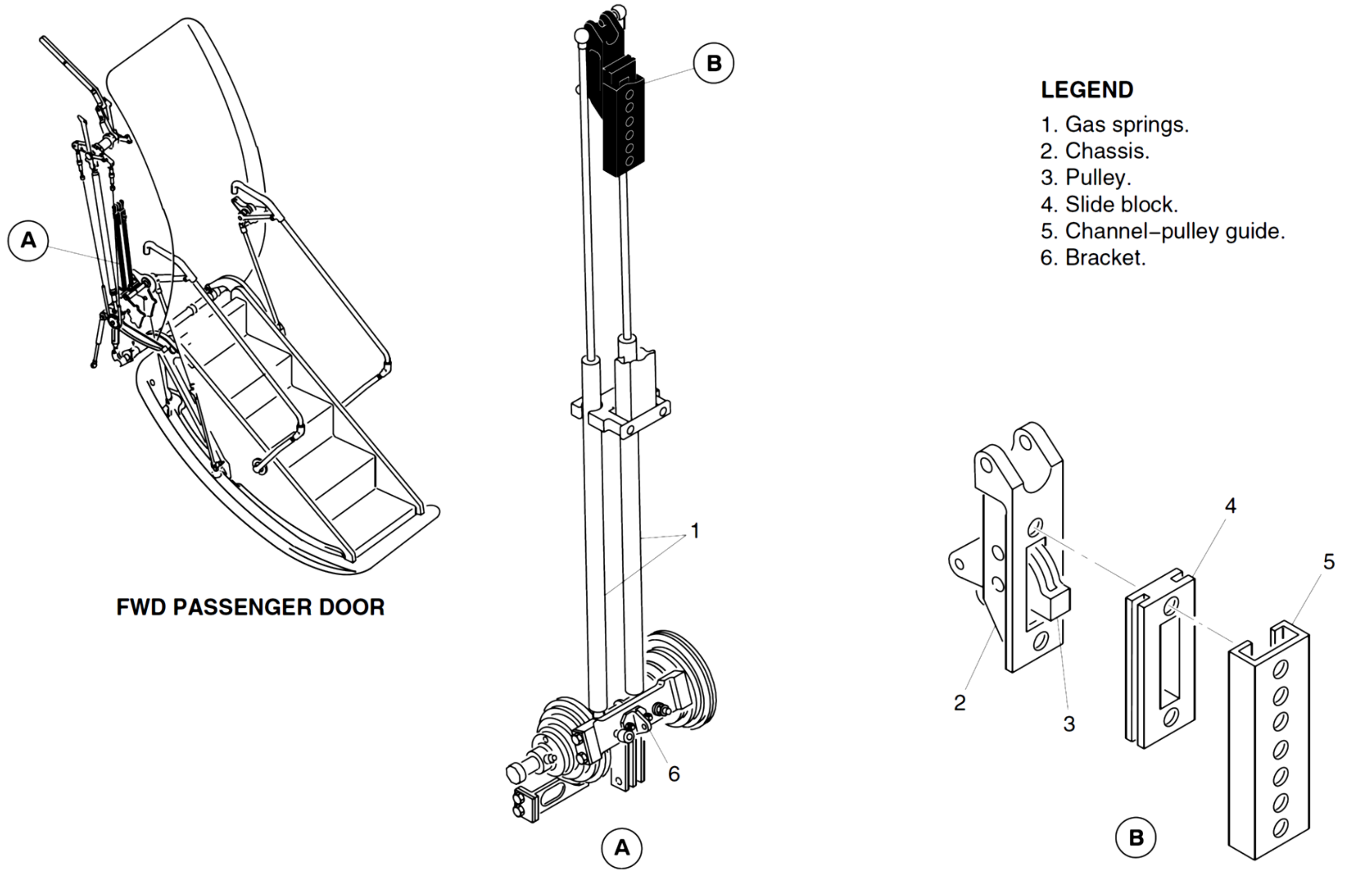


Figure 52-24 Forward Passenger Door Balance System

Refer Figure 52-25

A cable assembly attached to the fixed bracket runs up and over the slide block pulley and down to wind around a balance pulley attached to the shaft assembly. The shaft assembly is bolted to the counterbalance system mounting. The slide block and pulley assembly moves up and down in a fixed guide rail, under the action of the balance gas springs when the door is opened and closed. A support pulley system on the shaft assembly is connected by a cable to a cable support attached to the forward face of the door structure.

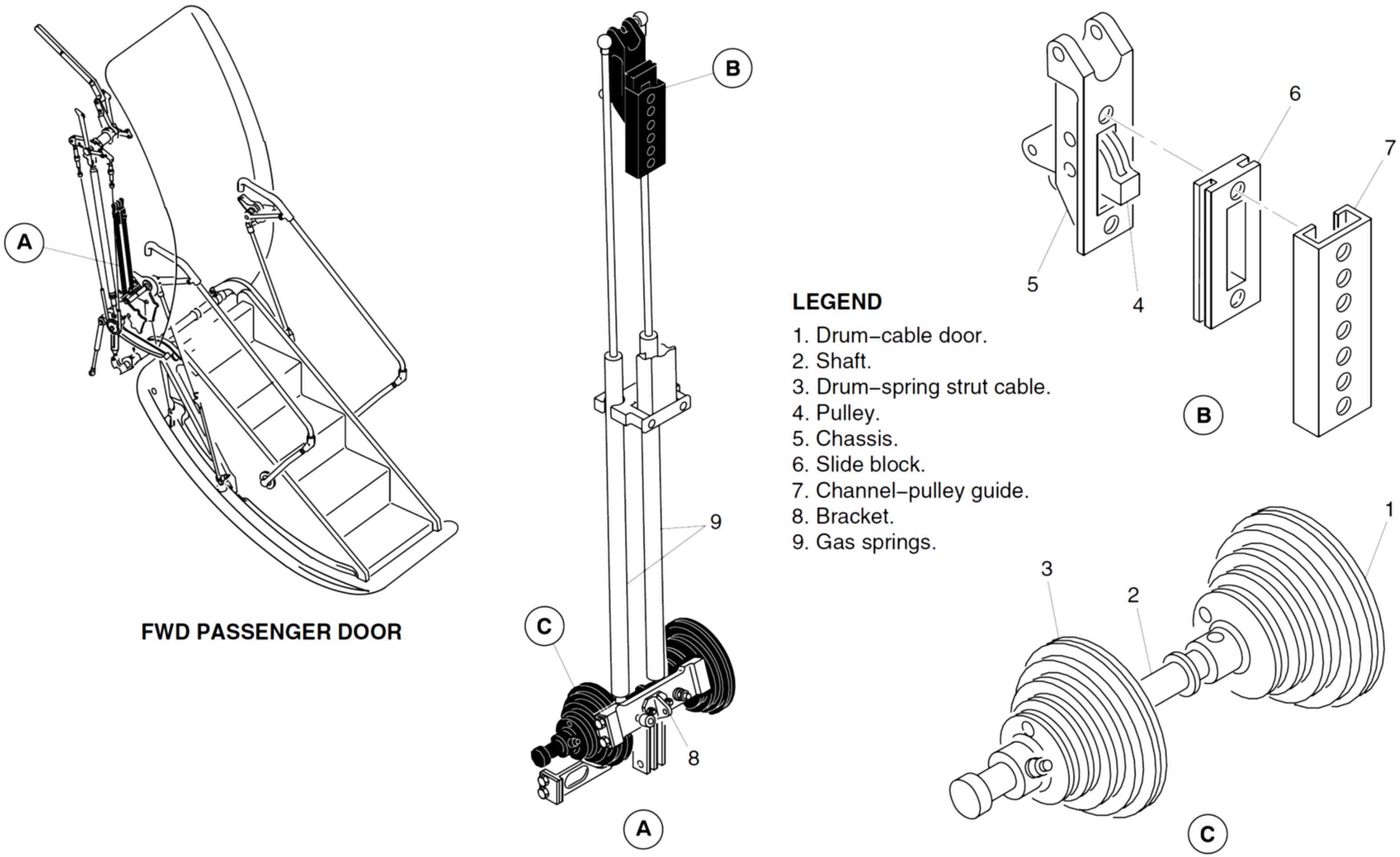
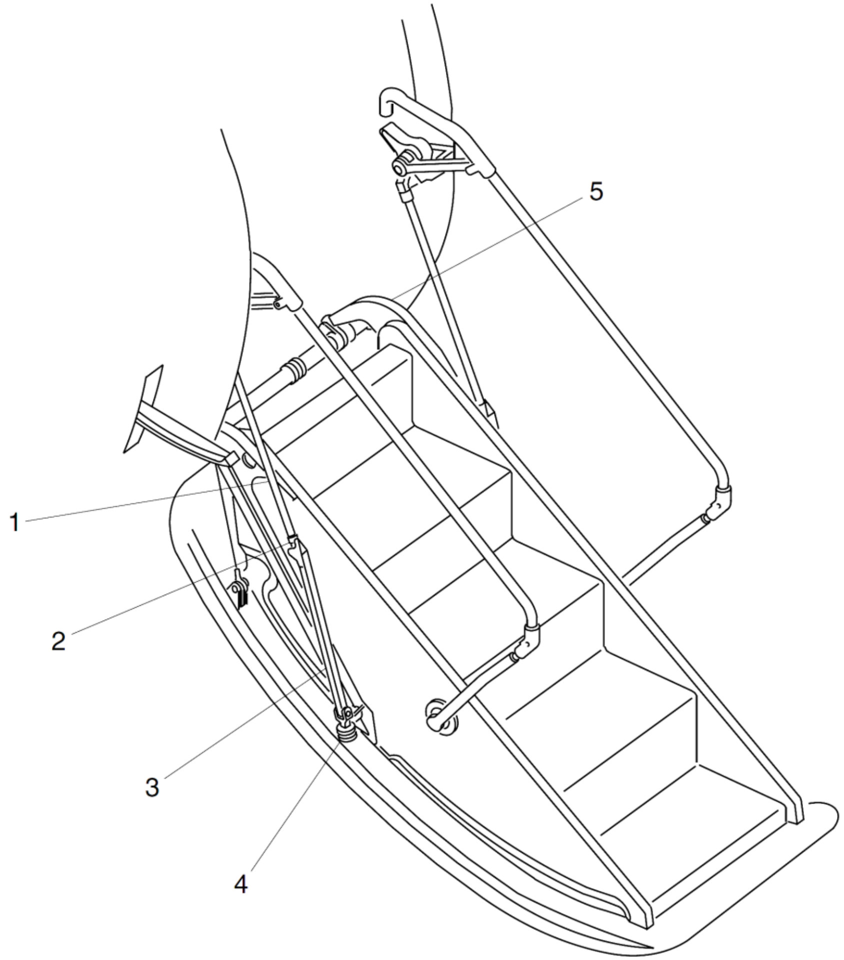


Figure 52-25 Forward Passenger Door Balance System - Detailed

Refer Figure 52-26 (Forward Passenger Door Struts - Detailed)

The door is supported on each side in the open position by the upper and the lower struts which fold automatically when the door is closed. Installed in each of the lower struts is a rubber bumper to protect the structure from impact damage if one of the balance gas springs is defective. Installed in each of the upper struts is an over travel positive stop to prevent the structure from excessive bending when people are standing on the stairs. The goosenecks are installed on the forward and aft sides of the forward passenger doors. On aircraft with ModSum IS4Q2400003 incorporated, the p–clip (MS2191WDG4) attachment on gooseneck frame is replaced with a low profile Tie mount (CB3019A3N)). The Tie mount is bonded to the gooseneck frame.



LEGEND

1. Upper strut.
2. Overtravel positive stop.
3. Lower strut.
4. Rubber bumper.
5. Gooseneck.

Figure 52-26 Forward Passenger Door Struts - Detailed

Refer Figure 52-27

The door has an inflatable rubber seal installed on the fuselage structure around the edge of the door opening. Pressurized air to and from the seal is controlled by a pressurization valve operated by a rod assembly in the door lift mechanism. An auxiliary weather seal installed on the upper part of door cut–out seals the door when air pressure is not available.

Two proximity sensors monitor the forward passenger door. One sensor is located at the top aft side of the door, to make sure that the door is closed and fully down. The other sensor, located on the fuselage structure adjacent to the pressurization valve, monitors the lift link position to make sure that the door mechanism is fully overcentre and in safety.

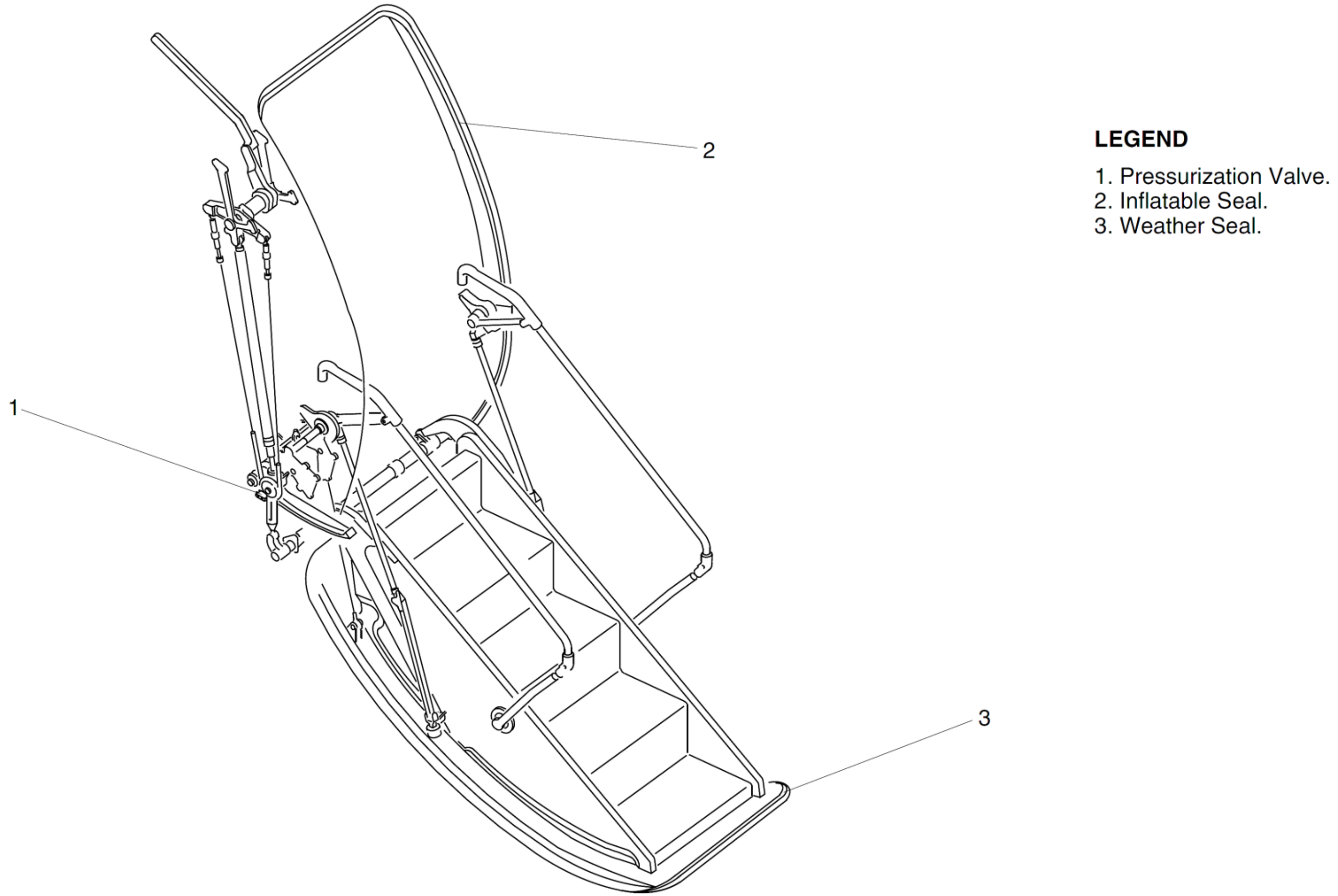


Figure 52-27 Forward Passenger Door Seal

Refer Figure 52-28

Opening of the forward passenger door is by operation of either the internal or external handle. The handle force builds up and then decreases when it passes the overcenter position. At the same time, the seal pressurization valve closes and releases seal pressure. The deflated seal vents any residual passenger compartment air pressure. The "locked" proximity switch also disconnects. Further handle travel, through the actuating rod and crank, moves the door upward and inward to clear the ten stop bolts from their stop brackets. The "closed" proximity switch disconnects as the door opens. The door is kept at the lifted position by the counterbalance system. Pushing on the handrails, or pulling on the external handhold then opens the door. The rate of opening is controlled by the gas springs of the counterbalance system. If a gas spring fails, rubber bumpers on each handrail lower strut gives free fall damage protection.

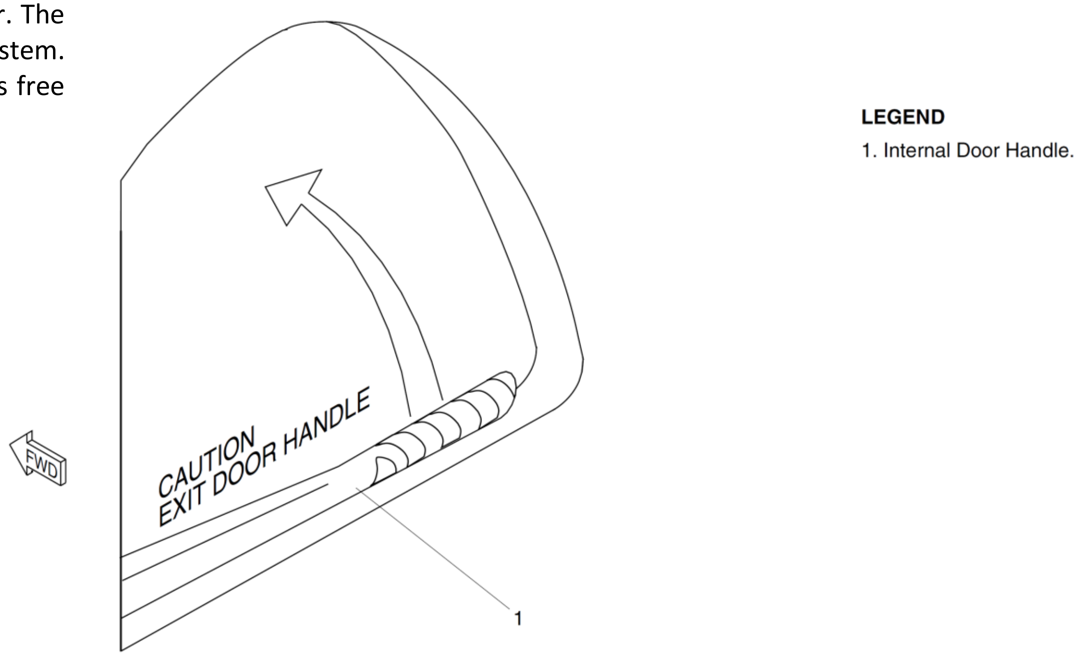


Figure 52-28 Forward Passenger Door Internal Handle

Refer Figure 52-29

Closing the door from the inside is done by pulling up on the handrail handles, the motion being assisted by the counter balance system. Operation of the internal handle to the closed position rotates the door hinge shaft, placing the door mechanism including both the actuating rod and the mechanical spring in an overcenter position and locating the stop bolts behind their related stop brackets. Final movement of the door mechanism opens the pressurization valve and inflates the seal, and the door is fully closed and locked. Closing the door from the outside is done by first pushing the handrails inboard to break the overcenter lock, then manually raising and closing the door. Moving the external handle flush with the fuselage locks the door in the same way as the internal handle. During the opening and closing sequences, the gas springs dampen any tendency for high impact at both ends of the door travel.

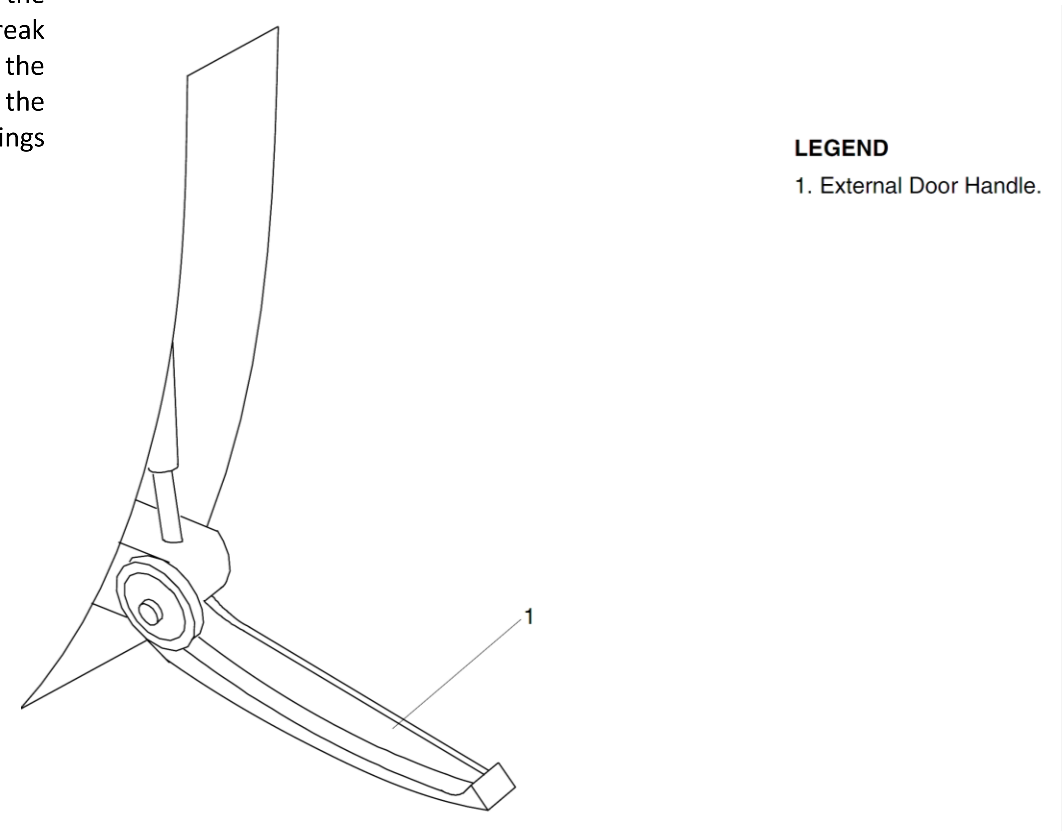


Figure 52-29 Forward Passenger Door External Handle

Refer Figure 52-30

If the door is unlocked, the FUSELAGE DOORS warning and the MASTER WARNING light flash. The Doors Systems page on the Multi-Function Display (MFD) shows the unlocked door as a red filled rectangle. If the door is not locked, the door seal will not inflate. If the door is not properly closed and locked when the aircraft leaves the ground, the cabin will not pressurize automatically.

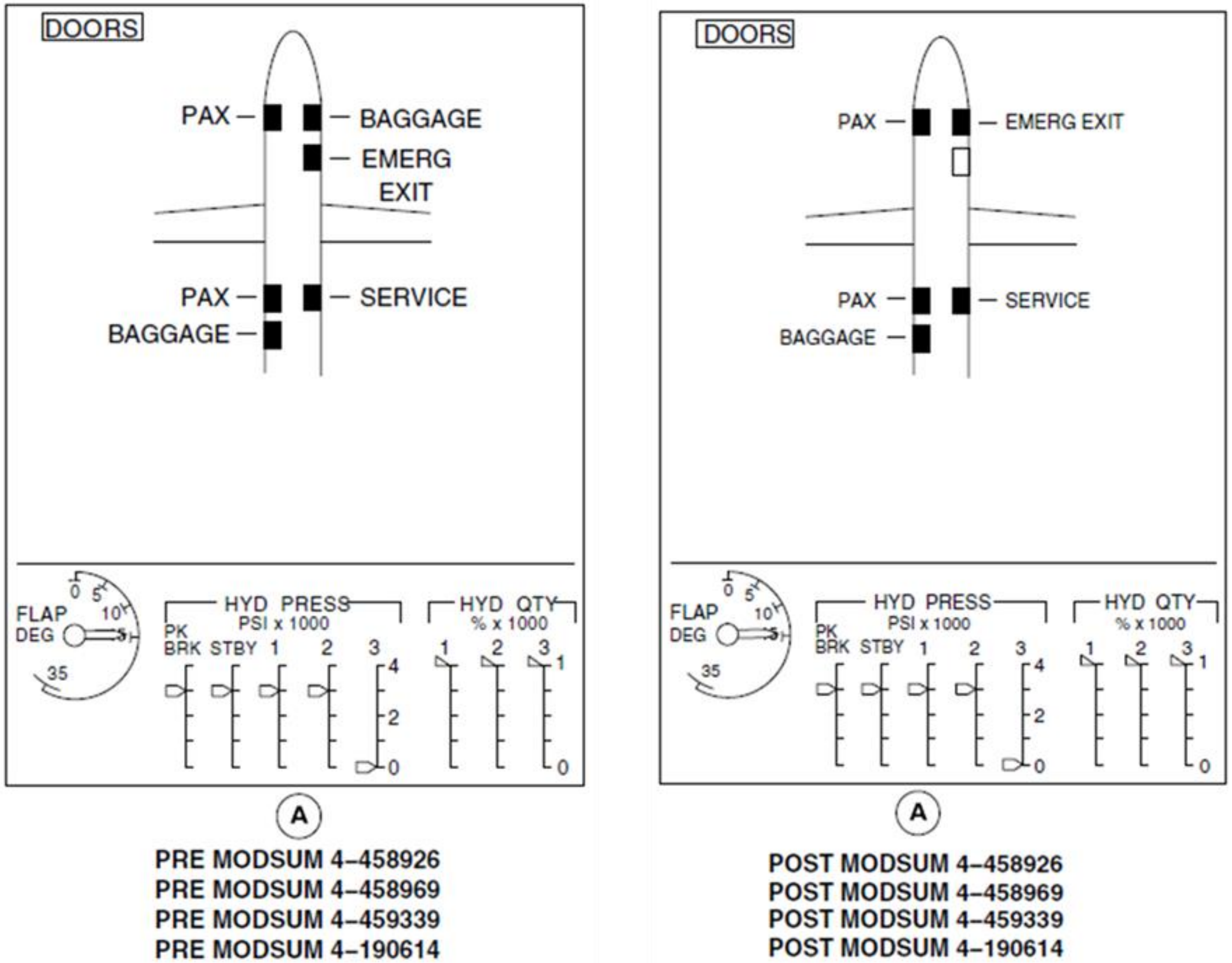


Figure 52-30 Door System Page

Mechanism, Door Lift

The door lift mechanism is operated by either the internal or external handle. The lift mechanism lifts the door on opening, and locks it on closing. At the locked position, the actuating rod engages the “locked" proximity switch and the pressurization valve. The geometry of this lift mechanism causes an over center condition in both door up and down positions, holding the door in either position. A mechanical spring attached to the crank and pinion assembly, and secured to the fuselage structure, adds positive over center locking.

Mechanism, Door Balance

The door balance mechanism balances the weight of the door using two gas springs, a shaft assembly, cables and a slide block and pulley assembly. The rate of opening and closing is controlled by the two gas springs, attached at the top to a slide block and pulley assembly and at the bottom to a fixed bracket. A cable assembly attached to the fixed bracket runs up and over the slide block pulley and down to wind around a balance pulley, mounted on a shaft assembly. The shaft assembly is bolted to the counterbalance system mounting. The slide block and pulley assembly goes up and down in a fixed guide rail, under the action of the gas springs when the door is opened and closed. A support pulley, installed on the shaft assembly, is connected by cable to a cable support attached to the forward face of the door structure.

Handrail, Door

Refer Figure 52-31 & *Refer Figure 52-32*

Two handrails, one on each side of the door, fold automatically when the door is closed and unfold when the door is opened. A rubber bumper is installed in each of the lower struts to protect the structure from impact damage if one gas spring fails.

On Airstair door with jetway compatibility (Modsum 4Q422100, 4Q458687, 4Q458845 or SB 84−52−79 or SB 84−52−86), there are two collapsible handrails, one on each side of the door. The handrails fold automatically when the door is closed and unfold when the door is opened. A retainer bracket on the handrail assembly prevents the removable pin from coming out of the handrail. To make the door jetway compatible, turn the retainer bracket, pull out the removable pin from the handrail assembly, and put the handrail assembly on the handrail holder assembly.

Struts

The struts go slightly overcenter against the stops to hold the door in the fully open position. The struts include an energy absorber, to protect the structure from impact damage, if one gas spring balance fails.

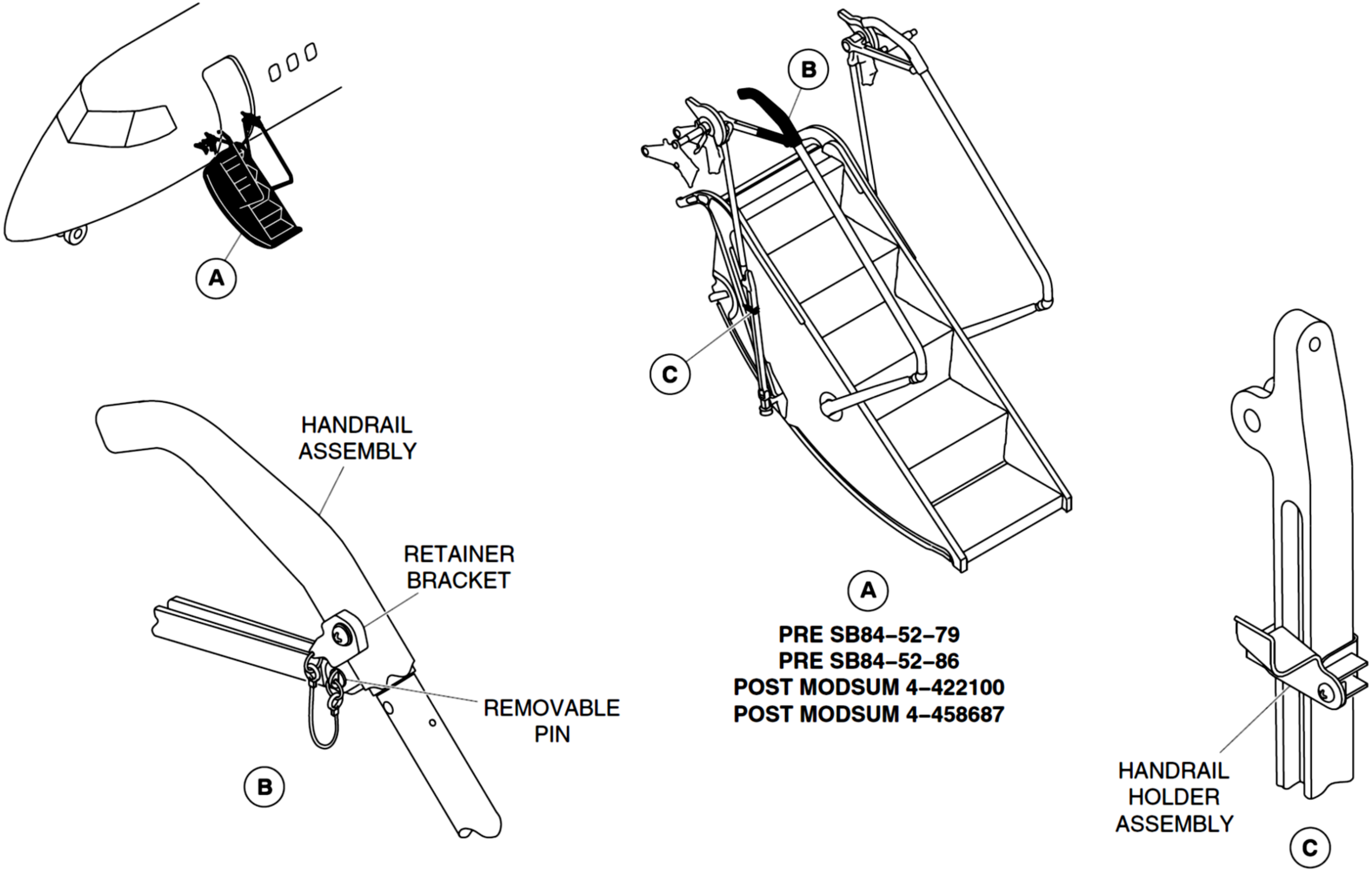


Figure 52-31Airstair Door with Jetway – Folding Handrail Detailed 1 of 2

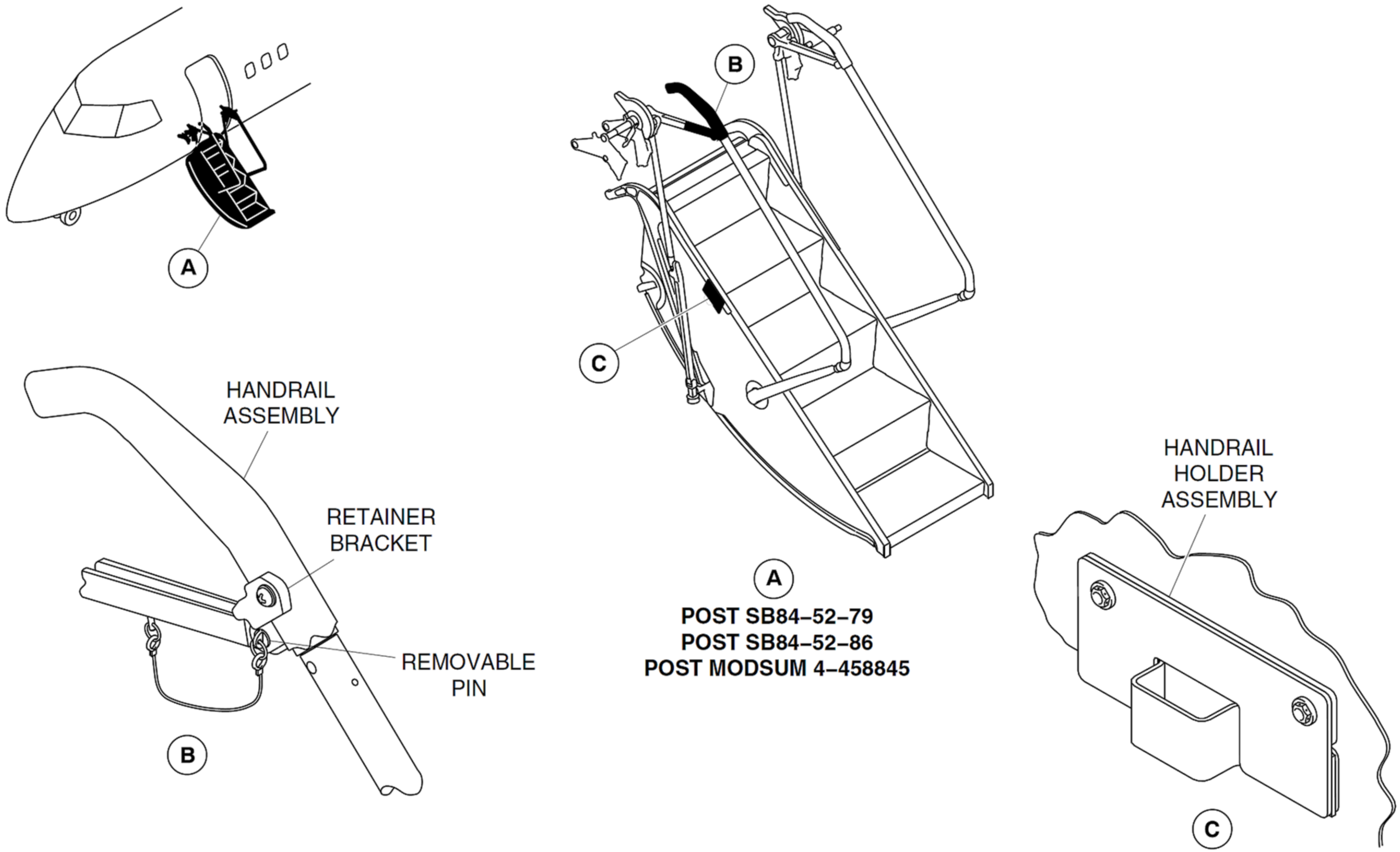


Figure 52-32 Airstair Door with Jetway – Folding Handrail Detailed 2 of 2

Airstair

The airstair has five steps and four tread lights. It is attached to the door shell by four attachment bolts.

Shell, Door

The door is made of an aluminum alloy external skin braced by cross beams and intercostal.

Seals, Door

A door seal is installed on the door surround structure and inflates with pressurized air. An auxiliary weather seal supplies sealing when air pressure is not available.

Valve, Control−Door Seal

The door seal pressurization valve is operated when the door is closed and the lift mechanism is overcenter. The valve prevents pressurization of the seal unless the door is fully closed and locked. The valve releases seal pressure when the door is unlocked.

Electro pneumatic shut−off valve

The solenoid operated valve is installed on the relief port line of the forward passenger door seal pressurization valve. The primary function of this electro pneumatic shut−off valve is to prevent depressurization of the door seal system if airspeed is more than 100 knots (airspeed > 100 knots). This feature prevents inadvertent door seal depressurization for the forward passenger door.

Drain plugs

Two drain plugs are installed on the channel adjacent to the hinge of the door. When the door is open, the drain plugs can be removed to drain accumulated water from the channel.

Door Seal Pressurization System

Introduction

The door seal pressurization system seals the forward passenger and aft baggage doors to prevent loss of cabin pressure.

General Description

The door seal pressurization system uses airframe deicing air to inflate seals on the forward passenger door and the aft baggage door.

The seals inflate when the door is closed and the door handle is in the closed position.

The seals deflate when the door handle is turned to the open position. If the aircraft engines are not operating, the air is supplied from a pressurized reservoir tank.

When the aircraft is on the ground, the tank is pressurized through a charging valve located in the wardrobe.

The door seal pressurization system has the components that follow:

- Tank, Reservoir
- Valve, Drain
- Check Valve, Heated
- Valve, Charging
- Lines
- Filter, Door Seal
- Valves, Control
- Electro pneumatic shut−off valve

Detailed Description

Refer Figure 52-33

The door seal pressurization system is connected to the airframe deicing system. When the deicing system is pressurized, air is routed through the heated check valve to pressurize the reservoir tank and the door seals.

When the engines are not operating, the reservoir tank supplies the door seals with pressurized air through one outlet. A pipe connected to this outlet divides into two lines, one for each door seal. Each door seal inflates when the door handle is moved to the locked position.

Air is distributed to the seal through the control valve. When the handle is moved to the unlocked position, the seal is depressurized through the control valve.

The inflatable seals are located on the door surround of the passenger door and on the aft baggage door itself. The seals for the two doors have the same cross−section. When the seals are pressurized, they extend up to 1 inch and form a seal by contacting the facing structure.

Each seal has a fabric core wrapped in silicon material. The passenger door seal has a polyester core. The aft baggage door seal has a NOMEX core for increased fire protection.

A drain valve is fitted to the system to allow depressurization and water draining of the system, for maintenance action.

After maintenance the reservoir can be pressurized on the ground, through the charging valve located in the wardrobe on aircraft with PRE MODSUM 4−153583. On aircraft with POST MODSUM 4−153583 through the charging valve located below the floor at the forward passenger door entrance.

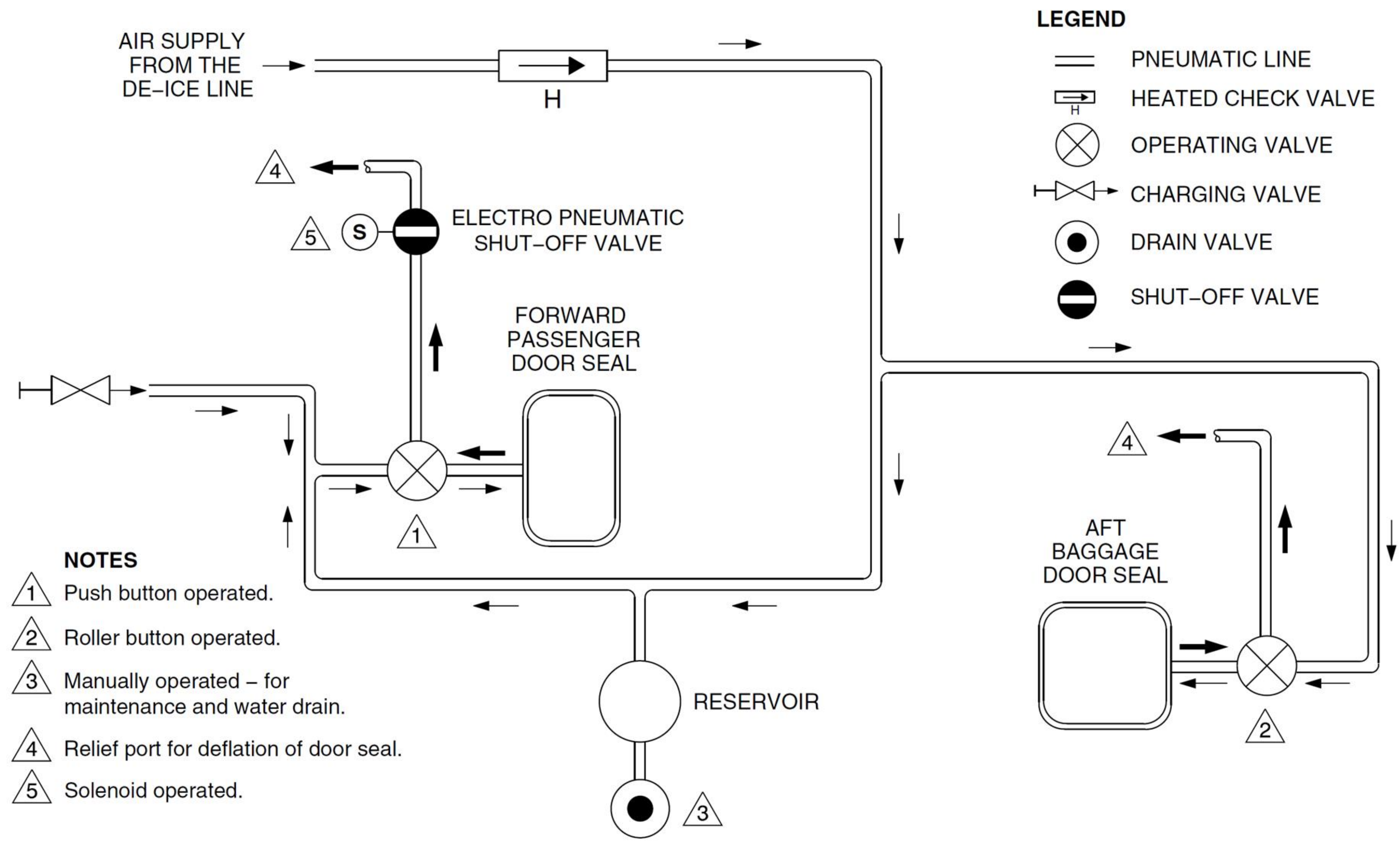


Figure 52-33 Door Seal Pressurization System

Tank, Reservoir

Refer Figure 52-34

The reservoir tank holds enough air for five door opening and closing sequences. It has a volume of 652 cubic inches (10.7 litres) when not pressurized by the deice system. It is located in the pressurized part of the centre fuselage under the cabin floor.

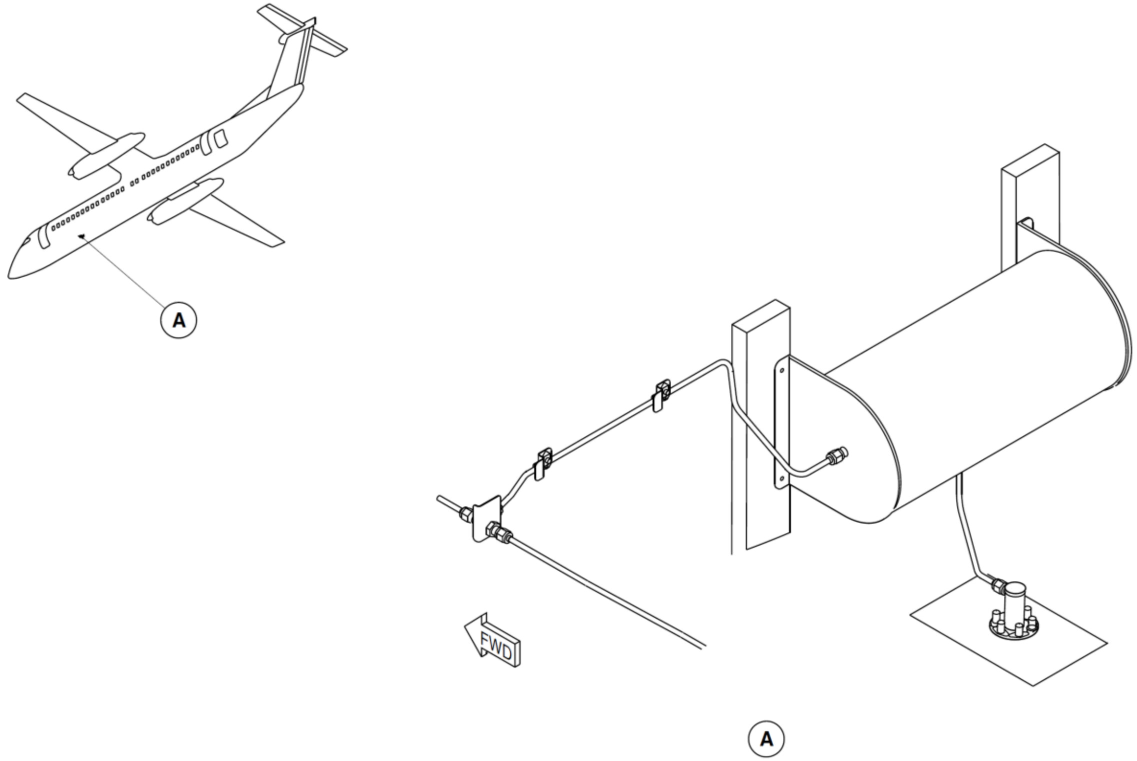
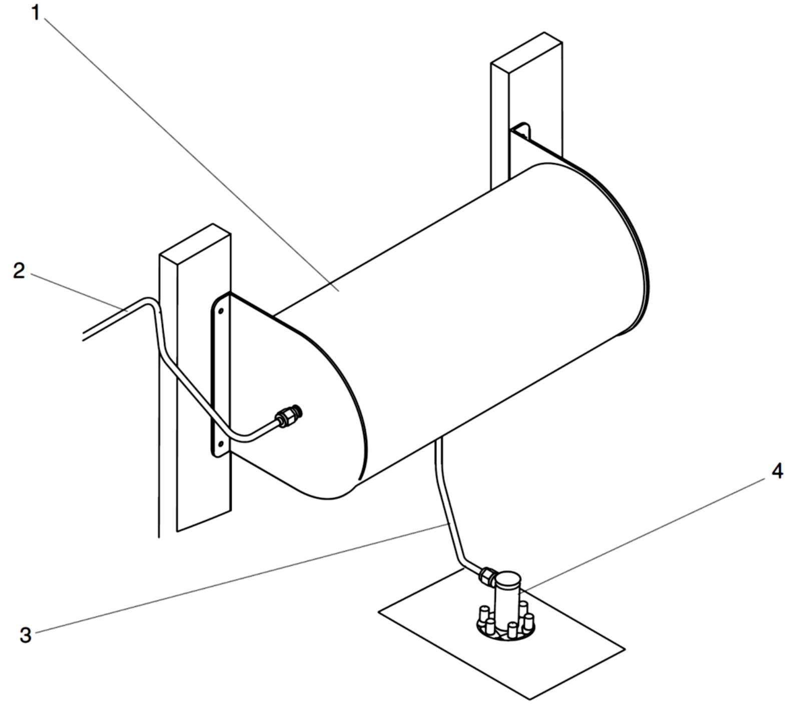


Figure 52-34 Reservoir Tank - Location

Valve, Drain

Refer Figure 52-35

The manually operated drain valve is used to depressurize the system before maintenance on the door seal system. It is also used to drain any water that may be in the reservoir. The valve is operated from the outside of the aircraft under the fuselage.



LEGEND
1. Reservoir Tank.
2. Door Seal Line.
3. Drain Line.
4. Drain Valve.

Figure 52-35 Drain Valve - Location

Check Valve, Heated

Refer Figure 52-36

The valve allows airframe deicing pressure to flow to the door seal system and prevents reverse flow of air, back to the deicing system. The valve is electrically heated to prevent moisture, which may be in the deicing air supply, from freezing inside the valve. The Timer Monitor Unit (TMU) of the deicing system controls the heater, with inputs from Static Air Temperature sensors 1 and 2. If both SAT sensors indicate that the temperature is greater than 9 °C, the heater is deactivated. The heater is activated when one SAT sensor indicates that the temperature is less than 5 °C. Failure of the heater or failure to activate the heater is detected by the TMU and the DEICE TIMER caution light comes on.

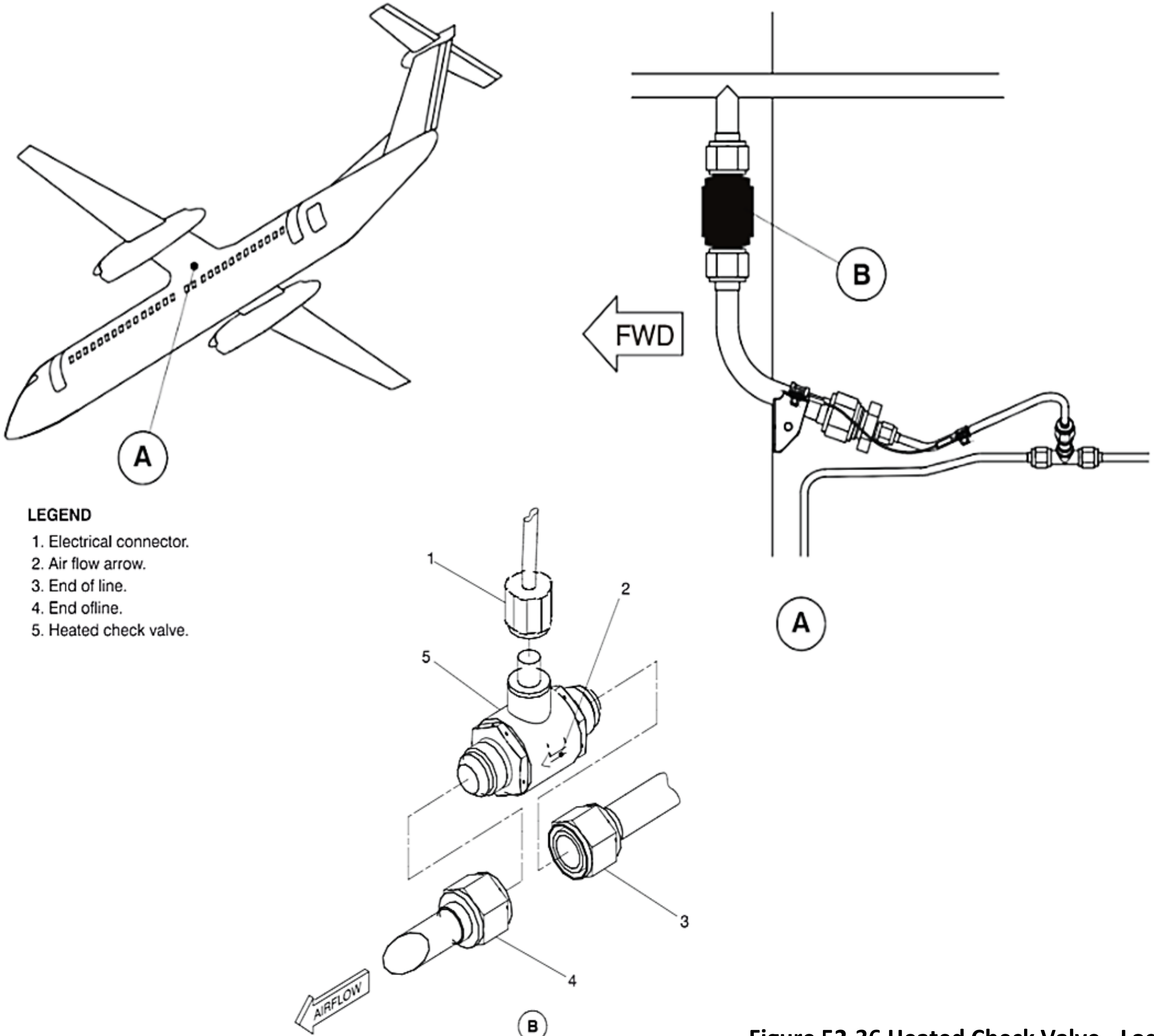


Figure 52-36 Heated Check Valve - Location

Valve, Charging

Refer Figure 52-37

The manually operated charging valve allows the system to be pressurized from an external source. This allows opening and closing of the two doors without operating an engine.

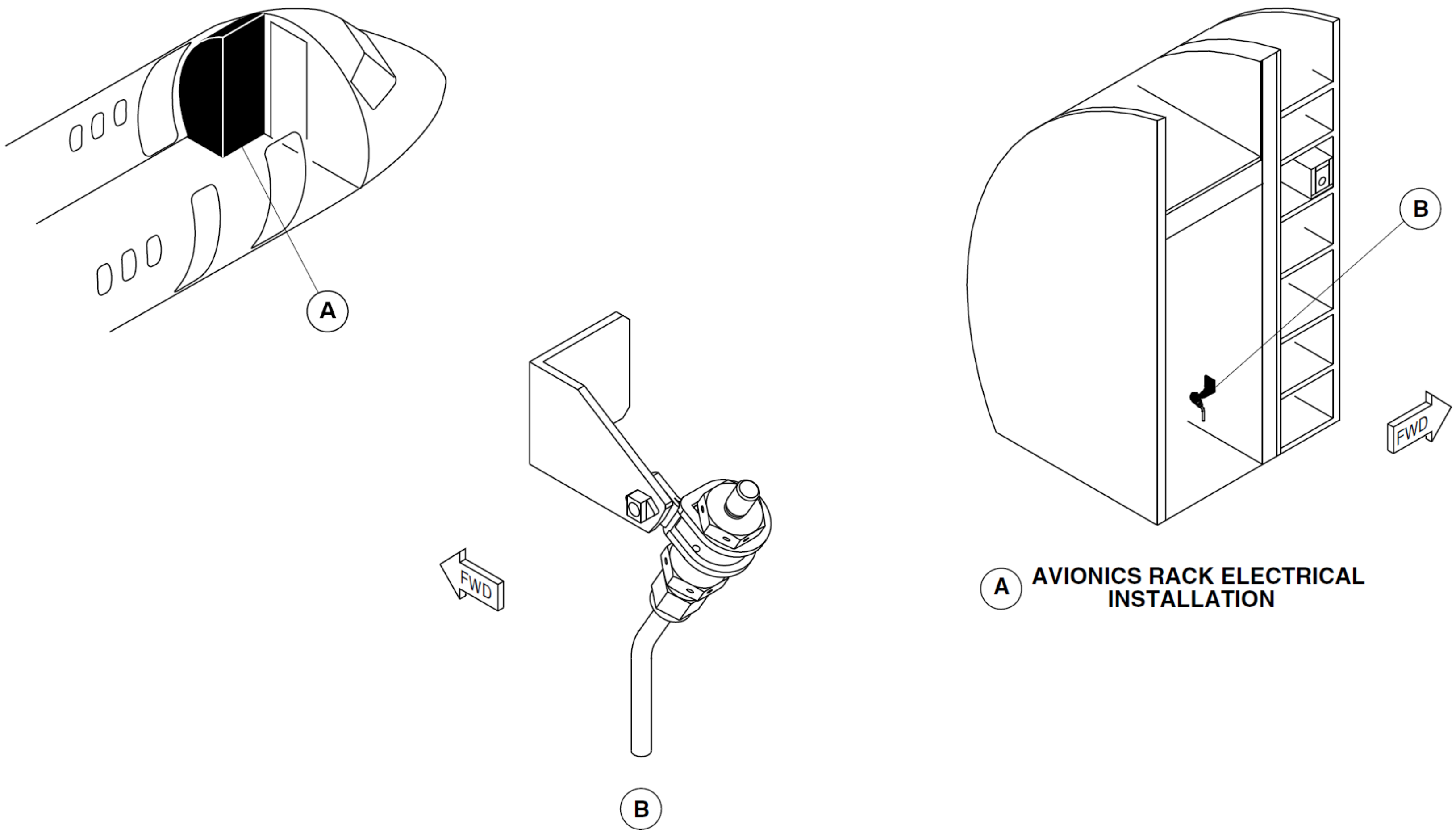


Figure 52-37 Charging Valve - Location

Lines

All the components in the system are connected by the aluminum plumbing lines.

Filter, Door Seal

On aircraft with MODSUM 4−113726 or SB 84−52−71 incorporated, a door seal filter, which contains a desiccant, is installed. The desiccant removes the moisture if present in the bleed air, before it enters the inflatable door seal of the aft baggage compartment and the air stair door.

On aircraft with SB84−52−91 or MODSUM 4−113942 incorporated, a door seal filter, which contains a desiccant, is not installed.

Valves, Control

The control valves allow inflation or deflation of the passenger and aft baggage door seals. Each control valve is installed on the appropriate doorframe, within the door mechanism.

The control valve is a three way mechanical valve. One port is connected to the pressure line, while the second port is connected to the seal. The third port is connected to the ambient air and acts as a relief port to deflate the seal.

The door handle mechanisms on the passenger and aft baggage doors operate their related control valves. When the handle is in the locked position, the seal port is connected to the pressure port and the seal is inflated.

When the handle is turned to open the door, the seal port is connected to the relief port and the seal deflates. The passenger door has a pushbutton which activates its control valve. The aft baggage door has a roller which activates its control valve.

Electro pneumatic shut−off valve

Refer Figure 52-38

The solenoid operated valve is installed on the relief port line of the forward passenger door seal pressurization valve. On the ground when the door is unlocked, the handle mechanism operates the control valve and connects the door seal port to the relief port inside the valve and the seal deflates. The seal deflation pressure passes through this shut−off valve to the ambient air.

The primary function of this electro pneumatic shut−off valve is to prevent depressurization of the door seal system if airspeed is more than 100 knots (airspeed > 100 knots). If the airspeed is 90 knots or on the ground the valve will allow the door seal system to be depressurized. This feature prevents inadvertent door seal depressurization for the forward passenger door. The airspeed signals to the valve are provided by the aircraft air data system.

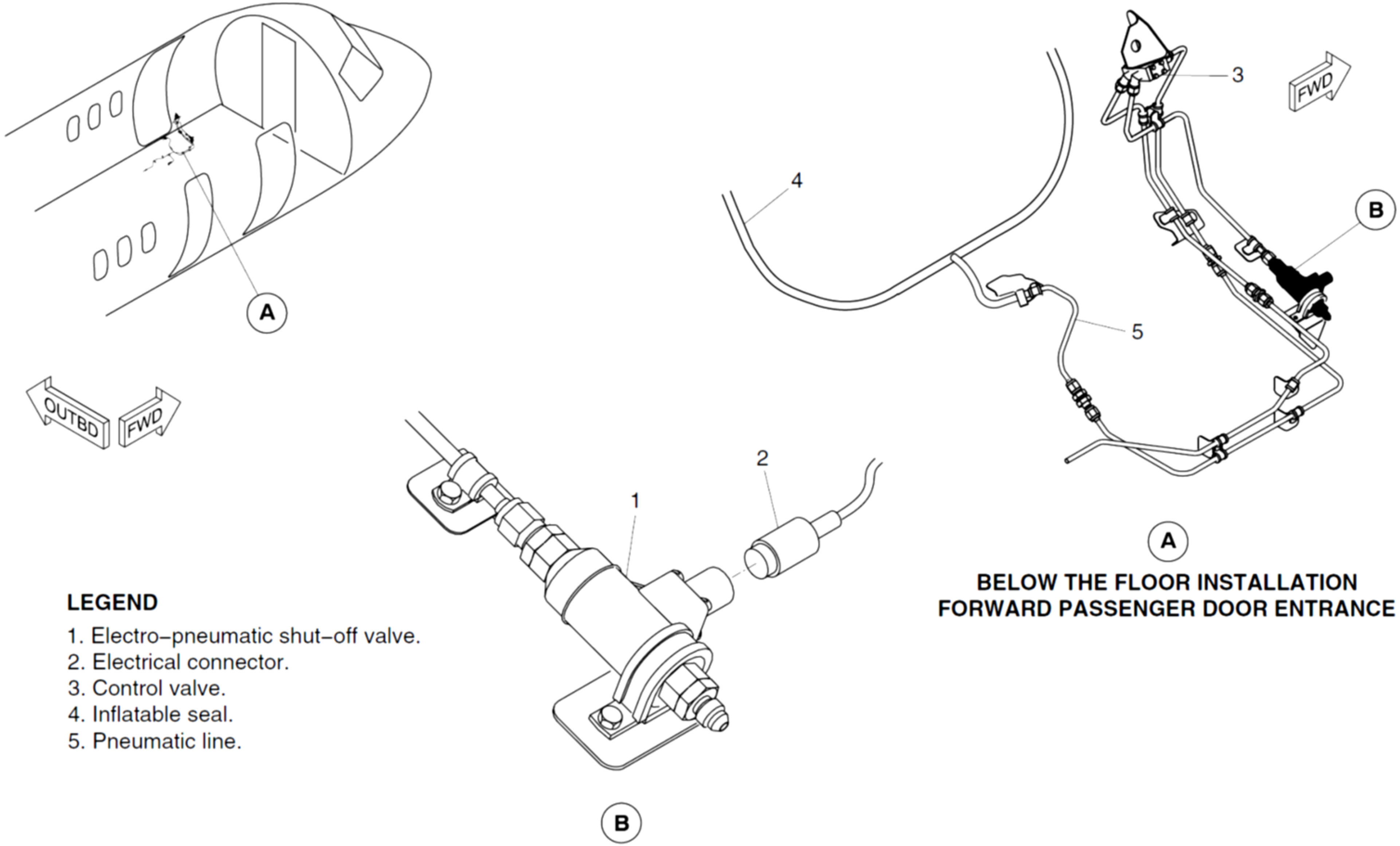


Figure 52-38 Electro Pneumatic Shut-off Valve - Location

Emergency Exits

Introduction

The emergency exits gives access to and from the passenger and flight compartments for emergency evacuation.

General Description

Refer Figure 52-39 & Refer Figure 52-40

There are four emergency exits located in the passenger compartment, three Type I and one Type II/III. On aircraft with ModSum 4−190561 incorporated, the type II/III emergency exit door is eliminated. The door warning system monitors emergency exit door status and supplies visual indication and warning in the flight compartment. There is an emergency exit located in the flight compartment roof.

The three Type I exits are:

- Forward Passenger door
- Aft Passenger door
- Aft Service door.

The emergency exits include the components that follow:

- Flight Compartment Escape Hatch
- Upper Forward Emergency Type II Exit Door
- Lower Forward Emergency Type III Exit Door

On aircraft with the ModSum 4−458926 or 4−459339 or 4−190614 incorporated there are four type−I emergency exits located in the passenger compartment. The door warning system monitors emergency exit door status and supplies visual indication and warning in the flight compartment. There is an emergency exit located in the flight compartment roof.

The four Type I exits are:

- Forward Passenger door
- Aft Passenger door
- Aft Service door
- Forward Emergency Exit Type I door

Detailed Description

All the Type I and the Type II/III emergency exits have both internal and external handles. The flight compartment escape hatch is the only emergency exit which does not have an external handle.

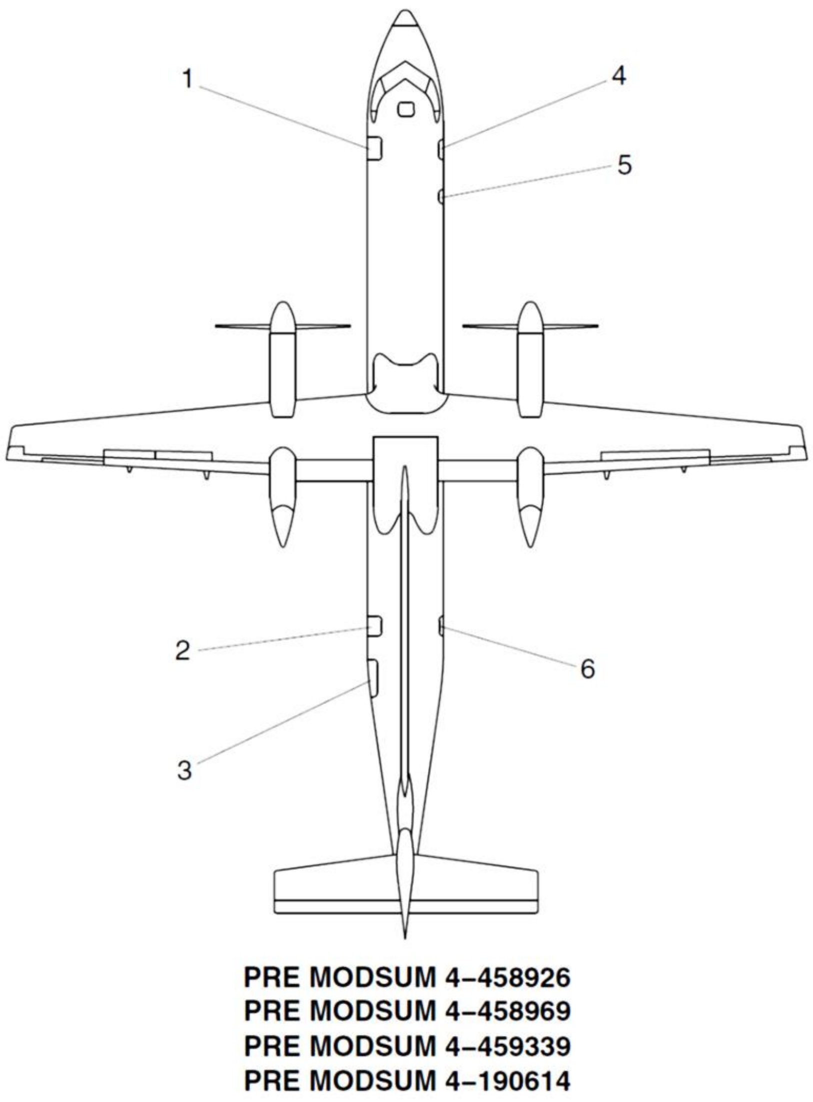
Type of Exits

The types of exits are defined as follows:

- **Type I**: Floor level exit with a rectangular opening of not less than 24 in. (610 mm) wide by 48 in. (1219 mm) high.

- **Type II:** Rectangular opening of not less than 20 in. (508 mm) wide by 44 in. (1118 mm) high. Type II exits must be floor level unless located over wing.

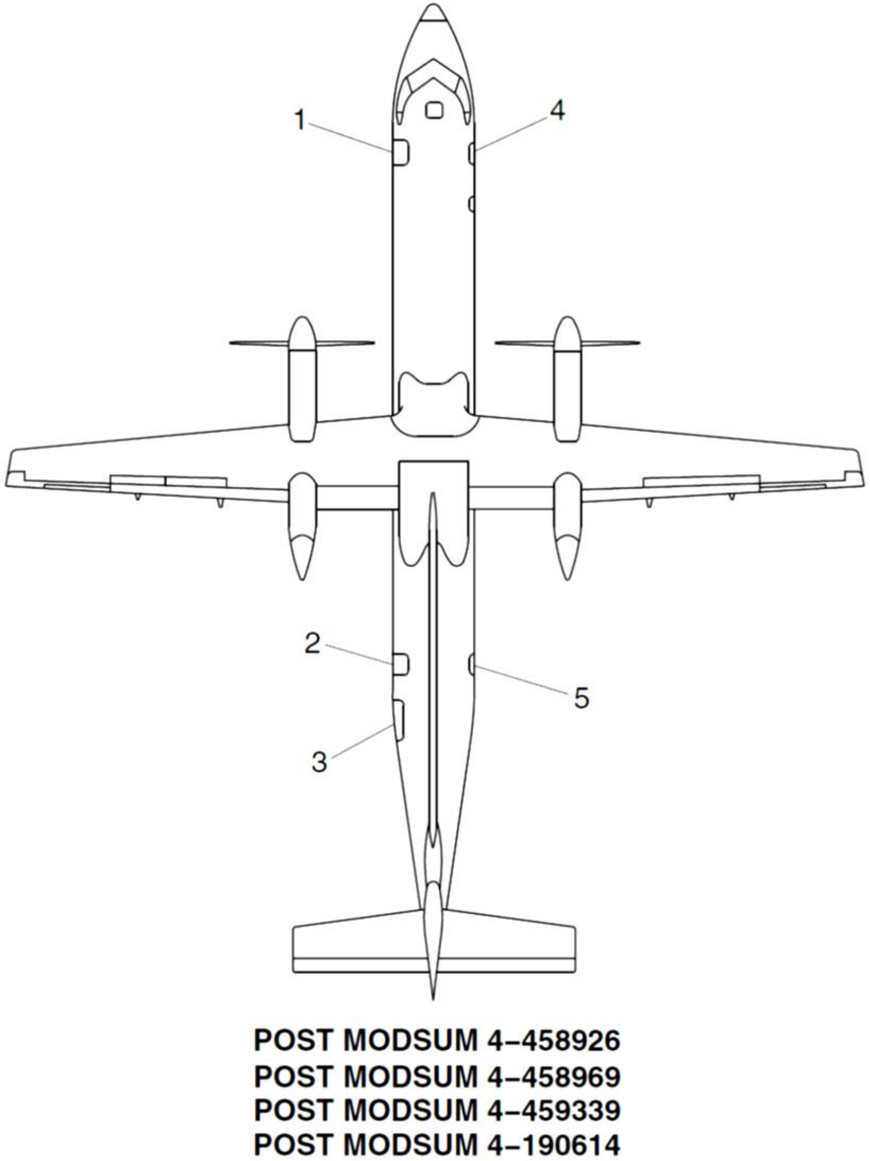
- **Type III:** Rectangular opening of not less than 20 in. (508 mm) wide by 36 in. (914 mm) high. The step up inside the airplane may not be more than 20 in. (508mm).



LEGEND

1. Forward Passenger Door.
   "Closed" Sensor and "Locked" Sensor.
2. Aft Passenger Door.
   "Closed" Sensor and "Locked" Sensor.
3. Aft Baggage Door.
   "Closed" Sensor and "Locked" Sensor.
4. Forward Baggage Door.
   "Closed" Sensor and "Locked" Sensor.
5. Upper (Type II) Emergency Exit.
   "Locked" Sensor.
6. Aft Service Door.
   "Closed" Sensor and "Locked" Sensor.

Figure 52-39 Emergency Exits – Detailed 1 of 2



LEGEND

1. Forward passenger door
   "Closed" sensor and "Locked" sensor.
2. Aft passenger door
   "Closed" sensor and "Locked" sensor.
3. Aft baggage door
   "Closed" sensor and "Locked" sensor.
4. Forward RHS type I
   "Closed" sensor and "Locked" sensor.
5. Aft service door
   "Closed" sensor and "Locked" sensor.

Figure 52-40 Emergency Exits – Detailed 2 of 2

Emergency Exits

Refer Figure 52-41, Refer Figure 52-42 & Refer Figure 52-43

The forward and aft passenger doors are located on the left side of the fuselage and are also used as emergency exits. The forward passenger door opens outwards and down. The aft passenger and aft service door open outward and forward. The aft service door is located on the right side of the aircraft at the rear of the passenger compartment.

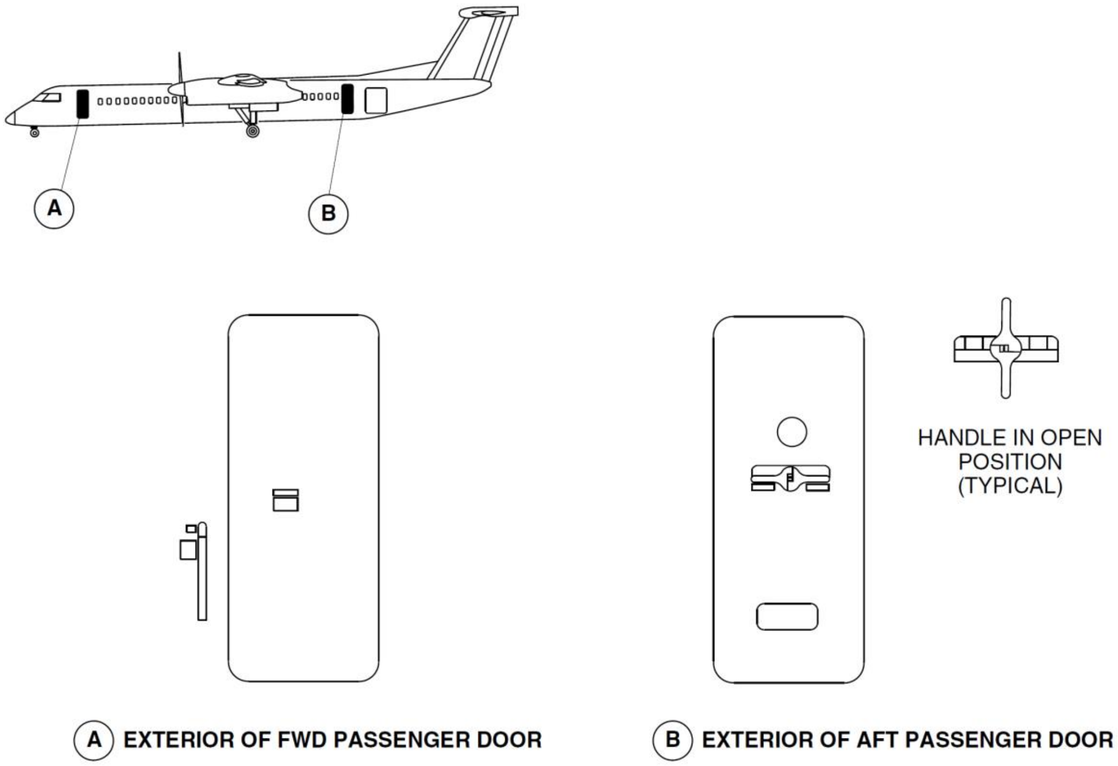


Figure 52-41 Emergency Exits, Passenger Doors – Locations

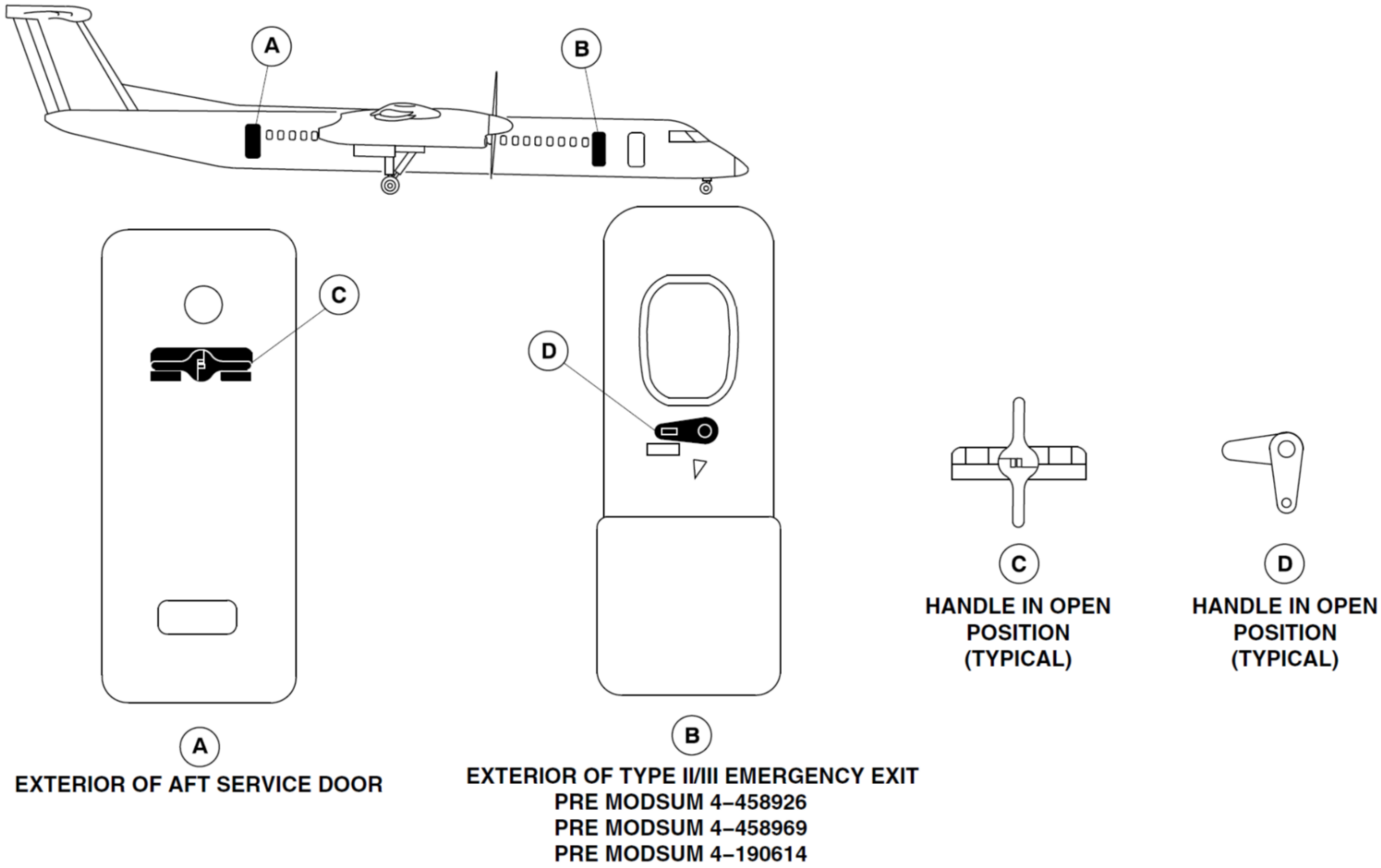


Figure 52-42 Aft Service, Type II/III Door

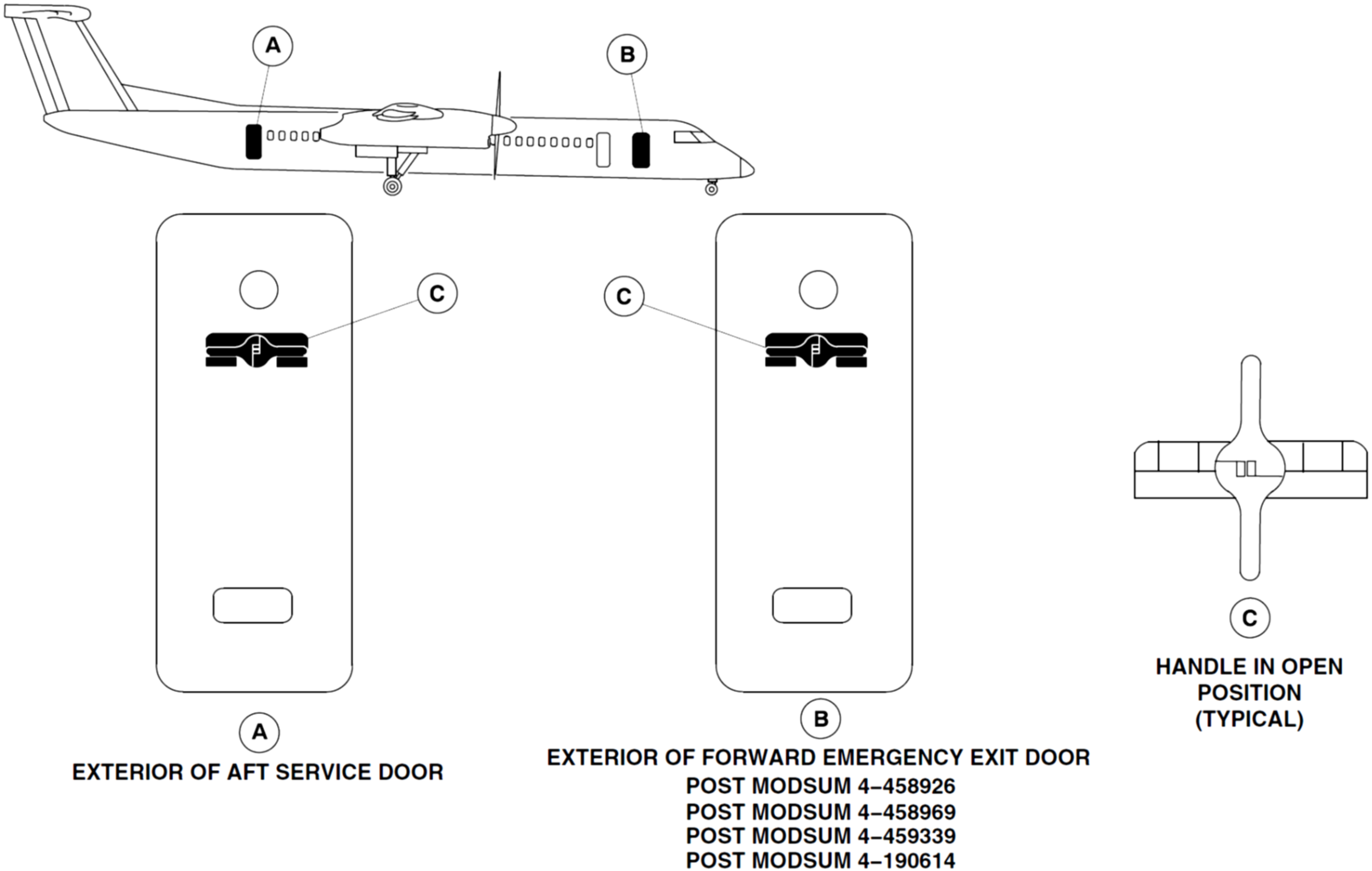


Figure 52-43 Aft Service, Type I Door

Refer Figure 52-44

The flight compartment escape hatch is located in the flight compartment roof, aft of the pilot seats. It is completely detachable, or can be partially opened to increase the airflow in the flight compartment when the aircraft is on the ground. A rope helps the flight crew exit the flight compartment aircraft when using the flight compartment emergency escape hatch.

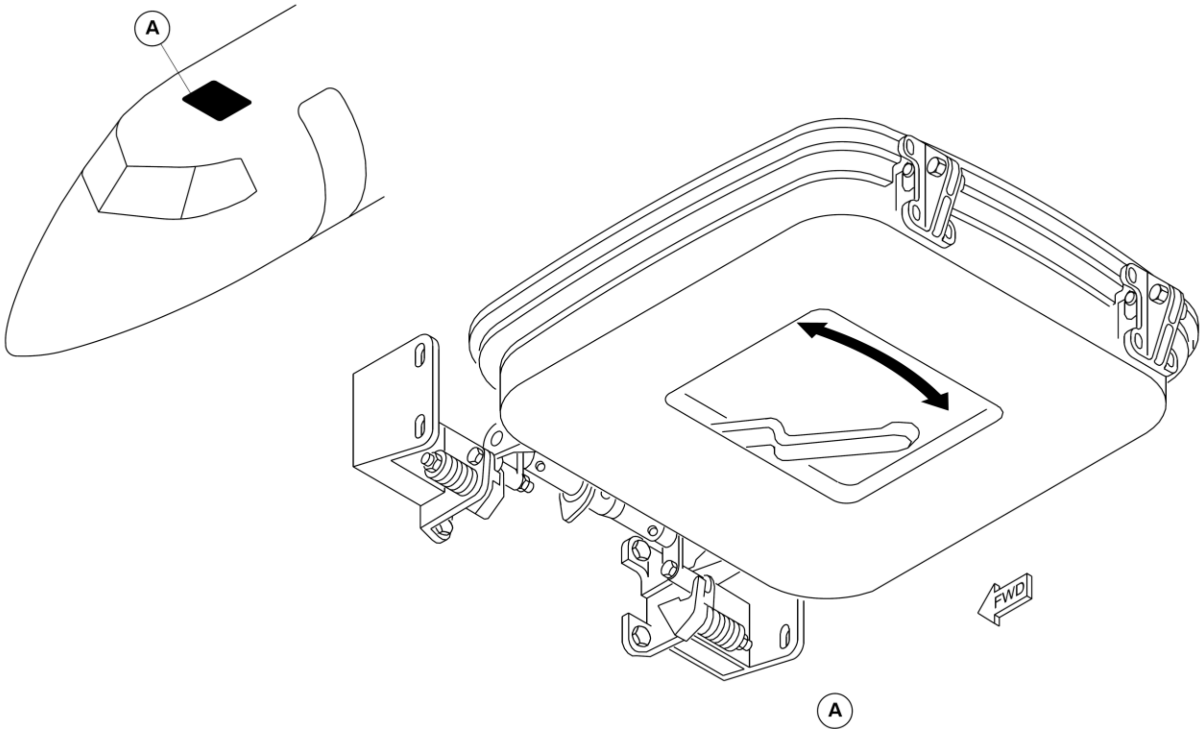


Figure 52-44 Flight Compartment Escape Hatch – Location

Refer Figure 52-45

The type II/III emergency exit doors are located on the forward right side of the passenger compartment. The emergency exit has an upper and lower door. The upper door is a plug type door which is removed completely for opening. The lower door is hinged at the bottom, and opens outward and downward from the floor level.

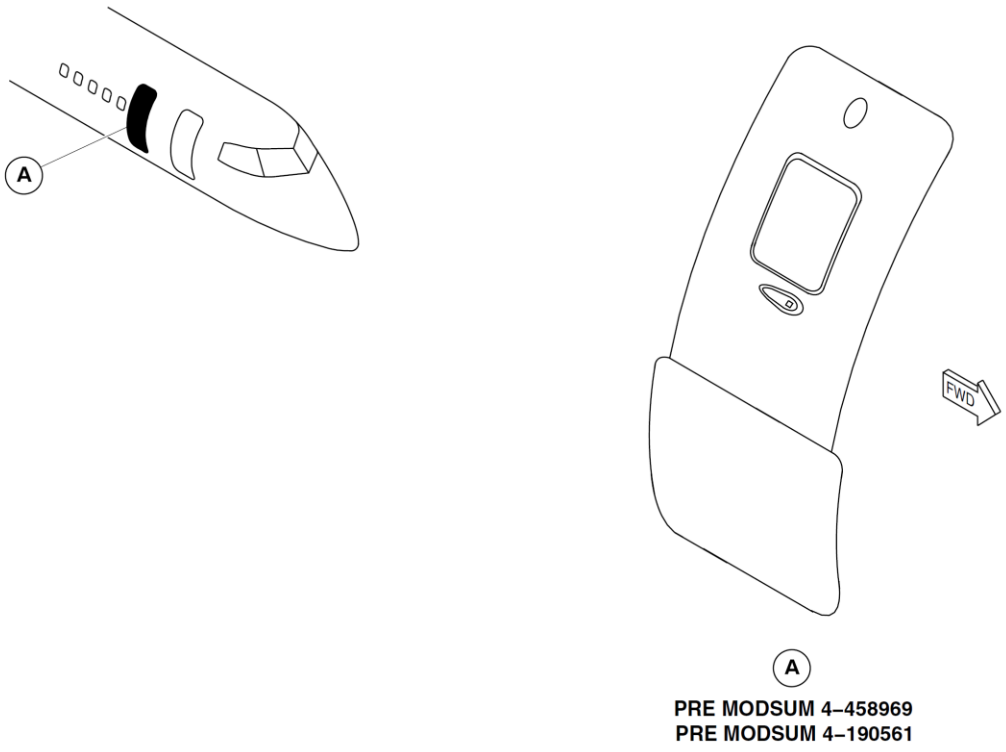


Figure 52-45 Type II/III Emergency Exit – Location

Refer Figure 52-46

On aircraft with ModSum 4−458926 or 4−459339 or 4−190614 incorporated, a type I emergency exit door is located on the forward right side of the passenger compartment. The emergency exit opens outward and forward.

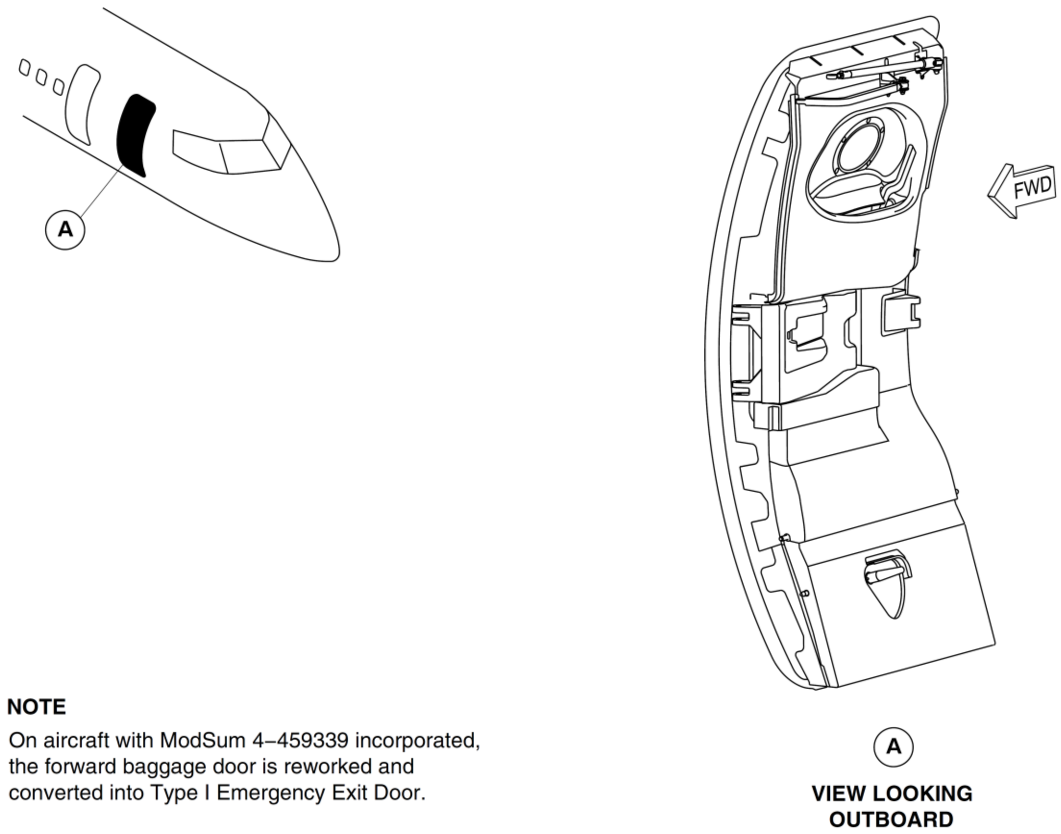


Figure 52-46 Forward Type I Emergency Exit – Location

Forward Passenger Door

Refer Figure 52-47

The forward passenger door is located at the left side of the fuselage at the front of the passenger compartment and is classified as a type I emergency exit.

The forward passenger door is hinged at the bottom and has five steps built into the inboard side of the door structure. The steps also strengthen the door. This door has an external aluminum alloy skin braced by cross beams and intercostal.

The door seal pressurization system prevents cabin air leakage through the forward passenger door.

Door pressurization loads are carried by ten adjustable pressure stop bolts, five on each side. The stop bolts engage with stop brackets on the door surround structure.

The forward passenger door is the semi−plug type. On opening, the initial movement of the door is upward and inward. The door then rotates outward about the door’s hinge.

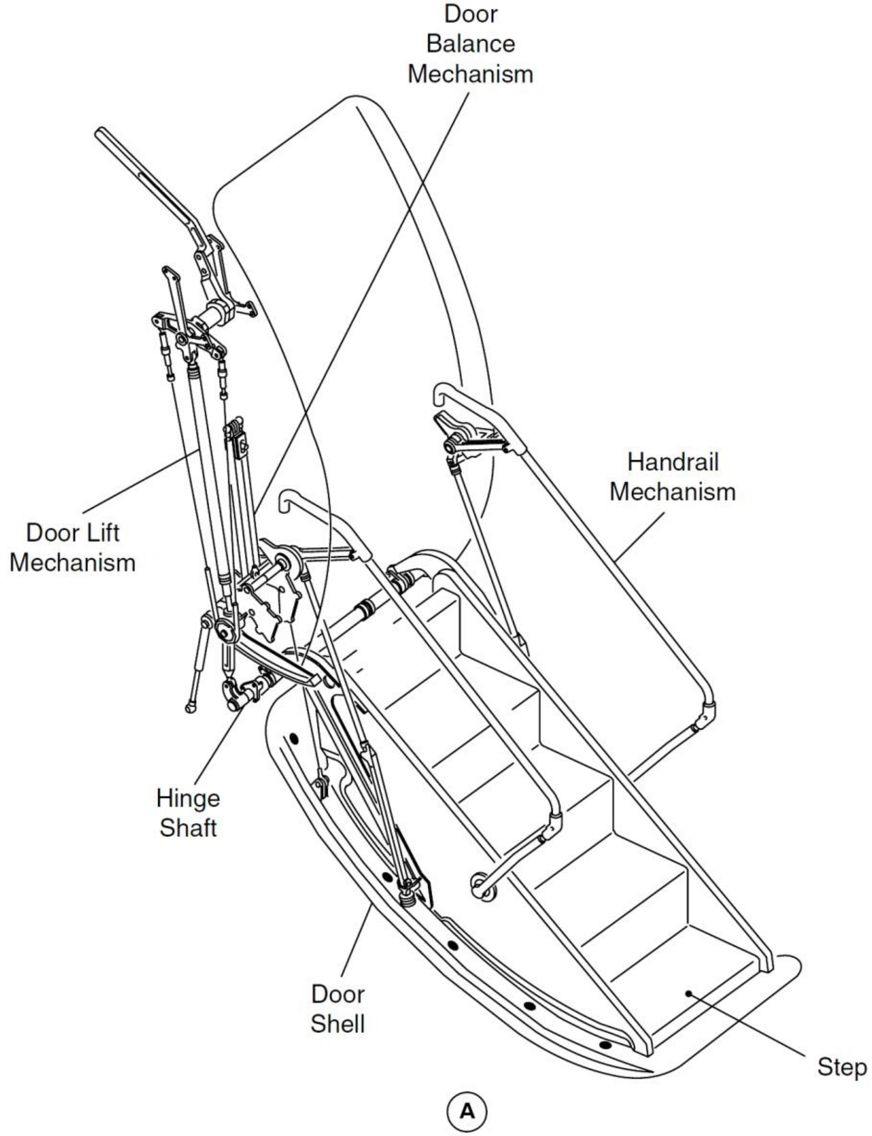


Figure 52-47 Forward Passenger Door

Aft Service Door

Refer Figure 52-48

The aft service door is located on the right side of the aircraft at the rear of the passenger compartment and is classified as a Type I emergency exit. It is a translating type secured against pressurization loads by five stops on each vertical edge, that engage with five related stop brackets installed on the door surround.

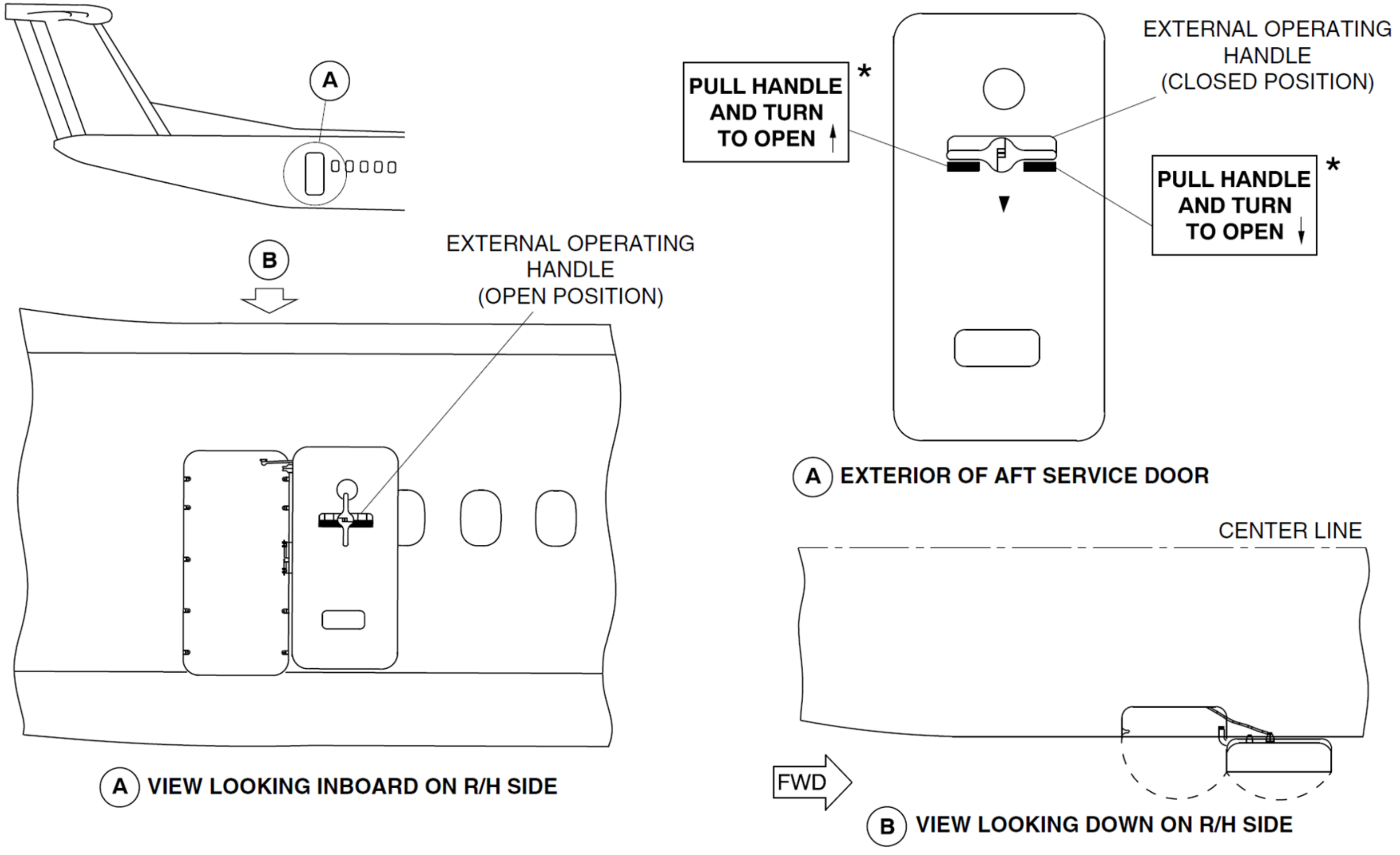


Figure 52-48 Aft Service Door - Operation

Forward Emergency Exit Type I Door

[Refer Figure 52-49]

On aircraft with ModSum 4−458926 or 4−459339 or 4−190614 incorporated a forward emergency exit type I door is located on the right side of the forward fuselage. The forward emergency exit type I door is classified as a Type I emergency exit. It is a translating type door and is secured against pressurization loads by five stops on each vertical edge that engage with five related stop brackets installed on the door surround structure.

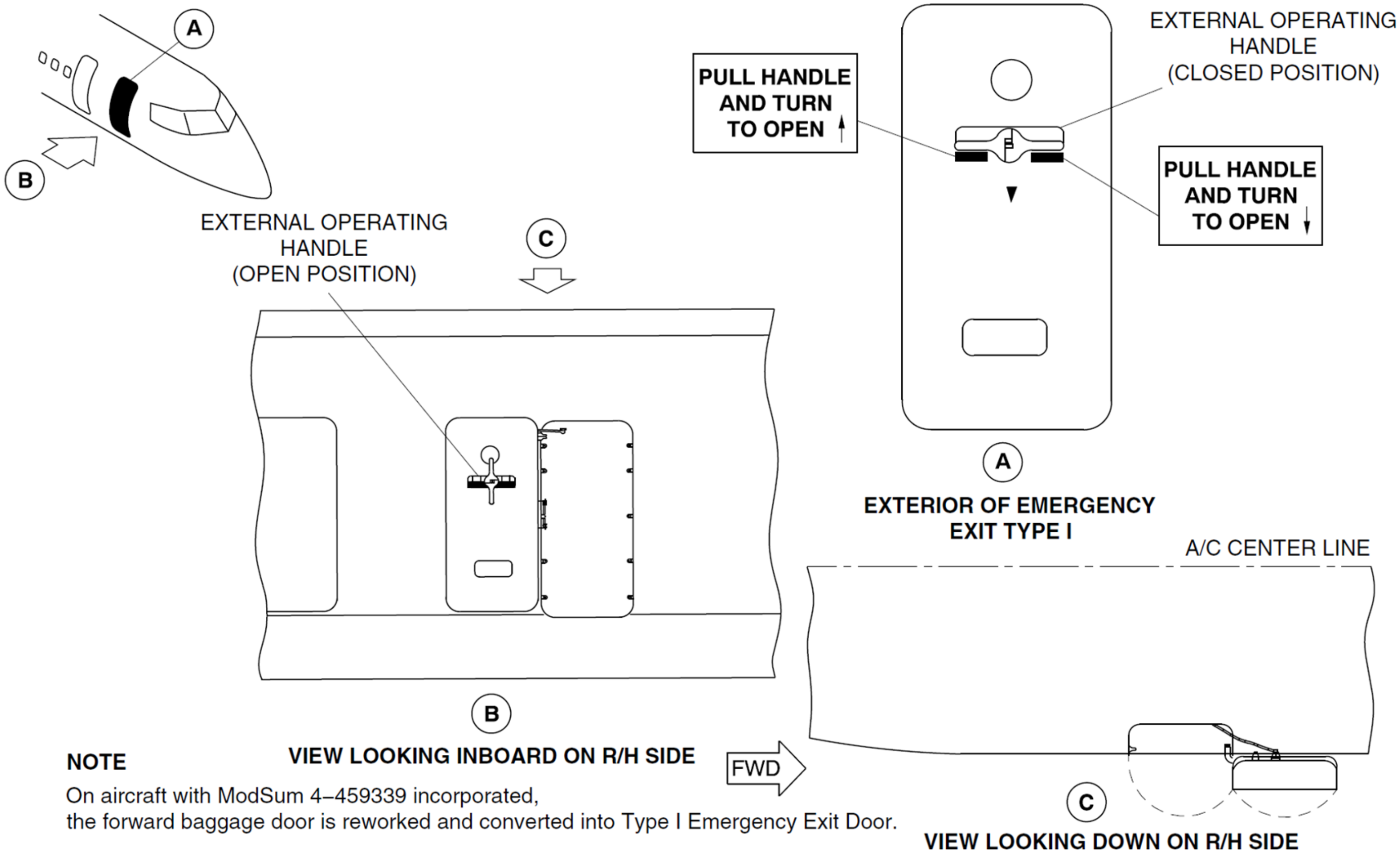


Figure 52-49 Forward Emergency Exit Type I Door - Operation

Flight Compartment Escape Hatch

Refer Figure 52-50

The flight compartment escape hatch is opened by turning the handle counterclockwise to the open position. The hatch is opened to permit depressurization or to increase airflow in the flight compartment when the aircraft is on the ground. The door movement is controlled by the geometry of the torque tube and rollers. The hatch pivots about the rear fittings and opens approximately one inch at the front.

A pull force on the handle of approximately 40 lb. (18.1 kg), releases the rollers against the action of the locking and release springs, and the hatch can be removed from the flight compartment roof.

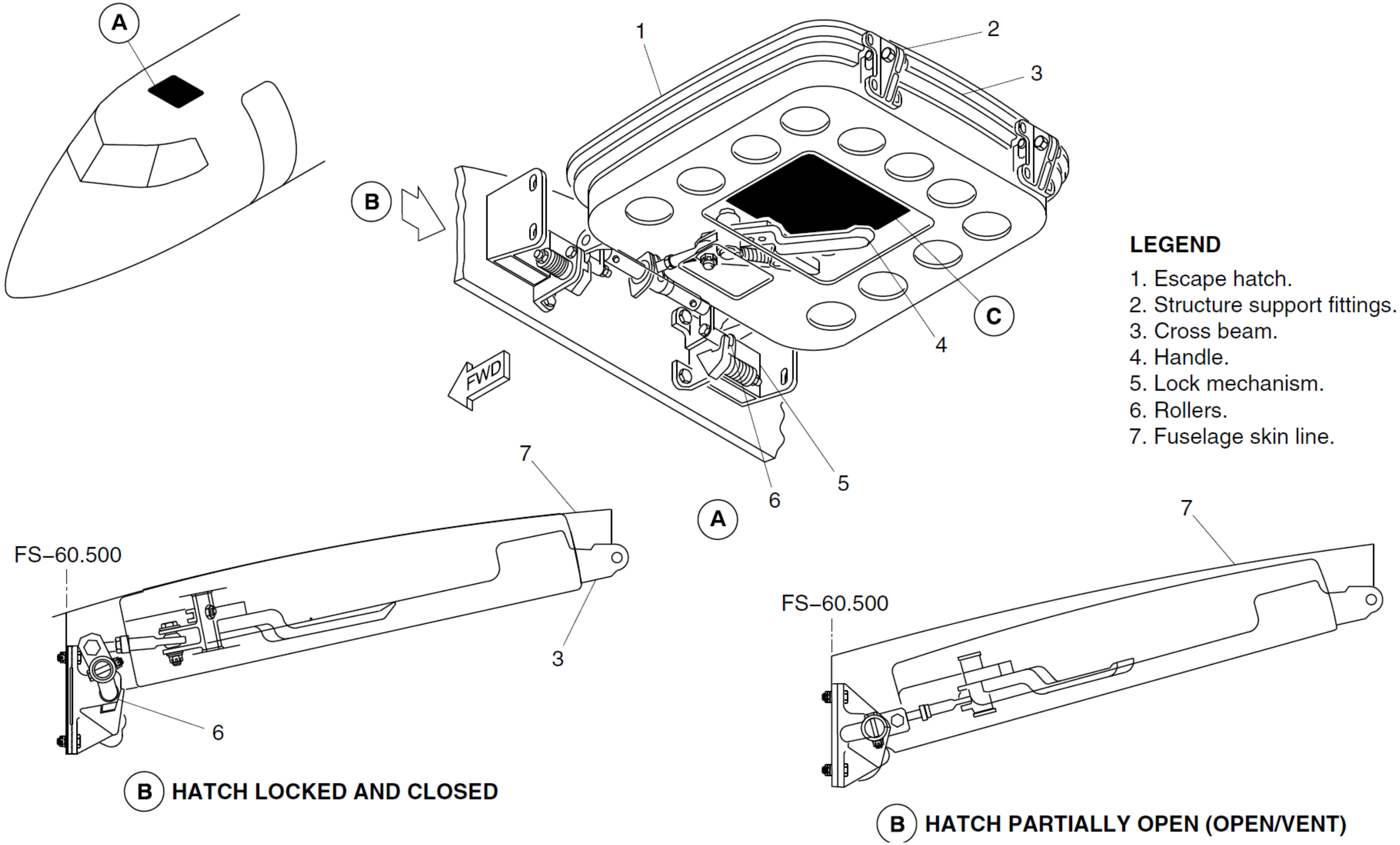


Figure 52-50 Flight Compartment Escape Hatch - Operation

Type II/III Exit Doors

Refer Figure 52-51

The upper emergency exit door is secured by a lock pin and three adjustable stops on each vertical edge that engage with three stop brackets installed on the sides of the door. To secure it against cabin pressure loads. The lower emergency exit door is installed on a piano hinge at the lower edge. It is kept in the closed position by four stops on the upper edge of the door that engage with their related stop brackets on the lower edge of the upper door. The lower door opens automatically when the upper door is removed, and the lower door handle is in the GROUND position. The lower door stays closed when the lower door handle is in the WATER position.

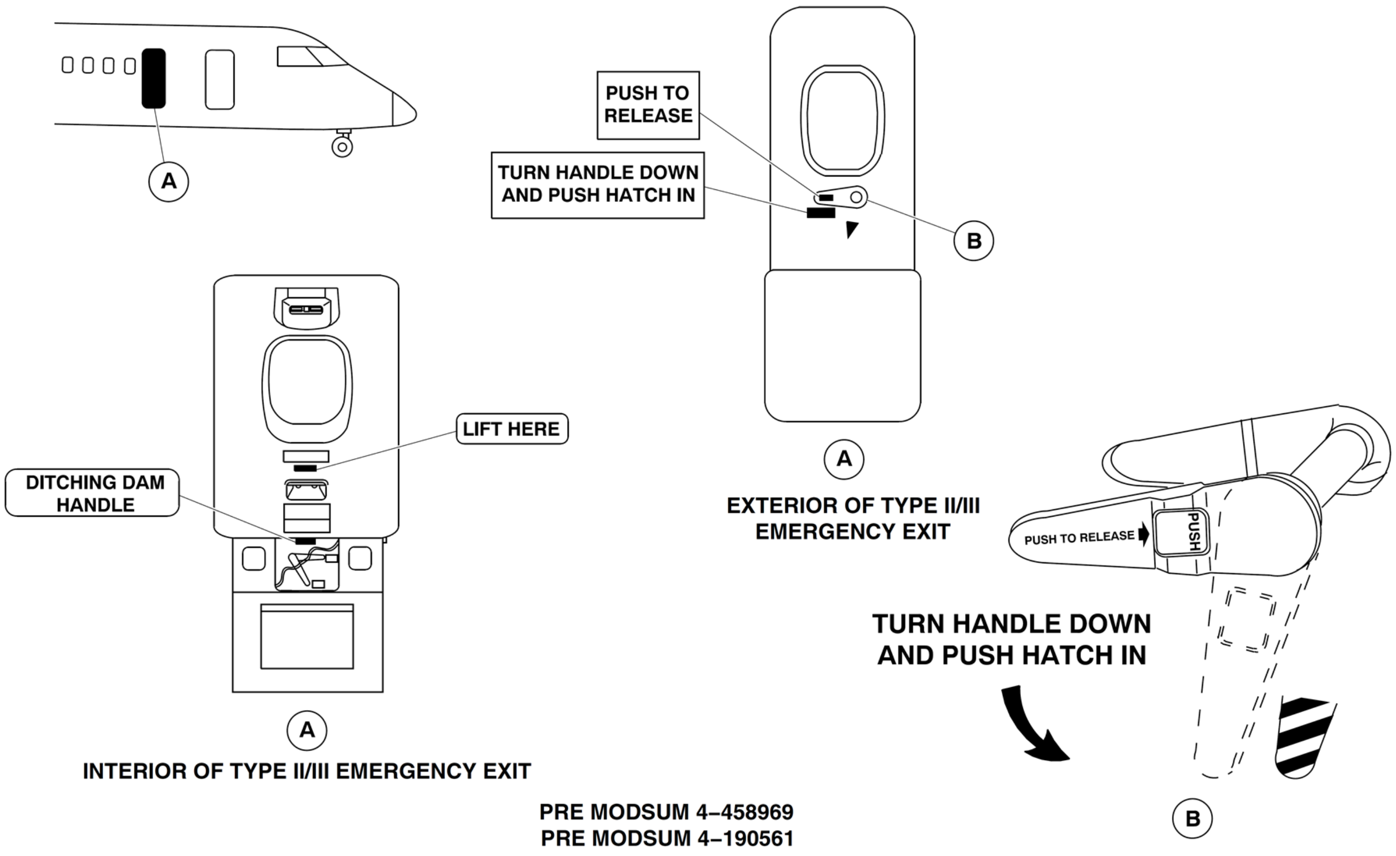


Figure 52-51 Type II/III Exit Doors - Operation

Door Warning System

Refer Figure 52-52

The door warning system monitors the position of the emergency exit doors that follow:

- Forward Passenger Door
- Aft Passenger Door
- Type II/III Emergency Exit Doors
- Aft Service Door

On aircraft with ModSum 4−458926 or 4−459339 or 4−190614 incorporated, the door warning system also monitors the emergency exit doors that follow:

- Forward Passenger Door
- Aft Passenger Door
- Forward Emergency Exit Type I Door
- Aft Service Door

When any of the Type I or TYPE II emergency exit doors are unlocked, both the FUSELAGE DOORS warning light and the MASTER WARNING light flash. The Door Systems page on the Multi-Function Display (MFD) shows the unlocked door as red filled rectangle with a related legend. There is no visual warning if the flight compartment hatch is unlocked.

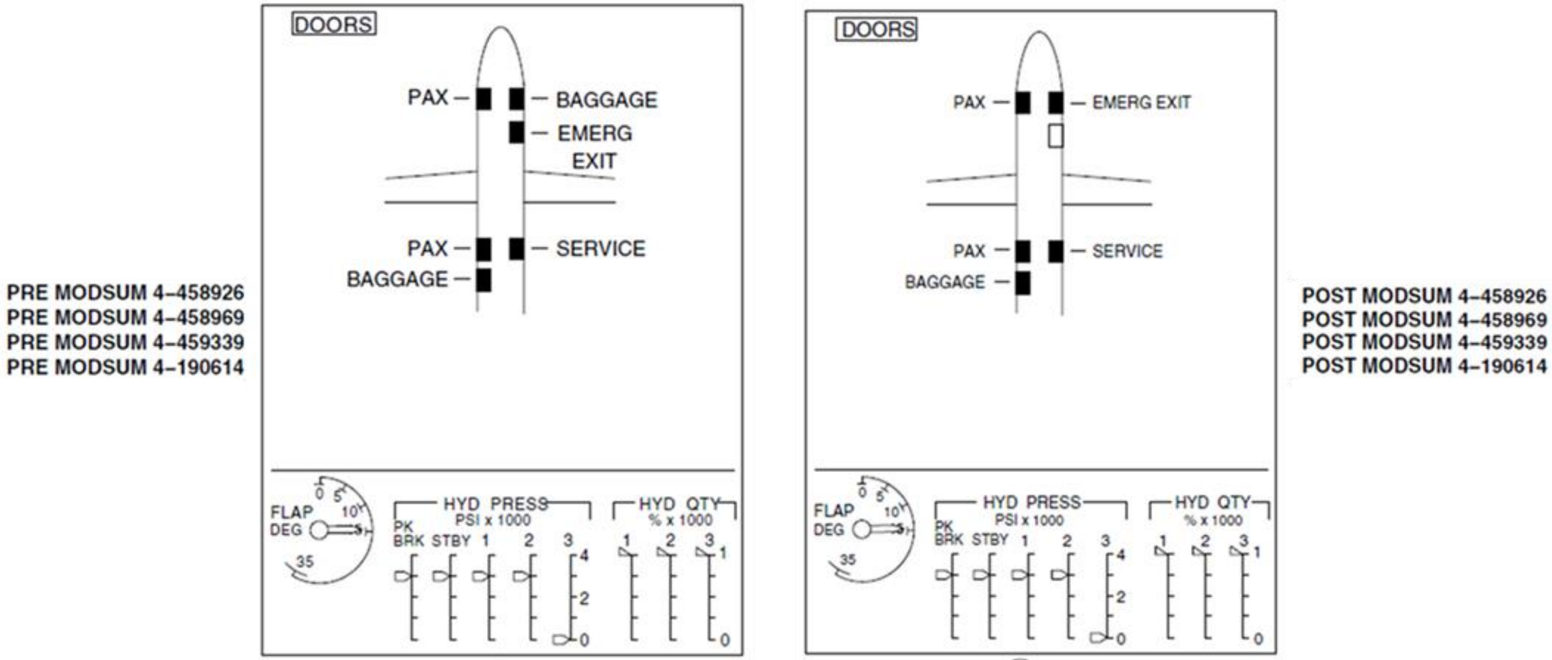


Figure 52-52 Doors System Page – Doors Opened

Flight Compartment Escape Hatch

Introduction

The flight compartment escape hatch gives emergency access and ventilation to the flight compartment.

General Description

The flight compartment escape hatch is located in the flight compartment roof, aft of the pilot seats. It is completely detachable, or can be partially opened to increase the airflow in the flight compartment when the aircraft is on the ground.

The hatch is installed in the flight compartment roof by two support fittings at the rear, and is secured at the front by two locking and release arms. An operating handle, located in the center of the hatch, is kept in an open or closed position by an overcentre spring.

A seal installed around the edge of the hatch to prevent pressurization leakage.

The flight compartment escape hatch includes the components that follow:

- Latch Mechanism
- Door Seal

Detailed Description

Refer Figure 52-53

The handle is turned counterclockwise to open the hatch. The hatch is opened to permit depressurization or to increase airflow in the flight compartment when the aircraft is on the ground.

The door movement is controlled by the geometry of the torque tube and rollers. The hatch pivots about the rear fittings and opens approximately 1 in. (25.4 mm) at the front.

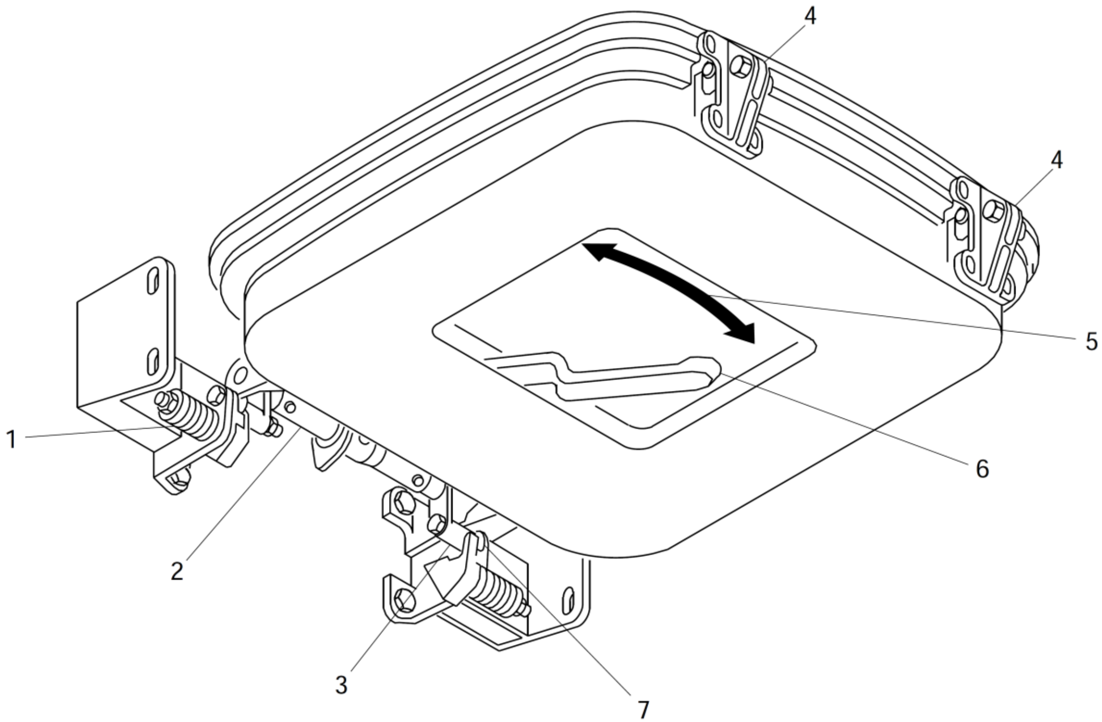
A pull force on the handle of approximately 40 lb (18.1 kg), releases the rollers against the action of the locking and release springs, and the hatch can be removed from the flight compartment roof.

Latch Mechanism

The latch mechanism controls opening and closing of the hatch. When the handle is moved to the open position, the mechanical linkage connected to the handle turns the torque tubes. The rollers then engage in detents on the structure to let the hatch door partially open.

Door Seal

The door seal helps to prevent pressurization leakage when the hatch is closed. The seal is installed around the edge of the hatch.



LEGEND

1. Arm/spring detent mechanism.
2. Torque shaft assembly.
3. Arm.
4. Rear support fittings.
5. Instructional label.
6. Operating handle.
7. Roller.

Figure 52-53 Flight Compartment Escape Hatch

Forward Emergency Type II Exit Door

Introduction

The type II emergency exit doors supply an exit for emergency evacuation of the passenger compartment.

General Description

On aircraft with Modsum 4−458969 incorporated, the type II/III door is disabled. The internal handles are removed and the external handle is locked to make the door inoperable.

The type II emergency exit doors are located on the forward right side of the passenger compartment. The emergency exit has an upper and lower door. The upper door is a plug type door which is removed completely for opening. The lower door is hinged at the bottom, and opens outward and downward from the floor level.

The door warning system monitors door status and supplies visual indication and warning in the flight compartment. The indications are shown in:

- Door Warning System

The upper and lower emergency Type II exit doors has the components that follow:

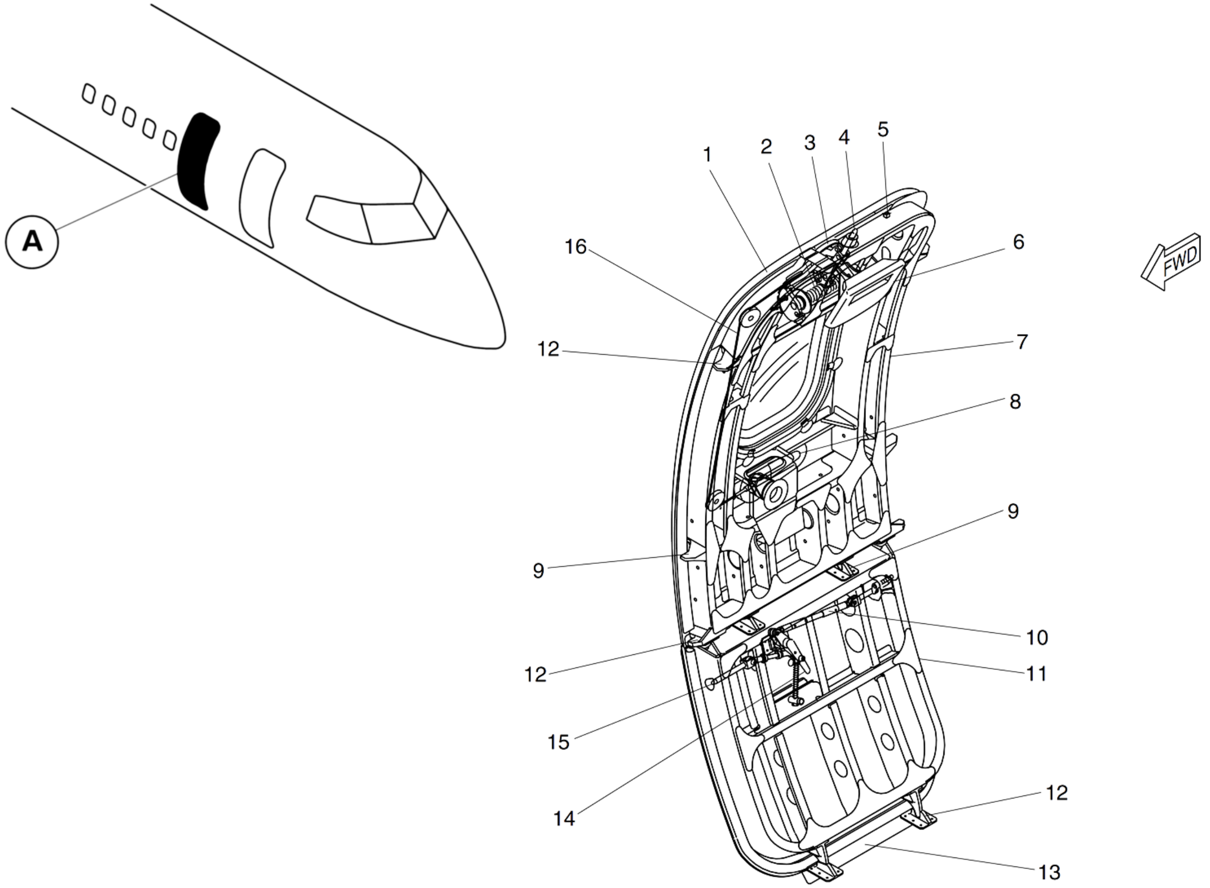
- Latch mechanism
- Door seal

Detailed Description

Refer Figure 52-54

The upper door is secured by a lock pin and three adjustable stops on each vertical edge that engage with three stop brackets installed on the sides of the door. To secure it against cabin pressure loads. The lower door is installed on a piano hinge at the lower edge. It is kept in the closed position by four stops on the upper edge of the door that engage with their related stop brackets on the lower edge of the upper door.

The lower door opens automatically when the upper door is removed, and the lower door handle is in the GROUND position. The lower door stays closed when the lower door handle is in the WATER position.

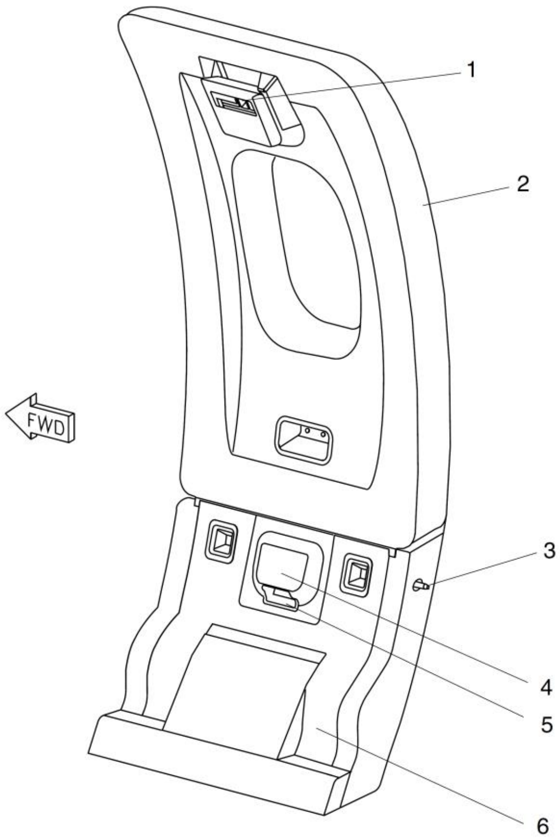


LEGEND

1. Door Seal.
2. Bias Spring.
3. Vent Dish.
4. Lock Pin.
5. Proximity Switch.
6. Internal Handle.
7. Upper Door.
8. External Handle.
9. Backup Stop.
10. Linkage at Lower Door
11. Lower Door.
12. Main Stop.
13. Hinge.
14. Lower Door Handle.
15. Shoot Bolt.
16. Cable Assembly.

Figure 52-54 Type II Emergency Exit Door

Refer Figure 52-55
The upper door is unlocked by operation of the internal or external handle. Both handles are spring−loaded and interconnected by cables and pulleys. When the internal handle of the upper door is pulled, the vent dish opens and pressure in the aircraft is released to the atmosphere. The locking pins retract and the door can then be removed.



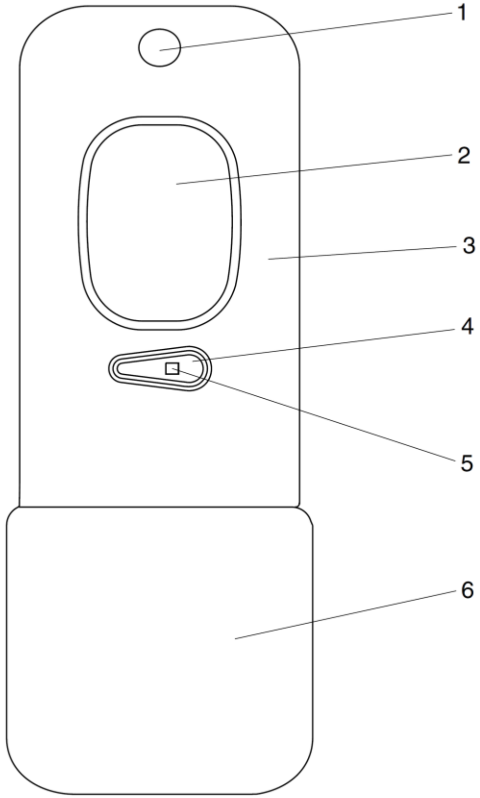
LEGEND

1. Upper Door Internal Handle.
2. Upper Door.
3. Lower Door Shoot Bolt.
4. Lower Door Handle Cover.
5. Lower Door Handle.
6. Lower Door.

Figure 52-55 Type II Emergency Exit Door – Internal Detailed

Refer Figure 52-56

To remove the upper door using the external handle, operate the pushbutton and the handle springs out. The handle is turned counterclockwise to operate the cable and shaft quadrant to open the vent dish and retract the locking pin. The door is pushed inward and removed.



LEGEND

1. Vent Dish.
2. Window.
3. Upper Door.
4. External Handle.
5. Push Button.
6. Lower Door.

Figure 52-56 Type II Emergency Exit Door – External Detailed

Refer Figure 52-57

The lower door handle has two positions, GROUND and WATER. The handle is normally in the GROUND position, and lets the lower door open automatically when the upper door is removed. When the handle is moved to the WATER position, the two shoot bolts engage with the fittings on the door structure to prevent the opening of the lower door.

The lower door handle has an overcentre spring to hold the handle in the engaged, or disengaged position. A stop cable, made of nylon rope, is installed between the forward face of the lower door and the door cut−out structure, to keep the door from opening too far and damaging the structure. The cable can also be used to pull the door up to the closed position.

The upper and lower doors are sealed by a passive type seal. A neoprene sponge seal is installed around the vent dish.

Mechanism, Latch (Upper door)

The lower end of the internal handle is connected to the handle shaft which is spring loaded to the locking direction of the lock pin. When the internal handle is pulled, the vent dish is opened and the lock pin is retracted, the upper door can now be removed. The lock pin is kept at the locked position by two independent bias springs. The internal handle has a cover to prevent the door from being accidentally opened.

Mechanism, Latch (Lower door)

The lower door handle is installed with an overcenter spring, to hold the handle at the engaged or disengaged position. Two shoot bolts engage to fittings on the door structure to keep the door closed.

Door seal (Upper and lower door)

The upper and lower door seals are a passive type. Several holes in the seals let them be inflated by cabin pressure.

Vent Dish seal (Upper door)

This seal is a compression seal made of neoprene sponge and fits around the periphery of the vent dish.

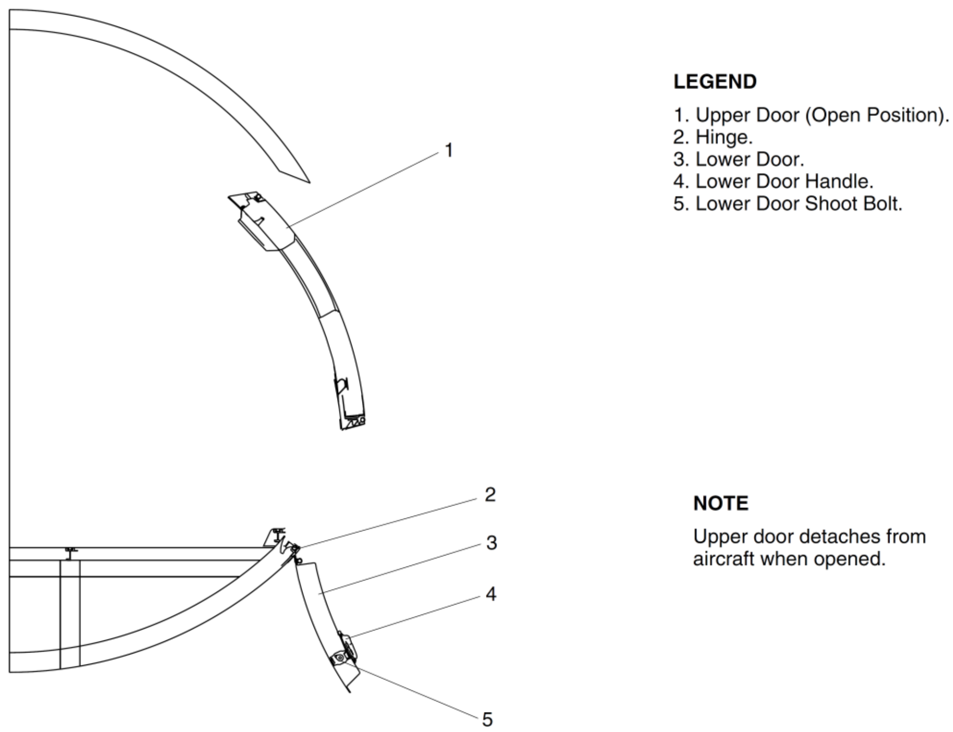


Figure 52-57 Type II Emergency Exit Door Operation

Refer Figures 52-57A & 52-57B

A proximity switch, installed on the upper edge of the upper door, monitors the lock position of the door. If the door is unlocked, the FUSELAGE DOORS warning and the MASTER WARNING light flash.

The Doors Systems page on the Multi-Function Display (MFD) shows the unlocked door as a red filled rectangle. If the door is not properly closed and locked when the aircraft leaves the ground, the cabin will not pressurize automatically.

LEGEND

1. Fuselage Doors (Red).

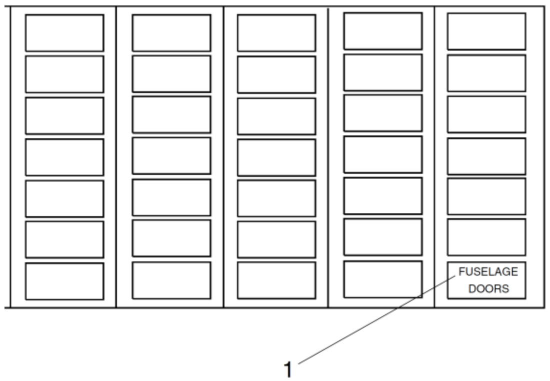


Figure 52-57A Fuselage Doors Warning Light

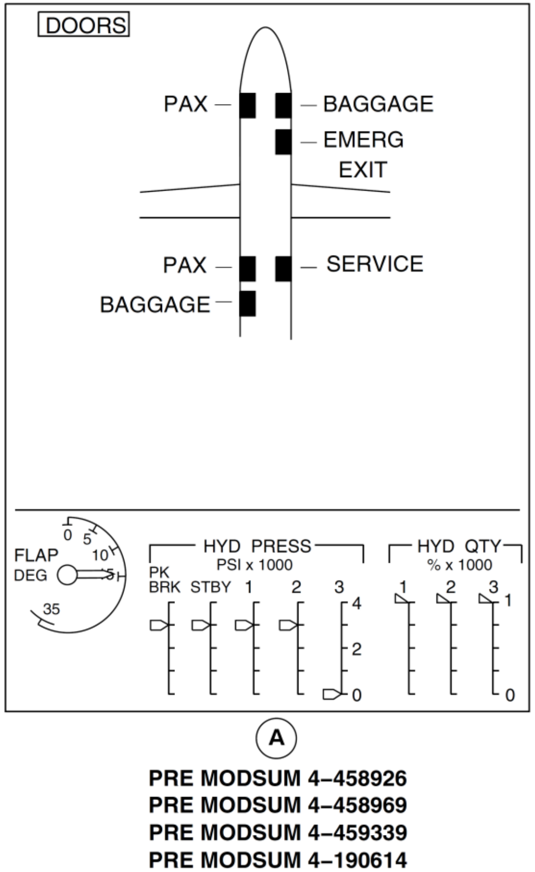


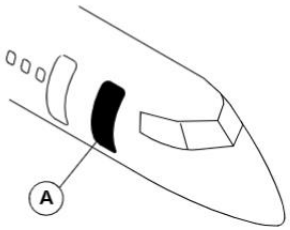
Figure 52-57B Fuselage Doors Page

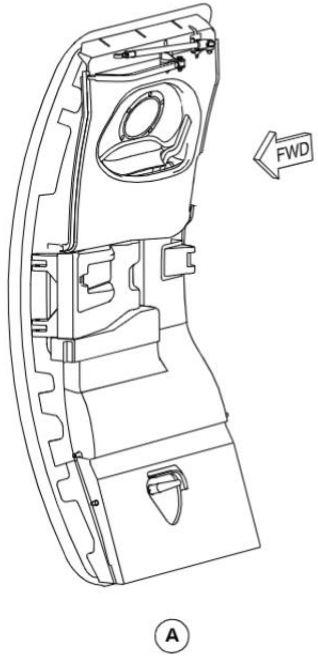
Forward Emergency Exit Type I Door

Introduction

On aircraft with the ModSum 4−458926 or 4−190614 incorporated a forward emergency exit type I door is installed. The forward emergency exit type I door gives access to the forward passenger compartment.

On aircraft with ModSum 4−459339 incorporated, the forward baggage door is reworked to add the internal handle, the window and the ditching dam guide pin bracketry and converted into the type I emergency exit door. The forward emergency exit type I door gives access to the forward passenger compartment.





General Description

The forward emergency exit type I door is located on the right side of the forward fuselage. It is a translating type door and is secured against pressurization loads by five stops on each vertical edge that engage with five related stop brackets installed on the door surround structure. The proximity sensor monitors the position of the upper fixed rollers and the door handle. A red warning light comes on in the flight compartment when the door is open and unlocked.

The forward emergency exit type I door system has the components that follow:

- Mechanism, Handle
- Mechanism, Vent door
- Mechanism, Lift and Latch
- Mechanism, Hinge
- Seal, Door
- Dam, Ditching

The forward emergency exit type I door interfaces with:

- Door Warning System
- Proximity Sensors

Detailed description

Refer Figure 52-58

The door operating and retention system is operated by the internal or external handle. The internal handle is pulled inwards and turned counter clockwise to unlock the handle. When the door handle is pulled inwards, the vent door opens to release the pressure in the cabin and the proximity switch opens. When the handle is turned counter clockwise, the lift/latch shaft pushes the lift/latch rollers down, and the door moves inwards and upwards to clear the door stops. At the uppermost lifted position, the hold−up hook is connected to the hold−up hook pin of the door hinge, and the door is moved outward. As the door moves outward, the interlock cam connects to a latch crank on the lift/latch shaft, to prevent the accidental handle operation.

The door has two fixed rollers located on each side around the upper and lower parts running in guides on the fuselage, which gives tangential movement of the door. In pressurized flight, the door is held by the stops against the pressurization and vertical loads. In unpressurized flight, the door is kept from upward movement by cam rollers on each side of the lift/latch shaft. If one cam roller assembly fails, the second cam roller assembly prevents the door from opening in flight. The lift/latch shaft is secured at the closed position by means of both the handle lock mechanism and the overcenter springs.

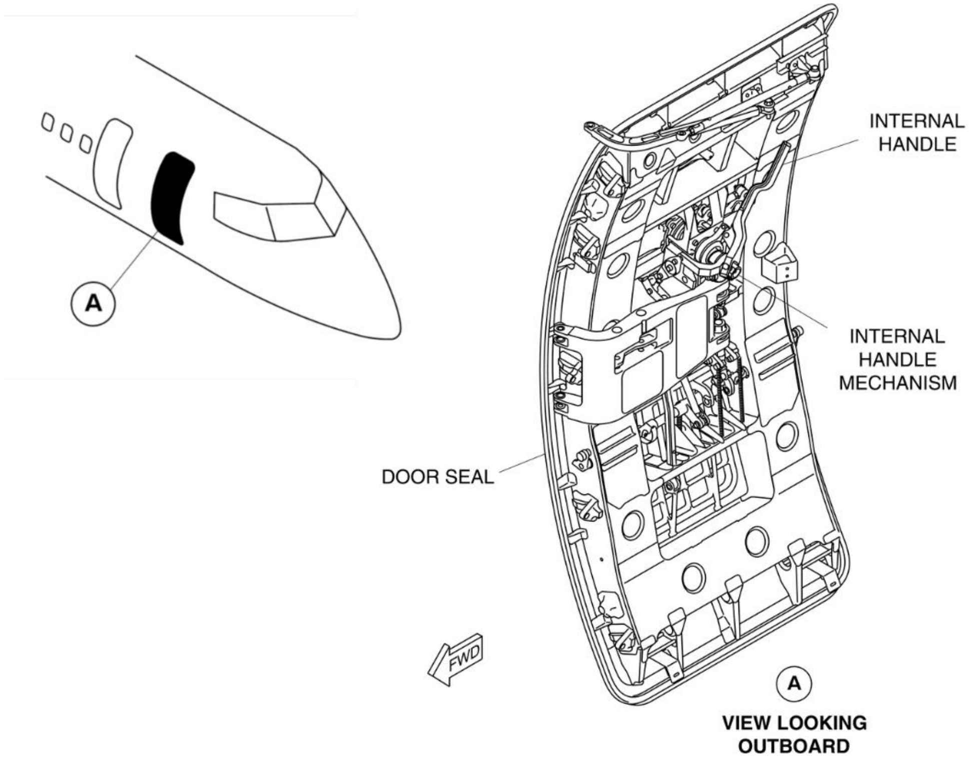


Figure 52-58 Forward Type I Emergency Exit Door

Refer Figure 52-59

The door is unlocked by operation of the spring loaded internal or external handle. The door opens or closes with the door face in parallel with the fuselage skin (this is done by the main and secondary hinges and a stabilizer). When the door reaches the fully opened position, the door open lock hook on the hinge arm is connected to the door open lock pin on the door to keep the door at its position. An interlock assembly prevents the handle operation when the door is opened, preventing the hold−up hook from being unlatched.

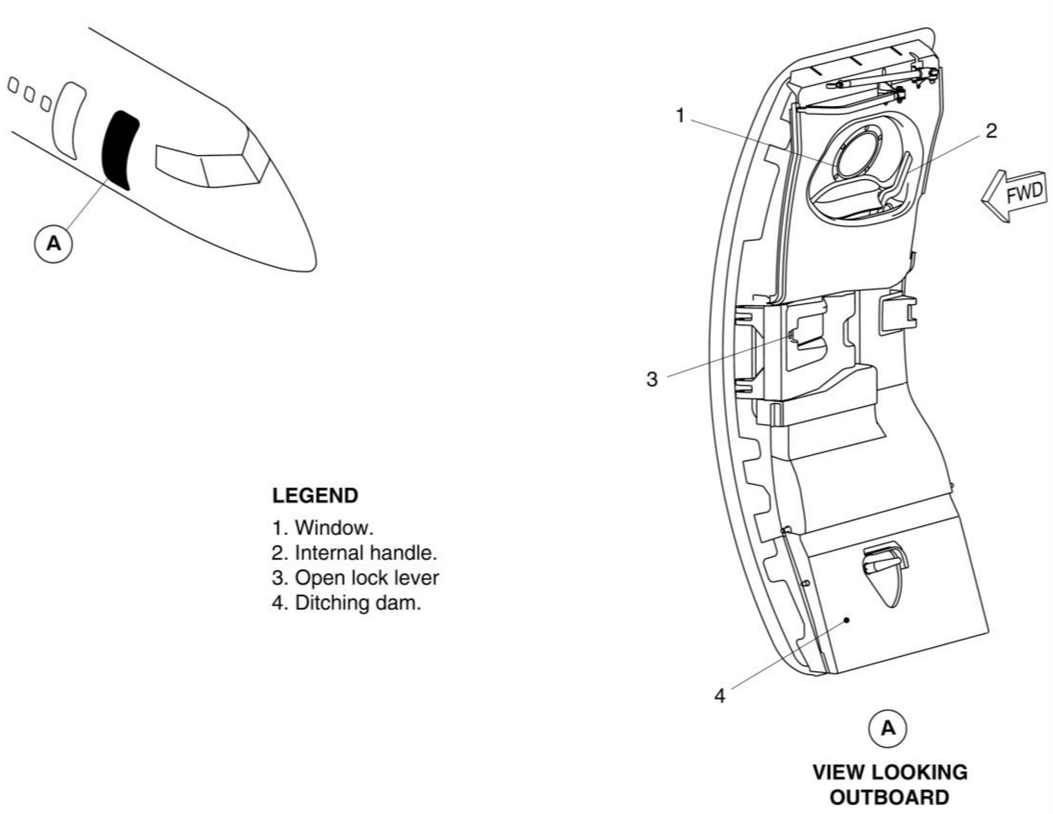


Figure 52-59 Forward Type I Emergency Exit Door – Internal View

Refer Figure 52-60 (Forward Type I Emergency Exit Door - Operation)
The external handle is kept in a recess in the outer skin and is connected to the same handle shaft as that of the internal handle. The handle can be accessed when the flap covers are pushed in. The handle is then pulled out and turned clockwise to open the door. For safety, a window in the door gives a view of the outside before the door is opened.

To close the door, the release lever adjacent to the lock hook is pulled to release the open lock, and the door is brought to the closed position. The internal handle is turned clockwise to disconnect the door hold−up hook from the hold−up hook pin. The door is moved to the closed position, and the stops align with their related stop bracket lugs. The handle is moved outward to lock the door and closes the proximity switch. The door can also be closed externally by using the external handle. The main door and vent door are sealed around their periphery. The seal is inflated by cabin pressure through holes in the seal to keep the gap around the closed door. Cabin pressurization is prevented when the vent door is not fully closed and locked.

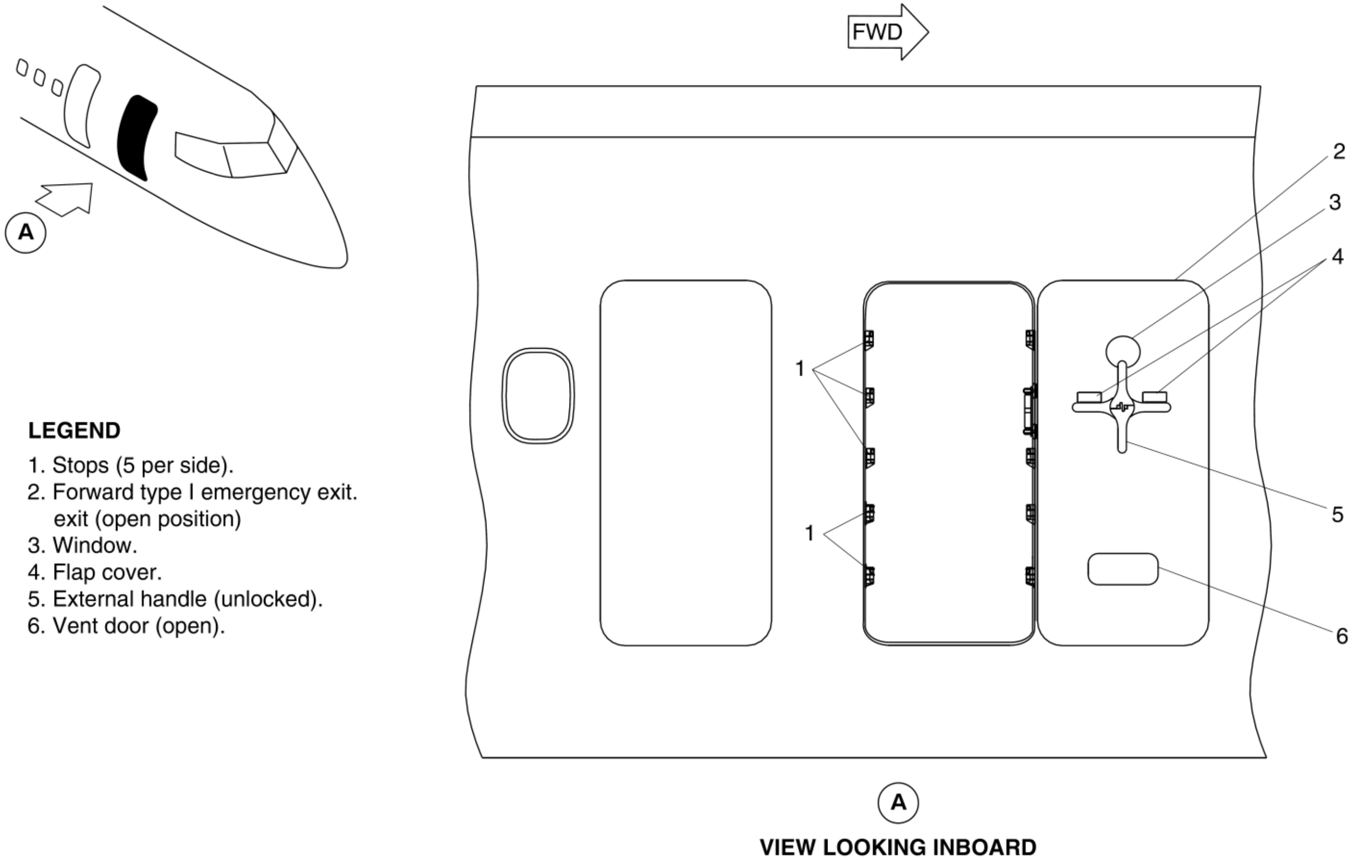


Figure 52-60 Forward Type I Emergency Exit Door - Operation

Mechanism, Handle

The internal handle operates the lift/latch shaft, and the vent door operating rod. The lower end of the internal handle is connected to the shaft of the handle mechanism. When either the internal or external handle is operated, the handle shaft can be turned to operate the lift/latch through cranks, rods and an idler link. The external handle is kept flush with the door skin and locked in the closed position.

Mechanism, Vent

The linkage from the handle transmits the handle input to the vent door mechanism. When the door handle is pulled, the vent door opens to release the pressure from the cabin. When the door is in the closed position, the final movement to lock the internal or external handle closes the vent door. The vent door is spring−loaded in the direction of the open position, so that it is kept open in case of fracture or disconnection of the vent door operating mechanism.

The primary functions of the vent door are as follows:

- Release residual cabin pressure to the atmosphere before the door is opened in order to make outward load on the door minimal
- Prevents the pressurization loads on the door in case of manual pressurization before the door is fully closed and locked.

Mechanism, Lift and Latch

The linkage from the handle to the lift/latch shaft transmits handle input to the shaft and locks the lift/latch shaft assembly when the handles are in their locked position. The bias spring assembly for the latch causes an overcenter against the lift/latch shaft, and locks it at the closed or opened position. The overcenter position and linkage from the handle mechanism gives dual locking of the lift/latch shaft.

The lift/latch shaft is installed to a cam roller assembly at both ends. When the door is opened, the lift/latch shaft turns and let the rollers push down the guide tracks, lifting the door for outward movement. The door operation sequence is reversed at door closing. When the door is in the closed position, the roller center is offset inward against the door lift−up plane, resulting in the rollers being kept in the door closing direction, against the upward movement of the door.

Mechanism, Hinge

The main and secondary hinge arms and the door makes a parallelogram, which keeps the door face in parallel with the fuselage skin during door opening or closing. During the travel of the door around the furthest point from the fuselage, the yawing movement of the door is suppressed by the stabilizer rod.

Seal, Door

The door seal is of the passive type which has several holes in the seal and is inflated by cabin pressure through the holes. This type of seal is installed on both the main and vent doors.

Dam, Ditching

The ditching dam is a composite panel installed in the bottom part of the type I emergency exit door. For ditching dam evacuation, the handle on the ditching dam rotated 60° clockwise to engage the lock pins on either side of the door and guide pins are retracted. This action will lock the ditching dam with the door surround structure and release it from the type I emergency exit door.

Baggage Doors

Introduction

The baggage doors give access to the forward and aft baggage compartments.

General Description

Refer Figure 52-61

The aft baggage/cargo compartment is accessible externally, through an outward and upward opening baggage door in the left side rear fuselage.

The forward baggage compartment is accessible externally through a door in the right front fuselage. The door opens inward and upward, and then translates outward and forward. The doors can be opened or closed manually by an external handle.

On aircraft with modsum 4−459324 or 4−458926 or 4−190614 incorporated, forward baggage door is removed and replaced with a Type 1 Emergency Exit Door.

On aircraft with modsum 4−459339 incorporated, the forward baggage door is reworked to add the internal handle, the window and the ditching dam guide pin bracketry and converted into Type I Emergency Exit Door.

Internal access to the forward baggage compartment is through a hinged door installed in the passenger compartment divider. The door has a key operated lock. There is no internal access to the aft baggage compartment.

The aft baggage door incorporates an inflatable door seal, and the forward baggage door a passive type seal.

Detailed Description

The baggage doors include the components that follow:

- Aft Baggage Door
- Forward Baggage Door

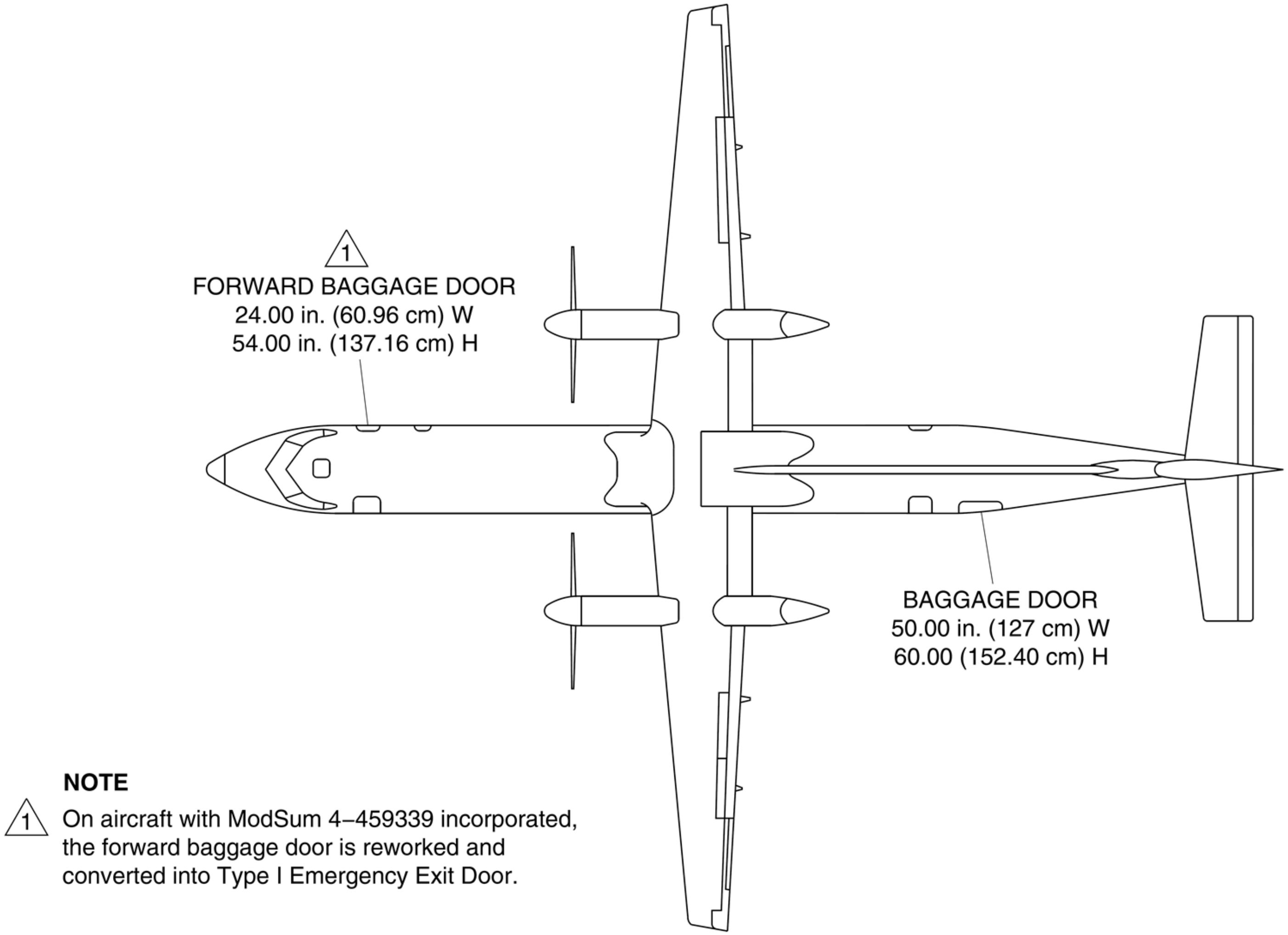


Figure 52-61 Baggage Doors – Location

Aft Baggage Door

Refer Figure 52-62 (Aft Baggage Door – Detailed)

The aft baggage door opens outward and upward about an upper hinge, and is held open by a telescopic strut. An external tee handle is used to lock and unlock the door from the outside. There is no internal handle for the baggage door. The door pressure seal inflates to prevent air leakage. The aft baggage door has an external, aluminum alloy skin braced by cross beams. The door warning system monitors the door status and supplies a visual indication and warning in the flight compartment.

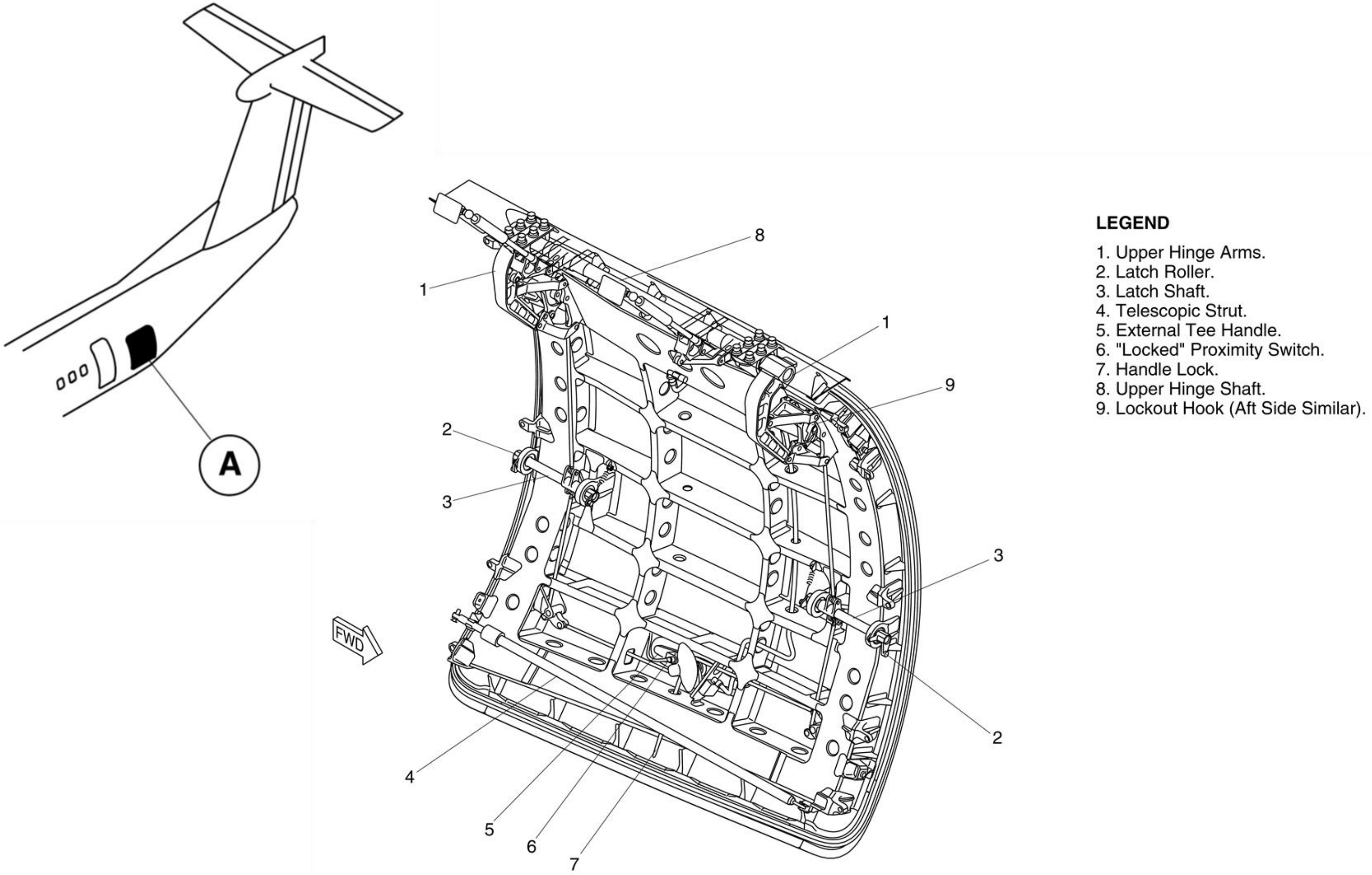


Figure 52-62 Aft Baggage Door – Detailed

Forward Baggage Door

Refer Figure 52-63 & Refer Figure 52-64

On aircraft with modsum 4−459324 or 4−458926 or 4−190614 incorporated, forward baggage door is removed and replaced with a Type 1 Emergency Exit Door.

On aircraft with modsum 4−459339 incorporated, the forward baggage door is reworked to add the internal handle, the window and the ditching dam guide pin bracketry and converted into Type I Emergency Exit Door.

The forward baggage door is unlocked by the operation of an external tee handle. The door opens or closes with the door face parallel with the fuselage skin. This is done by two hinges (main and secondary) and a stabilizer. Proximity sensors monitor the position of the upper fixed rollers and the door handle. A red warning light comes on in the flight compartment when the door is open or unlocked. The door warning system monitors door status and supplies visual indication and warning in the flight compartment.

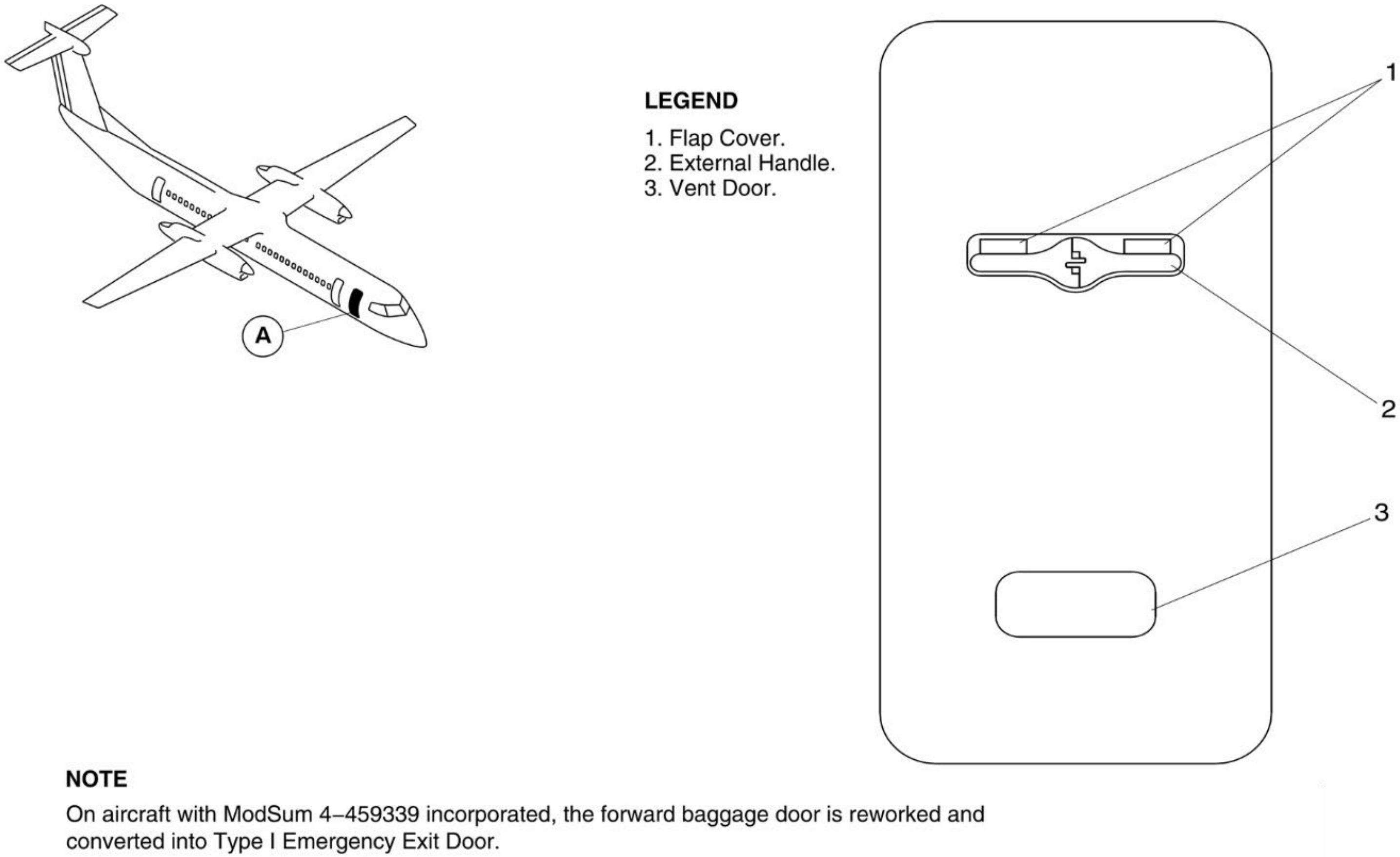
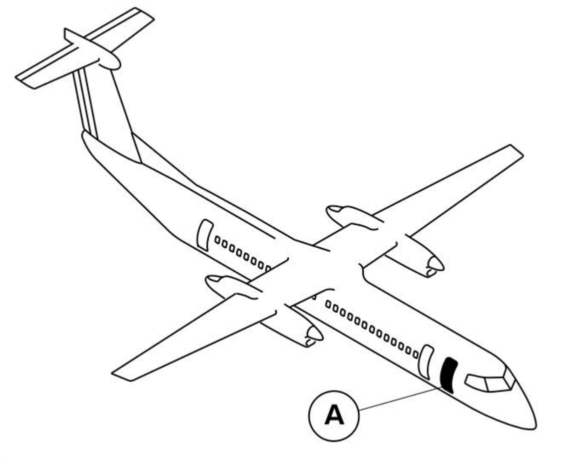
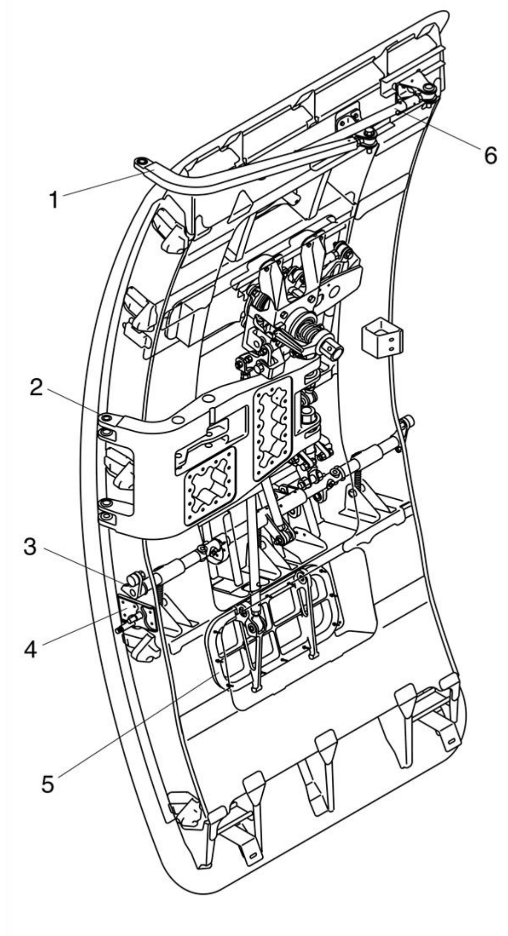


Figure 52-63 Forward Baggage Door – Detailed External View



NOTE

On aircraft with ModSum 4–459339 incorporated, the forward baggage door is reworked and converted into Type I Emergency Exit Door.



LEGEND

1. Secondary Hinge.
2. Main Hinge.
3. Lift And Latch Shaft.
4. Interlock Assembly.
5. Vent Door.
6. Stabilizer Rod.

Figure 52-64 Forward Baggage Door – Detailed Internal View

Aft Baggage Door

Introduction

The aft baggage door gives access to the aft baggage compartment.

General Description

Refer Figure 52-65 & Refer Figure 52-66

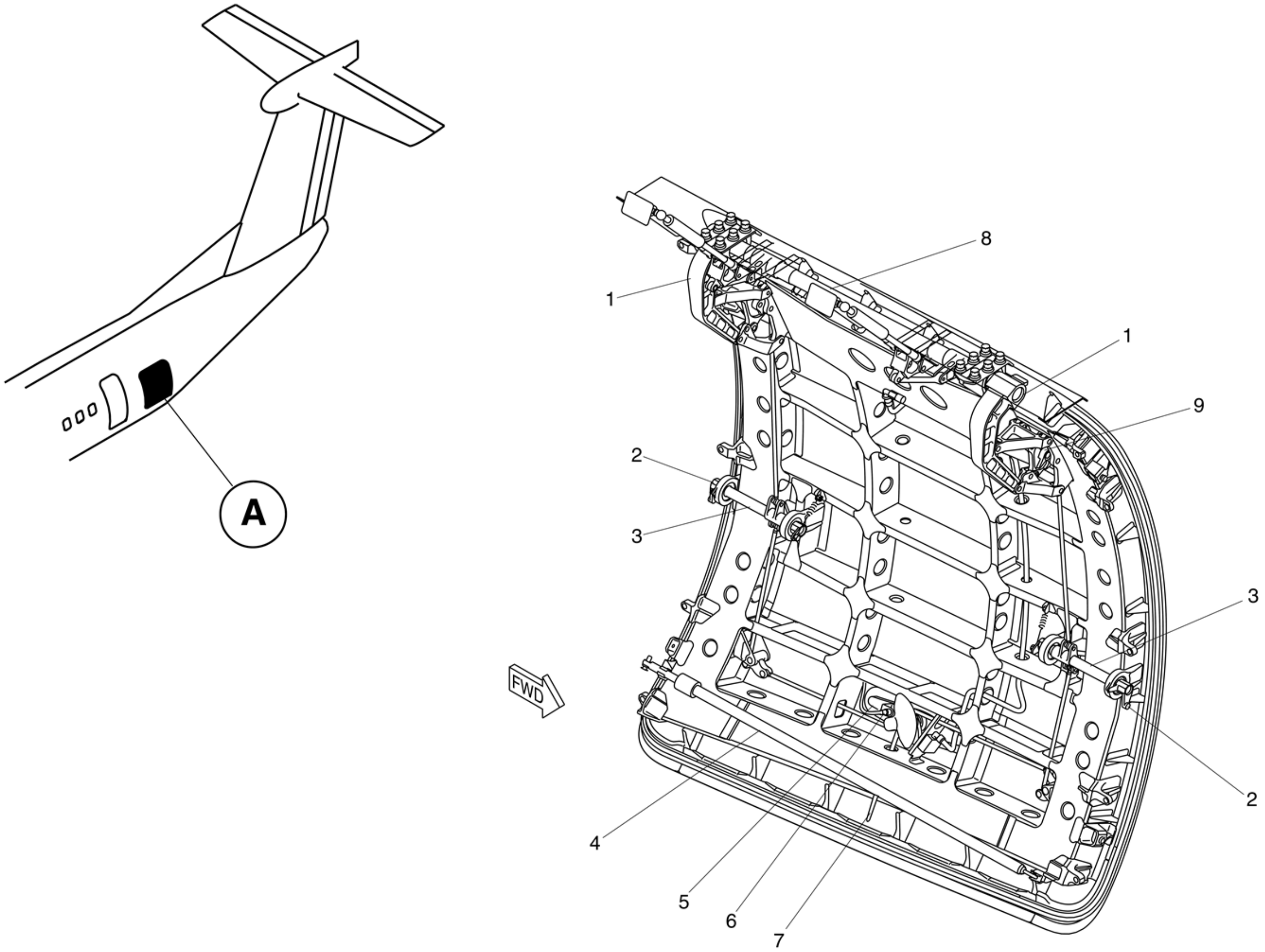
The aft baggage door is located on the left side of the fuselage at the rear baggage compartment. The door opens outward and upward about an upper hinge, and is held open by a telescopic strut.

An external tee handle is used to lock and unlock the door from the outside. There is no internal handle for the baggage door. The door pressure seal inflates to prevent air leakage.

The aft baggage door has an external, aluminum alloy skin braced by cross beams. The door warning system monitors the door status and supplies a visual indication and warning in the flight compartment.

The Aft Baggage Door System has the components that follow:

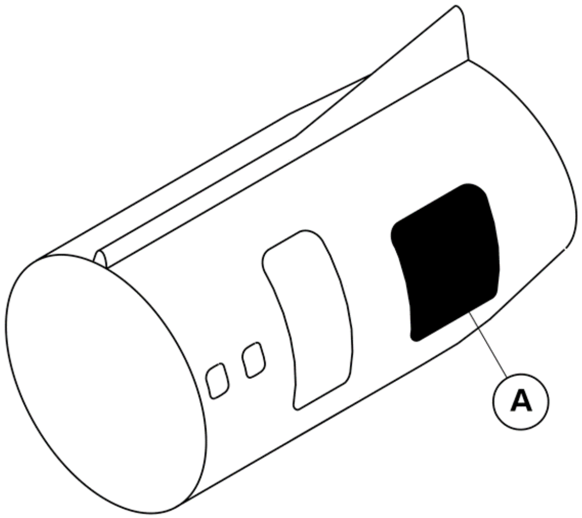
- Mechanism, Handle
- Mechanism, Hinge
- Mechanism, Latch
- Strut
- Mechanism, Door Balance
- Seal, Door
- Valve, Control−Door Seal



LEGEND

1. Upper Hinge Arms.
2. Latch Roller.
3. Latch Shaft.
4. Telescopic Strut.
5. External Tee Handle.
6. "Locked" Proximity Switch.
7. Handle Lock.
8. Upper Hinge Shaft.
9. Lockout Hook (Aft Side Similar).

Figure 52-65 Aft Baggage Door – Detailed



LEGEND

1. Telescopic strut.
2. Hinge.
3. External handle.

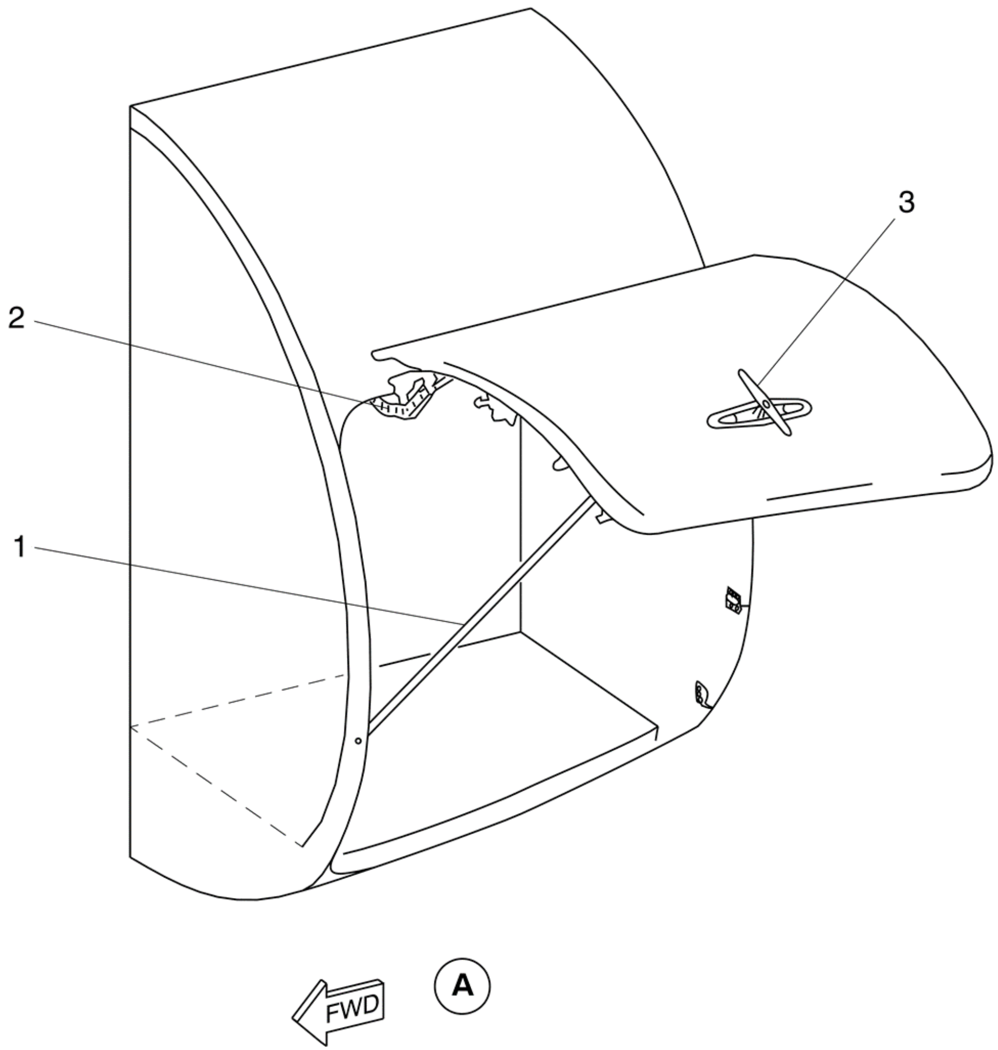


Figure 52-66 Aft Baggage Door – Open Position

Detailed Description

Refer Figure 52-67

The external tee handle, located in a recess in the lower part of the door, locks and unlocks the door. A push button in the handle releases it from its flush stowage position. The initial movement of the door is inward and downward, with roller guides guiding the latch rollers on each end of the latch shafts. The door then turn outwards about an upper hinge at the top of the door.

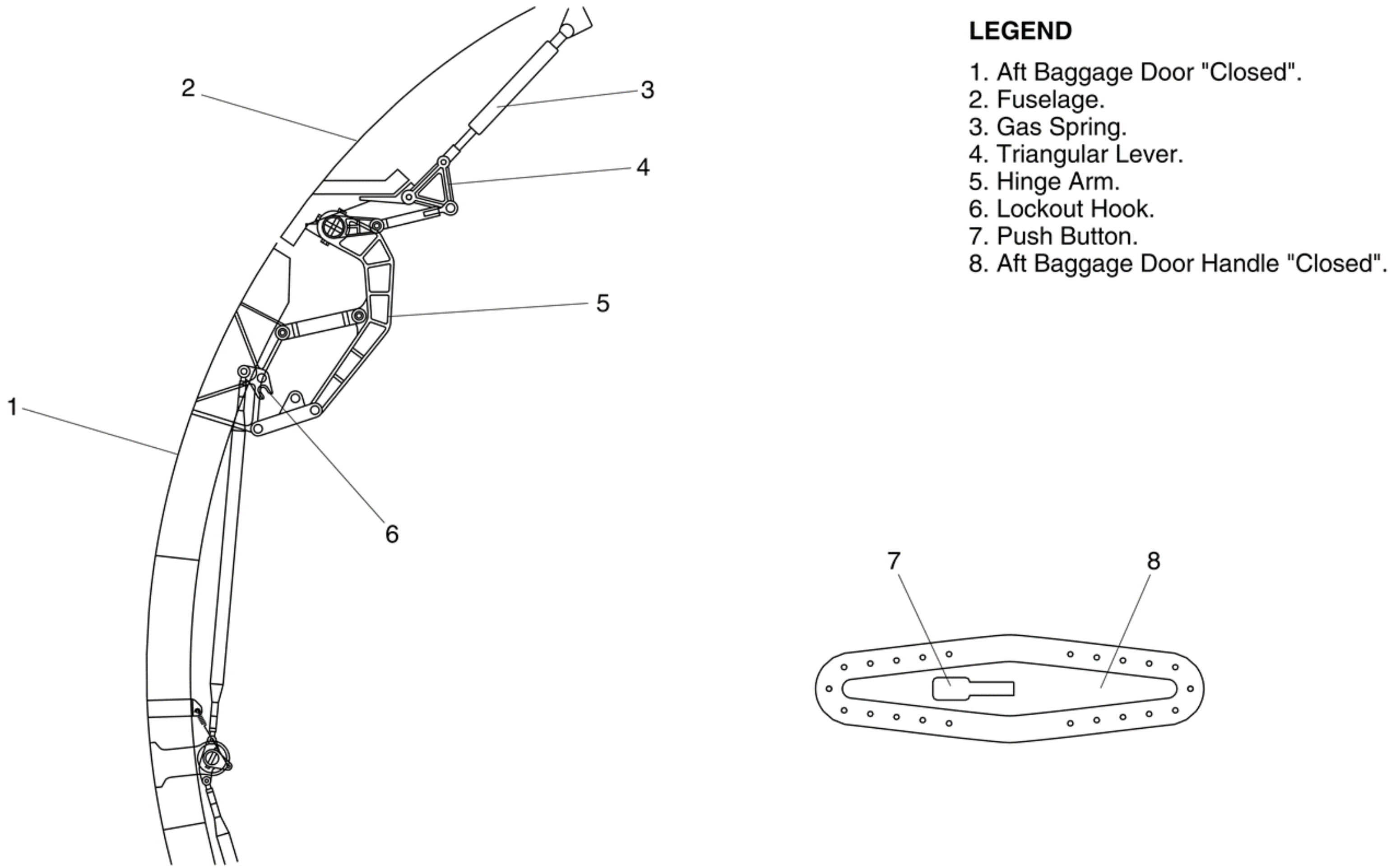


Figure 52-67 Aft Baggage Door – Closed Position

Refer Figure 52-68

Two upper hinge arms attach to an upper hinge shaft installed in the fuselage above the door. Roller guides on the door surround structure guide the latch rollers as the door moves to the fully lowered position. When the door reaches the fully lowered position, lockout hooks engage in the upper hinge arms to keep the door at a constant distance from the fuselage. A handle lock prevents operation of the tee handle, and keeps the lockout hook engaged. A telescopic strut helps hold the door in the open position.

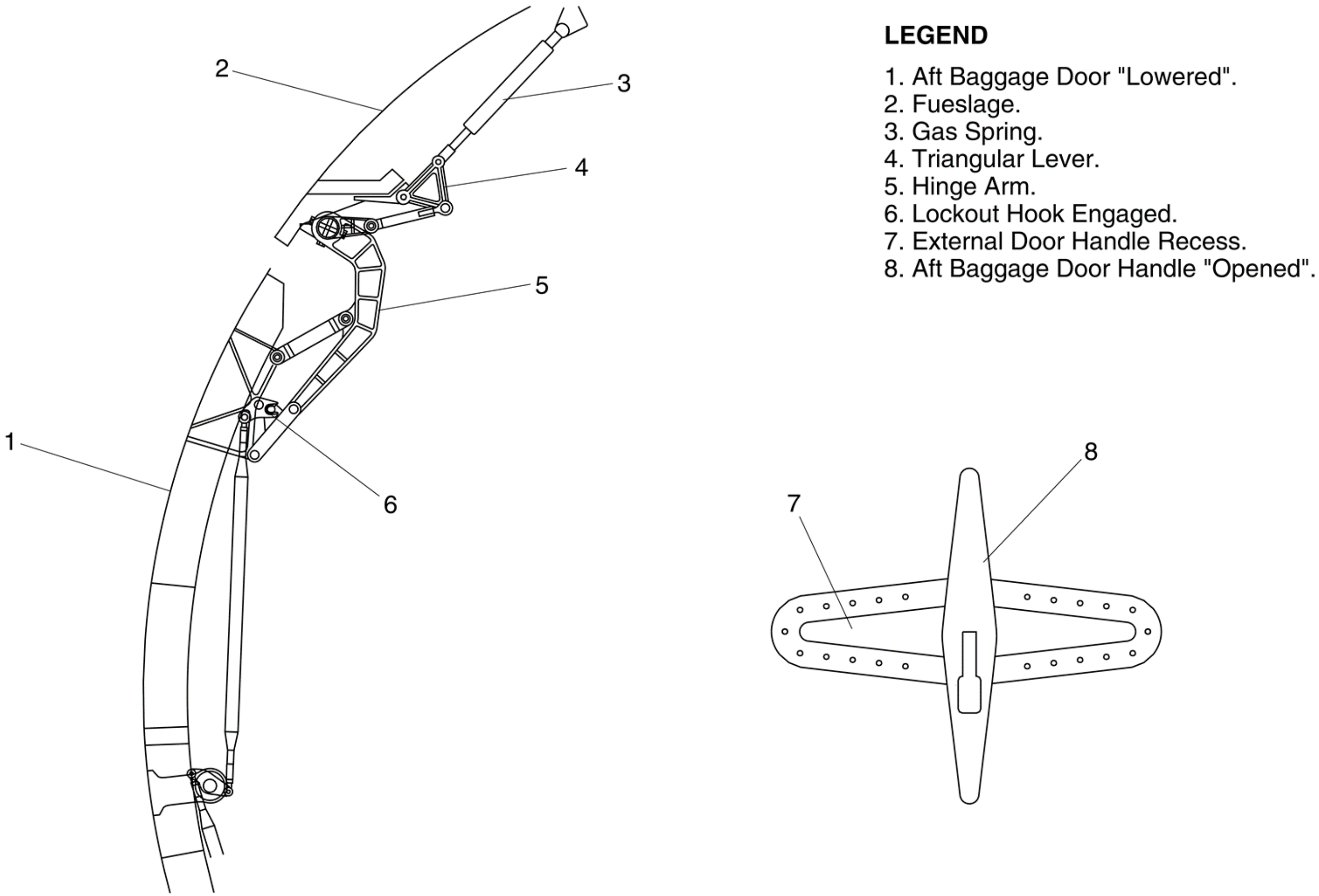


Figure 52-68 Aft Baggage Door – Lowered Position

Refer Figure 52-69

When the door is in the closed position, door pressurization loads are held by ten pressure stops, five on each side of the door. These stops engage with ten stop brackets on the door surround structure. The latch rollers also hold the aft baggage door and prevent vertical movement.

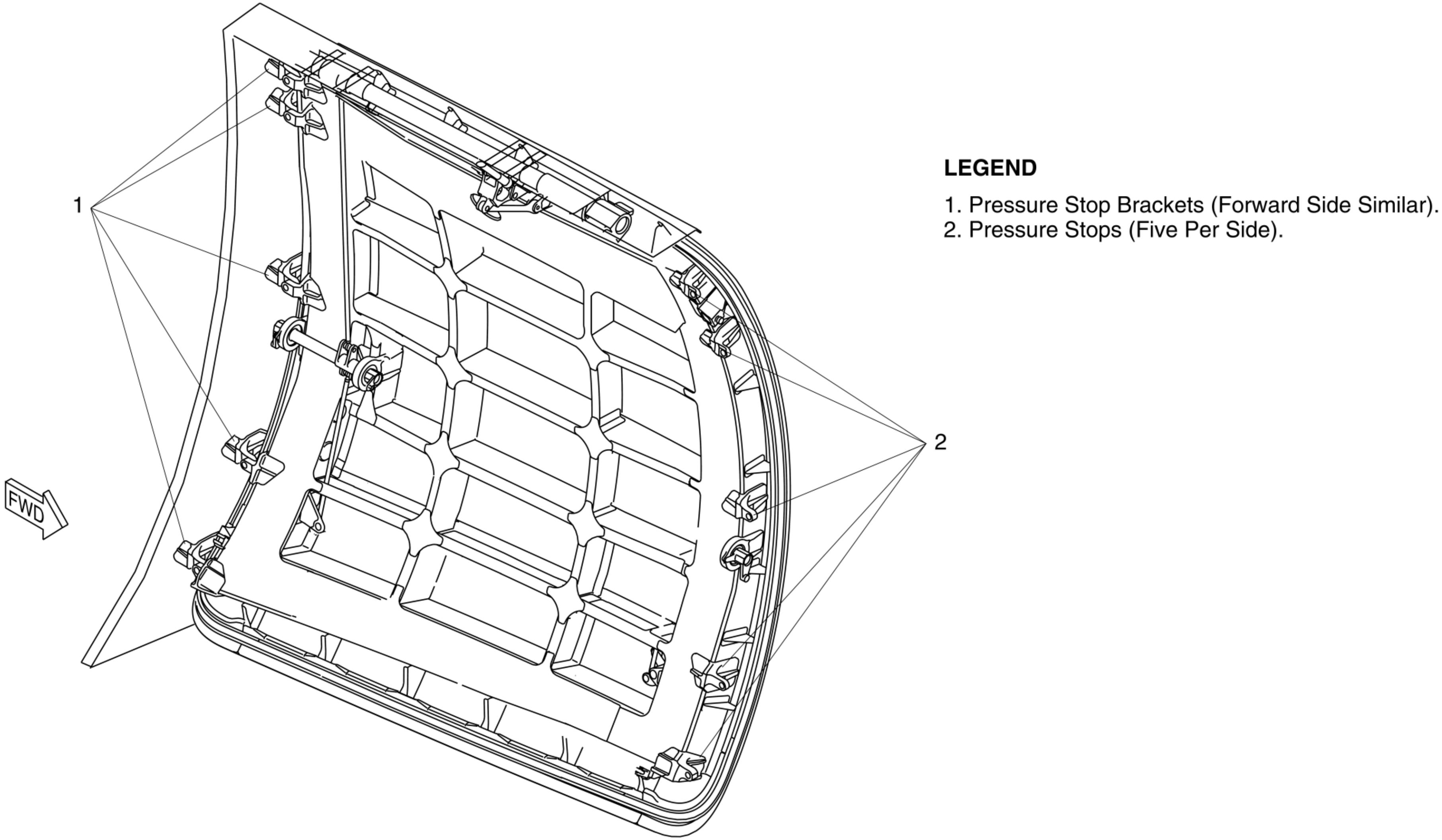


Figure 52-69 Aft Baggage Door – Stops

Refer Figure 52-70

A balance system helps to balance the weight of the door as the door is opened. The system is controlled by four gas springs, which apply a balance force to the upper hinge shaft through levers and rod assemblies. These springs also dampen and control the rate of door movement as the door is opened or closed.

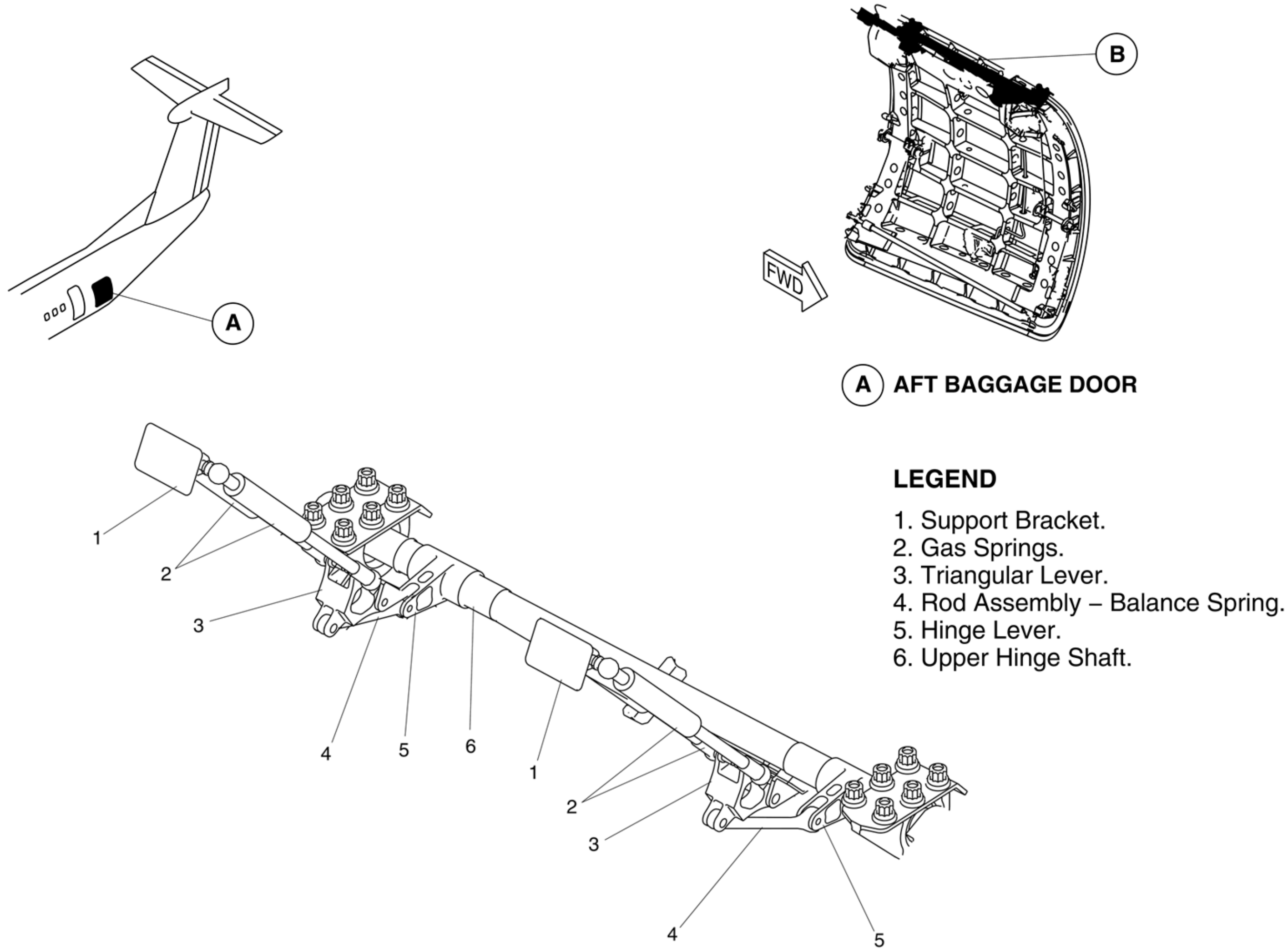


Figure 52-70 Aft Baggage Door – Balance System

Refer Figure 52-71

The door has an inflatable rubber seal installed around the edge of the door. Pressurized air is supplied to the seal and is controlled by a pressurization valve. The pressurization valve operates when the tee handle is extended or retracted. When the handle is extended, the seal deflates. When the handle is retracted, the seal inflates.

Two proximity sensors monitor the status of the aft baggage door. One sensor is located at the forward fixed−roller guide to monitor the open and closed positions. The other sensor, located in the door handle housing, monitors the handle retracted and extended position.

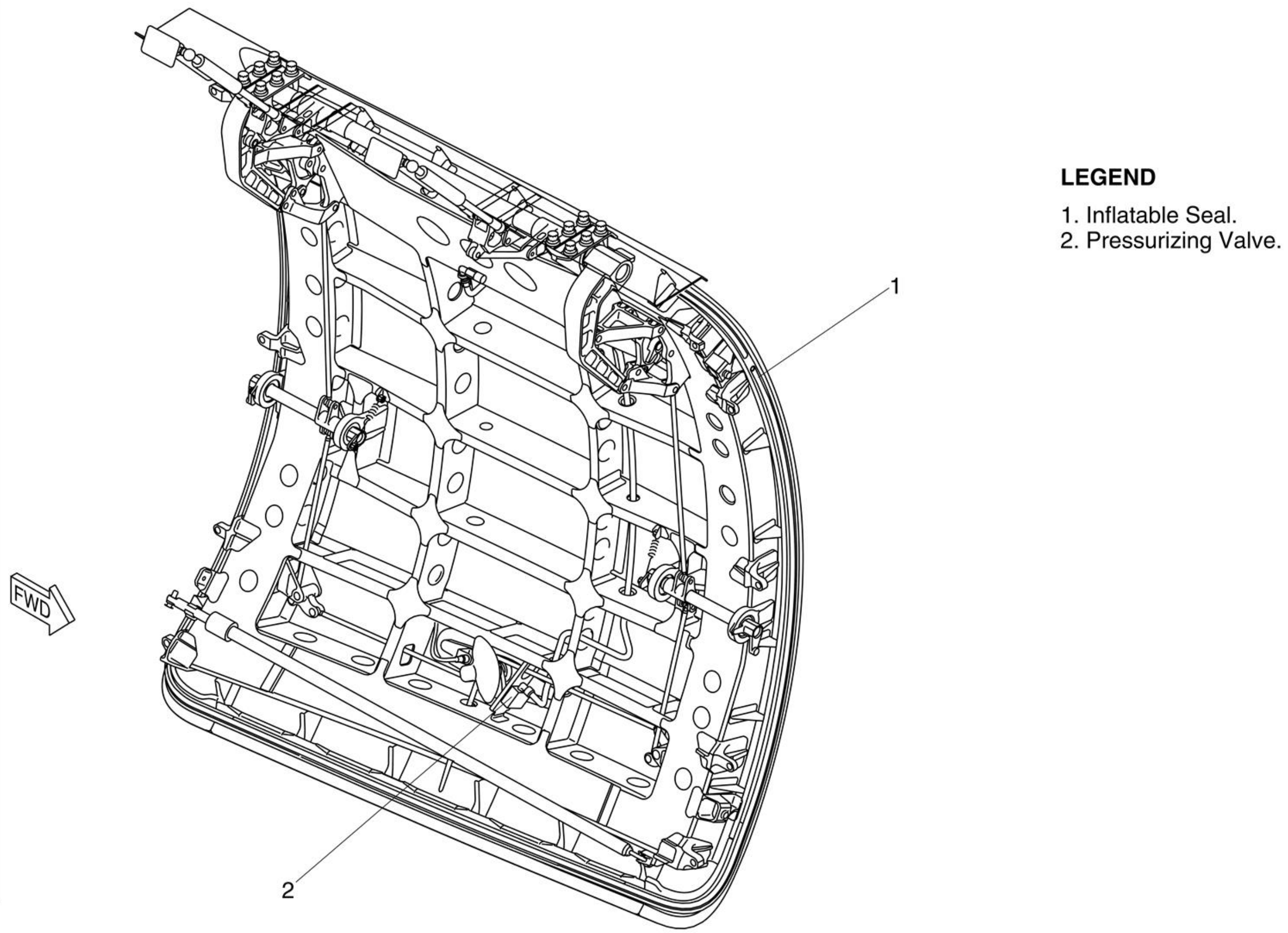


Figure 52-71 Aft Baggage Door – Seal and Pressurization Valve

Refer Figure 52-72

The door operating and retention system is linked by cams to the-external tee handle, and controls the door movement. The external-handle push–button is pushed to release the handle from the-recessed position. The handle is pulled out fully to release the lock-and to disconnect the seal pressurization valve and the "locked"-proximity switch. The handle is turned 90 degrees counterclockwise-to operate the latch rollers. The door then moves inward and-downward. The "closed" proximity switch opens, and the stops clear-their corresponding stop brackets on the fuselage. At the fully-lowered position, the lockout hooks latch to the upper hinge arms. The door then opens outward and upward about the door hinge,-helped by the counter balance assembly. As the door opens, the tee-handle locks in the open position by a handle lock mechanism. The-gas springs, located above the door hinge, helps to open and hold-the door open. The telescopic strut is then used to keep the door in-the opened position.

LEGEND

1. Pressurizing Valve.
2. Aft Baggage Door.
3. "Locked" Proximity Sensor.
4. External Handle.

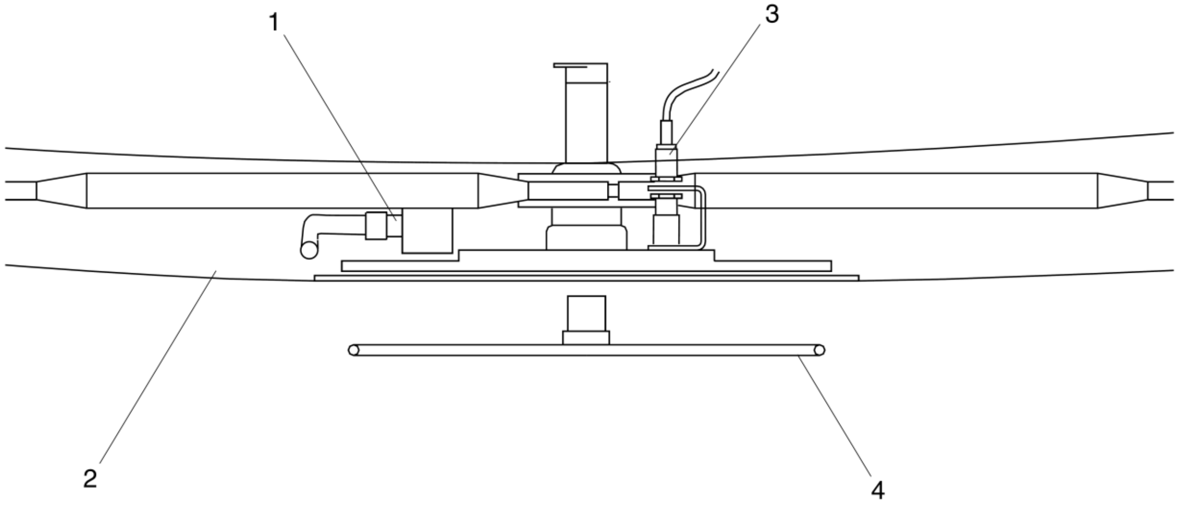


Figure 52-72 Aft Baggage Door – External Handle

Refer Figure 52-73

To close the aft baggage door, the telescopic strut is disengaged and-pulled down. The door movement is helped by the counter balance-system. When the door reaches the lowered position, the latch rollers-move into the roller guides, and the handle lock mechanism is-released. The external handle is turned clockwise to the horizontal-position to push the latch rollers down, and the door is lifted outward-and upward. This disconnects the lockout hook and puts the stop-bolts behind their related pressure stops, and connects the "closed"-proximity switch. The tee handle is pushed in to connect the "locked"-proximity switch and the pressurization valve inflates the door seal.

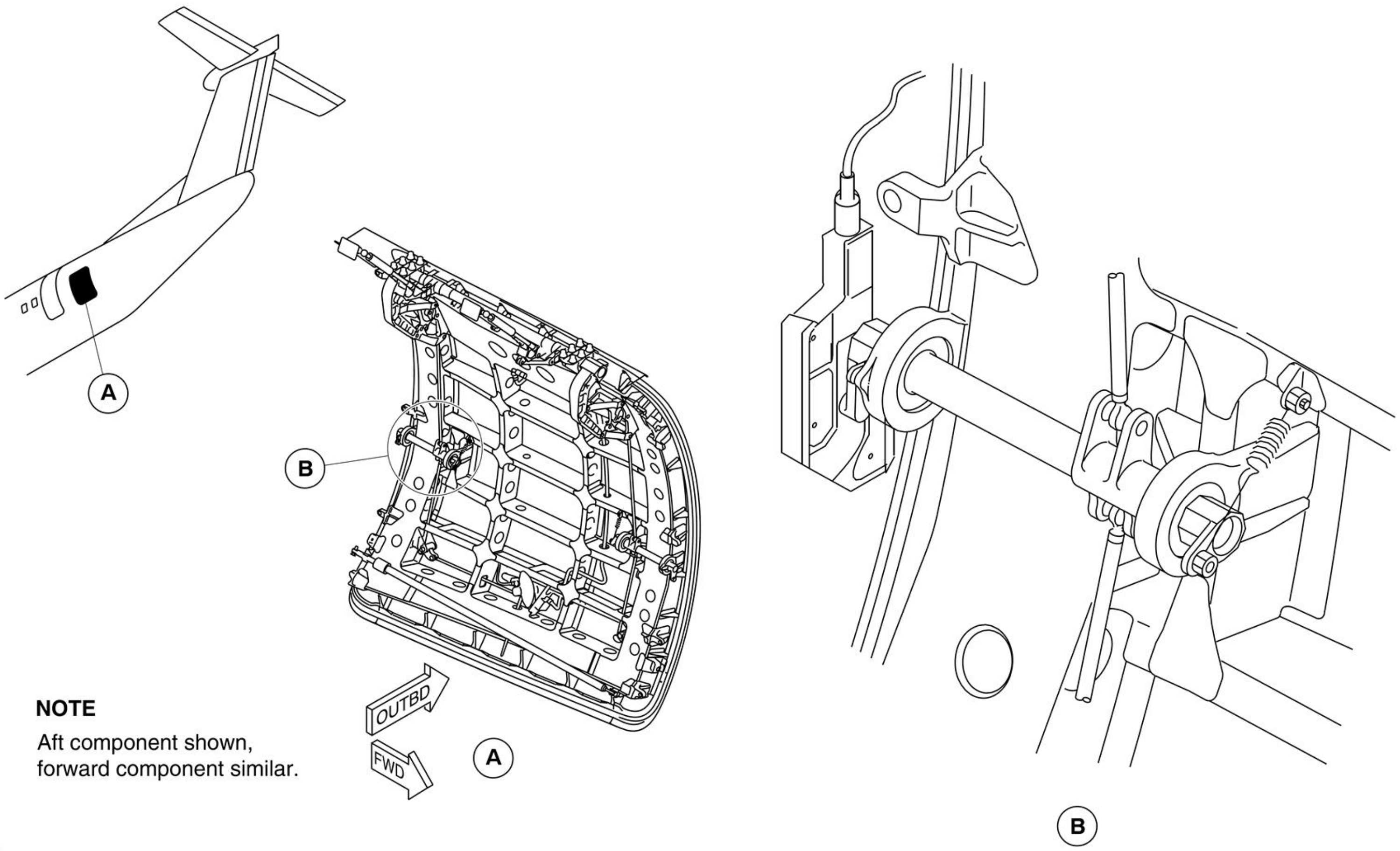


Figure 52-73 Aft Baggage Door – Roller Guide

Mechanism, Handle

The external handle mechanism is turned to operate both the latch-shaft and roller assemblies. The mechanism also operates the-movement of the linkages to the lockout hooks.

Mechanism, Hinge

The hinge mechanism is located above the aft baggage door and is-attached to the fuselage by two hinge support fittings. The balance-system rod assemblies are attached to the hinge. The two hinge-arms are attached to the door by upper and lower links.

Mechanism, Latch

The latch mechanism has latch shaft and roller assemblies located at-the forward and aft sides of the door. The tee handle is turned to-cause the latch rollers to move along the roller guides. In the fully-closed position, the latch roller is offset inward from the door-movement direction. The overcenter geometry keeps the roller at that-position, and against downward movement of the door. The latch-shafts are kept in the closed position by both the handle mechanism-and overcenter springs-

Strut

The telescopic strut keeps the door in the open position, and is used to close the door. One end of the strut is attached to a bracket on the aft baggage door, and the other end clamps to door when it is closed.

Mechanism, Door Balance

The door balance mechanism has four gas springs, triangular levers and rods, and helps to balance the door weight as the door turns about the hinge shaft. This reduces the force necessary to raise the door, and overbalances the door weight in the fully raised position.

Seal, Door

A door seal is installed on the door surround structure to prevent air leakage through the aft baggage door. The seal inflates with pressurized air from a reservoir tank or the deicing system.

Valve, Control–Door Seal

The door seal pressurization valve operates when the door handle is pushed in and the door is in the locked position. If the door is not fully closed and locked, the pressurization valve will not inflate the seal. The valve deflates the door seal when the door is unlocked by the external handle.

Forward Baggage Door

Introduction

The forward baggage door gives access to the forward baggage compartment.

On aircraft with modsum 4−459324 or 4−458926 or 4−190614 incorporated, forward baggage door is removed and replaced with a Type 1 Emergency Exit Door.

On aircraft with ModSum 4−459339 incorporated, the forward baggage door is reworked to add the internal handle, the window and the ditching dam guide pin bracketry and converted into the type I emergency exit door.

General Description

Refer Figure 52-74

The forward baggage door is located on the forward right side of the aircraft. The door is unlocked by the operation of an external tee handle. The door opens or closes with the door face parallel with the fuselage skin. This is done by two hinges (main and secondary) and a stabilizer.

Proximity sensors monitor the position of the upper fixed rollers and the door handle. A red warning light comes on in the flight compartment when the door is open or unlocked. The door warning system monitors door status and supplies visual indication and warning in the flight compartment. The indications are shown in:

- Door Warning System

The forward baggage door system has the components that follow:

- Handle mechanism
- Vent mechanism
- Lift and latch mechanism
- Hinge mechanism
- Door seal

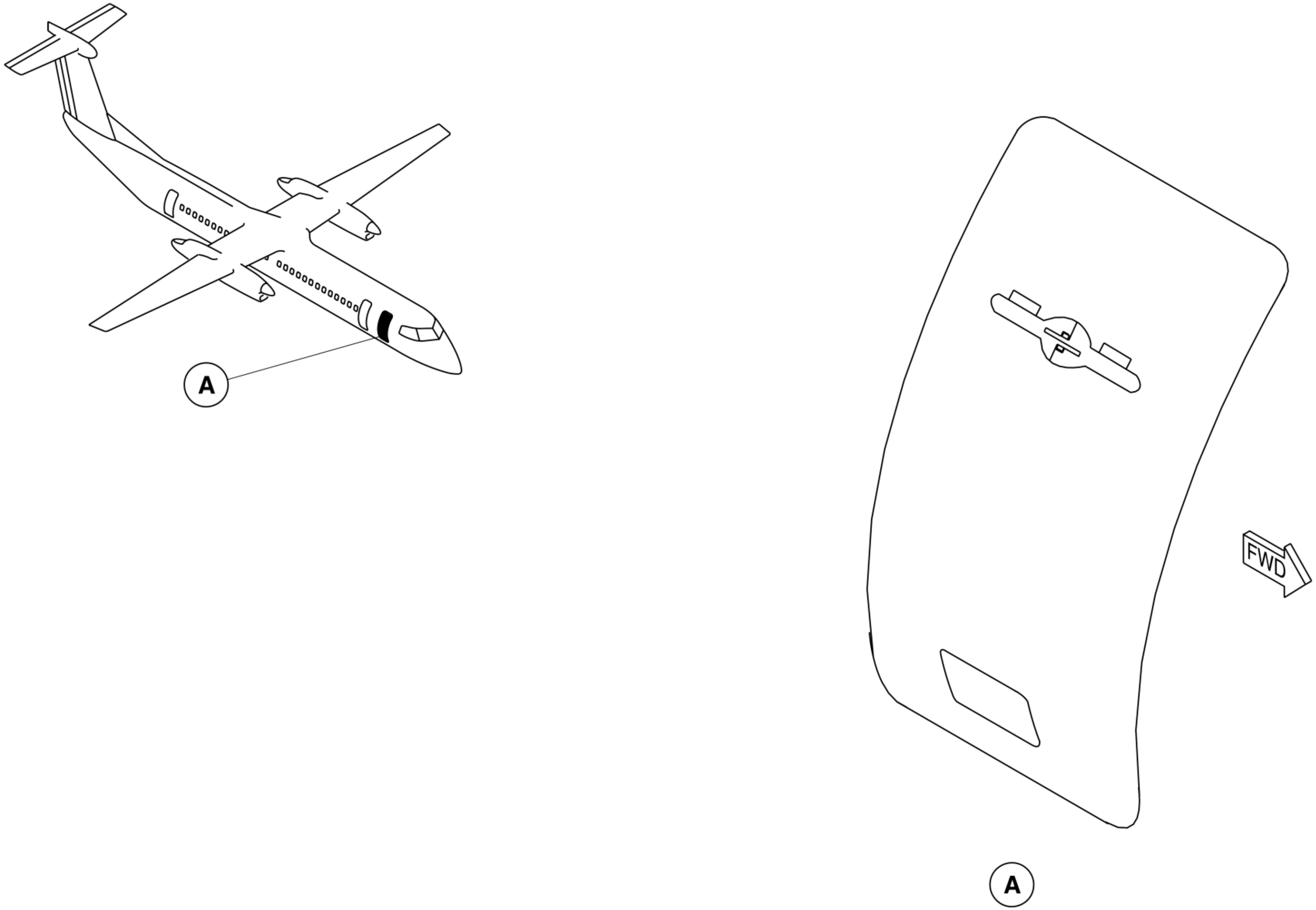


Figure 52-74 Forward Baggage Door – Location

Detailed Description

Refer Figure 52-75 & Refer Figure 52-76

The forward baggage door is of a translating type secured against-pressurization loads by five stops located on each vertical edge that-engages with five related stop brackets installed on the door-structure.

The door has two fixed rollers located on each side around the upper-and lower parts, that move in guides on the fuselage that give the-door tangential movement. In pressurized flight, the door is held by-the stops against pressurization and vertical loads.

In unpressurized-flight, the door is kept from upward movement by the cam rollers on-each side of the lift/latch shaft. If one cam roller assembly fails, the-second cam roller assembly prevents the door from opening in flight.

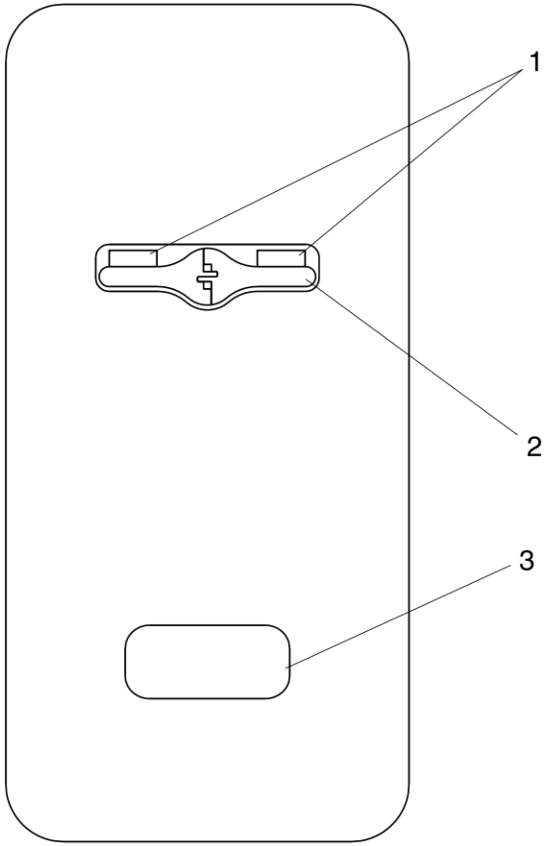
The lift/latch shaft is secured at the closed position by means of both-the handle lock mechanism and the over center springs.

The door opening and retention system is operated by the external-handle. The external handle is kept in a recess in the outer skin and-is connected to the handle shaft. The handle can be accessed when-the flap covers are pushed in. The handle is then pulled out and-turned counter clockwise. The first pulling operation opens the vent-door to release pressure in the cabin and the proximity switch opens.-Further operation makes the lift/latch turn pushing the lift/latch roller-down, and the door moves inwards and upwards to clear the door-stops.

At the uppermost lift position, the hold−up hook is connected to the-hold−up hook pin of the door hinge and the door is moved outward.-As the door moves outward, the interlock cam connects to a latch-crank on the lift/latch shaft, to prevent accidental handle operation.-When the door reaches the fully open position, the door open lock-hook on the hinge arm is connected to the door open lock pin to-keep the door in the open position.

To close the door, the release handle adjacent to the lock hook is-pulled to release the interlock mechanism. The door handle is turned-clockwise to disconnect the door hold−up hook from the hold−up-hook pin. The door is lowered to the closed position aligning the-stops behind their related stop bracket. Final inward movement of the-handle locks the door and closes the proximity switch. Automatic-cabin pressurization is prevented when the door is not fully closed-and locked.

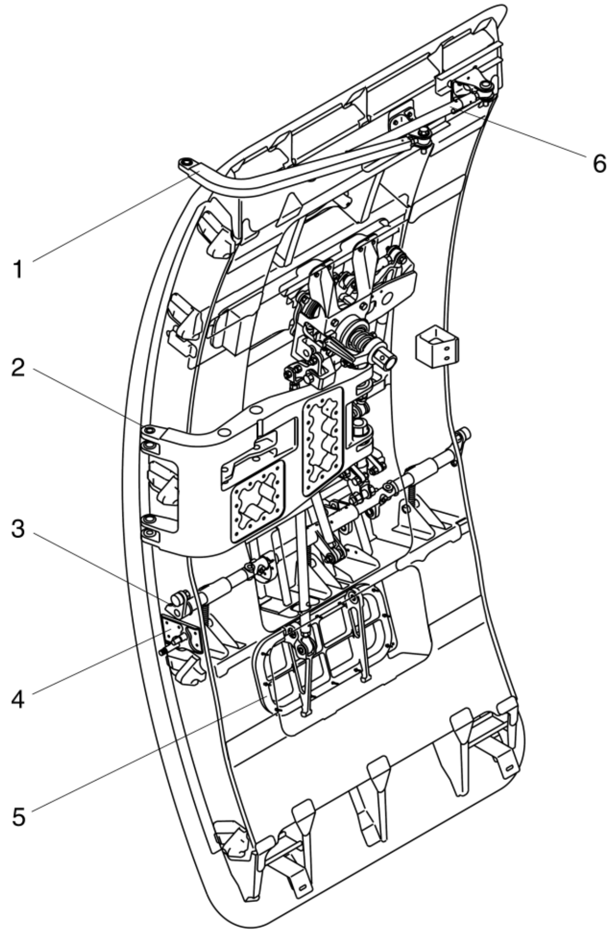
The baggage door and vent door have seals around their periphery.-The seals are inflated by cabin pressure through holes in the seals.



LEGEND

1. Flap Cover.
2. External Handle.
3. Vent Door.

Figure 52-75 Forward Baggage Door – Detailed External View



LEGEND

1. Secondary Hinge.
2. Main Hinge.
3. Lift And Latch Shaft.
4. Interlock Assembly.
5. Vent Door.
6. Stabilizer Rod.

Figure 52-76 Forward Baggage Door – Detailed Internal View

Mechanism, Handle

The external handle is connected to the shaft of the handle-mechanism. When the handle is pulled full travel, the handle shaft-turns the lift/latch shaft through cranks, rods, and an idler link.-

Mechanism, Vent door

The linkage from the handle transmits the handle input to the vent door mechanism. When the door handle is pulled, the vent door opens to release pressure from the cabin. When the door is in the closed position, the final movement to lock the handle closes the vent door. The vent door is spring−loaded in the direction of the open position, so that it is kept open in case of fracture or disconnection of the vent door operating mechanism.

The primary function of the vent door is as follows:

- To release residual cabin pressure to the atmosphere before the door is opened
- To prevent pressurization loads on the door before the door is fully closed and locked.

Mechanism, Lift and Latch

The linkage from the handle to the lift/latch shaft transmits handle input to the shaft, and locks the lift/latch shaft assembly when the handle is in the locked position.

The bias spring assembly for the latch causes an overcenter against the lift/latch shaft, and locks it at the closed or open position. The overcenter position and linkage from the handle mechanism, gives dual locking of the lift/latch shaft.

A cam roller assembly is installed at both ends of the lift/latch shaft. When the door is opened, the lift/latch shaft turns to let the rollers push down the guide tracks, lifting the door for outward movement. The door operation sequence is reversed at door closing. In the door closed position, the roller center is offset inward against the door lift−up plane, resulting in the rollers being kept in the door closing direction, against the upward movement of the door.

Mechanism, Hinge

The main and secondary hinge arms keeps the door face parallel with the fuselage skin when the door is opened or closed. During the travel of the door around the furthest position from the fuselage, the yawing movement of the door is suppressed by the stabilizer rod.

Door Seal

The door seal is of the passive type. Several holes in the seal let it be inflated by cabin pressure. This type of seal is also installed on the vent door.

Service Doors

Introduction

The aft service door gives access to the rear passenger- compartment, and also functions as a Type I emergency exit.

The aft- fuselage access door gives access to equipment in the- unpressurized area of the aft fuselage.

The Auxiliary Power Unit- (APU) access doors give access for ground maintenance of the- optional APU. The ground service and nose compartment doors give- access for maintenance and servicing purposes.-

General Description

Refer Figure 52-77 & Refer Figure 52-78

Service doors are installed at various locations on the aircraft.

The four nose compartment doors are located on the nose.

The aft service door is located on the right side of the aircraft at the- rear of the passenger compartment.

The Aft Fuselage Access Door is located flush with the underside of- the rear fuselage, through which the unpressurized equipment- compartment is accessed. The door is opened or closed from the- exterior of aircraft.

The APU access doors are located on the bottom of the tailcone section on either side of the fuselage centerline. The doors are hinged horizontally and kept in a locked and closed position by the two pin latches, hook latches and keepers, and the three shear pins.

The Refuel/Defuel Panel door is located on the lower side of the No. 2 engine nacelle, aft of the right main landing gear rear doors. The door gives access to the refuel/defuel adapter, Refuel/Defuel Indicator (RDI), and switches to carry out pressure refueling/defueling operations.

The dc electrical ground power door gives access to the 28 Vdc ground power receptacle on the left side of the aircraft nose. The door is attached to the nose section by a hinge assembly at the forward end and a latch assembly at the aft end. The ac electrical ground power door gives access to the ac ground power receptacle. The door is attached to the nacelle by a hinge at the top and two latch assemblies at the bottom.

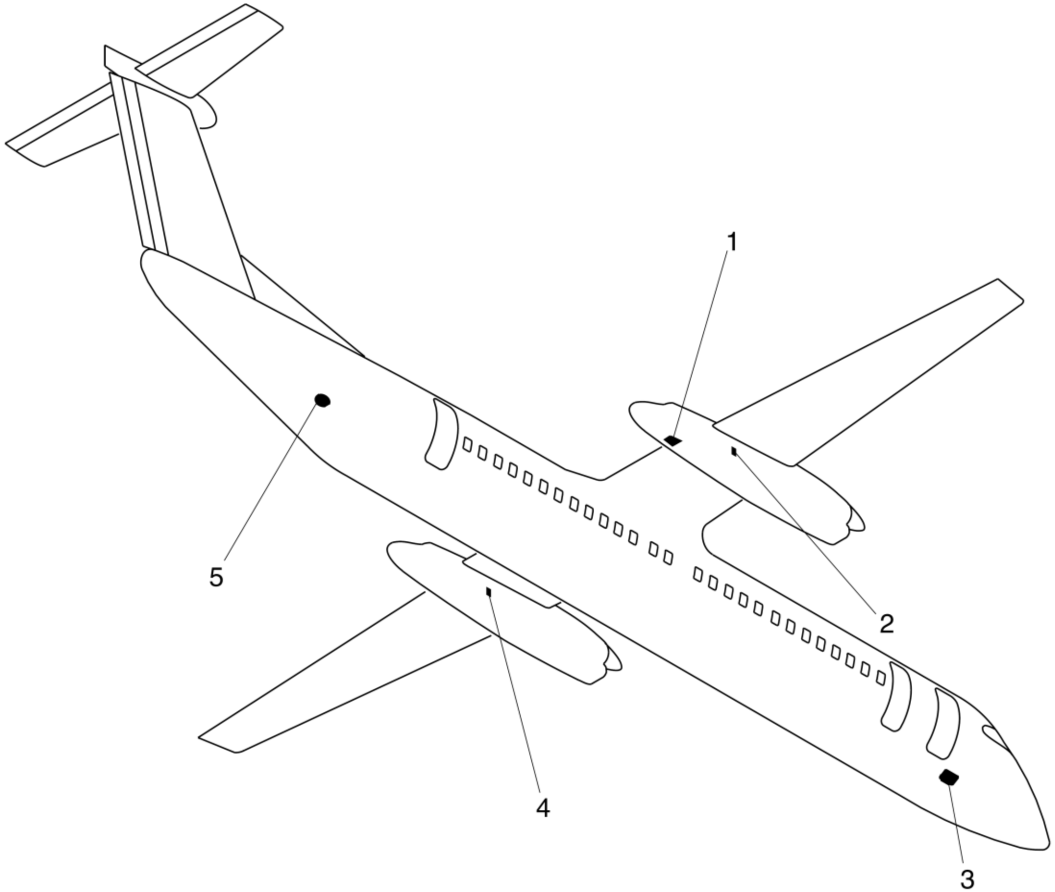
The hydraulic ground power doors, on the right side of both engine nacelles, give access to the hydraulic ground power connections. The doors are attached to the nacelle by a hinge at the top and two latch assemblies at the bottom.

The ground air conditioning door is located flush with the lower right side of the rear fuselage. The door opens outward and downward to give access to the air conditioning duct connection.

The lavatory service door is located flush with the lower right side of the forward fuselage. The door opens outward and downward to give access to the lavatory service port.

The Service Doors System consists of the components that follow:

- Aft Service Door
- Aft Fuselage Access Door
- APU Access Door
- Ground Service Doors
- Ground−Crew Interphone−Jack Door Assembly
- Nose Compartment Doors



LEGEND

1. Refuel/Defuel Panel Door.
2. Right Nacelle Hydraulic Ground Power Door.
3. Lavatory Service Door.
4. Left Nacelle Hydraulic Ground Power Door.
5. Ground Air Conditioning Connection Door.

Figure 52-77 Service Doors– Locations

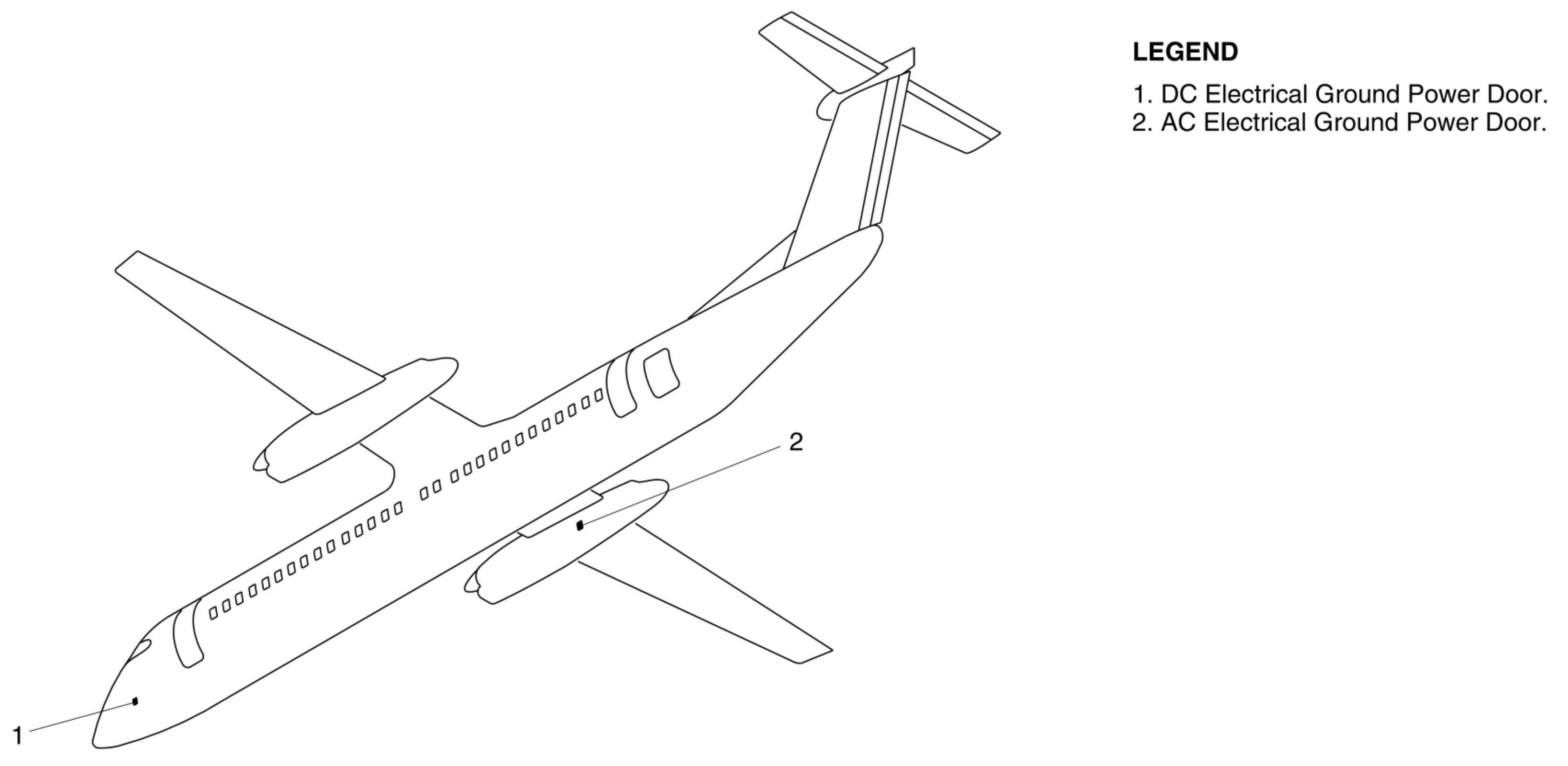


Figure 52-78 AC and DC Ground Power Doors – Locations

Detailed Description

The service door system includes an aft service door, an aft fuselage- access door and APU clamshell doors. It also includes several- ground service and nose compartment doors.

The ground service doors are identified as follows:

- Refuel/Defuel Panel Door
- DC Electrical Ground Power Door
- AC Electrical Ground Power Door
- Hydraulic Ground Power Door
- Ground Air Conditioning Connection Door
- Lavatory Service Door
- Ground−Crew Interphone−Jack Door

Aft Service Door

Refer Figure 52-79

The aft service door is classified as a Type I emergency exit. It is a translating type secured against pressurization loads by five stops on each vertical edge that engage with five related stop brackets installed on the door surround.

When the aft service door is unlocked, both the FUSELAGE DOORS warning light and the MASTER WARNING light flash.

The Door Systems page on the Multi−Function Display (MFD) shows the unlocked aft service door as a red−filled rectangle with a SERVICE legend.

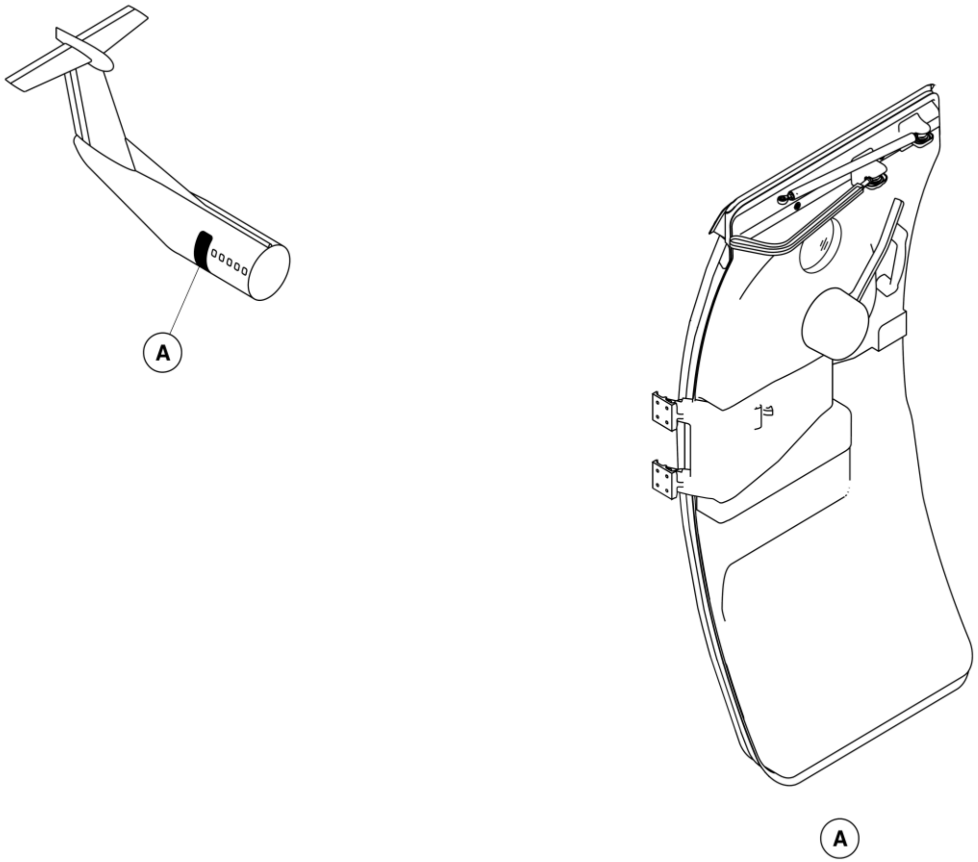


Figure 52-79 Aft Service Door – Location

Aft Fuselage Access Door

Refer Figure 52-80

The aft fuselage access door is installed in the fuselage structure at the right side by hinges, and at the left side by two latches. The door has a cast aluminum panel with a louvered opening to increase the airflow in the compartment. Moisture drain holes are located in the panel of the door to drain any accumulated water.

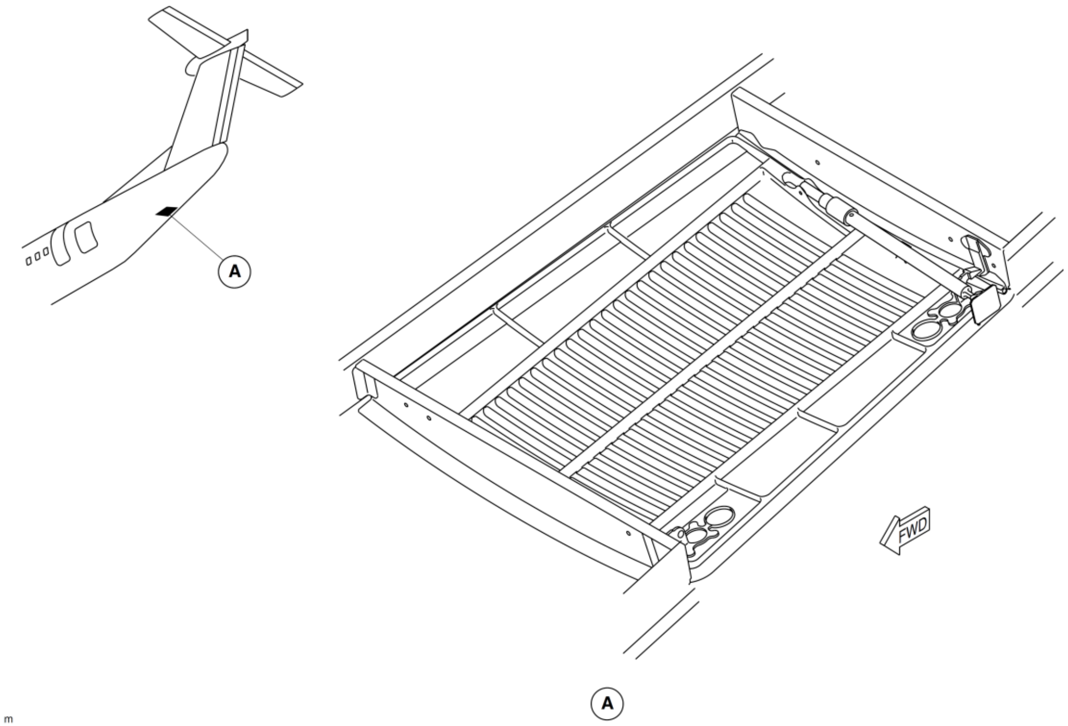


Figure 52-80 Aft Fuselage Access Door – Location

APU Access Door

Refer Figure 52-81

The APU access doors are made up of left and right two access door subassemblies. The left and right doors form a large part of the bottom of the tailcone and together make up a clamshell door system. The doors are hinged on the horizontal plane and open upward and outward, to give access to the APU. The doors are supported in the open position by telescopic struts.

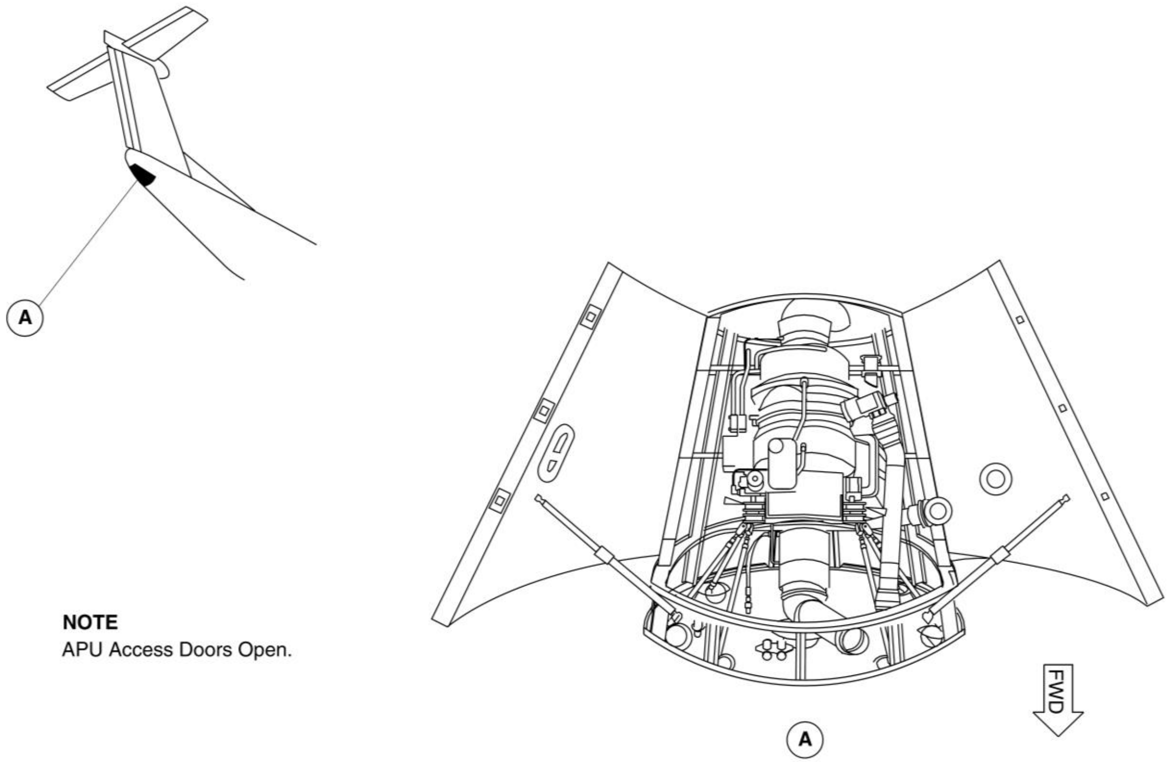


Figure 52-81 APU Access Door – Location

Refuel/Defuel Panel Door

Refer Figure 52-82

The Refuel/Defuel Panel door is located on the lower side of the No. 2 engine nacelle, aft of the right main landing gear rear doors. The door gives access to the refuel/defuel adapter, Refuel/Defuel Indicator (RDI), and switches to carry out pressure refueling/defueling operations.

The Refuel/Defuel Panel Door operates a door switch installed on the nacelle door structure. When the door is open, the switch causes the amber MASTER VALVE CLOSED light on the Refuel/Defuel panel and the FUELING ON caution light on the Caution and Warning Panel to come on. The service light, installed on the aft vertical section of the Refuel/Defuel compartment, also comes on. When the Refuel/Defuel Panel Door is closed and locked, it operates the door switch, which causes the FUELING ON caution light to go off. The MASTER VALVE CLOSED light and the service light also go off.

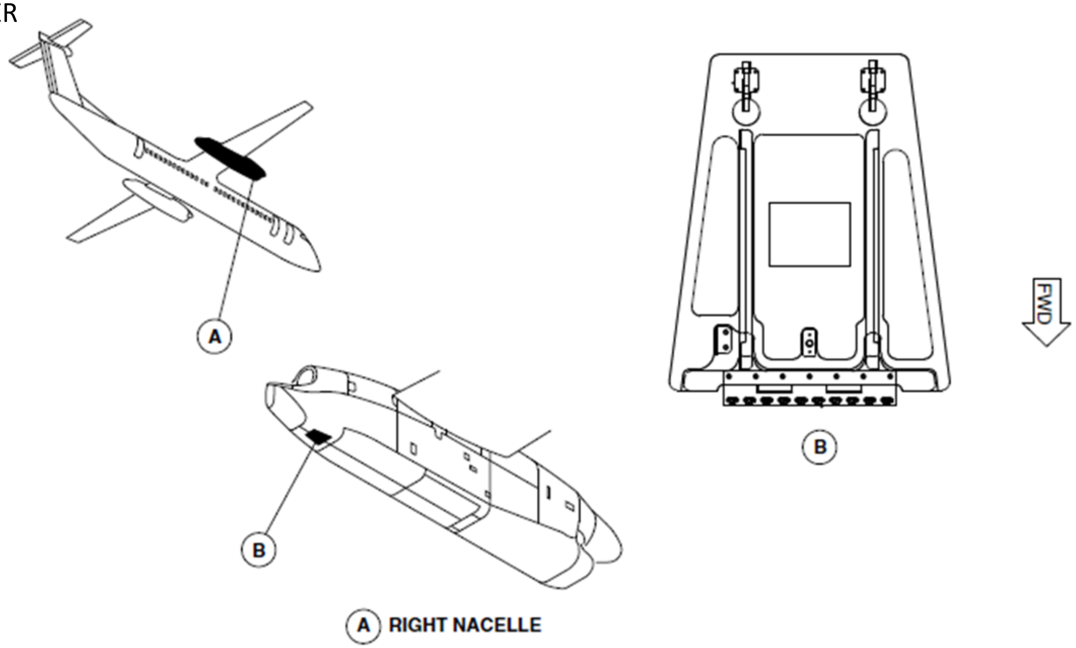


Figure 52-82 Refuel/Defuel Panel Door – Location

Ground Air Conditioning Door

Refer Figure 52-83

The ground air conditioning door, fabricated from alclad skin and stiffener, is located flush with the lower right side of the rear fuselage. The door opens outward and downward to give access to the airport. The door is secured to the fuselage structure at the lower side by a hinge and at the upper side by a latch. A gasket is fitted around the door cut−out, to provide weather sealing and prevent chafing of the door. The door can be opened and closed from outside the aircraft.

To open the door, press the latch release button to unlock the latch mechanism; the door will swing downward from the hinge on the door lower edge. To close the door, swing the door to the closed position and ensure the latch mechanism is engaged.

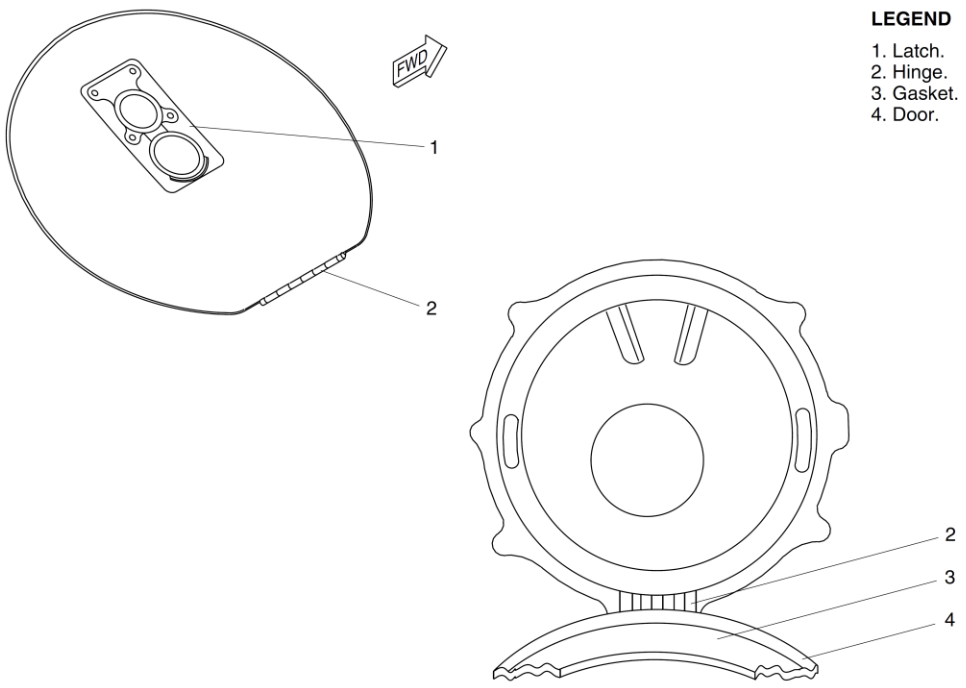


Figure 52-83 Ground Air Conditioning Door – Location

Lavatory Service Door

Refer Figure 52-84

The lavatory service door is attached to the fuselage structure, on the lower side by a hinge and on the upper side by two latches. The door has a gasket installed around the edge to help prevent chafing.

The latch release buttons are pushed to unlock the latch mechanism and the door swings downward from the hinge on the lower side. The door can be opened or closed from outside the aircraft. The door is moved to the closed position and the latch mechanism is engaged to close it.

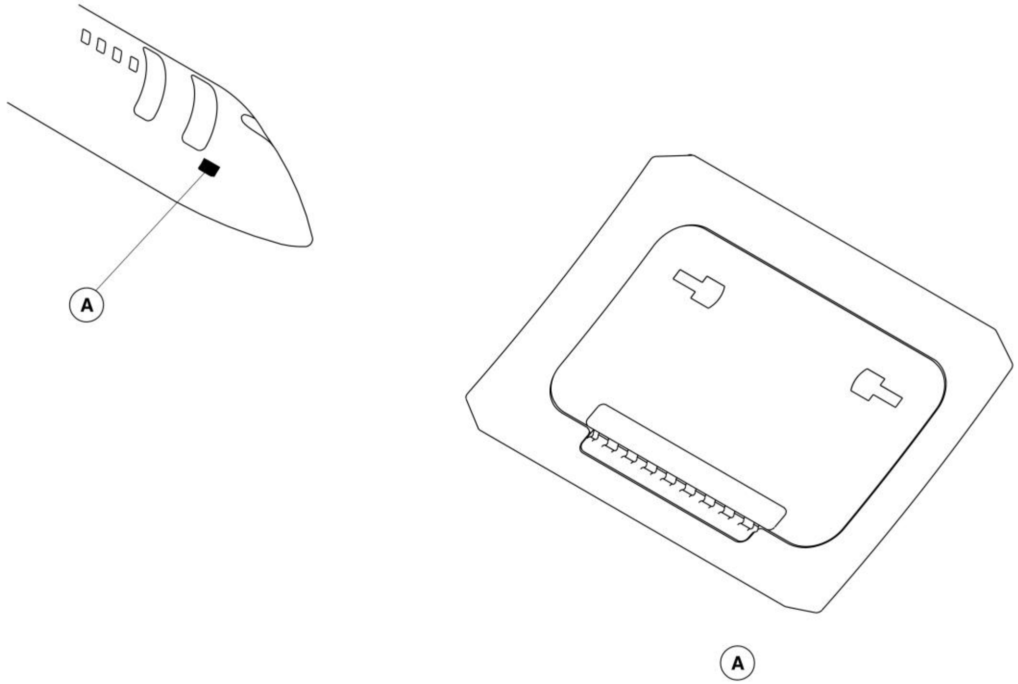


Figure 52-84 Lavatory Service Door – Location

Ground−Crew Interphone−Jack Door

Refer Figure 52-85

The ground−crew interphone−jack door gives access to the interphone jack for communication between the ground crew and the pilot. It is a circular door attached to the fuselage by one hinge and is retained in the closed position by a latch assembly.

The ground−crew interphone−jack door is located on the front fuselage, aft of the left lower door of the nose compartment. To open the door, push the latch lever and release the spring-loaded safety catch. The door housing has a seal to keep out water and contaminants.

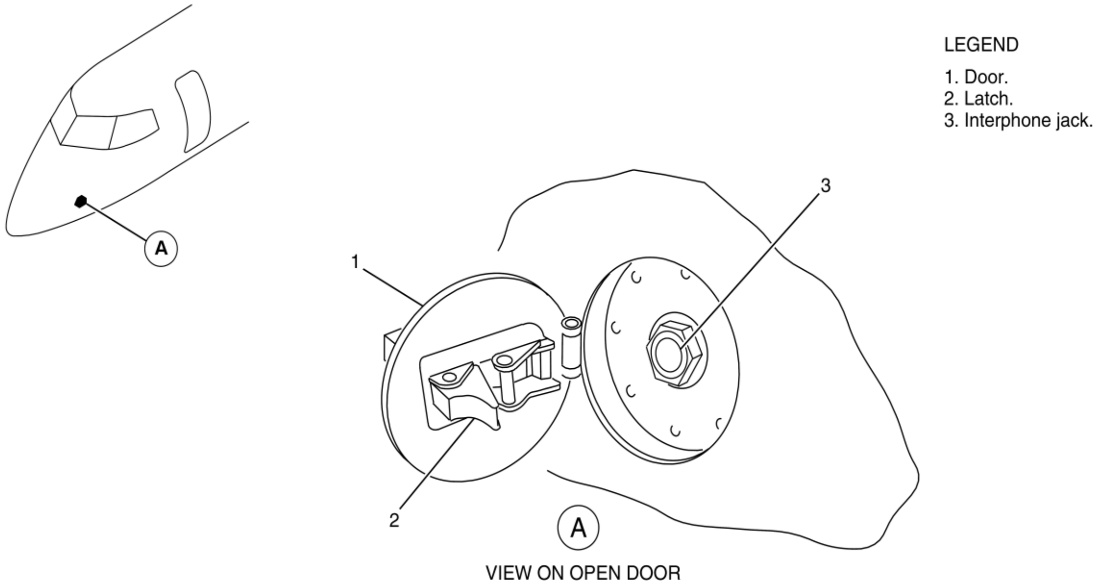


Figure 52-85 Ground-Crew Interphone Jack Door – Location

Nose Compartment Doors

Refer Figure 52-86 & Refer Figure 52-87

There are four nose compartment doors, which give access to the four nose compartments. Two upper nose compartment doors give access to both upper nose compartments. Two lower nose compartment doors give access to both lower nose compartments.

Each nose compartment door has two latches to lock the door. The upper doors are hinged at the top and open upward. The lower doors are hinged at the bottom and open downward. Each nose compartment door has a seal around the door structure to keep out water and contaminants.

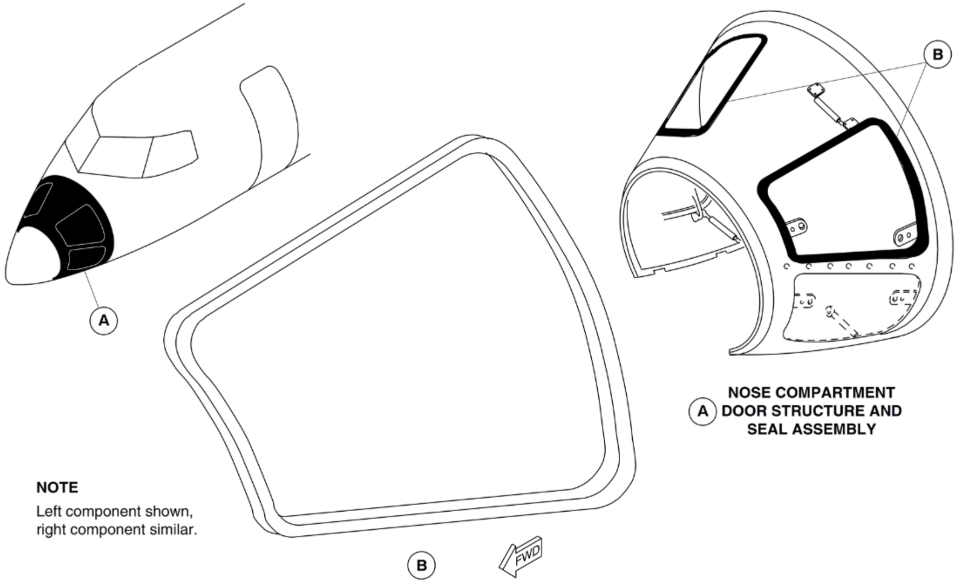


Figure 52-86 Upper Nose Compartment Door – Location

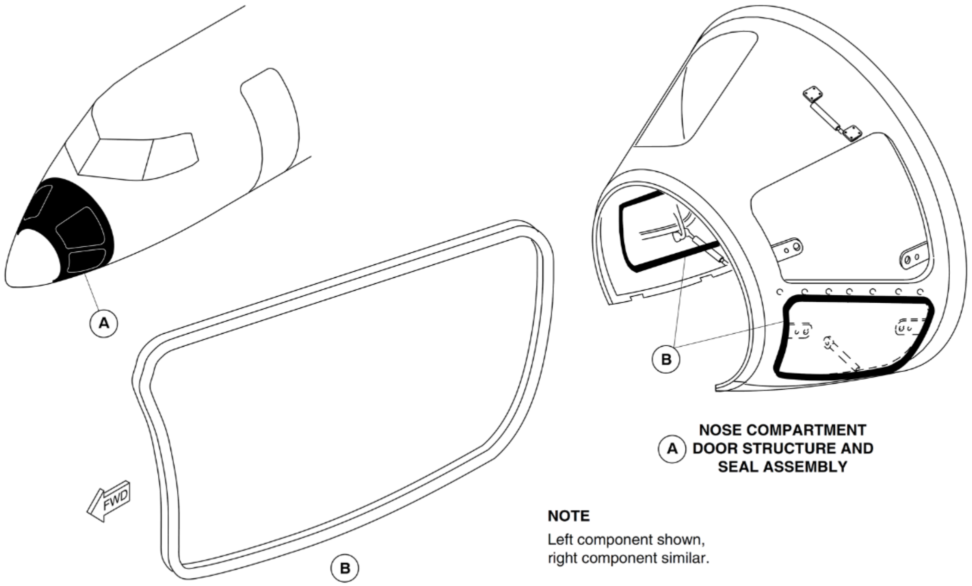


Figure 52-87 Lower Nose Compartment Door – Location

Aft Service Door

Introduction

The aft service door gives access to the rear passenger-compartment, and also functions as a Type I emergency exit.

General Description

Refer Figure 52-88

The aft service door is located on the right side of the aircraft at the-rear of the passenger compartment. The aft service door is classified-as a Type I emergency exit. It is a translating type secured against-pressurization loads by five stops on each vertical edge, that engage-with five related stop brackets installed on the door surround.

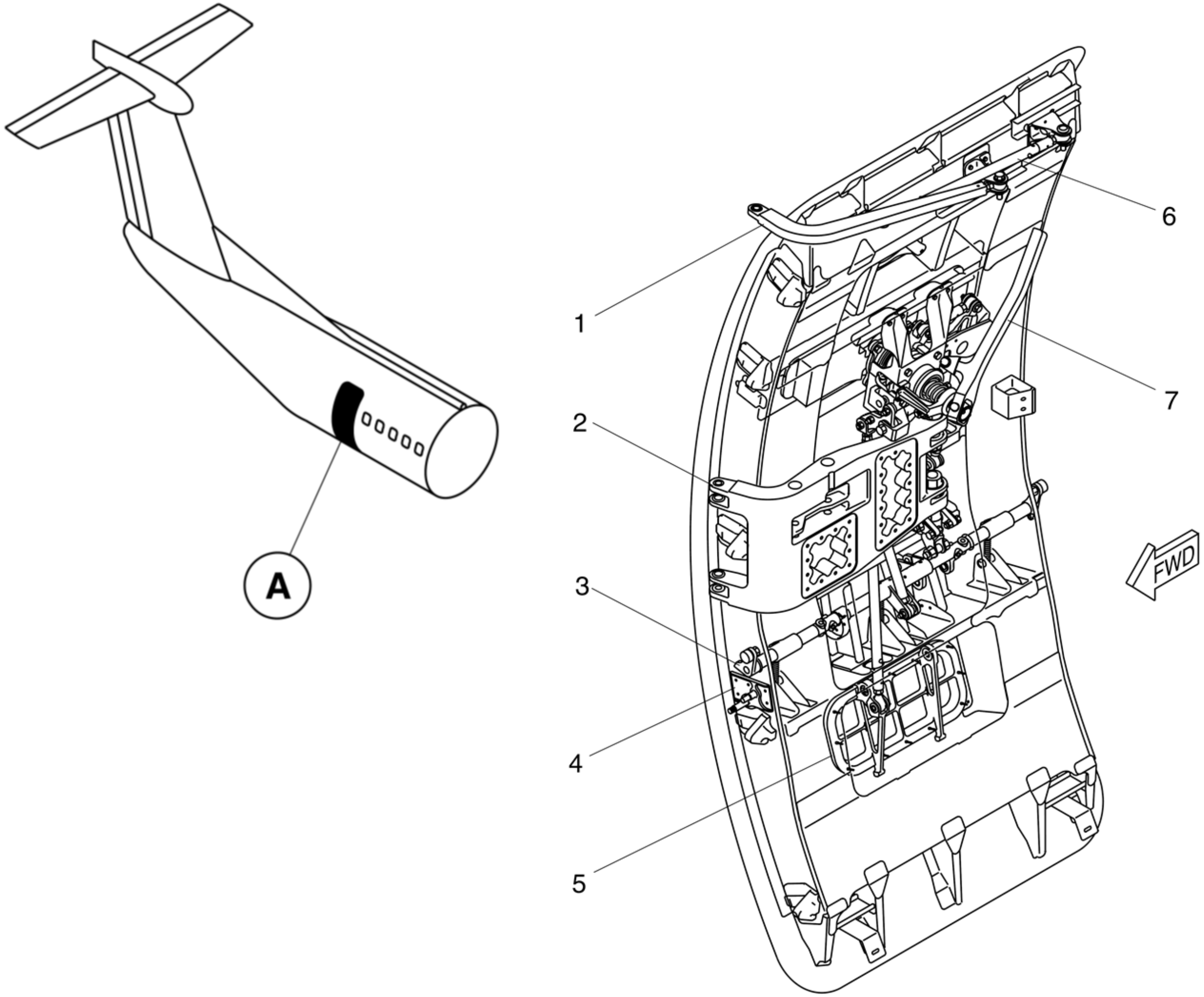
The proximity sensors monitor the position of the upper fixed rollers-and the door handle. A red warning light comes on in the flight-compartment when the door is open and unlocked.

The aft service door system has the components that follow:

- Handle mechanism
- Vent door mechanism
- Lift and latch mechanism
- Hinge mechanism
- Door seal

The Aft service door interfaces with:

- Door Warning System
- Proximity Sensors



LEGEND

1. Secondary Hinge.
2. Main Hinge.
3. Lift And Latch Shaft.
4. Interlock Assembly.
5. Vent Door.
6. Stabilizer Rod.
7. Handle Mechanism.

Figure 52-88 Aft Service Door – Location

Detailed Description

Refer Figure 52-89 & Refer Figure 52-90

The aft service door gives access to the rear passenger compartment and also function as a Type I emergency exit.

The door operating and retention system is operated by the internal or external handle. The internal handle is pulled inwards and turned counter clockwise to unlock the handle. When the door handle is pulled inwards the vent door opens to release pressure in the cabin and the proximity switch opens. When the handle is turned counter clockwise, the lift/latch shaft pushes the lift/latch rollers down, and the door moves inwards and upwards to clear the door stops.

At the uppermost lifted position, the hold−up hook is connected to the hold−up hook pin of the door hinge, and the door is moved outward. As the door moves outward, the interlock cam connect to a latch crank on the lift/latch shaft, to prevent accidental handle operation.

The door has two fixed rollers located on each side around the upper and lower parts, running in guides on the fuselage which give tangential movement of the door. In pressurized flight, the door is held by the stops against pressurization and vertical loads. In unpressurized flight, the door is kept from upward movement by cam rollers on each side of the lift/latch shaft. If one cam roller assembly fails, the second cam roller assembly prevents the door from opening in flight. The lift/latch shaft is secured at the closed position by means of both the handle lock mechanism and the overcenter springs.

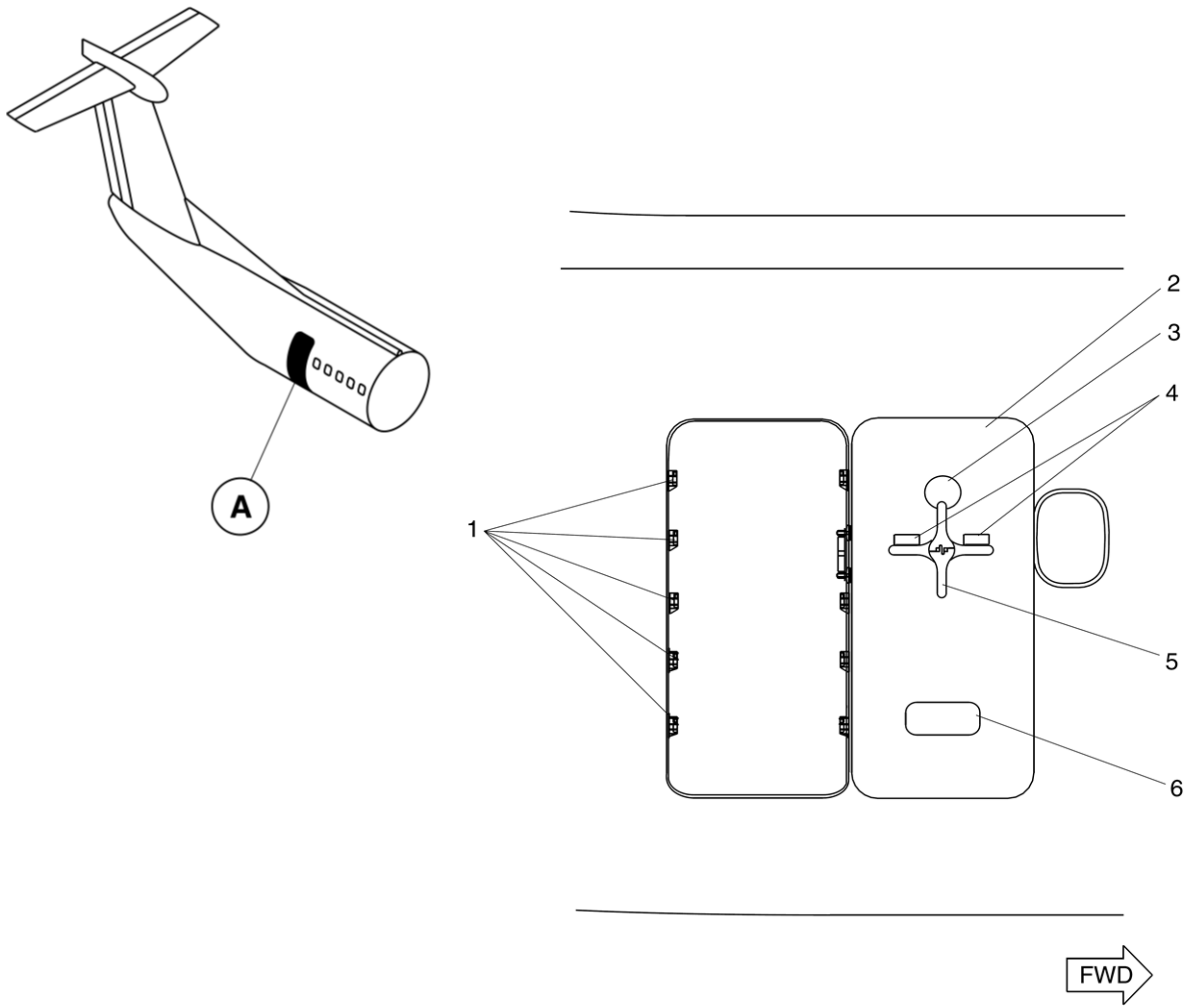
The door is unlocked by operation of the spring loaded internal or external handle. The door opens or closes with the door face in parallel with the fuselage skin. (This is done by two hinges main and secondary and a stabilizer).

When the door reaches the fully opened position, the door open lock hook on the hinge arm is connected to the door open lock pin on the door to keep the door at its position. An interlock assembly prevents handle operation when the door is opened, preventing the hold−up hook from being unlatched. The external handle is kept in a recess in the outer skin and is connected to the same handle shaft as that of the internal handle. The handle can be accessed when the flap covers are pushed in. The handle is then pulled out and turned clockwise to open the door.

For safety, a window in the door gives a view of the outside before the door is opened. To close the door, the release lever adjacent to the lock hook is pulled to release the open lock, and the door is brought to the closed position. The internal handle is turned clockwise to disconnect the door hold−up hook from the hold−up hook pin. The door is moved to the closed position, and the stops align with their related stop bracket lugs.

The handle is moved outward to lock the door and close the proximity switch. The door can also be closed externally by using the external handle.

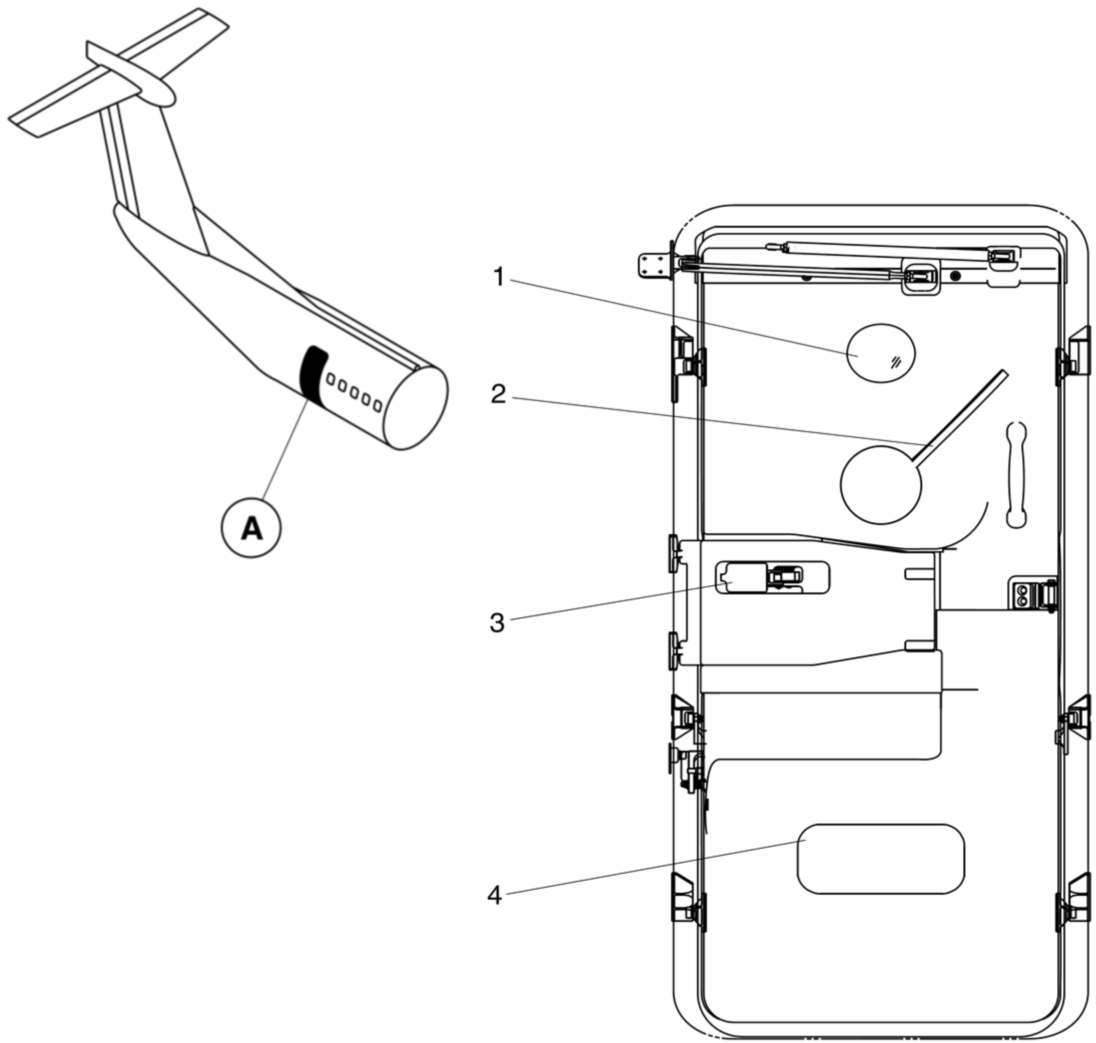
The main door and vent door are sealed around their periphery. The seal is inflated by cabin pressure through holes in the seal, to keep the gap around the door closed. Cabin pressurization is prevented when the vent door is not fully closed and locked.



LEGEND

1. Stops (5 Per Side).
2. Aft Service Door (Open Position).
3. Window.
4. Flap Cover.
5. External Handle (Unlocked).
6. Vent Door (Open).

Figure 52-89 Aft Service Door – Detailed External View



LEGEND

1. Window.
2. Internal Handle.
3. Open Lock Lever.
4. Vent Door.

Figure 52-90 Aft Service Door –Detailed Internal View

Mechanism, Handle

The internal handle operates the lift/latch shaft, and the vent door operating rod. The lower end of the internal handle is connected to the shaft of the handle mechanism.

When either the internal or external handle operated, the handle shaft can be turned to operate the lift/latch through cranks, rods and an idler link. The external handle is kept flush with the door skin and locked in the closed position.

Mechanism, Vent

The linkage from the handle transmits the handle input to the vent door mechanism. When the door handle is pulled, the vent door opens to release pressure from the cabin.

When the door is in the closed position, the final movement to lock the internal or external handle closes the vent door. The vent door is spring−loaded in the direction of the open position, so that it is kept open in case of fracture or disconnection of the vent door operating mechanism.

The primary function of the vent door is as follows:

- Release residual cabin pressure to the atmosphere before the door is opened, in order to make outward load on the door minimal
- Prevents pressurization loads on the door in case of manual pressurization before the door is fully closed and locked.

Mechanism, Lift and Latch

The linkage from the handle to the lift/latch shaft transmit handle input to the shaft and locks the lift/latch shaft assembly when the handles are in their locked position.

The bias spring assembly for the latch causes an overcenter against the lift/latch shaft, and locks it at the closed or open position. The overcenter position and linkage from the handle mechanism, gives dual locking of the lift/latch shaft.

The lift/latch shaft is installed to a cam roller assembly at both ends. When the door is opened, the lift/latch shaft turns and let the rollers push down the guide tracks, lifting the door for outward movement. The door operation sequence is reversed at door closing. When the door is in the closed position, the roller center is offset inward against the door lift−up plane, resulting in the rollers being kept in the door closing direction, against the upward movement of the door. –

Mechanism, Hinge

The main and secondary hinge arms and the door makes a parallelogram which keeps the door face in parallel with the fuselage skin during door opening or closing. During the travel of the door around the furthest point from the fuselage, the yawing movement of the door is suppressed by the stabilizer rod.

Seal, Door

The door seal is of the passive type which has several holes in the seal and is inflated by cabin pressure through the holes. This type of seal is installed on both the main and vent doors.

Aft Fuselage Access Door

Introduction

The aft fuselage access door gives access to equipment in the unpressurized area of the aft fuselage.

General Description

The Aft Fuselage Access Door is located flush with the underside of the rear fuselage, through which the unpressurized equipment compartment is accessed. The door is opened or closed from the exterior of aircraft.

The aft fuselage access door includes the components that follow:

- Latches
- Strut

Detailed Description

Refer Figure 52-91

The door is installed in the fuselage structure at the right side by hinges, and at the left side by two latches. The door has a cast aluminum panel with a louvered opening to increase the airflow in the compartment. Moisture drain holes are located in the panel of the door to drain any accumulated water.

The latch release buttons are pushed to unlock or lock the door. The door movement is outwards and then downwards. A telescopic strut keeps the door in the open position when the screw lock is tightened. A moulded rubber seal around the door cut out prevents door chafing.

To close the door, the screw lock is loosened to unlock the telescopic strut. The door is then pushed closed and the latch buttons pushed to lock the door.

Latches

The latch mechanism controls the opening and closing of the door. The release buttons are pushed to lock or unlock the latch mechanism.

Strut

The telescopic strut keeps the door in the open position.

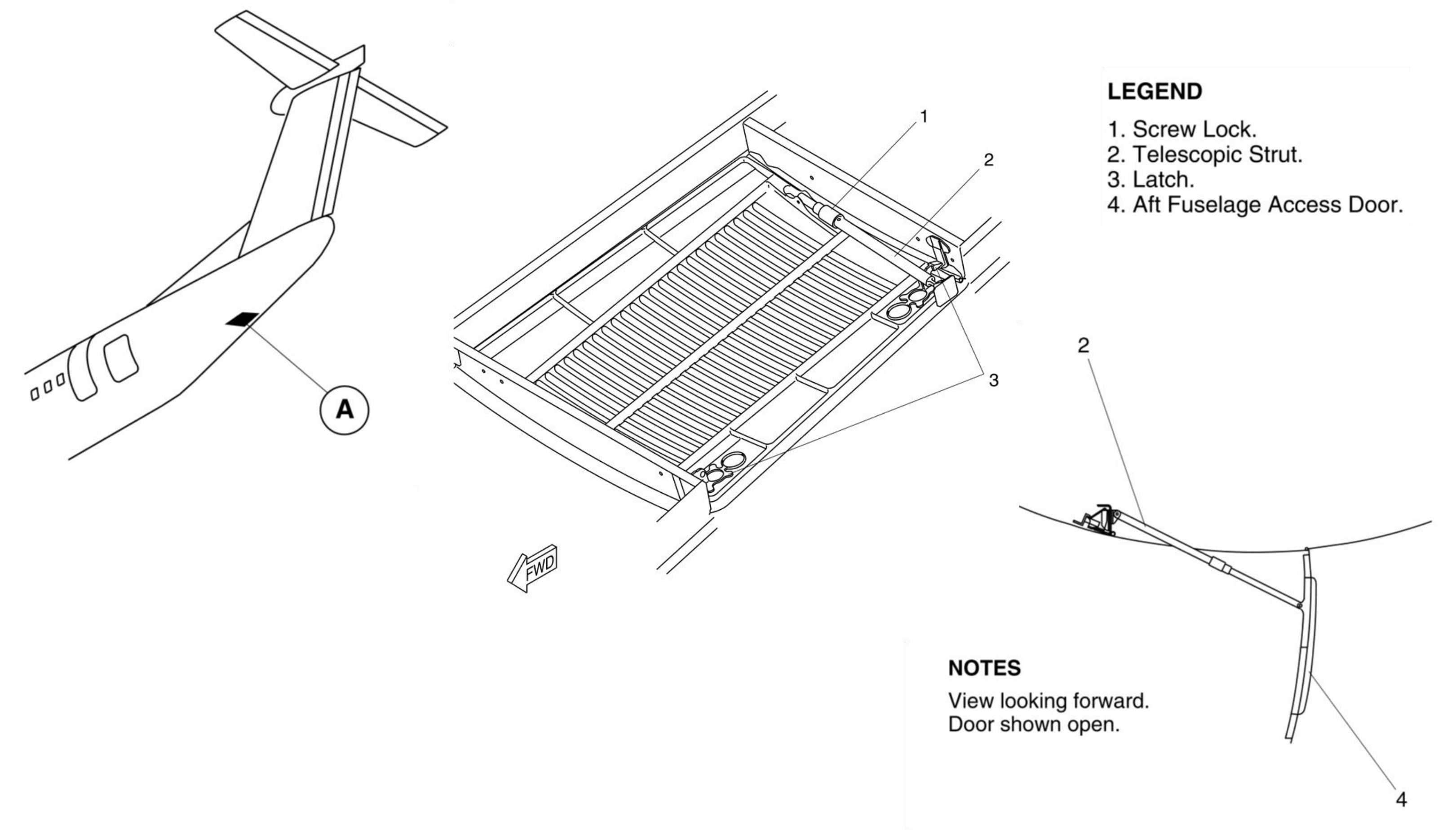


Figure 52-91 Aft Fuselage Access Door

APU Access Door

Introduction

The Auxiliary Power Unit (APU) access doors give access for ground maintenance of the optional APU.

General Description

The doors are located on the undersurface of the tailcone section on either side of the fuselage centerline. The left and right doors form a large part of the bottom of the tailcone, and together make up a clamshell door system.

The doors are hinged on the horizontal plane and open outward and upward, to give access to the APU engine assembly and components. The doors are held open by telescopic struts.

The APU access doors include the components that follow:

- Two Door Struts (one per door)
- Three Hook Latches
- Three Keepers
- Three Shear Pins
- Two Pin Latches
- Four Hinge Assemblies (two per door)
- Door and APU Drain Mast Seals.

Detailed Description

Refer Figure 52-92

The APU access doors are located on the bottom of the tailcone section on either side of the fuselage centerline. The doors are hinged horizontally and kept in a locked and closed position by the two pin latches, hook latches and keepers, and the three shear pins.

The APU access door structure is comprised of a thin skin external shell which is supported by longitudinal intercostals and transverse frames. The skin is 0.025 in. (0.0625 mm) thick commercially pure titanium, type CP–1.

The frame and intercostal members are also manufactured from type CP–1 titanium, 0.025 in. (0.0625 mm) thick. Inner skins along the upper and lower edges have been added to increase the torsional rigidity of the doors in the open position.

The two large clamshell–type doors are flush with the tailcone when in the closed and locked position. The doors are opened by unlatching the pin latches and the hook latches. The struts help the upward and outward movement of the doors. Once fully extended, the struts lock the doors in the open position. Closing of the doors is done by unlocking the strut push–release mechanisms, closing the doors and engaging the latches. The door will stay closed if both pin latches, and two of the three hook latches fail.

The fire proof seals give an air tight seal around the APU access doors and drain mast. These seals help prevent leakage of the fire extinguishing agent, if the agent is discharged.

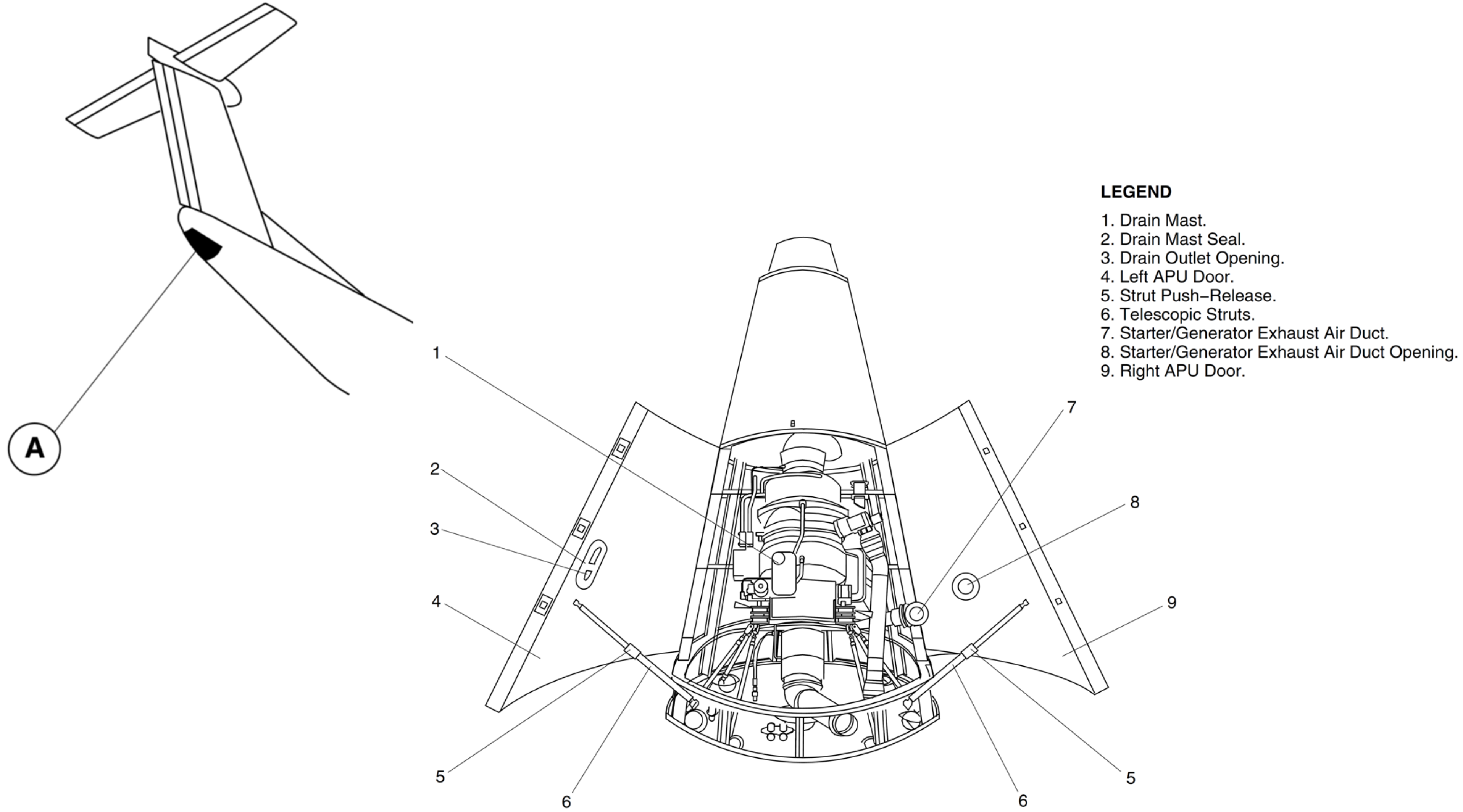


Figure 52-92 APU Access Door – Detailed

Door Struts

There is one telescoping, maintenance−free strut per door. The struts are attached to the canted station 994.1 frame and installed on each door by adjustable rod ends and stainless steel brackets. The struts are locked at the fully extended position and unlocked by push−release mechanisms. The struts can withstand a 1,000 lb. (453.6 kg) axial load or a 150 lb. (68 kg) side load.

Hook Latches, Keepers, and Shear Pins

The three hook latches on the left door, and keepers on the right door, keep pre−tension across the two doors at the door centerline. All hook latch components are made from either 17−4PH stainless steel, 17−4PH stainless steel, or CRES 304/305.

The main hook assembly is cad plated 4130 steel. A portion of the latch is painted dayglo orange and will be visible if the hook latch is not fully engaged. Three shear pins prevent relative movement between the doors along the door centerline. The shear pins are installed on the right door and engage with related strike plates, installed on the left door, when the doors are closed. The shear pins fit into close tolerance holes in the strike plates.

Pin Latches

The two pin latches are installed on the right door, one forward and one aft. The latches prevent relative movement between the doors and the tailcone structure. All pin latch components are made from either 17−4 CRES or 17−7 CRES cad plated.

Hinge Assemblies

Two hinge assemblies are installed on each door. A hinge assembly has a hinge lug fitting, a hinge arm, and a hinge base fitting. The hinge assembly components are made from 17−4PH stainless steel. The hinge assemblies give outward movement of the doors.

Door and Drain Mast Seals

There are seals on each door and the APU drain mast opening. The seals are made from silicone rubber (ZZ−R−765, Class 3B, Grade 50) covered with a protective nomex weave fabric. The seals meet the fire requirements of 2,000 °F (1080 °C) for 15 minutes. The door seals are kept in place by a titanium strap.

Training Information Points

Do not use permanent or temporary markers on titanium, plastics, composite materials, or painted surfaces. If you do this, you can cause damage.

Do not use chlorinated solvents. They can cause a hydrogen embrittlement of the titanium surfaces.

Titanium parts must not touch lead or zinc alloys. This can cause contamination damage to the parts surface.

Ground Service Doors

Introduction

The ground service doors give access for maintenance and servicing purposes.

General Description

The ground service doors are identified as follows:

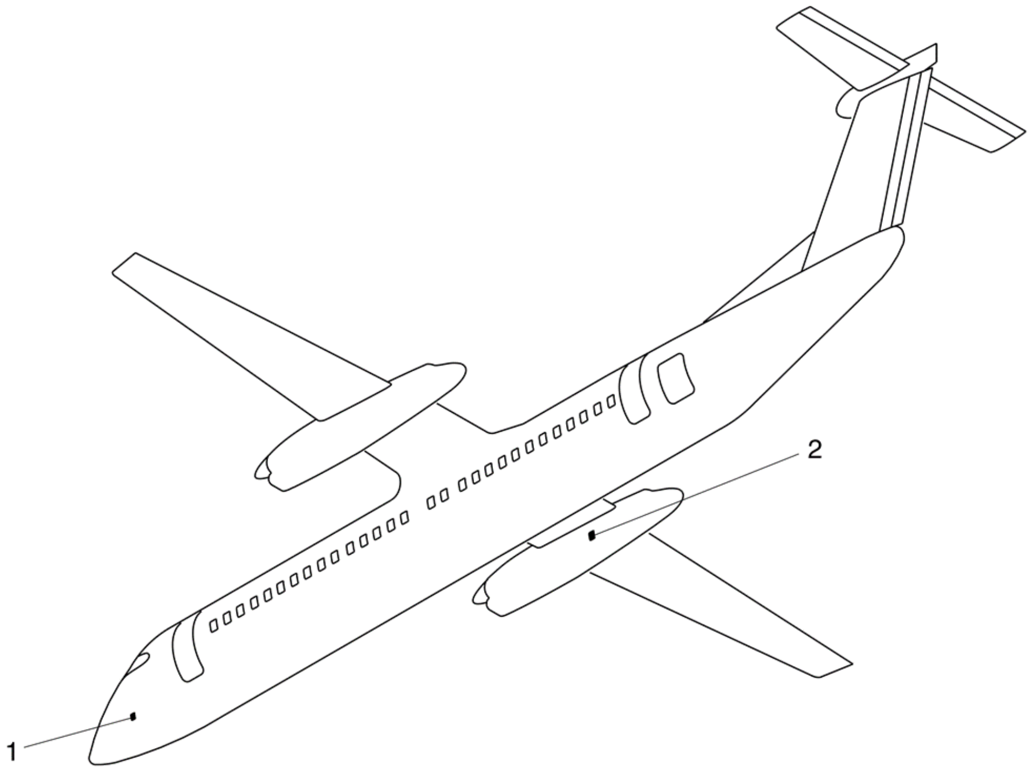
- Refuel/Defuel Panel Door
- DC Electrical Ground Power Door
- AC Electrical Ground Power Door
- Hydraulic Ground Power Door
- Ground Air Conditioning Connection Door
- Lavatory Service Door

The Refuel/Defuel Panel door is the only ground service door that will give an indication to the flight compartment, if open.

Detailed Description

Refer Figure 52-93 & *Refer Figure 52-94*

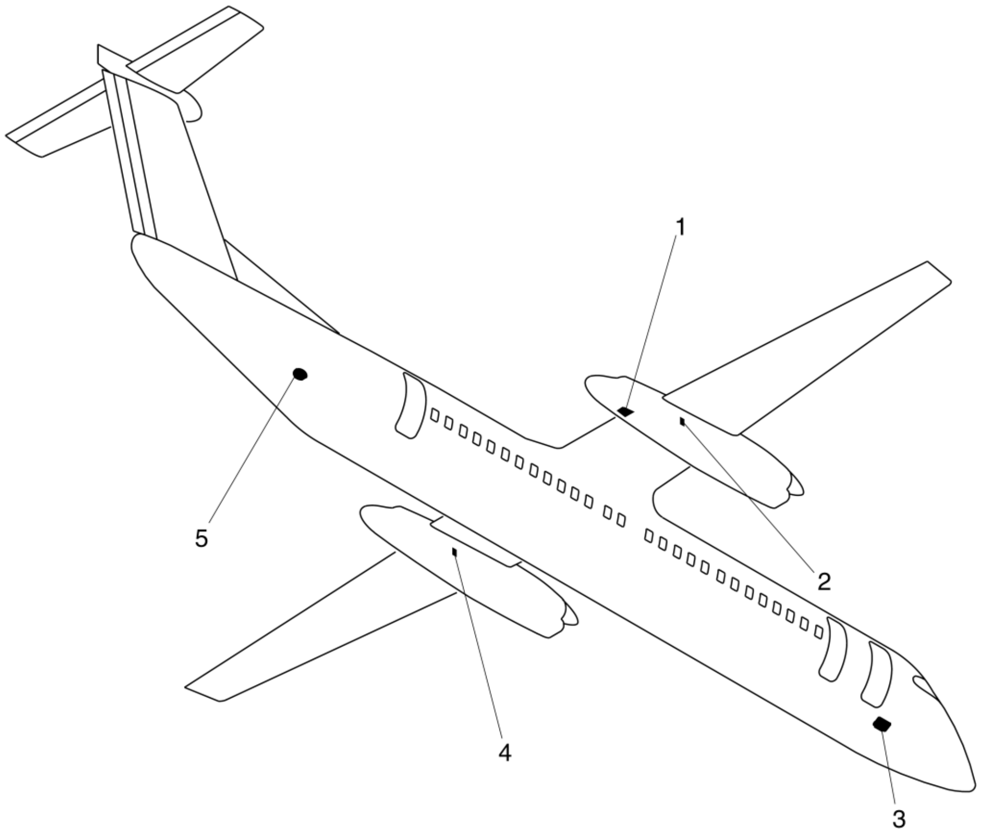
Service doors are installed at various locations on the aircraft to give access for servicing and maintenance purposes.



LEGEND

1. DC Electrical Ground Power Door.
2. AC Electrical Ground Power Door.

Figure 52-93 Electrical Ground Power Doors – Locations



LEGEND

1. Refuel/Defuel Panel Door.
2. Right Nacelle Hydraulic Ground Power Door.
3. Lavatory Service Door.
4. Left Nacelle Hydraulic Ground Power Door.
5. Ground Air Conditioning Connection Door.

Figure 52-94 Ground Service Doors – Locations

Refuel/Defuel Panel Door

Refer Figure 52-95

The Refuel/Defuel Panel door is located on the lower side of the No.-2 engine nacelle, aft of the right main landing gear rear doors. The-door gives access to the refuel/defuel adapter, Refuel/Defuel Indicator (RDI), and switches to carry out pressure refueling/defueling operations.

The front end of the door is attached to the lower right engine nacelle by a piano hinge assembly, and two latch assemblies at the aft end. The latch release buttons are pushed to unlock the latch release mechanism and the door moves downwards to the open position.

A steel braided cable assembly limits door travel in the open position. The door is moved upwards to engage the latches and to close and lock the door flush with the nacelle.

The Refuel/Defuel Panel Door operates a door switch, installed on the nacelle door structure. When the door is open, the switch causes the amber MASTER VALVE CLOSED light on the Refuel/Defuel panel, and the FUELING ON caution light on the Caution and Warning Panel to come on.

The service light, installed on the aft vertical section of the Refuel/Defuel compartment, also comes on. When the Refuel/Defuel Panel Door is closed and locked, it operates the door switch, which causes the FUELING ON caution light to go off. The MASTER VALVE CLOSED light and the service light also go off.

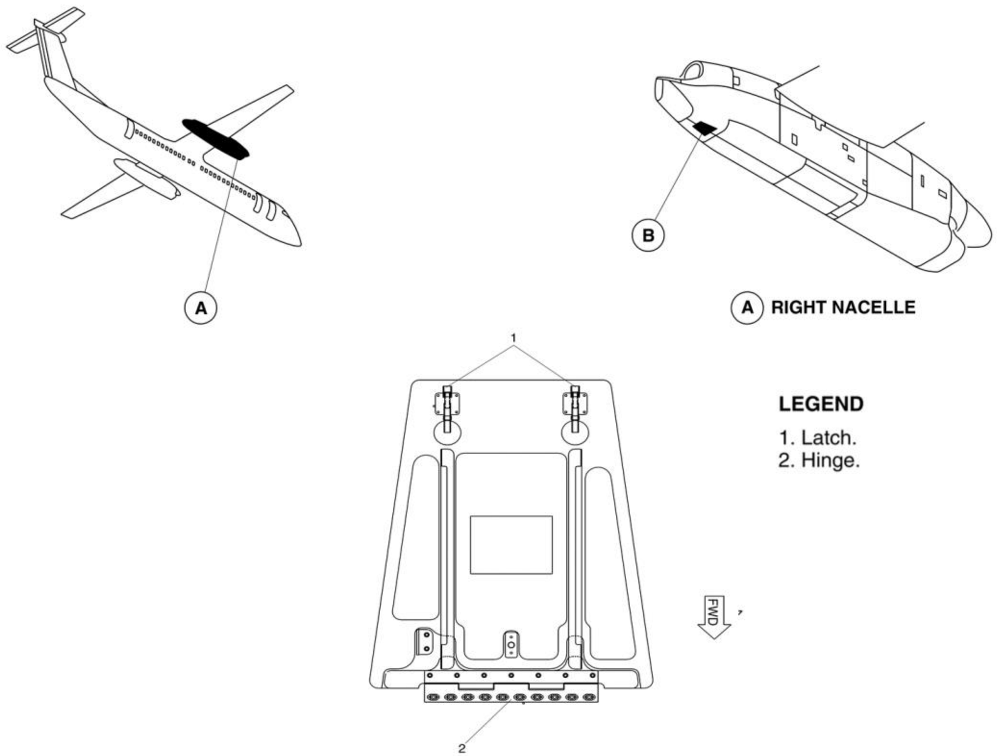


Figure 52-95 Refuel/Defuel Panel Door

DC Electrical Ground Power Door

Refer Figure 52-96

The DC electrical ground power door gives access to the 28 Vdc ground power receptacle on the left side of the aircraft nose. The door is attached to the nose section by a hinge assembly at the forward end and a latch assembly at the aft end.

To open the door, the latch release button is pushed to release the safety catch and the door is moved to the open position. To close the door, it is moved to the closed position and the latch is engaged.

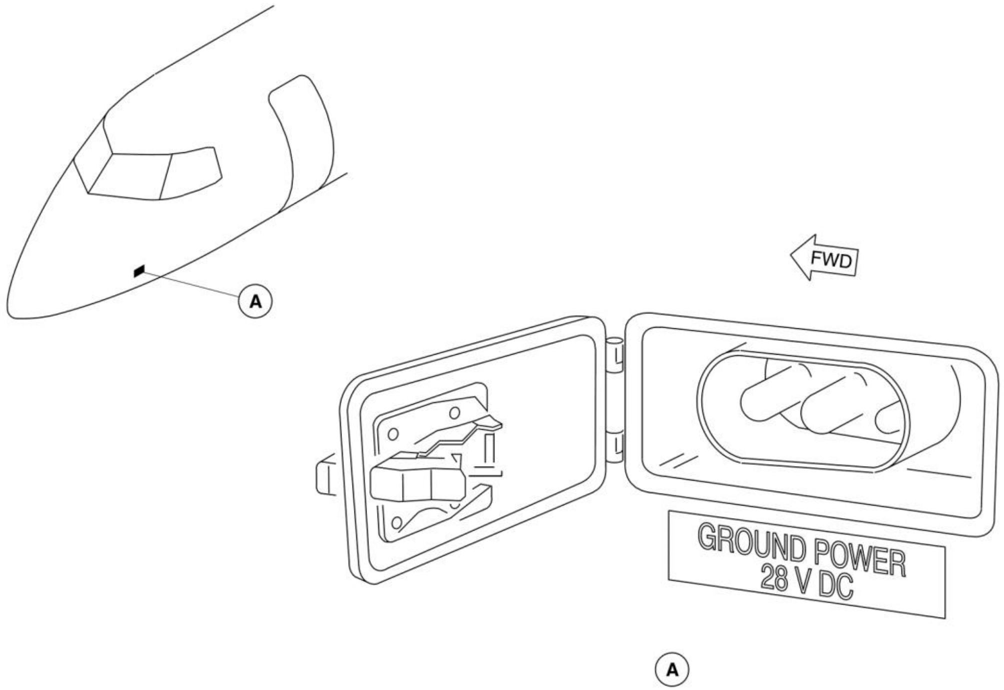


Figure 52-96 DC Electrical Ground Power Door

AC Electrical Ground Power Door

Refer Figure 52-97

The AC electrical ground power door gives access to the ac ground power receptacle on the left side of the No. 2 engine nacelle. The door is attached to the nacelle by a hinge at the top and two latch assemblies at the bottom. To open the door, the latch release buttons are pushed to release the safety catches. Both latches are engaged to lock the door.

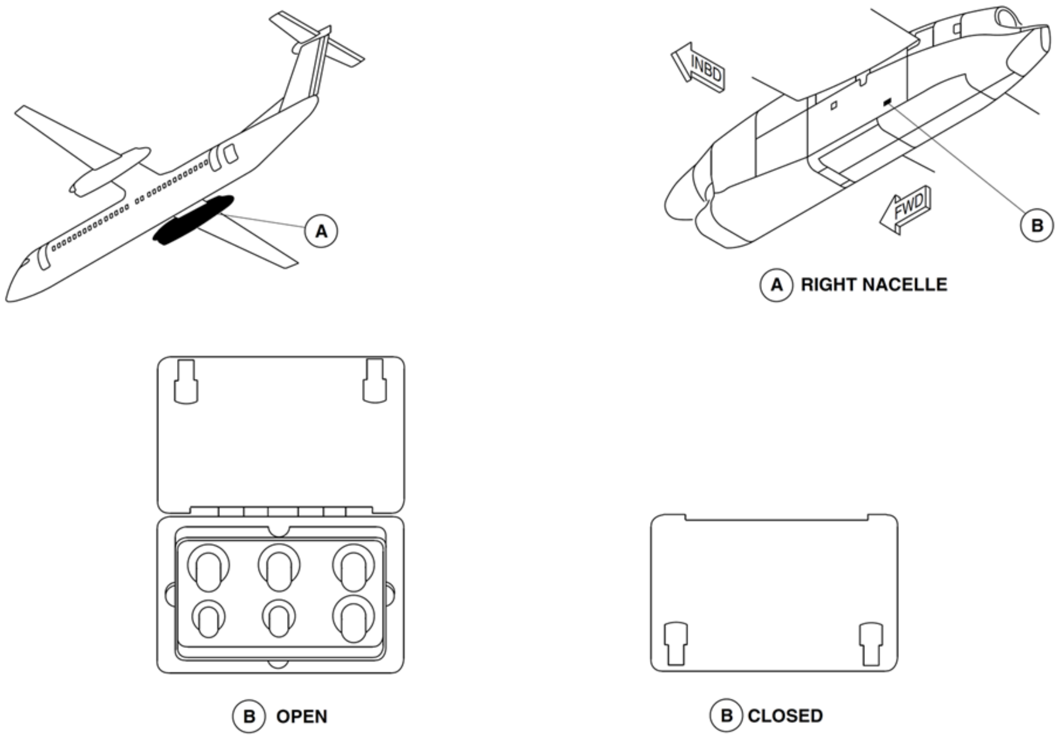


Figure 52-97 AC Electrical Ground Power Door – Detailed

Hydraulic Ground Power Door

Refer Figure 52-98

The hydraulic ground power doors, on the right side of both engine nacelles, give access to the hydraulic ground power connections. The doors are attached to the nacelle by a hinge at the top and two latch assemblies at the bottom. To open the door, the latch release buttons are pushed to release the safety catches. Both lathes are engaged to lock the door.

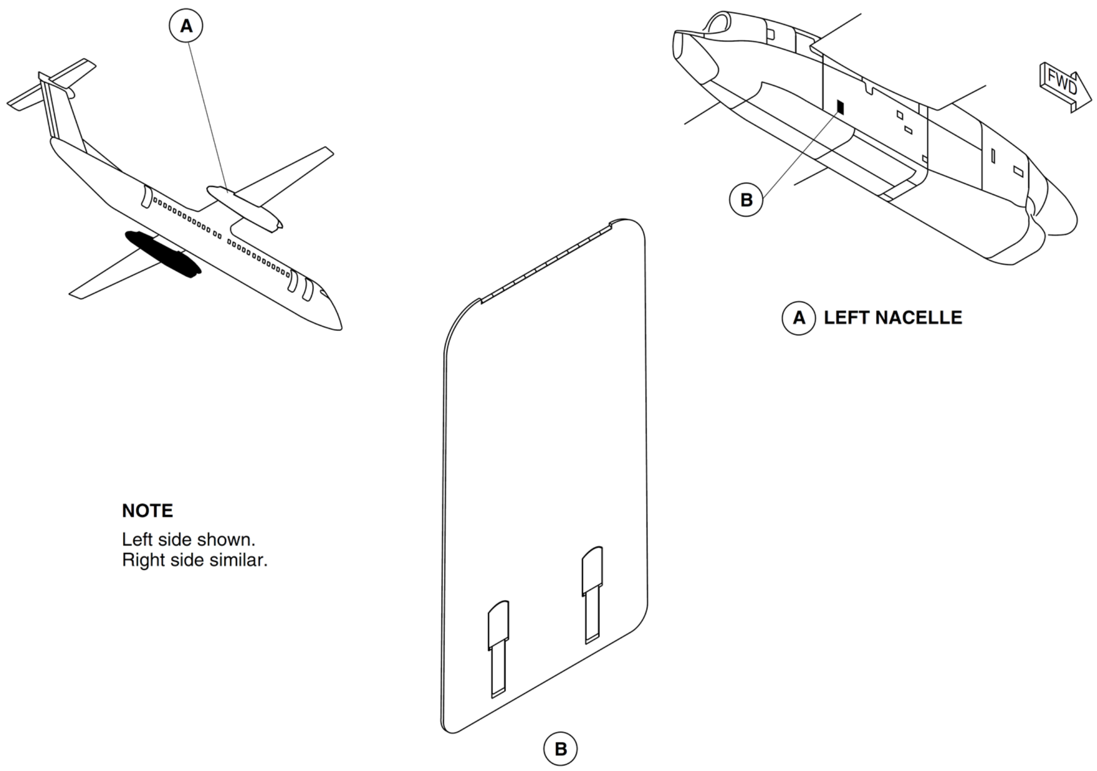


Figure 52-98 Hydraulic Ground Power Door

Ground Air Conditioning Connection Door

Refer Figure 52-99

The ground air conditioning door is located flush with the lower right side of the rear fuselage. The door opens outward and downward to give access to the air conditioning duct connection. The door is attached to the fuselage structure at the bottom by a hinge and at the top by a latch. A gasket is installed around the door to help prevent water from entering the area, and to help prevent chafing of the door.

The latch release button is pushed to unlock the latch mechanism and the door moves downward from the hinge on the door lower edge. The door is moved to the closed position and the latch mechanism is engaged to close it.

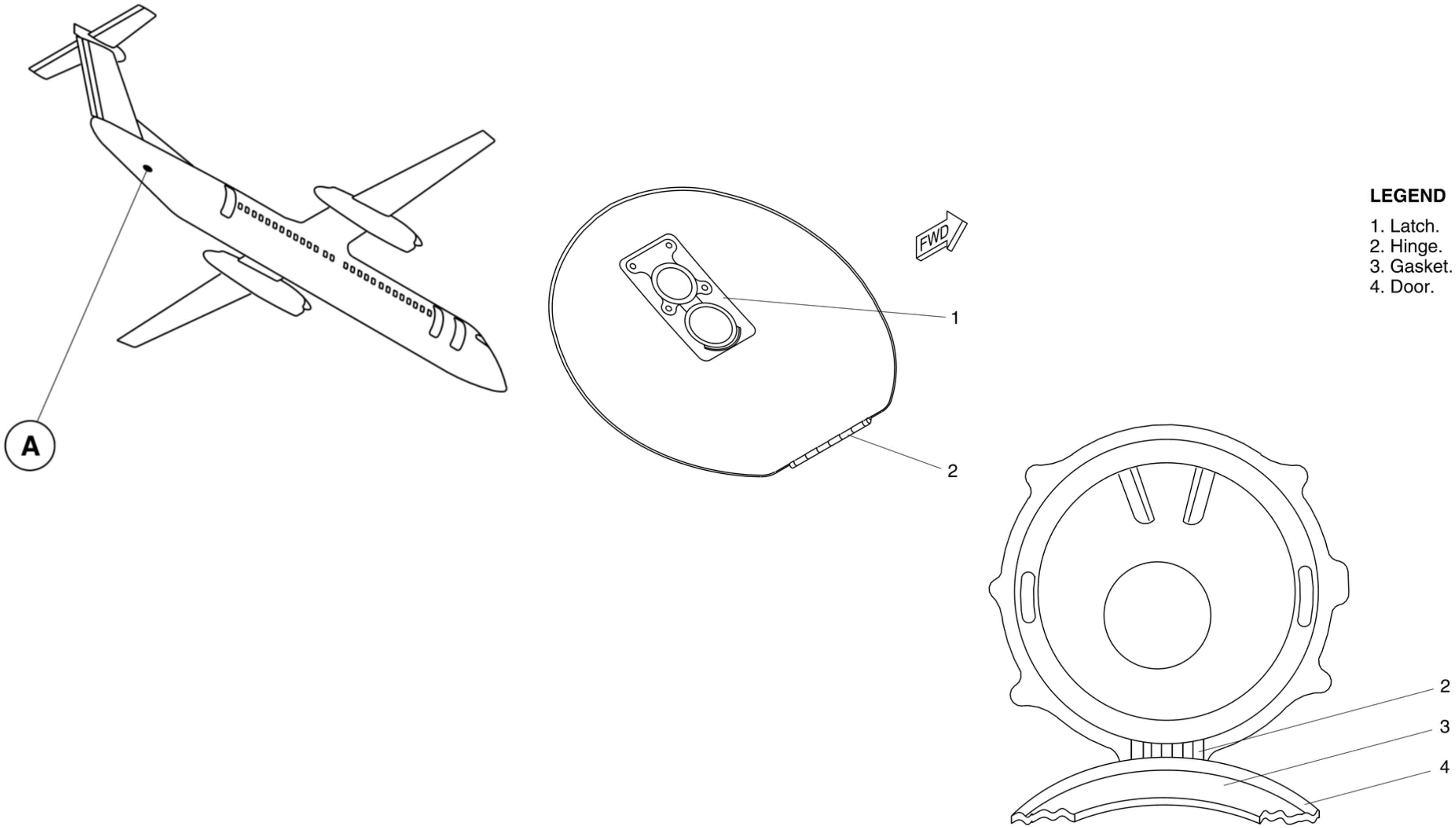


Figure 52-99 Ground Air Conditioning Connection Door

Lavatory Service Door

Refer Figure 52-100

The lavatory service door is located flush with the lower right side of the forward fuselage. The door opens outward and downward to give access to the lavatory service port. The door is attached to the fuselage structure, on the lower side by a hinge and on the upper side by two latches. The door has a gasket installed around the edge to help prevent chafing.

The latch release buttons are pushed to unlock the latch mechanism and the door swings downward from the hinge on the lower side. The door can be opened or closed from outside the aircraft. The door is moved to the closed position and the latch mechanism is engaged to close it.

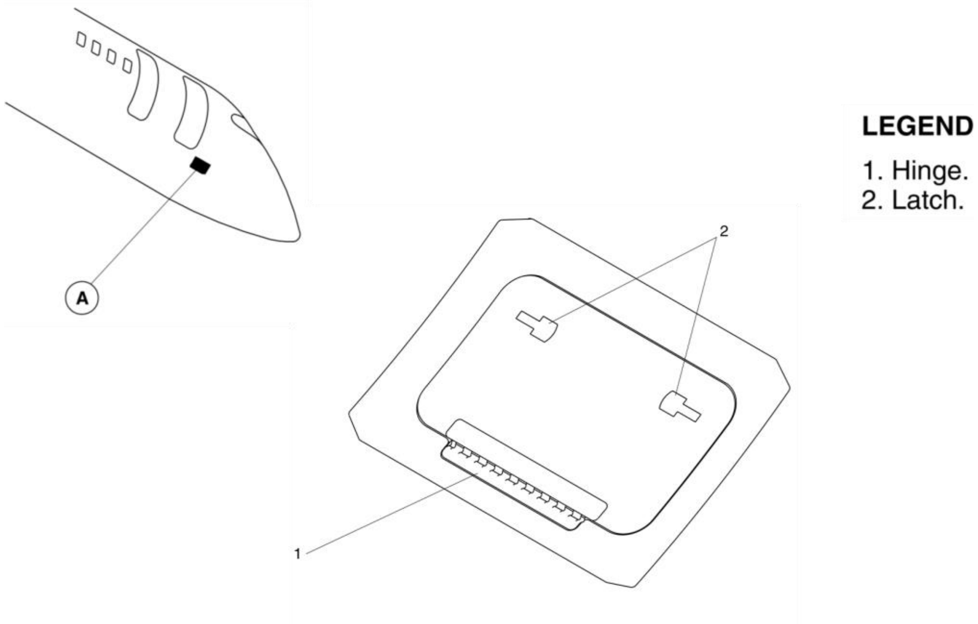


Figure 52-100 Lavatory Service Door

Interior Doors

Introduction

The door gives access to the flight compartment from the passenger compartment.

General Description

The flight compartment door separates the passenger compartment from the flight compartment.

The door can be locked to prevent passenger entry. It is hinged on the left side and swings aft to open.

The door includes a locking mechanism and a rotating handle on both sides of the door.

Detailed Description

Refer Figure 52-101

The handle on the flight compartment side opens the door if the door is locked or not locked. The lock is disengaged by operating the small lever on the flight compartment side of the door, or by the using a key on the passenger side. The key can only be removed when the door lock is engaged.

In an emergency, the door can be removed from its hinges, by removing the hinge pins from the upper and center hinges and lifting the door from the bottom hinge.

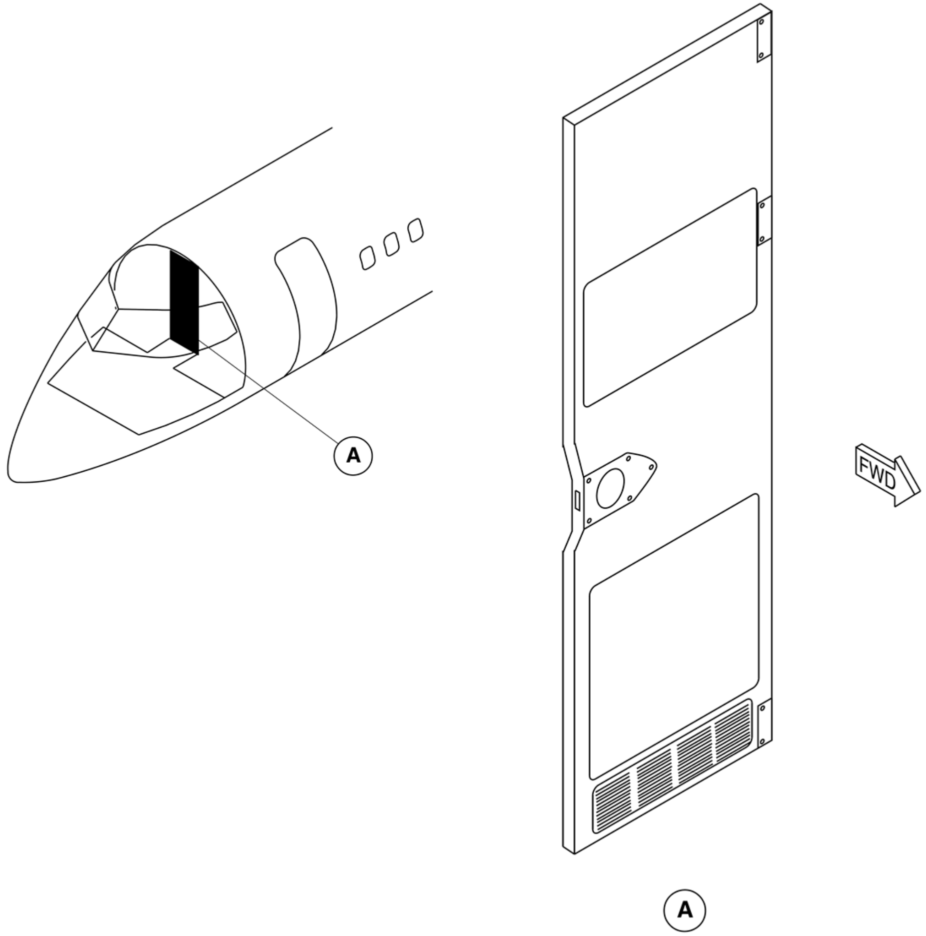


Figure 52-101 Flight Compartment Door

Fortified Flight Compartment Door and Remote Access System

Introduction

The Fortified Flight Compartment Door and Remote Access System are optional and consist of the following major components:

- Fortified Flight Compartment Door
- Internal Doors Control Panel
- Flight Attendant Emergency Access Push−button Annunciator.

General Description

The Fortified Flight Compartment Door and Internal Doors Control-Panel provide the flight crew control of access to the flight deck. A-Flight Attendant Emergency Access Push−button Annunciator-enable the cabin crew to gain access to the flight compartment-should the flight crew become incapacitated.

Fortified Flight Compartment Door

The Fortified Flight Compartment Door separates the flight-compartment from the passenger compartment. It is hinged on the-left side door post on the pilot's bulkhead and opens aft into the-passenger cabin.

The Fortified Flight Compartment Door meets ballistic, intrusion and-decompression requirements. In the case of decompression, a-barometrically actuated lock releases a vent panel located on the-lower door allowing the panel to open forward and down. The door includes a locking mechanism consisting of a slide latch-and a deadbolt lock.

Internal Doors Control Panel

The Internal Doors Control Panel is located on the overhead console-and has the label "INTERNAL DOORS" inscribed on the face plate. It-allows full control and indication of the flight compartment door-during flight operations. The "INTERNAL DOORS" caution light on the Caution and Warning-Panel monitors both the internal forward baggage door and the flight-compartment door.

Flight Attendant Emergency Access Push−button Annunciator

The emergency access mode consists of a Flight Attendant-Emergency Access Push−button Annunciator located on the-Maintenance Panel in the forward cabin. It permits access by the-cabin crew to the flight compartment in the event that there is no-response from the flight crew.

Detailed Description

The Fortified Flight Compartment Door, Internal Doors Control Panel-and Flight Attendant Emergency Access Push−button Annunciator-functions are described as follows:

Fortified Flight Compartment Door

Refer Figure 52-102 & Refer Figure 52-103

A slide latch, operable from the flight compartment side only, locks-the door.

To secure a vacant flight compartment, a deadbolt lock located-below the slide latch permits the flight compartment to be locked-from the cabin side using a key.

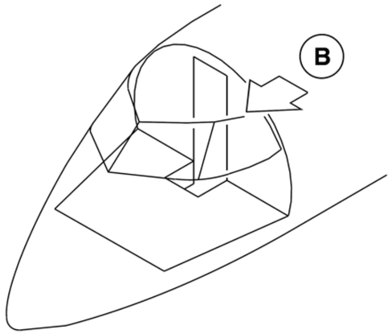
The deadbolt can be operated from the flight compartment by turning-a rotary action handle. When the dead bolt rotary knob is set to the-unlatch position, two red dots on the dead bolt plate behind the knob-will become visible. When the key is rotated in the lock, one of the-arms of the rotary action handle will rotate 90 degrees covering half-of each red dot. When the rotary knob is set to the latched position,-two green dots will become visible and actuation of the key access-feature will be disabled.

In the case of a door jam, three door hinge pins accessible only from-the flight compartment can be retracted to allow the door to be-removed as follows:

- Unlock the hinge by rotating the hinge bail outboard to release it from its lock position and push or step down on bottom hinge pin, then lock it in its retracted position.
- Unlock and pull down upper hinge pin.
- Unlock and lift middle hinge pin.
- Push flight compartment door at hinge side.
- Rotate the whole flight compartment door counter clockwise and stow against the lavatory wall.

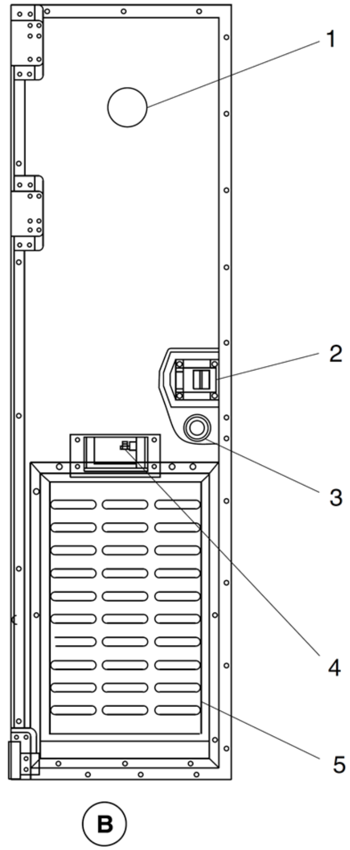
NOTE: It may require a large force to open the flight compartment door.

Upon forcing the flight compartment door open, it may fall straight aft and lay flat on the cabin floor.



LEGEND

1. Spy hole.
2. Slide latch (operable from flight compartment only).
3. Deadbolt lock.
4. Barometric vent panel lock.
5. Vent panel.



VIEW LOOKING FWD ON FLIGHT COMPARTMENT DOOR

Figure 52-102 Fortified Flight Compartment Door – Detailed 1 of 2

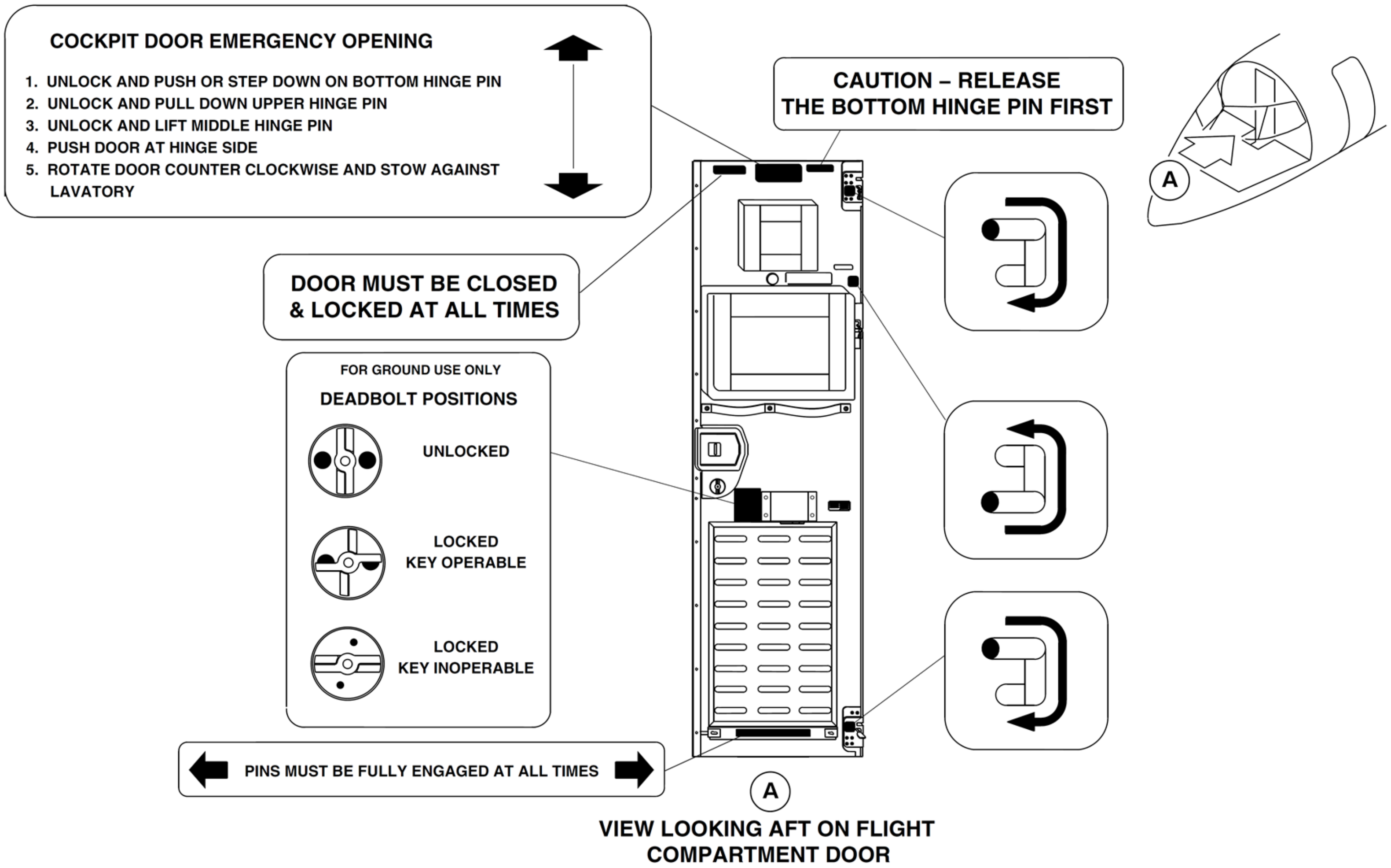


Figure 52-103 Fortified Flight Compartment Door – Detailed 2 of 2

Internal Doors Control Panel

Refer Figure 52-104

The Internal Doors Control Panel is located on the overhead console,-directly below the Fire Protection Panel and to the right of the DC-Control Panel. This panel consists of the following:

- Baggage Door / Cockpit Door light
- Rotary Knob
- Fail / Auto Unlock light
- Lock Isolate guarded switchlight

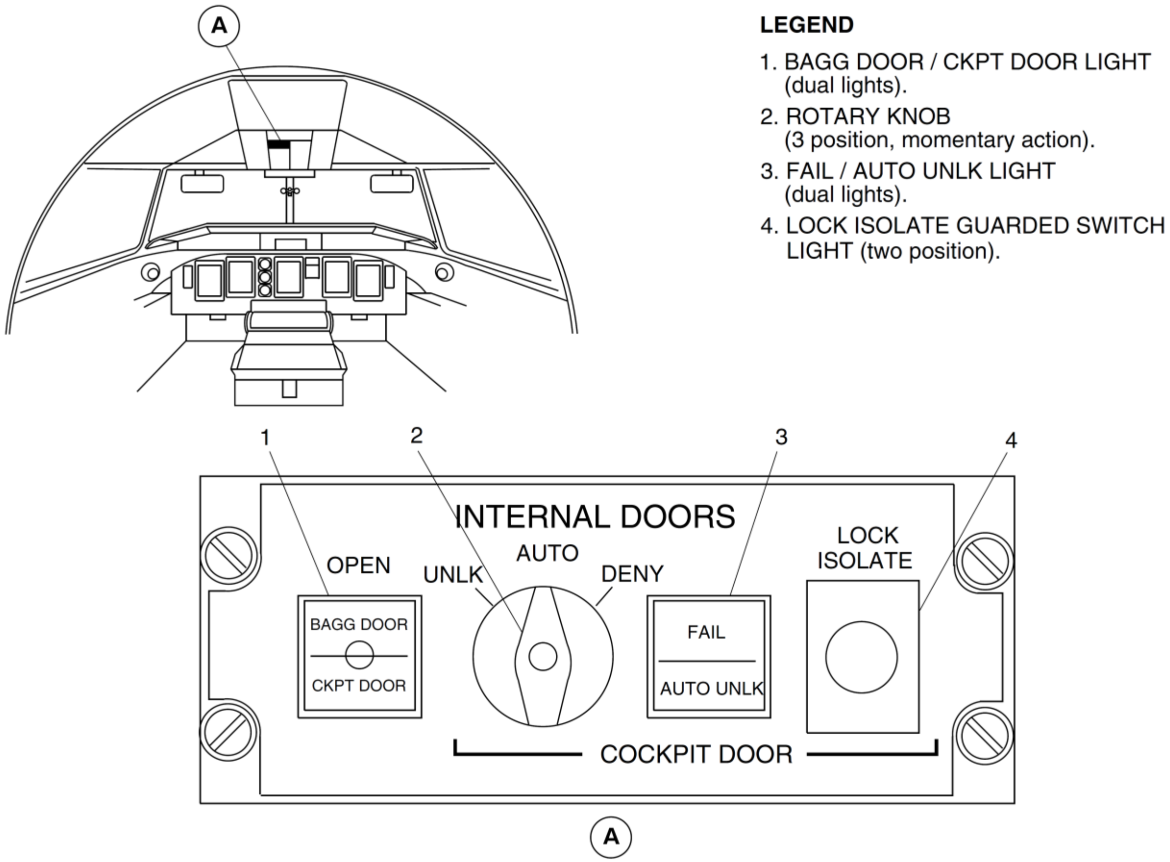


Figure 52-104 Internal Doors Control Panel – Detailed

The functions of these items are described in the following table:

ITEM	DESCRIPTION
1. BAGG DOOR/CKPT DOOR LIGHT (dual lights)	**BAGG DOOR** Amber light illuminates if internal forward baggage door is open. **CKPT DOOR** Amber light illuminates if flight compartment door is unlatched
2. ROTARY KNOB (3 position, momentary action)	**UNLK (momentary action)** Push knob in and rotate left Solenoid will de–activate and unlatch flight compartment door. **AUTO** Normal position Will enable the auto–unlock sequence to be activated by the cabin crew. **DENY (momentary action)** Rotate Knob Right Cancels auto–unlock sequence and terminates automatic unlatching of flight compartment door.
3. FAIL/AUTO UNLK LIGHT (dual lights)	**FAIL** Light illuminates if solenoid failure or LOCK ISOLATE switch is pushed Flight compartment door unlatches. **AUTO UNLK** Light illuminates when the cabin crew initiates the auto–unlock system from the cabin mounted switch.
4. LOCK ISOLATE GUARDED SWITCHLIGHT (two position)	**PUSH** • Light illuminates and power removed from lock solenoid • Flight compartment door unlatches • CKPT DOOR light illuminates • INTERNAL DOORS light on Caution and Warnings panel illuminates • FAIL light illuminates

Flight Attendant Emergency Access Push–button Annunciator

A Flight Attendant Emergency Access Push–button Annunciator is mounted directly below the optional DPAS (Digital Passenger Address System) on the wardrobe maintenance panel. A white light indicator illuminates on the upper half of the Emergency Access Push–button Annunciator when pressed.

A Master Caution tone sounds and the INTERNAL DOORS light on the Caution and Warning Panel illuminates. On the Internal Doors Control Panel, the amber AUTO UNLK advisory light illuminates indicating that the 40 second auto–unlock sequence has been initiated. The flight crew can either select UNLK or DENY. If there is no response by flight crew, the flight compartment door will automatically open after a timed delay of 40 seconds. An amber advisory light will illuminate on the lower half of the Flight Attendant Emergency Access Push–button Annunciator indicating that the flight compartment door is unlatched.

Door Warning System

Introduction

The door warning system monitors the position of the fuselage doors.

General Description

Refer Figure 52-105A, 105B

The door warning system supplies a visual indication, in the flight compartment, when a fuselage door open or unlocked warning condition occurs.

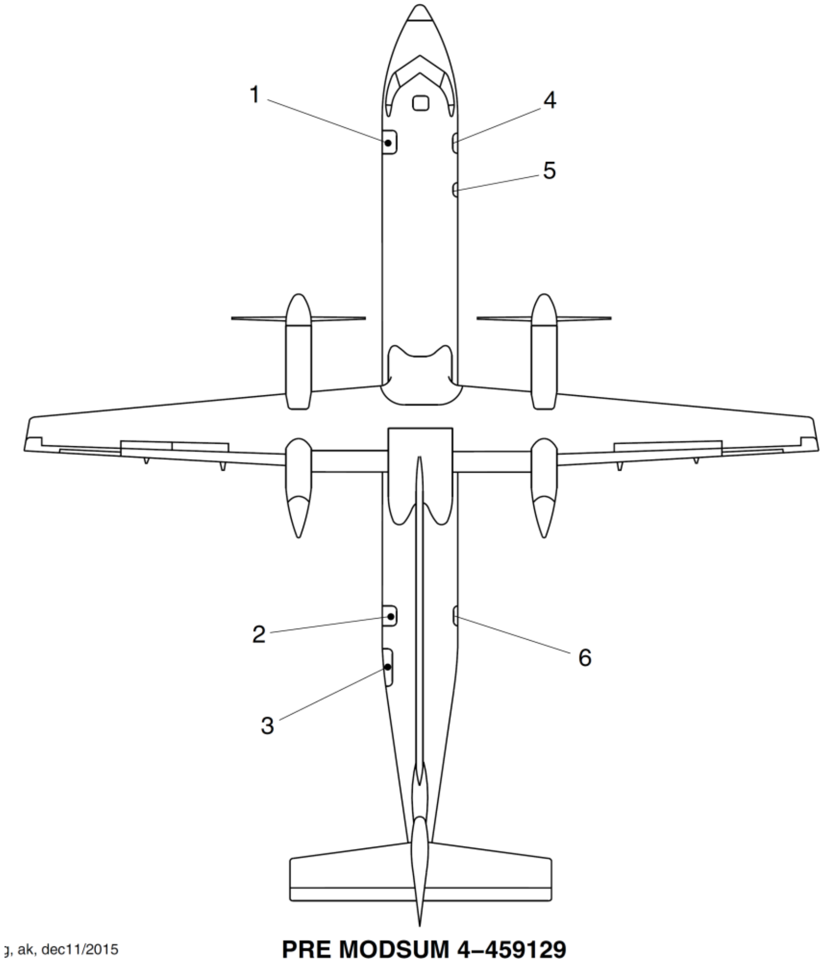
The door warning system sends door sensor outputs to the Proximity Sensor Electronic Unit (PSEU). The PSEU outputs data to the Input Output Processors (IOP 1 and IOP 2). This data is displayed in the flight compartment and is used by the Cabin Pressure Control System (CPCS).

The door warning system monitors the position of the fuselage doors that follow:

- Forward Passenger Door
- Aft Passenger Door
- Forward Baggage Door
- Aft Baggage Door
- Aft Service Door
- Type II Emergency Exit Door
- Type III Emergency Exit Door

The door warning system uses two sensors for each of the fuselage doors, and one sensor for the Type II emergency exit.

The system does not monitor the flight compartment emergency hatch.



LEGEND

1. Forward Passenger Door.
   "Closed" Sensor and "Locked" Sensor.
2. Aft Passenger Door.
   "Closed" Sensor and "Locked" Sensor.
3. Aft Baggage Door.
   "Closed" Sensor and "Locked" Sensor.
4. Forward Baggage Door.
   "Closed" Sensor and "Locked" Sensor.
5. Upper (Type II) Emergency Exit.
   "Locked" Sensor.
6. Aft Service Door.
   "Closed" Sensor and "Locked" Sensor.

Figure 52-105A Aircraft Doors and Sensors

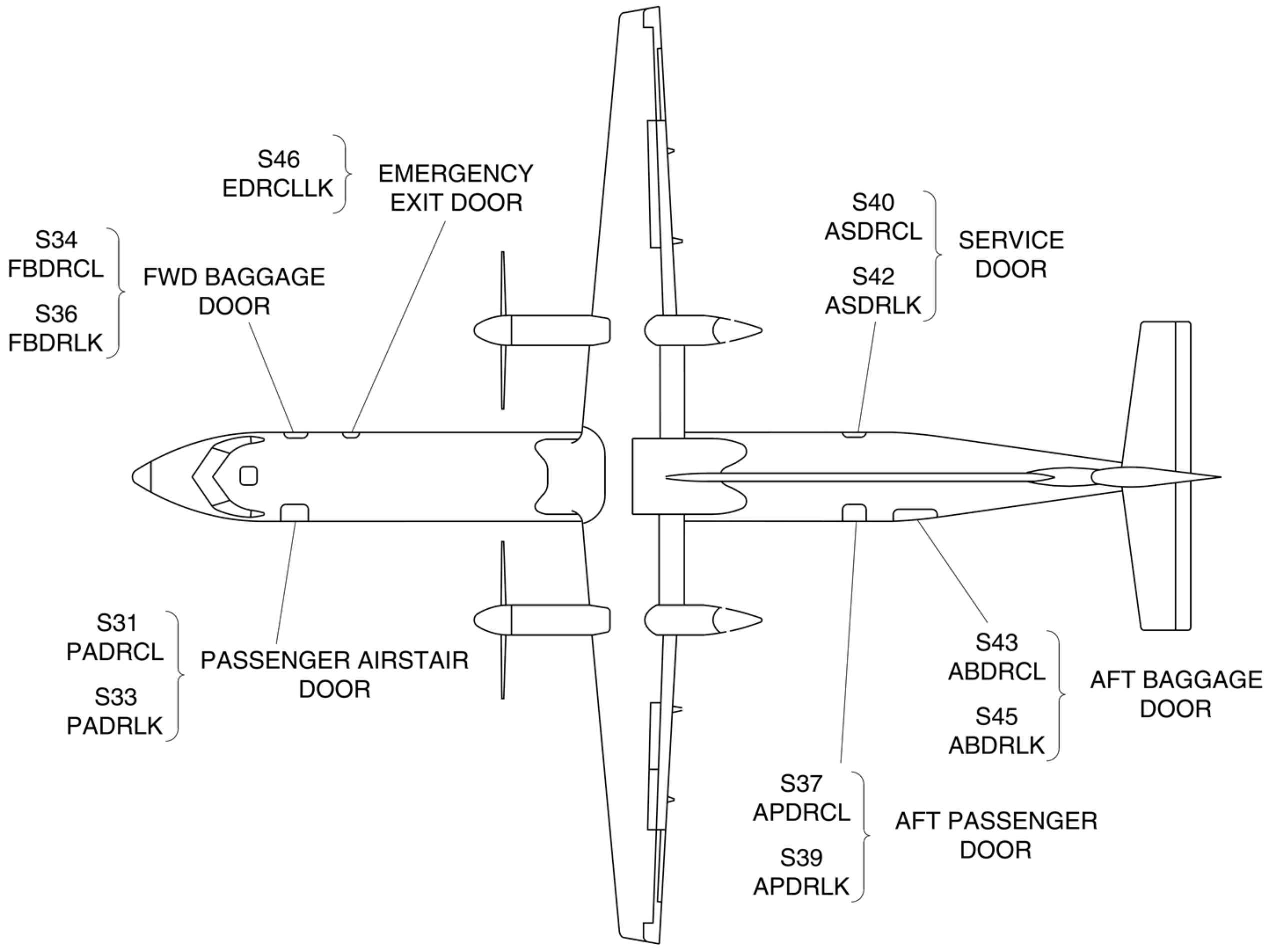


Figure 52-105B Aircraft Doors and Sensors – Detailed

Detailed Description

Refer Figure 52-106, Refer Figure 52-107 & Refer Figure 52-108

The fuselage door positions are monitored by 11 proximity sensors, the outputs are received by the Proximity Sensor Electronic Unit (PSEU). The PSEU door position outputs are sent to the Input Output Processors (IOP 1 & IOP 2) in the Integrated Flight Cabinets (IFC). The data is then sent to the systems that follow:

- The Electronic Flight Information System (EFIS)
- The Caution and Warning Control Panel (CWCP)
- The Cabin Pressure Control System (CPCS)

If a fuselage door is not fully closed and locked, the FUSELAGE DOORS warning light and the MASTER WARNING switchlight start flashing. Three tones also sound over the flight compartment speakers. When the MASTER WARNING switchlight is pushed, the light turns off. The FUSELAGE DOORS warning light remains on until the door is fully closed and locked.

The DOORS system page on the Multi−Function Display (MFD) shows all the fuselage doors and identifies them as:

- PAX for each of the passenger doors
- BAGGAGE for each of the baggage doors
- SERVICE for the aft service door
- EMERG EXIT for the type II emergency exit door
- EMERG EXIT for the type III emergency exit door.

If a fuselage door is not fully closed and locked, the DOORS system page identifies the door and shows it as a red filled rectangle. If the door is fully closed and locked, it is shown only as a green empty rectangle. If the data from the PSEU is invalid, the DOORS system page shows a global white INVALID DATA message.

The PSEU also transmits output signals to the Cabin Pressure Control System (CPCS). If the doors are not closed and locked, the Cabin Pressure Controller (CPC) prevents automatic pressurization of the aircraft.

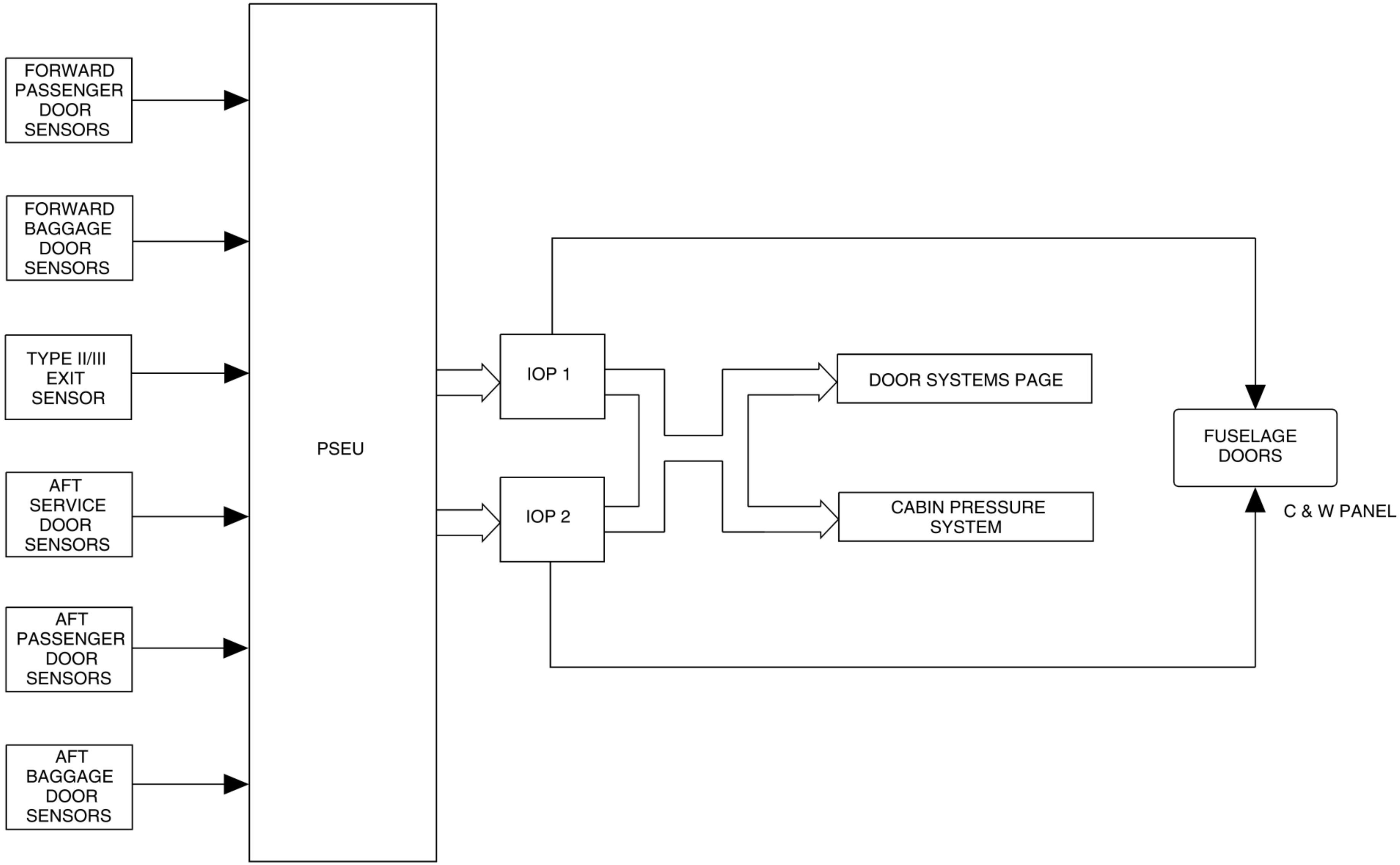


Figure 52-106 Door Proximity Sensor Inputs to PSEU – Detailed

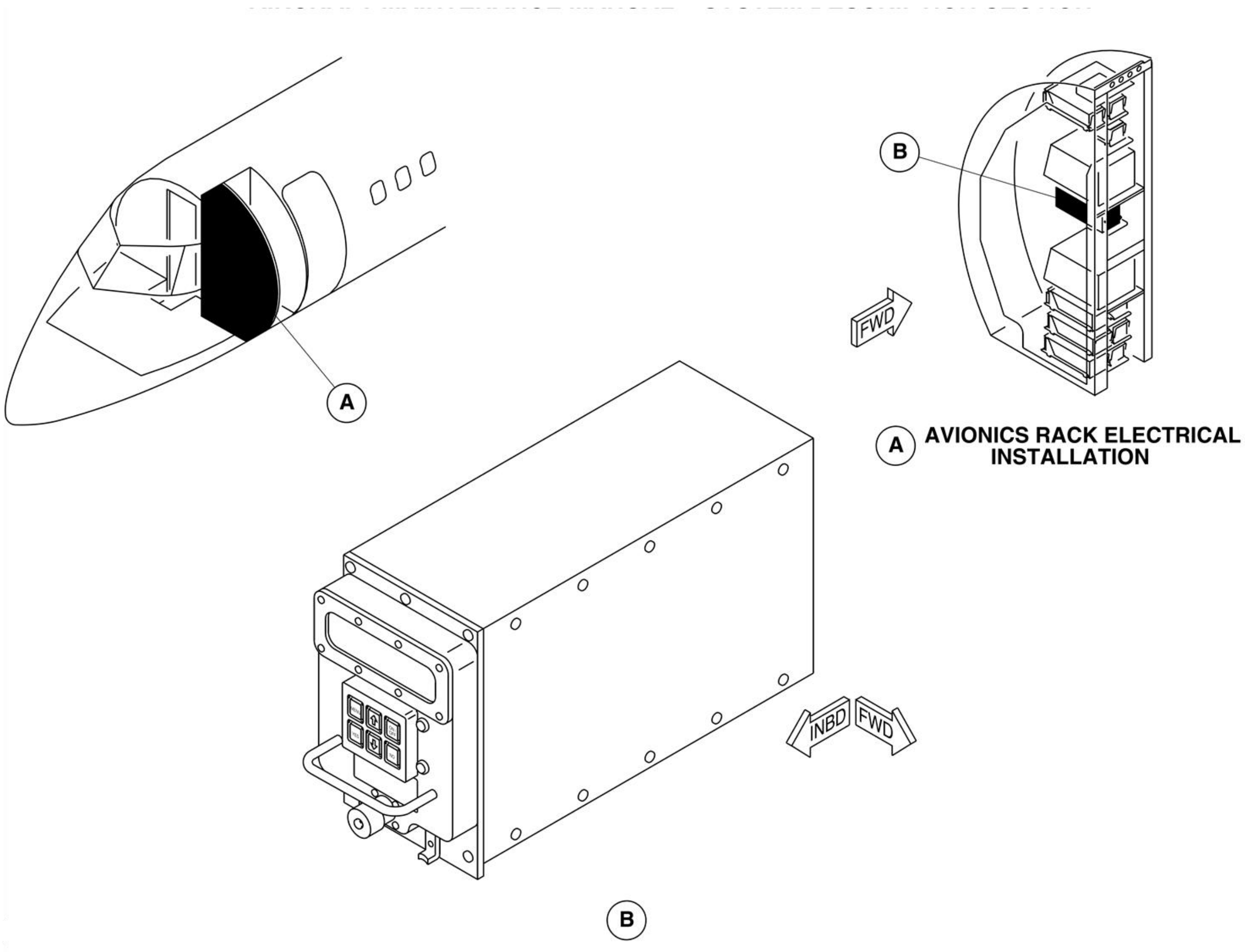


Figure 52-107 PSEU – Location

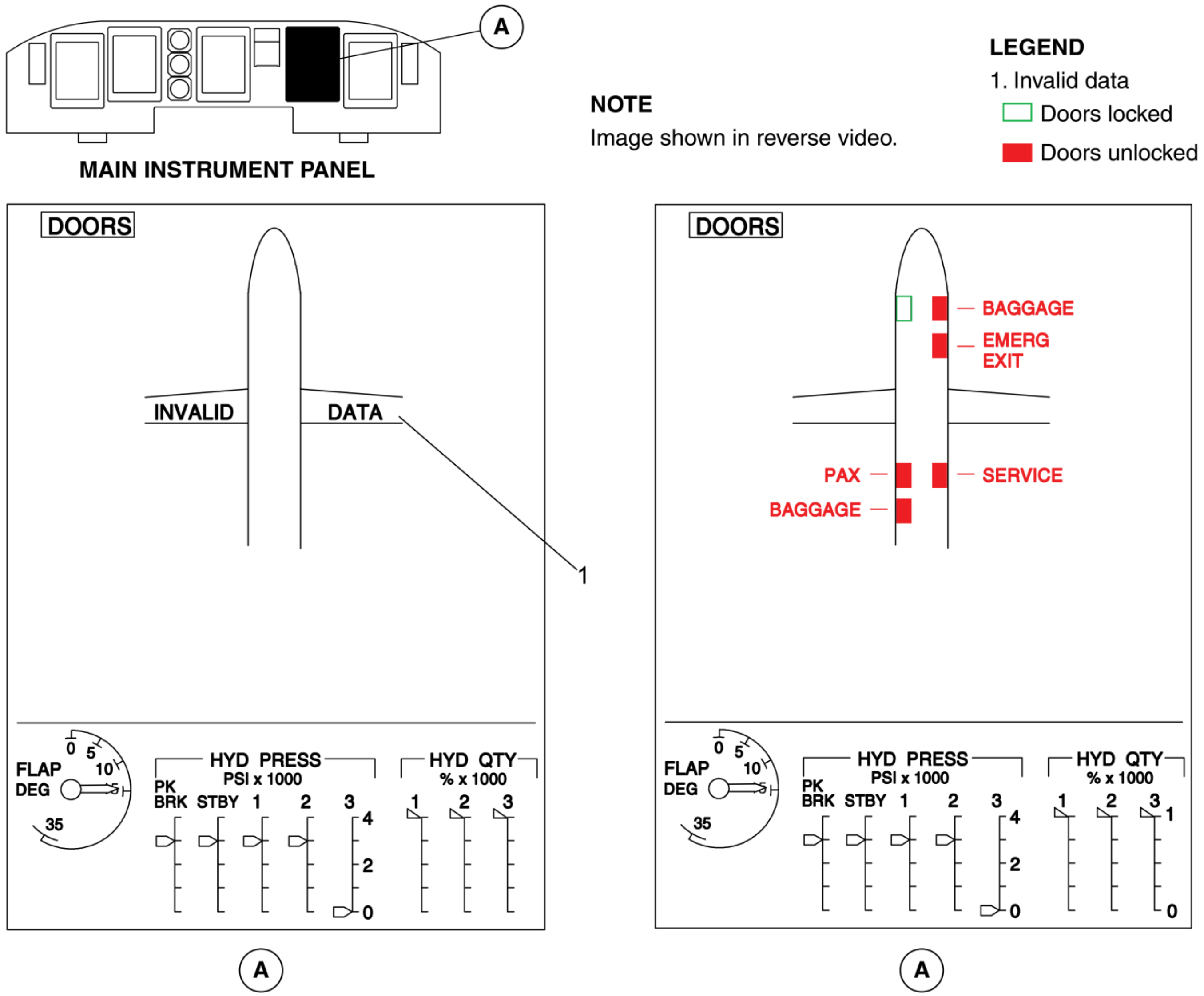


Figure 52-108 Doors System Page

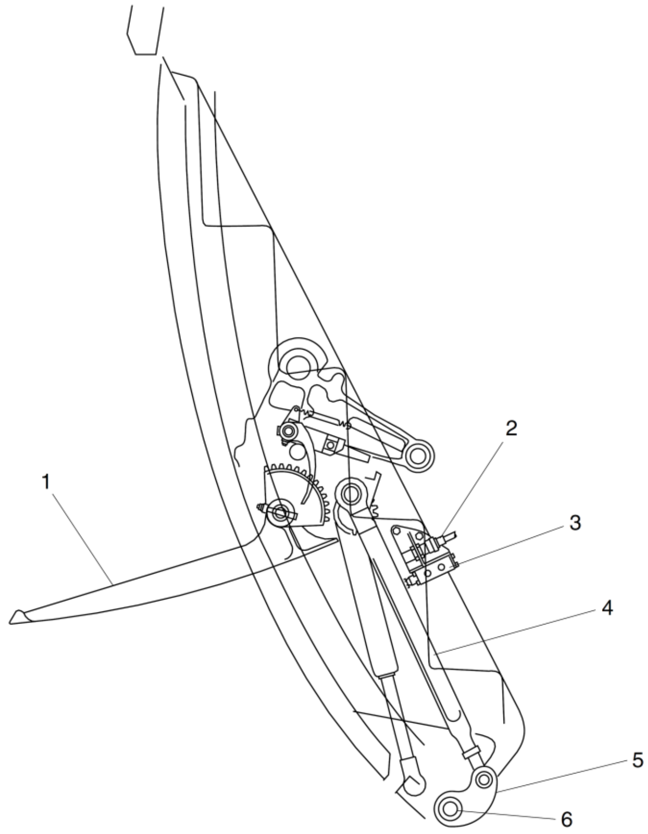
Forward Passenger Door Monitoring

Refer Figure 52-109 & Refer Figure 52-110

The position of the forward passenger door is monitored by two proximity sensors.

One sensor is located on the top aft side of the door, to verify that the door is closed by making sure that the door is fully down. The other is located adjacent to the door seal pressurizing valve and verifies that the door is locked, when the door closing mechanism is fully overcentred and in safety.

The forward passenger door can be visually inspected to make sure it is closed and locked, by viewing the position of the internal handle. If the door is not locked, the pressurization door seal will not inflate.



LEGEND
1. External handle.
2. Locked sensor.
3. Pressurizing valve.
4. Actuating rod.
5. Crank.
6. Door hinge shaft.

Figure 52-109 Forward Passenger Door – Locked Sensor

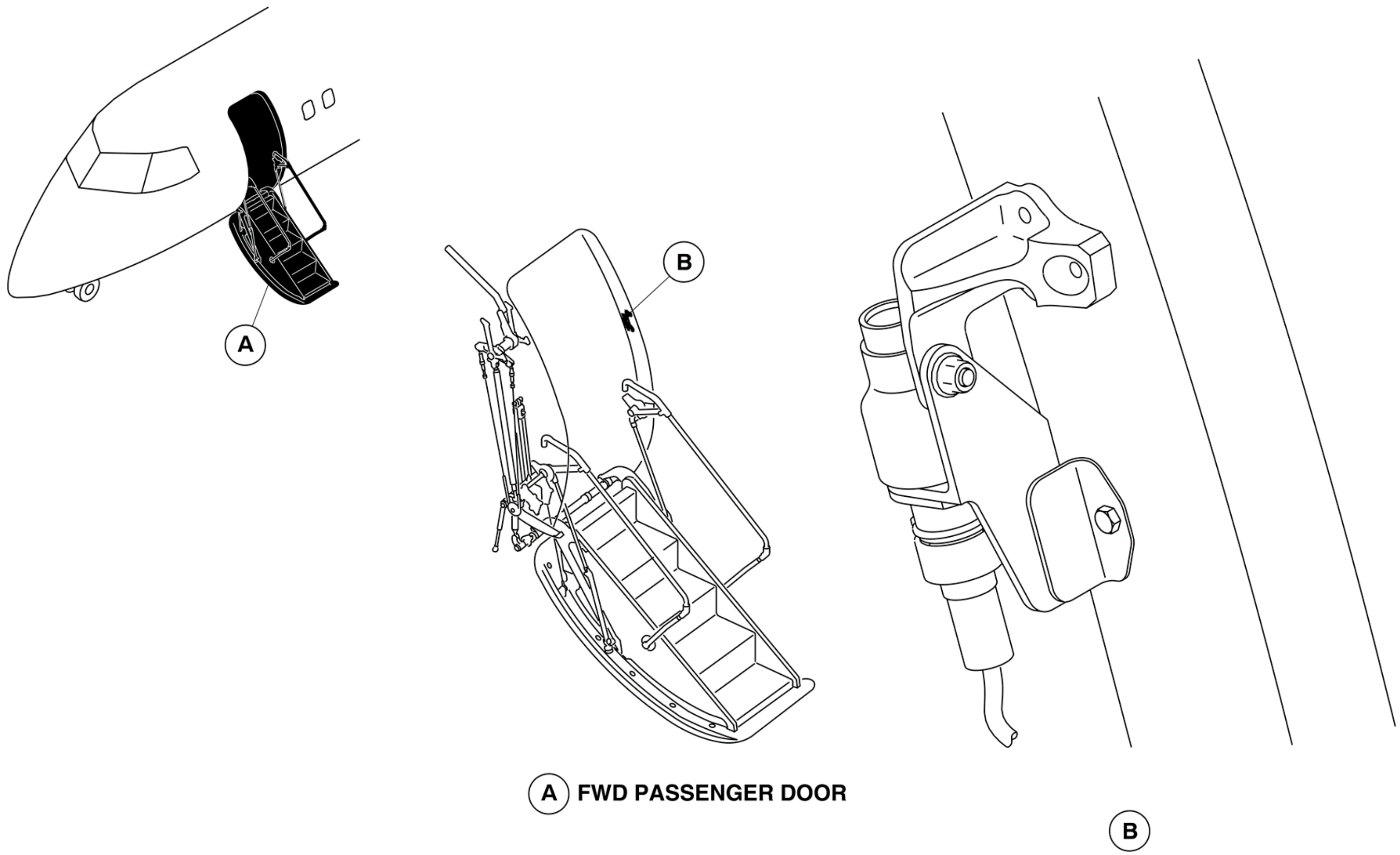


Figure 52-110 Forward Passenger Door – Closed Sensor

Aft Passenger Door Monitoring

Refer Figure 52-111

The position of the aft passenger door is monitored by two proximity sensors. One sensor is located at the aft upper roller guide and make sure that the door is closed. The other sensor is located at the door handle and make sure that the door is locked. The locking mechanism of the aft passenger door can be inspected to make sure that the door is closed and locked. The door can also be checked to be in the locked position, by seeing that the handle mechanism is pushed fully outward and the vent door is closed, or by checking the lift/latch shaft position indicator on the door.

The location of the two proximity sensors for the aft passenger door is similar to that of the aft service and forward baggage doors.

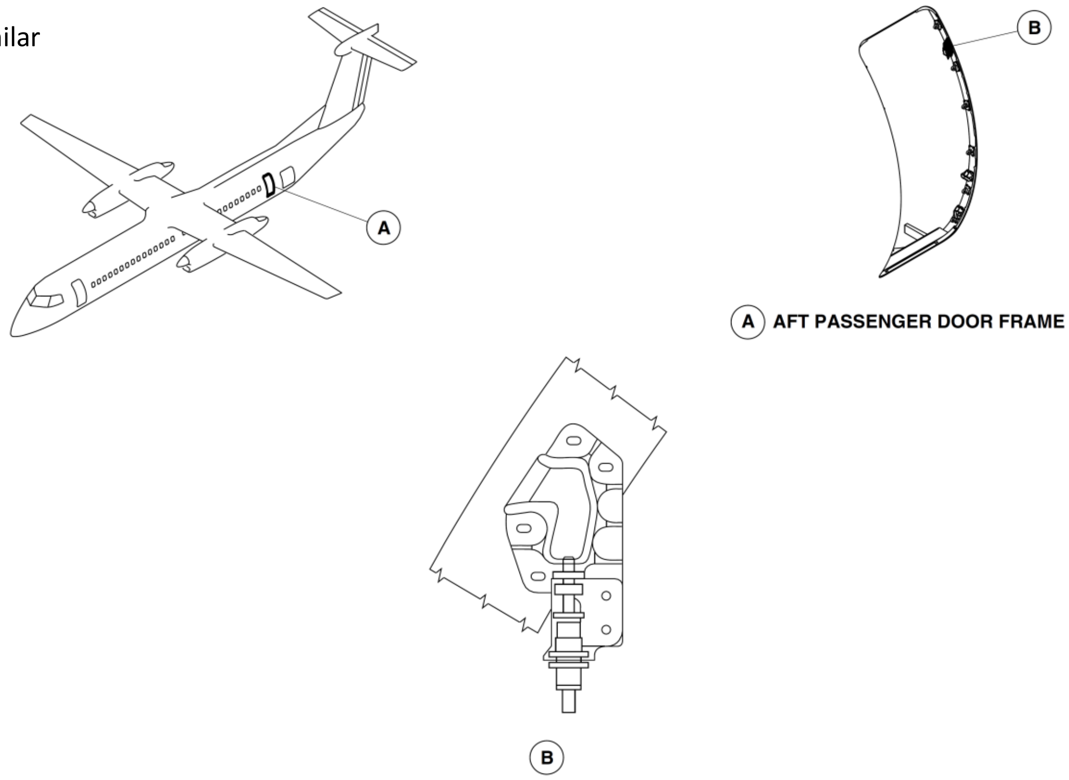


Figure 52-111 Aft Passenger Door – Closed Sensor

Forward Baggage Door Monitoring

The position of the forward baggage door is monitored by two proximity sensors. One sensor is located at the upper fixed roller and make sure that the door is closed. The other sensor is located at the handle and makes sure that the door is locked.

The lift/latch shaft position indicator on the door can be visually inspected to make sure that the door is closed and locked (Refer Aft Passenger Door).

Aft Baggage Door Monitoring

Refer Figure 52-112 & *Refer Figure 52-113*

The position of the aft baggage door is monitored by two proximity sensors. One sensor is located on the lower forward fuselage frame and verifies that the door is closed. The other sensor, is located at the door handle and verifies that the door is locked.

The door must be closed and locked, with the external handle in the recess for the door seal to inflate.

Aft Service Door Monitoring

The position of the aft service door is monitored by two proximity sensors. One sensor is located at the upper fixed roller position and monitors the door closed position. The other sensor located within the handle mechanism verifies that the door is locked.

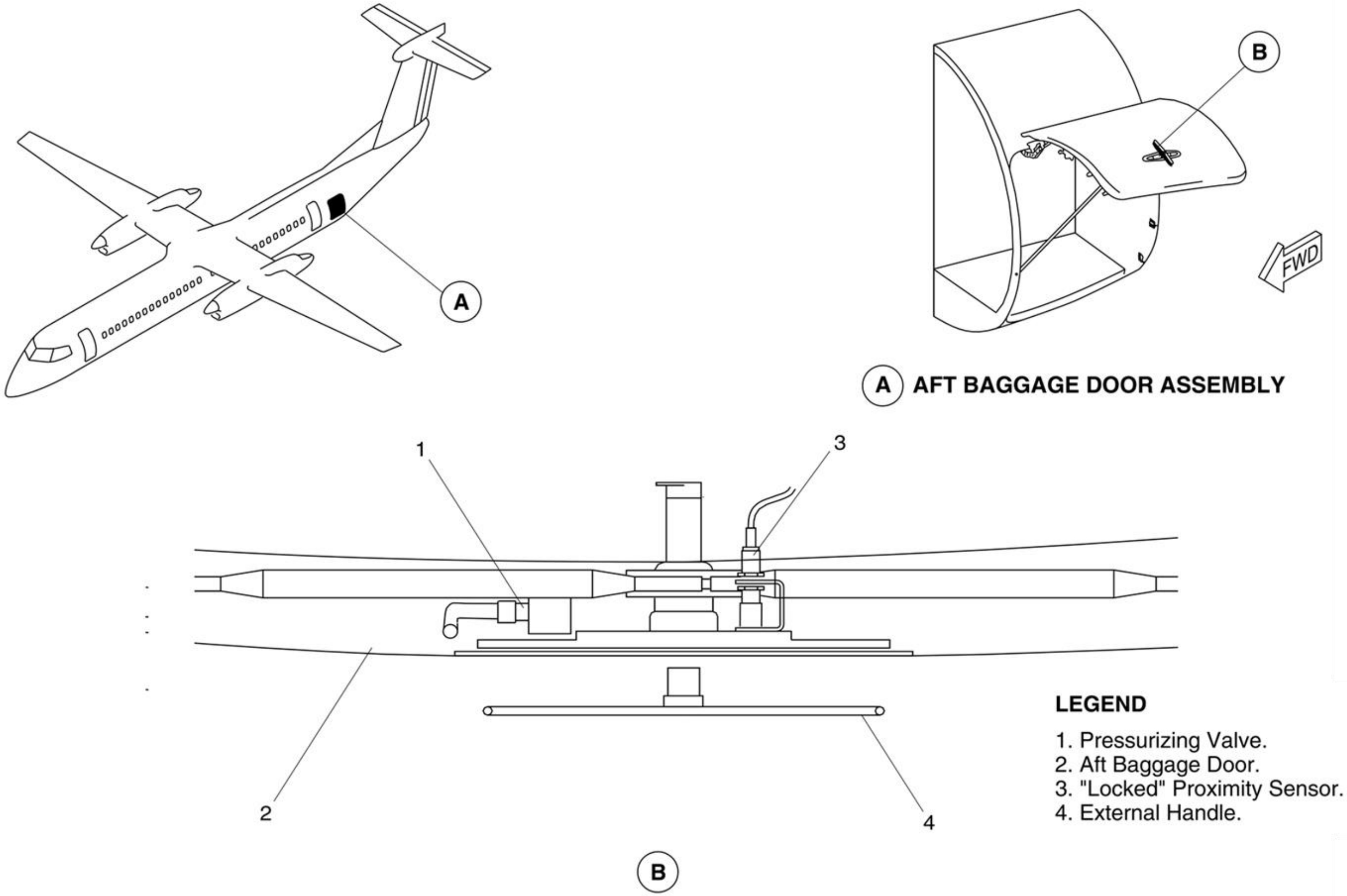


Figure 52-112 Aft Passenger Door – Locked Sensor

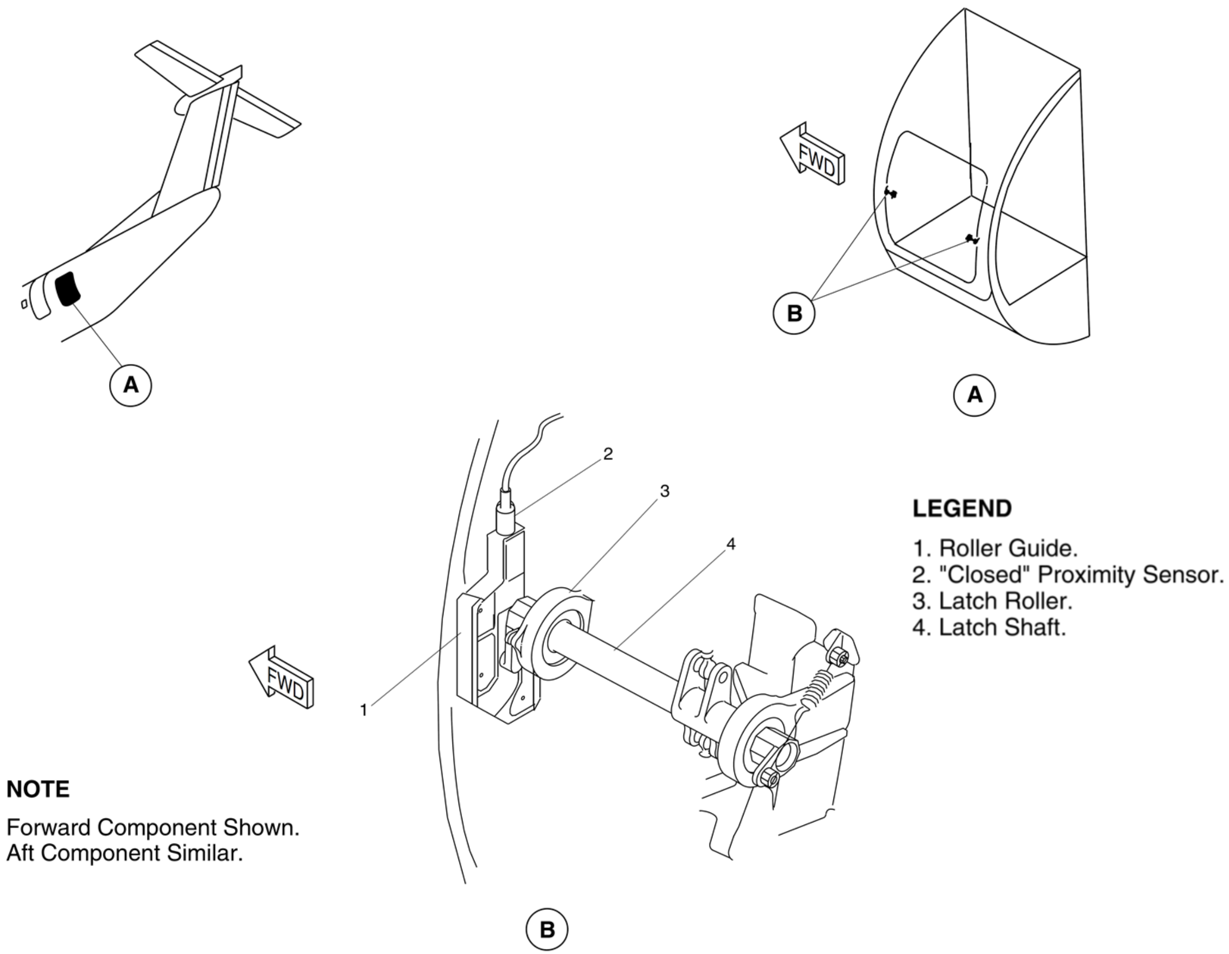


Figure 52-113 Aft Baggage Door – Closed Sensor

Type II Emergency Exit Door Monitoring

Refer Figure 52-114

The position of the emergency exit door is monitored by one proximity sensor, located on the upper part of the door frame.

The door may be visually confirmed closed and locked, by the position of the internal handle and the physical position of the door.

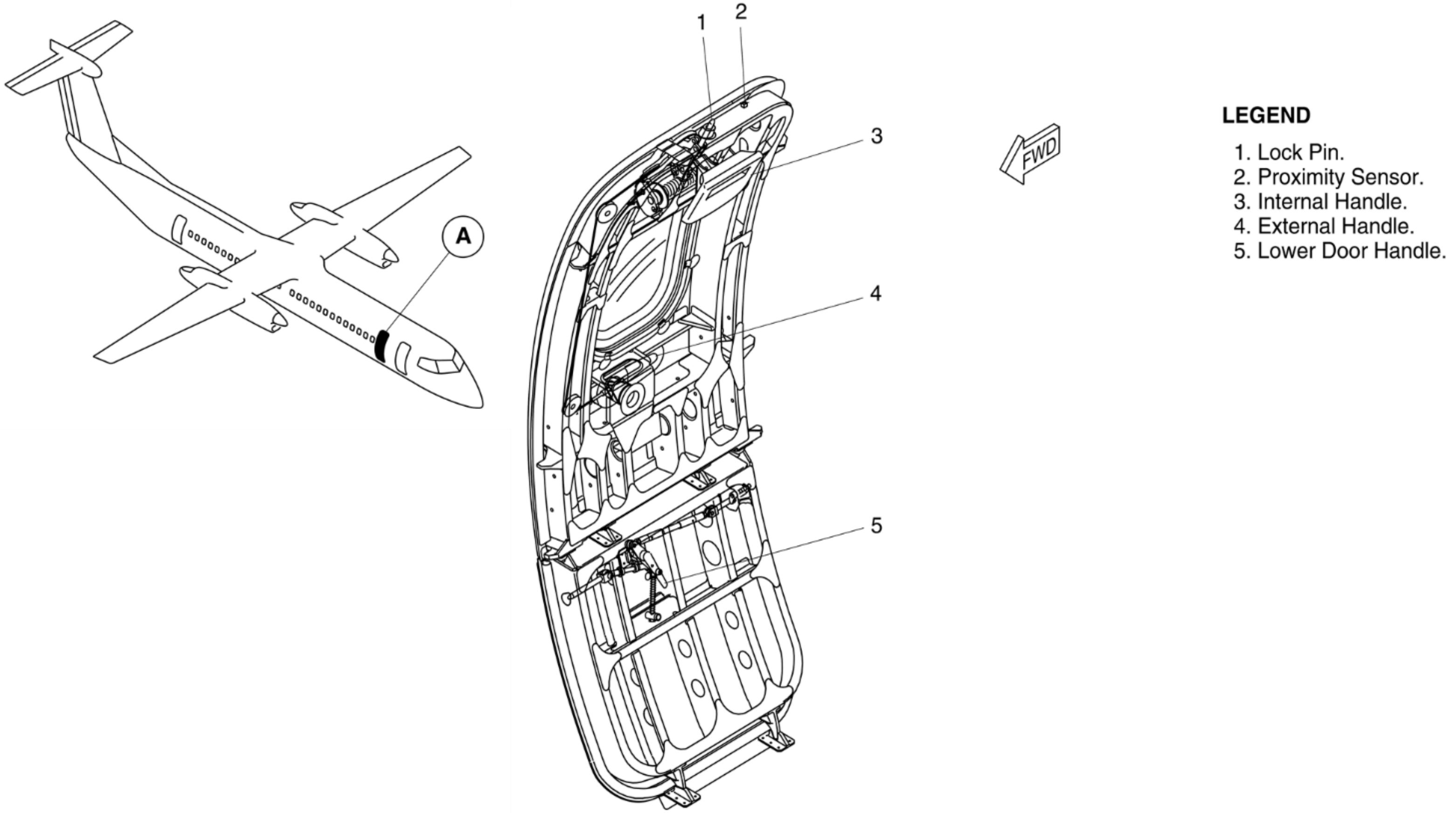


Figure 52-114 Type II Emergency Exit Door – Locked Sensor

Door Warning System Input to the Pressurization System

When the aircraft is on the ground, weight on wheels, and all the doors are closed and locked, the CPC starts normal ground sequences. When a door is open or unlocked, the CPC starts a door open sequence, which prevents automatic pressurization. Manual pressurization is still available.

The door position signals are stored in the CPCS, when the aircraft leaves the ground.

If the memorized signal indicates that the doors are closed and locked, the CPC starts the normal flight operational sequence regardless of the door position signal during flight. A door open signal during flight will not affect the CPCS.

If the memorized signal indicates that the doors are not closed and locked, the CPC will start the door open operational sequence regardless of door position signals during flight. Cabin pressurization will not take place when the CPCS is in the automatic mode. –

NOTE: Discrete inputs to the CPC are continuously monitored by the CPC when the AUTOMATIC mode is engaged.

On aircraft with SB84−32−66 or ModSum 4−126420 incorporated cabin pressurization inhibit to exclude translating doors which are equipped with vent panels.

Door Warning System Input to the Caution and Warning Control Panel. (CWCP)

The PSEU transmits an output signal to the Caution and Warning Control Panel (CWCP) the door position signal is transmitted to the CWCP through two independent paths.

The position signals received by the CWCP can be one of the two:

- Electrical ground, door closed and locked
- Open circuit, door open and/or unlocked

Door Warning System input to the Integrated Flight Cabinet

All door proximity sensor output signals are sent to the PSEU to-generate one output signal for each door, in the closed and locked-position. This signal is sent to the Input Output Processors (IOP 1-and IOP 2) in the Integrated Flight Cabinets (IFC). The data is-processed and displayed on the Doors System page.

The output signal is divided to supply each of the IOPs with an-independent signal.

When the FUSELAGE DOORS warning light is-on, the fuselage Doors System page shows any open or unlocked-door as a solid red rectangle.

A single failure will not prevent indication of the fuselage door status.

Doors Warning System Indication

When any door is open and unlocked the indications that follow will be shown:

- The Doors Systems page shows the open door as a red filled rectangle
- The FUSELAGE DOORS warning light flashes
- The MASTER WARNING light flashes

A closed and locked door is shown as green empty rectangle on the-Doors System page.

An open door is shown as a red filled rectangle-with a legend identifying the door in red T1 font.

The doors are identified by the legend that follows:

- “PAX” for the Passenger doors
- “BAGGAGE” for the Baggage doors
- “SERVICE” for the Aft Service door

- "EMERG EXIT" for the Type II emergency exit.

If data from both IOPs is invalid, the Doors System page will show a-global white "INVALID DATA" message.

When the FUSELAGE DOORS warning light on the Caution and-Warning Panel flashes, the MASTER WARNING light also flashes.-The MASTER WARNING light is pushed to acknowledge the warning-and the MASTER WARNING light goes off. The FUSELAGE-DOORS warning light then stays on continuously as long as the door-is open or unlocked.

If a subsequent door open/unlocked condition is sensed the-FUSELAGE DOORS and MASTER WARNING lights will start-flashing again.

Proximity Sensors

The proximity sensors are hermetically sealed two wire devices, used to sense the distance to targets installed on the fuselage doors.

The inductance at the sensor changes with the sensor's proximity to the target. As the target gets closer to the sensor face, the inductance increases and transmits a high signal. High and low signals, for target far or near conditions, are transmitted to the PSEU.

Proximity sensors can fail in one of two modes:

- An open circuit with no output
- A short circuit output

When any proximity sensor fails, a signal is sent to the PSEU, and the FUSELAGE DOORS warning light flashes.

The proximity sensors cannot be repaired if they fail.

Two proximity sensors are located near each door to indicate when the door is closed and when it is locked. One proximity sensor is installed on the door surround structure and senses when the forward roller guide is in the proper position for latching the door. When this occurs, the door is in the closed position. The other proximity sensor is installed in the door handle housing. It senses whether the handle is retracted (door locked) or extended (door unlocked).